STA301 Quiz-1 Lecture 9-15

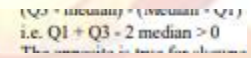
with merged Handouts

Junaid and moaz #Midterm mcqs

ORANGE MONKEY TEAM

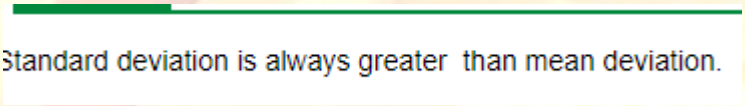
1. **In a case of positively skewed distribution**

   Q1+Q2-2median>0



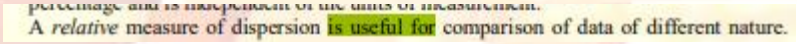
2. **The standard deviation is always _____ than mean deviation**

   Greater…..confirm



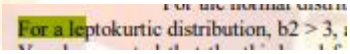
3. **Which of the following measure is useful for comparison of data of different nature**

   Coefficient of variation(Relative measure)…..confirm



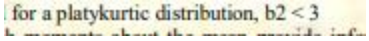
4. **For a leptokurtic distribution b2(moment ratio) will be**

   Greater than 3….confirm



5. **For a playkutric distribution b2(moment ratio) will be**

   Less than 3….confirm



6. **For the normal distribution b2(moment ratio) will be**

Equal to…..confimr

For the normal distribution, b2 = 3.

7. **The geometric mean of -2, -4, 3, 6, 0 will be**

Cannot be computed….confirm

a. -3
b. 0
c. 3
d. Cannot be Computed
e. None of these

ANSWER: Cannot be Computed

No explanation is available for this question!

8. **Larger the quartile deviation**

Greater is the scatter of values…..confirm

shareholders.)
From the above example, it is obvious that the larger the quartile deviation, the greater is the scatter of values within the series. The quartile deviation is superior to range as it is not affected by extremely large or small observations. It is

9. **The variance of a sample of 81 observations is equal to 64. The standard deviation of these observations will be**

8…..confirm

About 8,630,000 results (0.61 seconds)

8

Answer and Explanation:

Thus, the standard deviation is 8.

10. **Given the least squares regression line Y^=2+0.3x.**

The relationship between x and y is positive…..confirm

1. Given the least squares regression line $\hat{y} = 5 - 2x$:
   a. the relationship between x and y is positive.
   b. the relationship between x and y is negative.
   c. as x decreases, so does y.
   d. None of these choices.

MULTIPLE CHOICE

1. ANS: B PTS: 1 REF: SECTION 16.1-16.2

11. According to this empirical rule, approximately how much values will fall within (Mean – 3S.D, Mean + 3S.D)?

100%....confirm

- Approximately 100% (practically all) of the measurements will fall within 3 standard deviations of the mean i.e. within the interval ($\bar{X} - 3S$, $\bar{X} + 3S$).

12. According to this empirical rule, approximately how much values will fall within (Mean – 2S.D, Mean + 2S.D)?

95%.....confirm

- Approximately 95% of the measurements will fall within 2 standard deviations of the mean, i.e. within the interval ($\bar{X} - 2S$, $\bar{X} + 2S$).

13. According to this empirical rule, approximately how much values will fall within (Mean – S.D, Mean + S.D)?

68%.....confirm

- Approximately 68% of the measurements will fall within 1 standard deviation of the mean, i.e. within the interval ($\bar{X} - S$, $\bar{X} + S$)

14. In measure of relative dispersion, unit of measurement remains

Free of units…..confirm

About 6,710,000 results (0.51 seconds)

Since **relative measures of dispersion are free from the units in which the values have been expressed**, they can be compared even across different groups having different units of measurement. 02-Jan-2022

15. If 10% is added to each value of variable the geometric mean of new variable is added by

10%.....confirm



16. $^{5}C_{5}$ is equal to

1….confirm

17. Sometimes mean deviation can also be calculated around

Median

18. For a positively skewed distribution median is _ to/than the mid quartile range

Greater…..confirm

What happens to the median in a positively skewed distribution

In a positively skewed distribution, the mean is greater than the median as the data is more towards the lower side and the mean average of all the values. In contrast, **the median is the middle value of the data**. So, if the data is more bent

19. For a negatively skewed distribution median is _ to/than the mid quartile range

Less……..confirm

20. Relative dispersion is expressed in terms of

Ratio…..confirm

Relative measures of dispersion are calculated as **ratios or percentages**;

21. Large the quartile deviation

Greater is the scatter of values…..confirm

From the above example, it is obvious that the larger the quartile deviation, the greater is the scatter of values within the series. The quartile deviation is superior to range as it is not affected by extremely large or small observations. It is

22. In a left skewed distribution

X0-Q1 grater than Q3-Xm….confirm

Similarly, in left-skewed distributions, the distance from X0 to Q1 greatly exceeds the distance from Q3 to Xm.

23. In a right skewed distribution

Xm-Q3 grater than Q1-X0….confirm

On the other hand, for non-symmetrical distributions, the following would be true:
1. In right-skewed distributions the distance from Q_3 to X_m greatly exceeds the distance from X_0 to Q_1.

24. In right-skewed distribution

Median<mid-quartile range<midrange….confirm

2. in right-skewed distributions,
median < mid-quartile range < midrange:
THE POSITIVELY SKEWED CURVE

25. In left-skewed distribution

Midrange<mid-quartile range<median…..confirm

left-skewed distributions, midrange < mid-quartile range < median.

26. The distribution will be positively skewed if

Q1is closer to Q3…..confirm

AMOUNT by which each one deviates will give an indication of skewness. Where the distribution is positively skewed, Q1 will be closer to the median than Q3.In other words, the distance between Q3 and the median will be greater than

27. Relative measures of dispersion can be used for

Comparison of two data sets…….confirm

The relative measures of dispersion are used to **compare the distribution of two or more data sets**. This measure compares values without units.

28. Relative measure of dispersion corresponding to mean deviation is:

Coefficient of mean deviation……confirm

29. In a five number summary, which of the following is not used for data summarization?

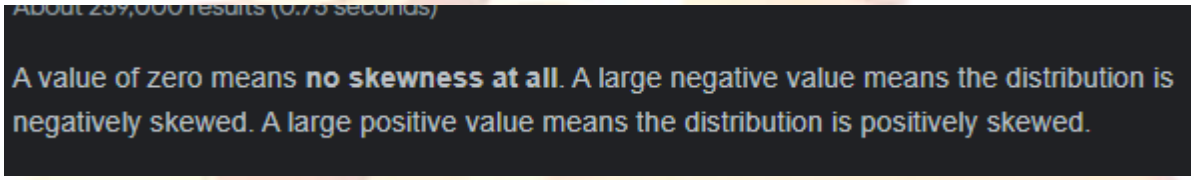
The 25 percentile

30. In a linear regression, best fitted line is obtained through

Method of least square

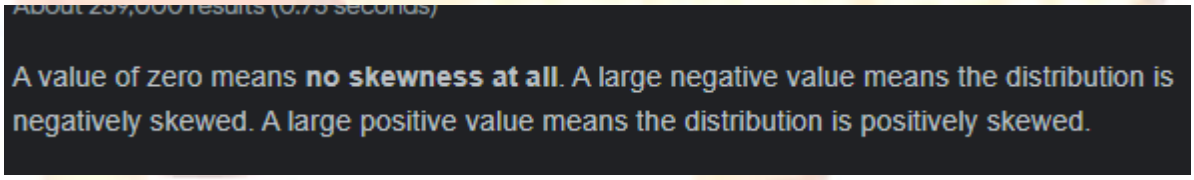
31. For a particular data set the pearson's coefficient of skewness is less then zero. What will be the shape of distribution?

negatively skewed…..confirm



32. For a particular data set the pearson's coefficient of skewness is greater then zero. What will be the shape of distribution?

Positively skewed…..confirm



33. In a box and whisker plot, box is divided into

Two parts….confirm



34. The geometric mean of -2, -4, 3, 6, 0

Cannot be computed

35. In a box and whisker plot if the median line is closer to the left of the box then distribution will be

right skewed…..confirm

Since the median line is closer to the *left* end of the box, hence the data are SKEWED to the RIGHT.

36. In a box and whisker plot if the median line is closer to the right of the box then distribution will be

Left skewed……confirm

37. Measure of dispersion is used to measure.

Interdependence between variables

38. For the co-efficient of quartile deviation is

Q3-Q1/Q3+Q1……confirm

COEFFICIENT OF QUARTILE DEVIATION

$$= \frac{Quartile\ Deviation}{Mid - Quartile\ Range}$$

$$= \frac{\frac{Q_3 - Q_1}{2}}{\frac{Q_3 + Q_1}{2}} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

39. Quartile deviation is superior to range as it is not affected by extremely

Both large and small values……confirm

s. The quartile deviation is superior to range as it is not affected by extremely large or small observations.

40. Which of the rule gives no information about the probability of observing a value within one standard deviation of the mean?

Chebychev's rule…..confirm

concentrated near the mean than for a set of data with a large standard deviation.

A limitation of the Chebychev's theorem is that it gives no information at all about the probability of observing a val within one standard deviation of the mean, since $1 - 1/k2 = 0$ when $k = 1$. Also, it should be noted that the Chebychev theorem provides weak information for our variable of interest. For many random variables, the probability

41. In a box whisker plot right end of the box is referred to as

Third quartile….confirm

2. A BOX is drawn in the space above the horizontal axis in such a way that the left end of the box aligns with the first quartile Q1 and the right end of the box is aligned with the third quartile Q3.

42. In a box whisker plot left end of the box is referred to as

First quartile…..confirm

2. A BOX is drawn in the space above the horizontal axis in such a way that the left end of the box aligns with the first quartile Q1 and the right end of the box is aligned with the third quartile Q3.

43. If we want to calculate average speed of a car then which of the following average will be used?

Harmonic mena….confirm

which is not the same as 42 miles per hour.
Let us now try the harmonic mean to find the average speed of the car.

$$H.M. = \frac{n}{\sum\left(\frac{1}{x}\right)}$$

44. If an observation in the data set is zero then its geometric mean will be

Zero…..confirm

If any value in the data set is zero, the geometric mean is **zero**.

45. Bowleys coefficient of skewness is expressed in terms of

Independent of unit….not sure

46. Which average gives the more weightage to the smaller values?

Harmonic mean…..confirm

The harmonic mean

The harmonic mean necessarily includes all the entries in a series, and it allows a more significant weighting to be given to smaller values. Harmonic mean can be calculated for a series that includes negative values, although it is impossible to compute when the series includes a value of zero.

47. In a box and whisker, if the median line is closer to the right of the box then distribution will be

Negative skewed….confirm

48. In a box and whisker, if the median line is closer to the left of the box then distribution will be

Positively skewed…..confirm

Another way of saying it is that if the median is to the left of the center of the box, then the distribution of the data values is positively skewed.

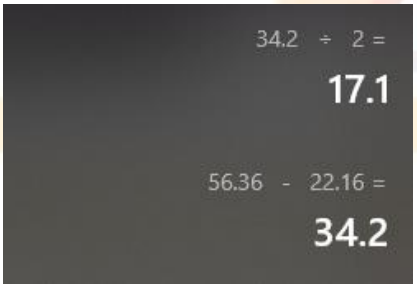
49. The measure used to calculate the variation present among the observations relative to their average is called

Coefficient of kurtosis……confirm

Explanation: The measures used to calculate the variation present among the observations relative to their average is called: **Coefficient of kurtosis**. 05-Apr-2021

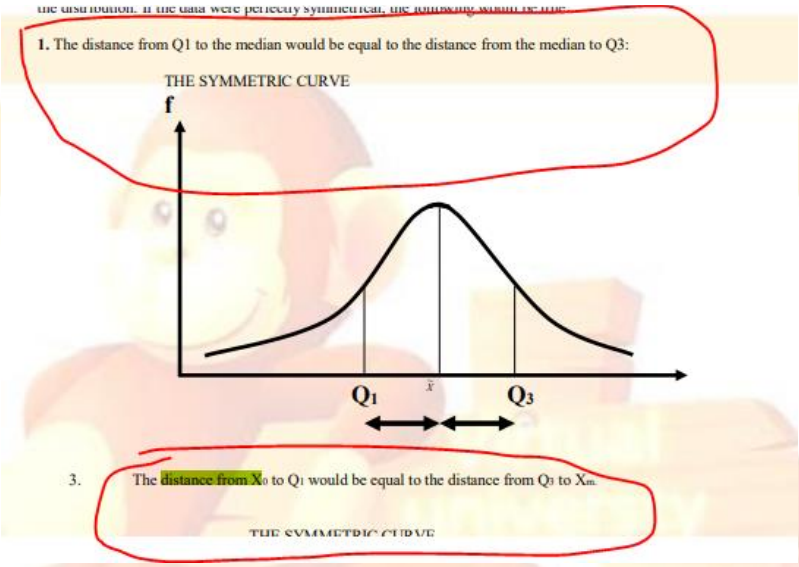
50. If the first and third quartile are 22.16 and 56.36 respectively, then the quartile deviation is

17.1….confirm



51. In a five number summary for a symmetrical distribution

All option …..confirm



52. Pearson's coefficient of skewness cannot be used when standard deviation is

0…..not sure

53. Deviations taken from mean can be

None of the above……confirm

- The sum of the deviations from the mean is zero.

54. Co-efficient of standard deviation is

None…..confirm

distribution. The standard deviation of an exponential distribution is equal to its mean, so its coefficient of variation is equal to 1. Distributions with

55. Quartile deviation is also defined as

Semi interquartile range…..confrim

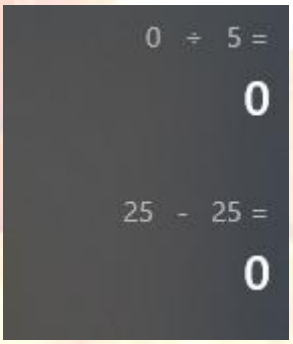
the upper quartile and Q_1 indicates the lower quartile. Quartile Deviation is also known as the **Semi Interquartile range**.

56. If data is collected in rupees like income, then absolute measure of dispersion will be in the units of

rupees

57. The mean of a distribution is 25, the mode is 25 and the standard deviation is 5, then the coefficient of skewness will be

Equal to zero…..cofnirm



58. Which of the following techniques is used to predict the value of one variable on the basis of other variables?

Regression analysis……confirm

Which of the following techniques is used to predict the value of one variable on the basis of other variables? **Regression analysis**.

59. Given the N values in a series, the geometric mean is.

The Nth root of the product og N positive values…..confirm

The Geometric Mean (G.M) of a series containing n observations is the nth root of the product of the values.

60. Quartile deviation is based on

No all values….confirm

61. Which of the rule is applied to the frequency distribution that is mound-shaped and symmetric.?

Emprical rule…..confirm

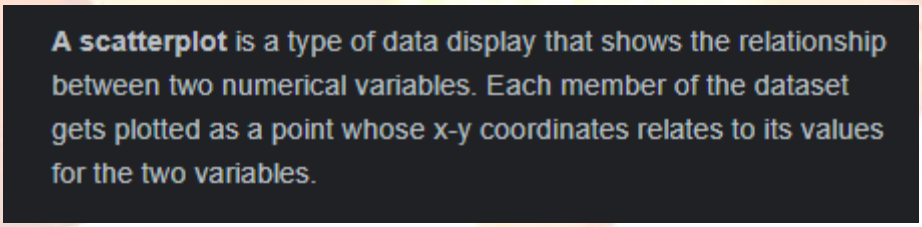


62. For a symmetrical data set mean value is 150 and standard deviation 25.95% values will lie between.

125,175….not sure

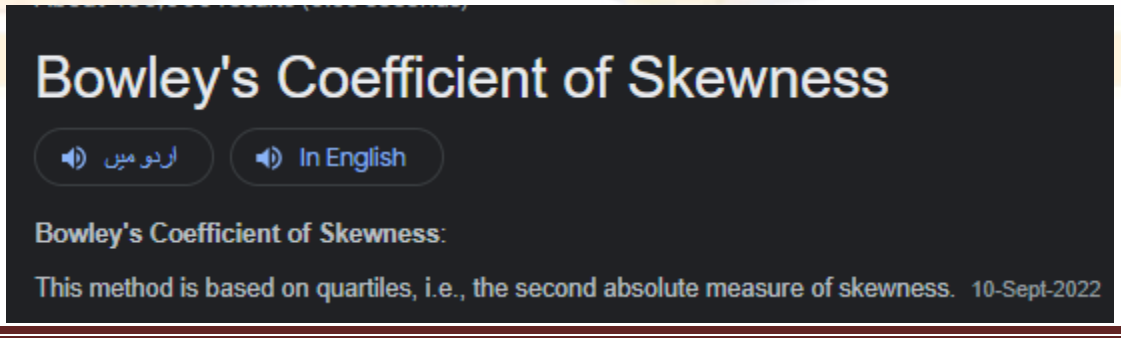
63. A graph of plotted points which show the relationship between two sets of data is known as

Scatter diagram……confirm



64. Which of the following method of skewness based on quartile?

Bowleys….confirm



65. Formula for quartile deviation is:

Q.D= q3-q1/2……confirm

QUARTILE DEVIATION

The quartile deviation is defined as half of the difference between the third and first quartiles i.e.

$$Q.D. = \frac{Q_3 - Q_1}{2}$$

66. When a researcher want to compare intensity of symptoms when different doses are administered. In this case different doses will be treated as

Dependent variable……confirm

The **dependent variable** is the response that is measured. For example: In a study of how different doses of a drug affect the severity of symptoms, a researcher could compare the frequency and intensity of symptoms when different doses are administered. 16-Nov-2022

67. IF the GM of a set of two observation is 10 and its HM is 8, then the AM of the set of observations is

12.5…….confirm

If the GM of a set of two observations is 10 and its HM is 8, then the AM of the set of observations is

a. 100

b. 12.5

c. 64

d. 7.5

68. For which of the following measures, we take deviations arount mean in square form

Standard deviation

69. If all the points in a scatter diagram lie on the least squares regression line, then the coefficient of coorelation.

1……confirm

Answer and Explanation:

The answer is **b.** 1.

70. In a left skewed distribution.

Xm-Q3 less than Q1-x0

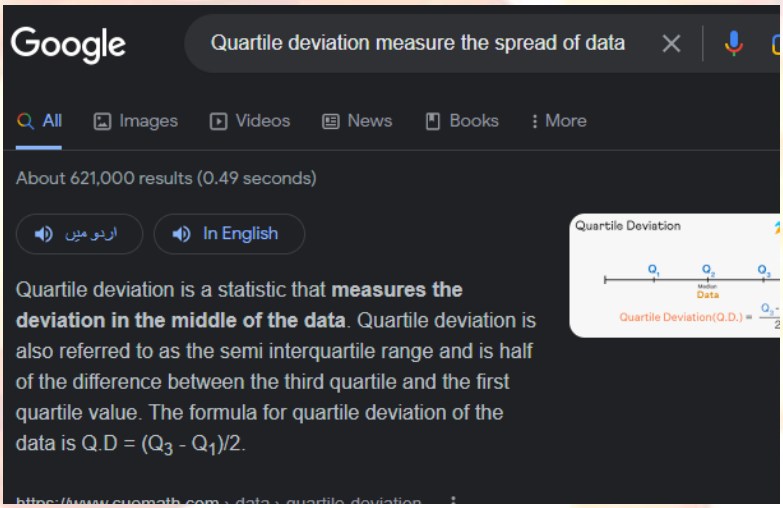
71. Second moment about mean will be equal to :

Variance…..confirm

What about the dispersion and the centre of our data-set? Do you not remember that the second moment about the mean is exactly the same thing as the variance, the positive square root of which is the standard deviation --- the most

72. Quartile deviation measures the spread of data around.

Median…..confirm



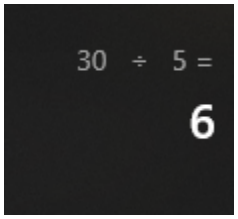
73. Five number summary gives good idea about the _____ of the distribution.

Shape….confirm

A five-number summary consists of X0,Q1, Median, Q3, and Xm ; It provides us quite a good idea about the shape of the distribution. If the data were perfectly symmetrical, the following would be true:

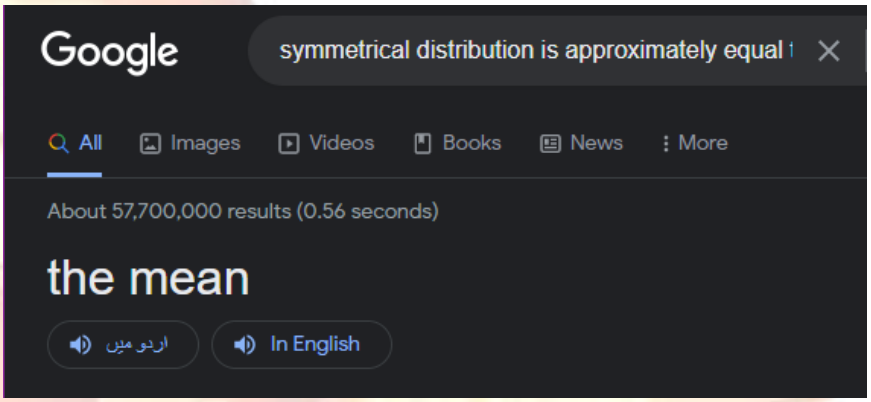
74. Find the mean of the values 2,4,7,8,9

6…..confirm



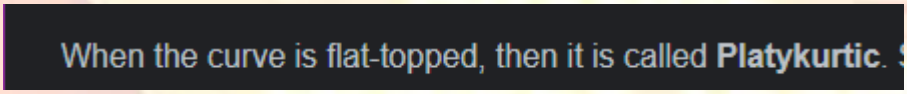
75. Range i.e (maximum value – minimum value) for a symmetrical distribution is approximately equal to

σ



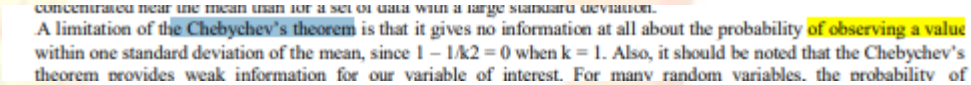
76. When the curve is Flat topped it is called.

Playkurtic….confrim



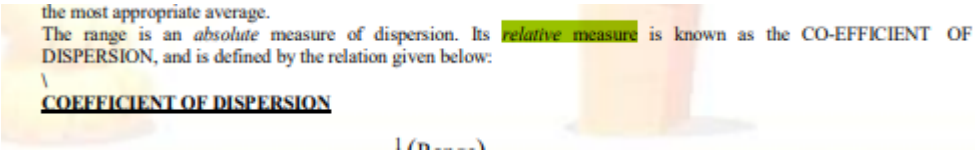
77. Which of the rule gives no information about the probability of observing a value within one standard deviation of the mean?

Chebychev's rule…..confirm


concentrated near the mean than for a set of data with a large standard deviation.
A limitation of the Chebychev's theorem is that it gives no information at all about the probability of observing a value within one standard deviation of the mean, since $1 - 1/k2 = 0$ when $k = 1$. Also, it should be noted that the Chebychev's theorem provides weak information for our variable of interest. For many random variables, the probability of

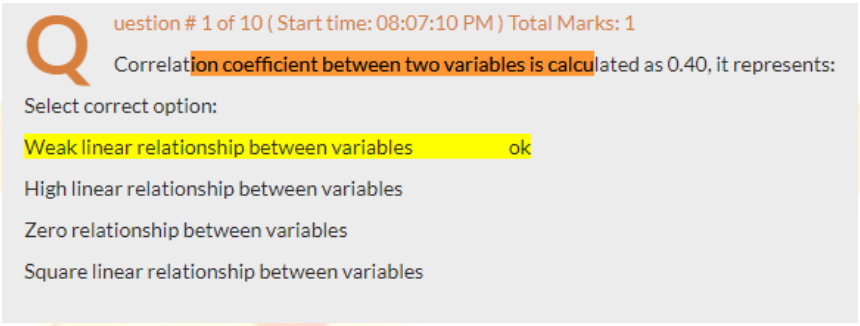
78. Relative measure of dispersion of range is :

Co-efficient of dispersion……confirm


the most appropriate average.
The range is an *absolute* measure of dispersion. Its *relative* measure is known as the CO-EFFICIENT OF DISPERSION, and is defined by the relation given below:
\
COEFFICIENT OF DISPERSION

79. Correlation coefficient between two variables is calculated as 0.40. it represents.

Weak linear relationship between variables……confirm



80. We use the general rule of multiplication to combine

Mutually exclusive events

81. If we roll three fair dice then the total number of outcomes are:

6*6*6 = 216…confirm

82. If an experiment produces different results even. If it is required a large number of times under similar conditions is known as

Random Experiment

83. P(|X|<1.5)=

P(-1.5 < X < 15)……confirm

84. A discrete probability function f(x) is always:

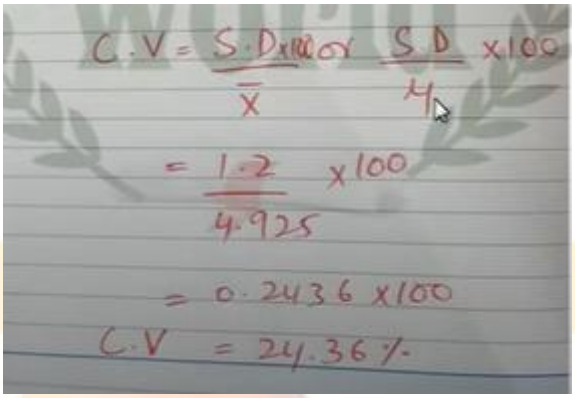
Non-negative…….confirm

85. Which of the following statements is an example of a continuous random variable

All options…confirm

86. If u = 4.925 and S.D= 1.2, then C.V(x) will be

24.36%.....confirm

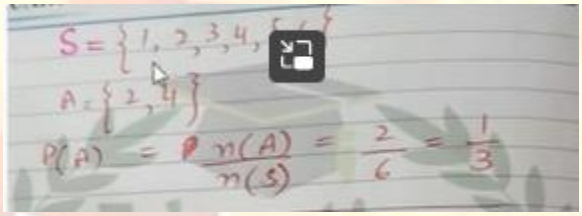


87. Two discrete r.b.s X and Y are said to be statistically independent if and only if for all possible pairs of values (Xi, Xj) the joint probability function f(x,y) can be expressed as the _____ of the two marginal probability function

Product……confirm

88. A fair die is rolled. Probablity of getting even force given that force is less than 5 is given by:

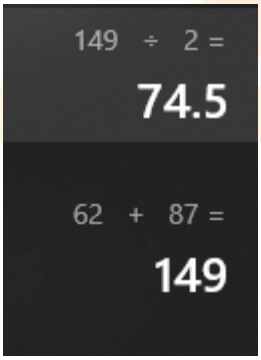
½



89.

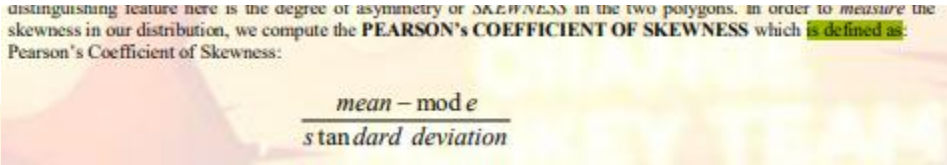
90. If Q1=62 and Q3=87, then the mid-quartile range will be

74.5…..confirm



91. Pearson coefficient of skewness is defined as

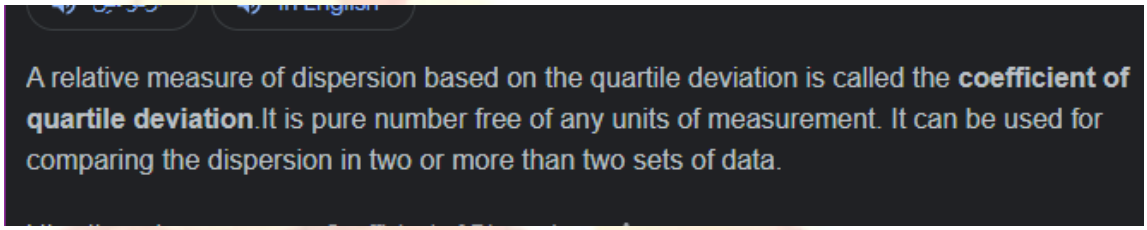
(Mean-Mode)/SD……confirm


distinguishing feature here is the degree of asymmetry or SKEWNESS in the two polygons. In order to *measure* the skewness in our distribution, we compute the **PEARSON's COEFFICIENT OF SKEWNESS** which is defined as:
Pearson's Coefficient of Skewness:

$$\frac{mean - \text{mod}\, e}{s\,\text{tan}\,dard\ \ deviation}$$

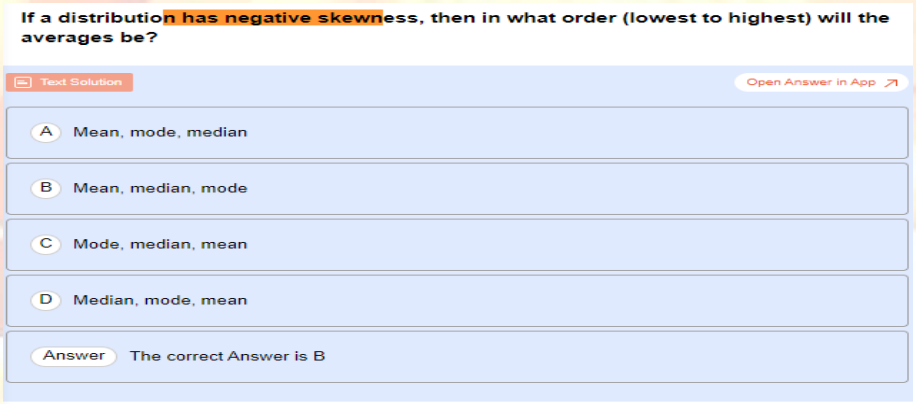
92. Relative measure of dispersion for quartile deviation is.

Co-efficient of Quartile deviation…..confirm


A relative measure of dispersion based on the quartile deviation is called the **coefficient of quartile deviation**.It is pure number free of any units of measurement. It can be used for comparing the dispersion in two or more than two sets of data.

93. If a distribution has negative skewness, in what order (lowest to highedt)will the average be

Mean, median, mode….confirm


If a distribution has negative skewness, then in what order (lowest to highest) will the averages be?

Text Solution | Open Answer in App

A Mean, mode, median

B Mean, median, mode

C Mode, median, mean

D Median, mode, mean

Answer The correct Answer is B

94. Dividing mean deviation by mean we get a pure number known as

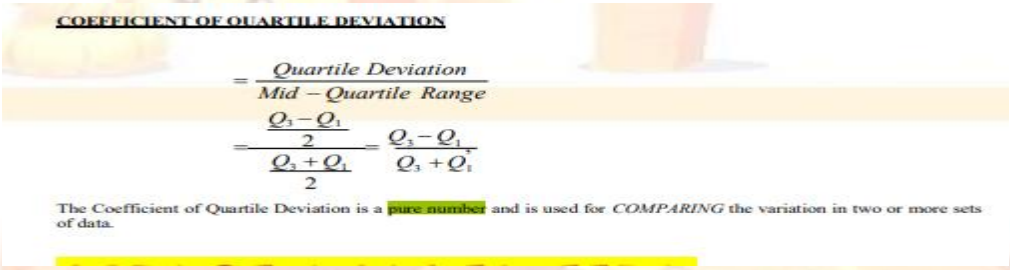
Co-efficient of mean deviation…..confirm

20. **Dividing mean deviation by mean we get a pure number known as**

1. Co-efficient of mean deviation
2. Co-efficient
3. Co-efficient of quartile deviation
4. None of the given

95. Dividing Quartile deviation by mean we get a pure number known as

Co-efficient of quartile deviation……confirm



COEFFICIENT OF QUARTILE DEVIATION

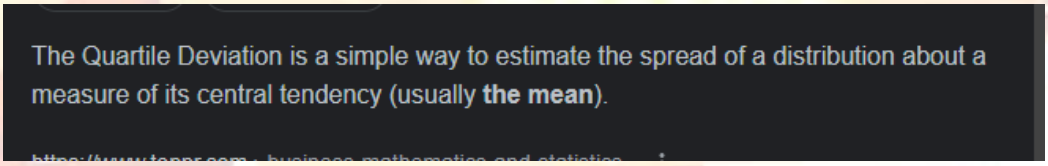
$$= \frac{Quartile\ Deviation}{Mid - Quartile\ Range}$$

$$= \frac{\frac{Q_3 - Q_1}{2}}{\frac{Q_3 + Q_1}{2}} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

The Coefficient of Quartile Deviation is a pure number and is used for *COMPARING* the variation in two or more sets of data.

96. Quartile deviation is used as a measure of disersion when we use _____ as a measure of central tendency

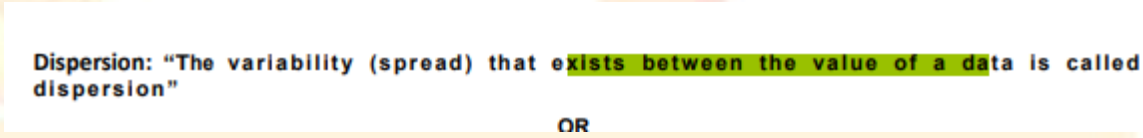
Mean…..confirm



The Quartile Deviation is a simple way to estimate the spread of a distribution about a measure of its central tendency (usually **the mean**).

97. Dispersion means the _____ that exists in a data set

Variability……confirm



Dispersion: "The variability (spread) that exists between the value of a data is called dispersion"

OR

98. Co-efficient of Quartile Deviation is a _____.

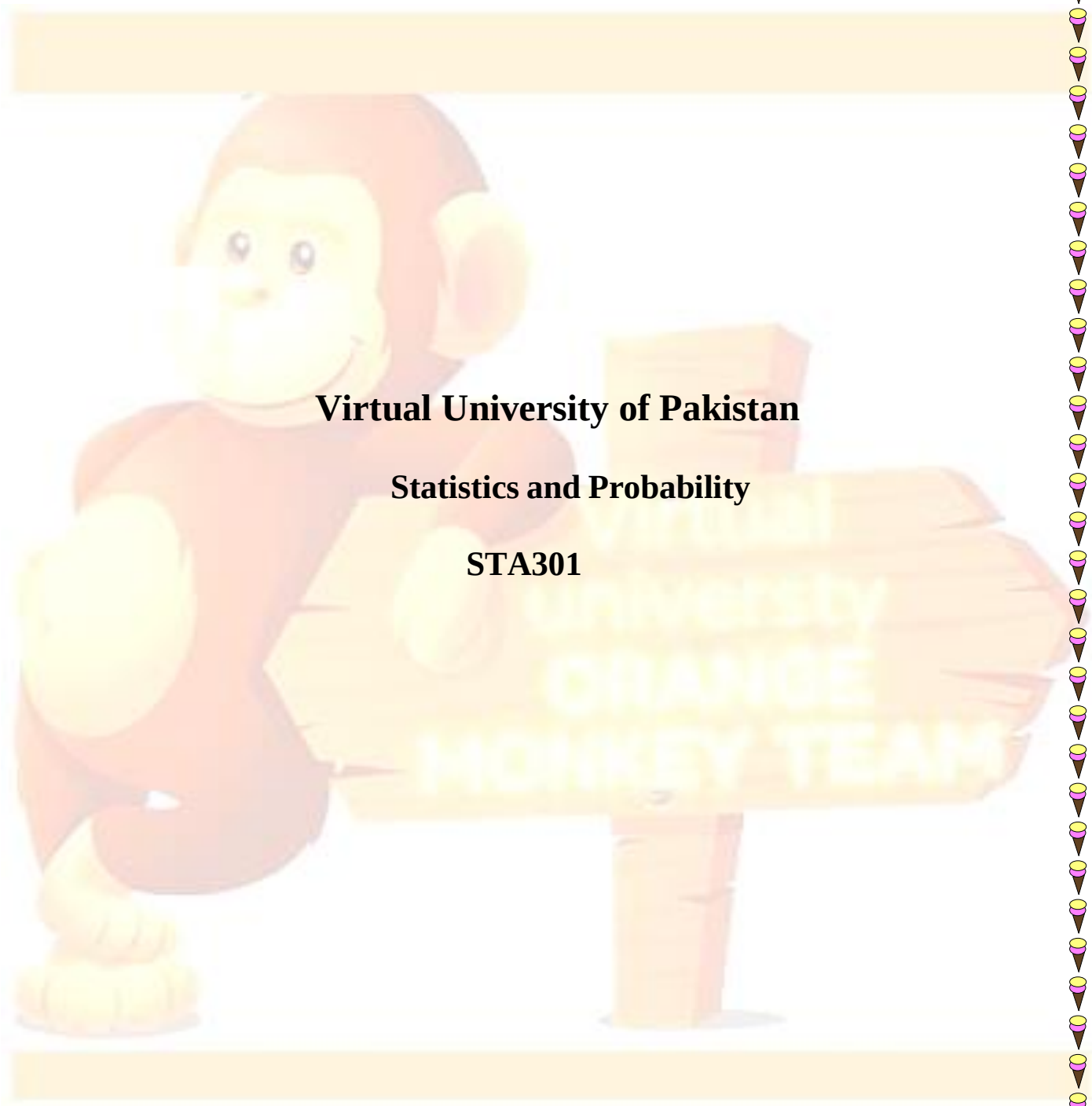
Pure number…confirm

The Coefficient of Quartile Deviation is a pure number and is used for *COMPARING* the variation in two or more sets of data.

99. The mean of a distribution is 24. The mode is 25 and the standard deviation is 5, then the coefficient of skewness will be

Less than zero….confirm

$$\frac{mean - \text{mod}\, e}{SD} \quad \frac{24-25}{5} = \frac{-1}{5} = -0.2$$



Virtual University of Pakistan

Statistics and Probability

STA301

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ORANGE MONKEY TEAM

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LECTURE NO. 1

WHAT IS STATISTICS?

- That science which enables us to draw conclusions about various phenomena on the basis of real data collected on sample-basis
- A tool for data-based research
- Also known as Quantitative Analysis
- A lot of application in a wide variety of disciplines Agriculture, Anthropology, Astronomy, Biology, Economic, Engineering, Environment, Geology, Genetics, Medicine, Physics, Psychology, Sociology, Zoology …. Virtually every single subject from Anthropology to Zoology …. A to Z!
- Any scientific enquiry in which you would like to base your conclusions and decisions on real-life data, you need to employ statistical techniques!
- Now a day, in the developed countries of the world, there is an active movement for of Statistical Literacy.

THE NATURE OF THIS DISCIPLINE

DESCRIPTIVE STATISTICS

↓

PROBABILITY

↓

INFERENTIAL STATISTICS

MEANINGS OF 'STATISTICS'

The word "Statistics" which comes from the Latin words *status*, meaning a political state, originally meant information useful to the state, for example, information about the sizes of population sand armed forces. But this word has now acquired different meanings.

- In the *first place*, the word *statistics* refers to "numerical facts systematically arranged". In this sense, the word statistics is always used in plural. We have, for instance, statistics of prices, statistics of road accidents, statistics of crimes, statistics of births, statistics of educational institutions, etc. In all these examples, the word statistics denotes a set of numerical data in the respective fields. This is the meaning the man in the street gives to the word *Statistics* and most people usually use the word *data* instead.
- In the *second place*, the word *statistics* is defined as a discipline that includes procedures and techniques used to collect process and analyze numerical data to make inferences and to research decisions in the face of

uncertainty. It should of course be borne in mind that uncertainty does not imply ignorance but it refers to the incompleteness and the instability of data available. In this sense, the word statistics is used in the singular. As it embodies more of less all stages of the general process of learning, sometimes called *scientific method*, statistics is characterized as a science. Thus the word *statistics* used in the plural refers to a set of numerical information and in the singular, denotes the science of basing decision on numerical data. It should be noted that statistics as a subject is mathematical in character.

- *Thirdly*, the word statistics are numerical quantities calculated from sample observations; a single quantity that has been so collected is called a *statistic*. The mean of a sample for instance is a statistic. The word statistics is plural when used in this sense.

CHARACTERISTICS OF THE SCIENCE OF STATISTICS

Statistics is a discipline in its own right. It would therefore be desirable to know the characteristic features of statistics in order to appreciate and understand its general nature. Some of its important characteristics are given below:

- Statistics deals with the behaviour of aggregates or large groups of data. It has nothing to do with what is happening to a particular individual or object of the aggregate.
- Statistics deals with aggregates of observations of the same kind rather than isolated figures.
- Statistics deals with variability that obscures underlying patterns. No two objects in this universe are exactly alike. If they were, there would have been no statistical problem.
- Statistics deals with uncertainties as every process of getting observations whether controlled or uncontrolled, involves deficiencies or chance variation. That is why we have to talk in terms of probability.
- Statistics deals with those characteristics or aspects of things which can be described numerically either by counts or by measurements.
- Statistics deals with those aggregates which are subject to a number of random causes, *e.g.* the heights of persons are subject to a number of causes such as race, ancestry, age, diet, habits, climate and so forth.
- Statistical laws are valid *on the average* or in the long run. There is n guarantee that a certain law will hold in all cases. Statistical inference is therefore made in the face of uncertainty.
- Statistical results might be misleading the incorrect if sufficient care in collecting, processing and interpreting the data is not exercised or if the statistical data are handled by a person who is not well versed in the subject mater of statistics.

THE WAY IN WHICH STATISTICS WORKS

As it is such an important area of knowledge, it is definitely useful to have a fairly good idea about the way in which it works, and this is exactly the purpose of this introductory course.
The following points indicate some of the main functions of this science:

- Statistics assists in summarizing the larger set of data in a form that is easily understandable.
- Statistics assists in the efficient design of laboratory and field experiments as well as surveys.
- Statistics assists in a sound and effective planning in any field of inquiry.
- Statistics assists in drawing general conclusions and in making predictions of how much of a thing will happen under given conditions.

IMPORTANCE OF STATISTICS IN VARIOUS FIELDS

As stated earlier, Statistics is a discipline that has finds application in the most diverse fields of activity. It is perhaps a subject that should be used by everybody. Statistical techniques being powerful tools for analyzing numerical data are used in almost every branch of learning. In all areas, statistical techniques are being increasingly used, and are developing very rapidly.

- A modern administrator whether in public or private sector leans on statistical data to provide a factual basis for decision.
- A politician uses statistics advantageously to lend support and credence to his arguments while elucidating the problems he handles.
- A businessman, an industrial and a research worker all employ statistical methods in their work. Banks, Insurance companies and Government all have their statistics departments.
- A social scientist uses statistical methods in various areas of socio-economic life a nation. It is sometimes said that "a social scientist without an adequate understanding of statistics, is often like the blind man groping in a dark room for a black cat that is not there".

THE MEANING OF DATA

The word "data" appears in many contexts and frequently is used in ordinary conversation. Although the word carries something of an aura of scientific mystique, its meaning is quite simple and mundane. It is Latin for "those that are given" (the singular form is "datum"). Data may therefore be thought of as the *results of observation.*

EXAMPLES OF DATA

- Data are collected in many aspects of everyday life.
- Statements given to a police officer or physician or psychologist during an interview are data.
- So are the correct and incorrect answers given by a student on a final examination.
- Almost any athletic event produces data.
- The time required by a runner to complete a marathon,
- The number of errors committed by a baseball team in nine innings of play.
- And, of course, data are obtained in the course of scientific inquiry:
- the positions of artifacts and fossils in an archaeological site,
- The number of interactions between two members of an animal colony during a period of observation,
- The spectral composition of light emitted by a star.

OBSERVATIONS AND VARIABLES

In statistics, an *observation* often means any sort of numerical recording of information, whether it is a physical measurement such as height or weight; a classification such as heads or tails, or an answer to a question such as yes or no.

VARIABLES

A characteristic that varies with an individual or an object is called a *variable*. For example, age is a variable as it varies from person to person. A variable can assume a number of values. The given set of all possible values from which the variable takes on a value is called its Domain. If for a given problem, the domain of a variable contains only one value, then the variable is referred to as a *constant*.

QUANTITATIVE AND QUALITATIVE VARIABLES

Variables may be classified into quantitative and qualitative according to the form of the characteristic of interest. A variable is called a *quantitative variable* when a characteristic can be expressed numerically such as age, weight, income or number of children. On the other hand, if the characteristic is non-numerical such as education, sex, eye-colour, quality, intelligence, poverty, satisfaction, etc. the variable is referred to as a *qualitative variable*. A qualitative characteristic is also called an *attribute*. An individual or an object with such a characteristic can be counted or enumerated after having been assigned to one of the several mutually exclusive classes or categories.

DISCRETE AND CONTINUOUS VARIABLES

A quantitative variable may be classified as discrete or continuous. *A discrete* variable is one that can take only a discrete set of integers or whole numbers, which is the values, are taken by jumps or breaks. A discrete variable represents *count* data such as the number of persons in a family, the number of rooms in a house, the number of deaths in an accident, the income of an individual, etc.

A variable is called a *continuous* variable if it can take on any value-fractional or integral—within a given interval, *i.e.* its domain is an interval with all possible values without gaps. A continuous variable represents measurement data such as the age of a person, the height of a plant, the weight of a commodity, the temperature at a place, etc.

A variable whether countable or measurable, is generally denoted by some symbol such as X or Y and X_i or X_j represents the ith or j^{th} value of the variable. The subscript i or j is replaced by a number such as 1,2,3, … when referred to a particular value.

MEASUREMENT SCALES

By *measurement*, we usually mean the assigning of number to observations or objects and scaling is a process of measuring. The four scales of measurements are briefly mentioned below:

NOMINAL SCALE

The classification or grouping of the observations into mutually exclusive qualitative categories or classes is said to constitute a *nominal scale*. For example, students are classified as male and female. Number 1 and 2 may also be used to identify these two categories. Similarly, rainfall may be classified as heavy moderate and light. We may use number 1, 2 and 3 to denote the three classes of rainfall. The numbers when they are used only to identify the categories of the given scale carry no numerical significance and there is no particular order for the grouping.

ORDINAL OR RANKING SCALE

It includes the characteristic of a nominal scale and in addition has the property of *ordering* or *ranking* of measurements. For example, the performance of students (or players) is rated as excellent, good fair or poor, etc. Number 1, 2, 3, 4 etc. are also used to indicate ranks. The only relation that holds between any pair of categories is that of "greater than" (or more preferred).

INTERVAL SCALE

A measurement scale possessing a constant interval size (distance) but not a true zero point, is called an *interval scale*. Temperature measured on either the Celsius or the Fahrenheit scale is an outstanding example of interval scale because the same difference exists between 20º C (68º F) and 30º C (86º F) as between 5º C (41º F) and 15º C (59º F). It cannot be said that a temperature of 40 degrees is twice as hot as a temperature of 20 degree, i.e. the ratio 40/20 has no meaning. The arithmetic operation of addition, subtraction, etc. is meaningful.

RATIO SCALE

It is a special kind of an interval scale where the sale of measurement has a true *zero* point as its origin. The ratio scale is used to measure weight, volume, distance, money, etc. The, key to differentiating interval and ratio scale is that the zero point is meaningful for ratio scale.

ERRORS OF MEASUREMENT

Experience has shown that a continuous variable can never be measured with perfect fineness because of certain habits and practices, methods of measurements, instruments used, etc. the measurements are thus always recorded correct to the nearest units and hence are of limited accuracy. The *actual* or *true* values are, however, assumed to exist. For example, if a student's weight is recorded as 60 kg (correct to the nearest kilogram), his true weight in fact lies between 59.5 kg and 60.5 kg, whereas a weight recorded as 60.00 kg means the true weight is known to lie between 59.995 and 60.005 kg. Thus there is a difference, however small it may be between the measured value and the true value. This sort of departure from the true value is technically known as the *error of measurement*. In other words, if the observed value and the true value of a variable are denoted by x and $x + \varepsilon$ respectively, then the difference $(x + \varepsilon) - x$, i.e. ε is the error. This error involves the unit of measurement of x and is therefore called an *absolute error*. An absolute error divided by the true value is called the *relative error*. Thus the relative error $= \dfrac{\varepsilon}{x + \varepsilon}$, which when multiplied by 100, is *percentage error*. These errors are independent of the units of measurement of x. It ought to be noted that an error has both magnitude and direction and that the word *error* in statistics does not mean mistake which is a chance inaccuracy.

BIASED AND RANDOM ERRORS

An error is said to be *biased* when the observed value is consistently and constantly higher or lower than the true value. Biased errors arise from the personal limitations of the observer, the imperfection in the instruments used or some other conditions which control the measurements. These errors are not revealed by repeating the measurements. They are cumulative in nature, that is, the greater the number of measurements, the greater would be the magnitude of error. They are thus more troublesome. These errors are also called *cumulative* or *systematic errors*.

An error, on the other hand, is said to be unbiased when the deviations, i.e. the excesses and defects, from the true value tend to occur equally often. Unbiased errors and revealed when measurements are repeated and they tend to cancel out in the long run. These errors are therefore *compensating* and are also known as *random errors* or *accidental errors*.



LECTURE NO. 2

Steps involved in a Statistical Research-Project

- Collection of Data:
  - Primary Data
  - Secondary Data
- Sampling:
  - Concept of Sampling
  - Non-Random Versus Random Sampling
  - Simple Random Sampling
  - Other Types of Random Sampling

STEPS INVOLVED IN ANY STATISTICAL RESEARCH

- Topic and significance of the study
- Objective of your study
- Methodology for data-collection
  - Source of your data
  - Sampling methodology
  - Instrument for collecting data

As far as the objectives of your research are concerned, they should be stated in such a way that you are absolutely clear about the goal of your study --- EXACTLY WHAT IT IS THAT YOU ARE TRYING TO FIND OUT?
As far as the *methodology for DATA-COLLECTION* is concerned, you need to consider:

- Source of your data (the statistical population)
- Sampling Methodology
- Instrument for collecting data

COLLECTION OF DATA

The most important part of statistical work is perhaps the collection of data. Statistical data are collected either by a COMPLETE enumeration of the whole field, called CENSUS, which in many cases would be too costly and too time consuming as it requires large number of enumerators and supervisory staff, or by a PARTIAL enumeration associated with a SAMPLE which saves much time and money.

PRIMARY AND SECONDARY DATA

Data that have been originally collected (raw data) and have not undergone any sort of statistical treatment, are called *PRIMARY data.* Data that have undergone any sort of treatment by statistical methods at least ONCE, i.e. the data that have been collected, classified, tabulated or presented in some form for a certain purpose, are called *SECONDARY data.*

COLLECTION OF PRIMARY DATA

One or more of the following methods are employed to collect primary data:

- Direct Personal Investigation
- Indirect Investigation
- Collection through Questionnaires
- Collection through Enumerators
- Collection through Local Sources

DIRECT PERSONAL INVESTIGATION

In this method, an investigator collects the information personally from the individuals concerned. Since he interviews the informants himself, the information collected is generally considered quite accurate and complete. This method may prove very costly and time-consuming when the area to be covered is vast. However, it is useful for laboratory experiments or localized inquiries. Errors are likely to enter the results due to personal bias of the investigator.

INDIRECT INVESTIGATION

Sometimes the direct sources do not exist or the informants hesitate to respond for some reason or other. In such a case, third parties or witnesses having information are interviewed. Moreover, due allowance is to be made for the personal bias. This method is useful when the information desired is complex or there is reluctance or indifference on the part of the informants. It can be adopted for extensive inquiries.

COLLECTION THROUGH QUESTIONNAIRES

A questionnaire is an inquiry form comprising of a number of pertinent questions with space for entering information asked. The questionnaires are usually sent by mail, and the informants are requested to return the questionnaires to the investigator after doing the needful within a certain period. This method is cheap, fairly expeditious and good for extensive inquiries. But the difficulty is that the majority of the respondents (i.e. persons who are required to answer the questions) do not care to fill the questionnaires in, and to return them to the investigators. Sometimes, the questionnaires are returned incomplete and full of errors. Students, in spite of these drawbacks, this method is considered as the STANDARD method for routine business and administrative inquiries.

It is important to note that the questions should be few, brief, very simple, and easy for all respondents answer, clearly worded and not offensive to certain respondents.

COLLECTION THROUGH ENUMERATORS

Under this method, the information is gathered by employing trained enumerators who assist the informants in making the entries in the schedules or questionnaires correctly. This method gives the most reliable information if the enumerator is well-trained, experienced and tactful. Students, it is considered the BEST method when a large-scale governmental inquiry is to be conducted. This method can generally not be adopted by a private individual or institution as its cost would be prohibitive to them.

COLLECTION THROUGH LOCAL SOURCES

In this method, there is no formal collection of data but the agents or local correspondents are directed to collect and send the required information, using their own judgment as to the best way of obtaining it. This method is cheap and expeditious, but gives only the estimates.

COLLECTION OF SECONDARY DATA

The secondary data may be obtained from the following sources:

- Official, e.g. the publications of the Statistical Division, Ministry of Finance, the Federal and Provincial Bureaus of Statistics, Ministries of Food, Agriculture, Industry, Labour, etc.
- Semi-Official, e.g., State Bank of Pakistan, Railway Board, Central Cotton Committee, Boards of Economic Inquiry, District Councils, Municipalities, etc.
- Publications of Trade Associations, Chambers of Commerce, etc
- Technical and Trade Journals and Newspapers
- Research Organizations such as universities, and other institutions

Let us now consider the POPULATION from which we will be collecting our data. In this context, the first important question is: Why do we have to resort to Sampling?
The answer is that: If we have available to us every value of the variable under study, then that would be an ideal and a perfect situation. But, the problem is that this ideal situation is very rarely available --- very rarely do we have access to the entire population.
The census is an exercise in which an attempt is made to cover the entire population. But, as you might know, even the most developed countries of the world cannot afford to conduct such a huge exercise on an annual basis!
More often than not, we have to conduct our research study on a sample basis. In fact, the goal of the science of Statistics is to draw conclusions about large populations on the basis of information contained in small samples.

'POPULATION'

A statistical population is the collection of every member of a group possessing the same basic and defined characteristic, but varying in amount or quality from one member to another.

EXAMPLES

- **Finite population:**
  - IQ's of all children in a school.
- **Infinite population:**
  - Barometric pressure:
    (There are an indefinitely large number of points on the surface of the earth).
  - A flight of migrating ducks in Canada

(Many finite pops are so large that they can be treated as effectively infinite). The examples that we have just considered are those of existent populations.
A hypothetical population can be defined as the aggregate of all the conceivable ways in which a specified event can happen.

For Example:

- 1)All the possible outcomes from the throw of a die – however long we throw the die and record the results, we could always continue to do so far a still longer period in a theoretical concept – one which has no existence in reality.
- 2) The No. of ways in which a football team of 11 players can be selected from the 16 possible members named by the Club Manager.

We also need to differentiate between the sampled population and the target population. Sampled population is that from which a sample is chosen whereas the population about which information is sought is called the target population thus our population will consist of the total no. of students in all the colleges in the Punjab.

Suppose on account of shortage of resources or of time, we are able to conduct such a survey on only 5 colleges scattered throughout the province. In this case, the students of all the colleges will constitute the target pop whereas the students of those 5 colleges from which the sample of students will be selected will constitute the sampled population. The above discussion regarding the population, you must have realized how important it is to have a very well-defined population.

The next question is: How will we draw a sample from our population?

The answer is that: In order to draw a random sample from a finite population, the first thing that we need is the complete list of all the elements in our population.

This list is technically called the **FRAME.**

SAMPLING FRAME

A sampling frame is a complete list of all the elements in the population. For example:

- The complete list of the BCS students of Virtual University of Pakistan on February 15, 2003
- Speaking of the sampling frame, it must be kept in mind that, as far as possible, our frame should be free from various types of defects:
- does not contain inaccurate elements
- is not incomplete
- is free from duplication, and
- Is not out of date.

Next, let's talk about the sample that we are going to draw from this population.

As you all know, a sample is only a part of a statistical population, and hence it can represent the population to only to some extent. Of course, it is intuitively logical that the larger the sample, the more likely it is to represent the population. Obviously, the limiting case is that: when the sample size tends to the population size, the sample will tend to be identical to the population. But, of course, in general, the sample is much smaller than the population.

The point is that, in general, statistical sampling seeks to determine how accurate a description of the population the sample and its properties will provide. We may have to compromise on accuracy, but there are certain such advantages of sampling because of which it has an extremely important place in data-based research studies.

ADVANTAGES OF SAMPLING

1. Savings in time and money.

- Although cost per unit in a sample is greater than in a complete investigation, the total cost will be less (because the sample will be so much smaller than the statistical population from which it has been drawn).
- A sample survey can be completed faster than a full investigation so that variations from sample unit to sample unit over time will largely be eliminated.
- Also, the results can be processed and analyzed with increased speed and precision because there are fewer of them.

2. More detailed information may be obtained from each sample unit.

3. Possibility of follow-up:

(After detailed checking, queries and omissions can be followed up --- a procedure which might prove impossible in a complete survey).

4. Sampling is the only feasible possibility where tests to destruction are undertaken or where the population is effectively infinite.

The next two important concepts that need to be considered are those of sampling and non-sampling errors.

SAMPLING & NON-SAMPLING ERRORS

1. SAMPLING ERROR

The difference between the estimate derived from the sample (i.e. the statistic) and the true population value (i.e. the parameter) is technically called the sampling error. For example,

Sampling error = $\bar{X} - \mu$

Sampling error arises due to the fact that a sample cannot exactly represent the pop, even if it is drawn in a correct manner

2. NON-SAMPLING ERROR

Besides sampling errors, there are certain errors which are not attributable to sampling but arise in the process of data collection, even if a complete count is carried out.
Main sources of non sampling errors are:

- The defect in the sampling frame.
- Faulty reporting of facts due to personal preferences.
- Negligence or indifference of the investigators
- Non-response to mail questionnaires.

These (non-sampling) errors can be avoided through

- Following up the non-response,
- Proper training of the investigators.
- Correct manipulation of the collected information,

Let us now consider exactly what is meant by 'sampling error': We can say that there are two types of non-response --- partial non-response and total non-response. *'Partial non-response'* implies that the respondent refuses to answer some of the questions. On the other hand, '*total non-response'* implies that the respondent refuses to answer any of the questions. Of course, the problem of late returns and non-response of the kind that I have just mentioned occurs in the case of HUMAN populations. Although refusal of sample units to cooperate is encountered in interview surveys, it is far more of a problem in mail surveys. It is not uncommon to find the response rate to mail questionnaires as low as 15 or 20%.The provision of INFORMATION ABOUT THE PURPOSE OF THE SURVEY helps in stimulating interest, thus increasing the chances of greater response. Particularly if it can be shown that the work will be to the ADVANTAGE of the respondent IN THE LONG RUN.
Similarly, the respondent will be encouraged to reply if a pre-paid and addressed ENVELOPE is sent out with the questionnaire. But in spite *of* these ways of reducing non-response, we are bound to have some amount of non-response. Hence, a decision has to be taken about how many RECALLS should be made.
The term 'recall' implies that we approach the respondent more than once in order to persuade him to respond to our queries.
Another point worth considering is:
How long the process of data collection should be continued? Obviously, no such process can be carried out for an indefinite period of time! In fact, the longer the time period over which the survey is conducted, the greater will be the potential VARIATIONS in attitudes and opinions of the respondents. Hence, a well-defined cut-off date generally needs to be established. Let us now look at the various ways in which we can select a sample from our population. We begin by looking at the difference between non-random and RANDOM sampling. First of all, what do we mean by non-random sampling?

NONRANDOM SAMPLING

'Nonrandom sampling' implies that kind of sampling in which the population units are drawn into the sample by using one's personal judgment. This type of sampling is also known as purposive sampling. Within this category, one very important type of sampling is known as Quota Sampling.

QUOTA SAMPLING

In this type of sampling, the selection of the sampling unit from the population is no longer dictated by chance. A sampling frame is not used at all, and the choice of the actual sample units to be interviewed is left to the discretion of the interviewer. However, the interviewer is restricted by quota controls. For example, one particular interviewer may be told to interview ten married women between thirty and forty years of age living in town X, whose husbands are professional workers, and five unmarried professional women of the same age living in the same town. Quota sampling is often used in commercial surveys such as consumer market-research. Also, it is often used in public opinion polls.

ADVANTAGES OF QUOTA SAMPLING

- There is no need to construct a frame.
- It is a very quick form of investigation.
- Cost reduction.

DISADVANTAGES

- It is a subjective method. One has to choose between objectivity and convenience.
- If random sampling is not employed, it is no longer theoretically possible to evaluate the sampling error.
- (Since the selection of the elements is not based on probability theory but on the personal judgment of the interviewer, hence the precision and the reliability of the estimates can not be determined objectively i.e. in terms of probability.)
- Although the purpose of implementing quota controls is to reduce bias, bias creeps in due to the fact that the interviewer is FREE to select particular individuals within the quotas. (Interviewers usually look for persons who either agree with their points of view or are personally known to them or can easily be contacted.)
- Even if the above is not the case, the interviewer may still be making unsuitable selection of sample units.
- (Although he may put some qualifying questions to a potential respondent in order to determine whether he or she is of the type prescribed by the quota controls, some features must necessarily be decided arbitrarily by the interviewer, the most difficult of these being social class.)

If mistakes are being made, it is almost impossible for the organizers to detect these, because follow-ups are not possible unless a detailed record of the respondents' names, addresses etc. has been kept.
Falsification of returns is therefore more of a danger in quota sampling than in random sampling. In spite of the above limitations, it has been shown by F. Edwards that a well-organized quota survey with well-trained interviewers can produce quite adequate results.
Next, let us consider the concept of random sampling.

RANDOM SAMPLING

The theory of statistical sampling rests on the assumption that the selection of the sample units has been carried out in a random manner.
By random sampling we mean sampling that has been done by adopting the lottery method.

TYPES OF RANDOM SAMPLING

- Simple Random Sampling
- Stratified Random Sampling
- Systematic Sampling
- Cluster Sampling
- Multi-stage Sampling, etc.

In this course, I will discuss with you the simplest type of random sampling i.e. simple random sampling.

SIMPLE RANDOM SAMPLING

In this type of sampling, the chance of any one element of the parent pop being included in the sample is the same as for any other element. By extension, it follows that, in simple random sampling, the chance of any one sample appearing is the same as for any other. There exists quite a lot of misconception regarding the concept of random sampling. Many a time, haphazard selection is considered to be equivalent to simple random sampling.
For example, a market research interviewer may select women shoppers to find their attitude to brand X of a product by stopping one and then another as they pass along a busy shopping area --- and he may think that he has accomplished simple random sampling!

Actually, there is a strong possibility of bias as the interviewer may tend to ask his questions of young attractive women rather than older housewives, or he may stop women who have packets of brand X prominently on show in their shopping bags!.
In this example, there is no suggestion of INTENTIONAL bias! From experience, it is known that the human being is a poor random selector --- one who is very subject to bias.
Fundamental psychological traits prevent complete objectivity, and no amount of training or conscious effort can eradicate them. As stated earlier, random sampling is that in which population units are selected by the lottery method.

As you know, the traditional method of writing people's names on small pieces of paper, folding these pieces of paper and shuffling them is very cumbersome!
A much more convenient alternative is the use of *RANDOM NUMBERS TABLES.*
A random number table is a page full of digits from zero to 9. These digits are printed on the page in a TOTALLY random manner i.e. there is no systematic pattern of printing these digits on the page.

ONE THOUSAND RANDOM DIGITS

2	3	1	5	7	5	4	8	5	9	0	1	8	3	7	2	5	9	9	3	7	6	2	4	9	7	0	8	8	6	9	5	2	3	0	3	6	7	4	4
0	5	5	4	5	5	5	0	4	3	1	0	5	3	7	4	3	5	0	8	9	0	6	1	1	8	3	7	4	4	1	0	9	6	2	2	1	3	4	3
1	4	8	7	1	6	0	3	5	0	3	2	4	0	4	3	6	2	2	3	5	0	0	5	1	0	0	3	2	2	1	1	5	4	3	8	0	8	3	4
3	8	9	7	6	7	4	9	5	1	9	4	0	5	1	7	5	8	5	3	7	8	8	0	5	9	0	1	9	4	3	2	4	2	8	7	1	6	9	5
9	7	3	1	2	6	1	7	1	8	9	9	7	5	5	3	0	8	7	0	9	4	2	5	1	2	5	8	4	1	5	4	8	8	2	1	0	5	1	3
1	1	7	4	2	6	9	3	8	1	4	4	3	3	9	3	0	8	7	2	3	2	7	9	7	3	3	1	1	8	2	2	6	4	7	0	6	8	5	0
4	3	3	6	1	2	8	8	5	9	1	1	0	1	6	4	5	6	2	3	9	3	0	0	9	0	0	4	9	9	4	3	6	4	0	7	4	0	3	6
9	3	8	0	6	2	0	4	7	8	3	8	2	6	8	0	4	4	9	1	5	5	7	5	1	1	8	9	3	2	5	8	4	7	5	5	2	5	7	1
4	9	5	4	0	1	3	1	8	1	0	8	4	2	9	8	4	1	8	7	6	9	5	3	8	2	9	6	6	1	7	7	7	3	8	0	9	5	2	7
3	6	7	6	8	7	2	6	3	3	3	7	9	4	8	2	1	5	6	9	4	1	9	5	9	6	8	6	7	0	4	5	2	7	4	8	3	8	8	0
0	7	0	9	2	5	2	3	9	2	2	4	6	2	7	1	2	6	0	7	0	6	5	5	8	4	5	3	4	4	6	7	3	3	8	4	5	3	2	0
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6	1	5	7	0	0	6	3	6	0	0	6	1	7	3	6	3	7	7	5	6	3	1	4	8	9	5	1	2	3	3	5	0	1	7	4	6	9	9	3
3	1	3	5	2	8	3	7	9	9	1	0	7	7	9	1	8	9	4	1	3	1	5	7	9	7	6	4	4	8	6	2	5	8	4	8	6	9	1	9
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0	9	2	4	3	4	4	2	0	0	6	8	7	2	1	0	7	1	3	7	3	0	7	2	9	7	5	7	3	6	0	9	2	9	8	2	7	6	5	0
9	7	9	5	5	3	5	0	1	8	4	0	8	9	4	8	8	3	2	9	5	2	2	3	0	8	2	5	2	1	2	2	5	3	2	6	1	5	8	7
9	3	7	3	2	5	9	5	7	0	4	3	7	8	1	9	8	8	8	5	5	6	6	7	1	6	6	8	2	6	9	5	9	9	6	4	4	5	6	9
7	2	6	2	1	1	1	2	2	5	0	0	9	2	2	6	8	2	6	4	3	5	6	6	6	5	9	4	3	4	7	1	6	8	7	5	1	8	6	7
6	1	0	2	0	7	4	4	1	8	4	5	3	7	1	2	0	7	9	4	9	5	9	1	7	3	7	8	6	6	9	9	5	3	6	1	9	3	7	8
9	7	8	3	9	8	5	4	7	4	3	3	0	5	5	9	1	7	1	8	4	5	4	7	3	5	4	1	4	4	2	2	0	3	4	2	3	0	0	0
8	9	1	6	0	9	7	1	9	2	2	2	2	3	2	9	0	6	3	7	3	5	0	5	5	4	5	4	8	9	8	8	4	3	8	1	6	3	6	1
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8	1	4	4	3	3	1	7	1	9	0	5	0	4	9	5	4	8	0	6	7	4	6	9	0	0	7	5	6	7	6	5	0	1	7	1	6	5	4	5
1	1	3	2	2	5	4	9	3	1	4	2	3	6	2	3	4	3	8	6	0	8	6	2	4	9	7	6	6	7	4	2	2	4	5	2	3	2	4	5

Actually, Random Number Tables are constructed according to certain mathematical principles so that each digit has the same chance of selection. Of course, nowadays randomness may be achieved electronically. Computers have all those programmes by which we can generate random numbers.

EXAMPLE

The following frequency table of distribution gives the ages of a population of 1000 teen-age college students in a particular country.
Select a sample of 10 students using the random numbers table. Find the sample mean age and compare with the population mean age.

Student-Population of a College

Age (X)	No. of Students (f)
13	6
14	61
15	270
16	491
17	153
18	15
19	4
	1000

How will we proceed to select our sample of size 10 from this population of size 1000?

The first step is to allocate to each student in this population a sampling number. For this purpose, we will begin by constructing a column of cumulative frequencies.

AGE X	No. of Students f	Cumulative Frequency cf
13	6	6
14	61	67
15	270	337
16	491	828
17	153	981
18	15	996
19	4	1000
	1000	

Now that we have the cumulative frequency of each class, we are in a position to allocate the sampling numbers to all the values in a class. As the frequency as well as the cumulative frequency of the first class is 6, we allocate numbers 000 to 005 to the six students who belong to this class.

AGE X	No. of Students f	cf	Sampling Numbers
13	6	6	000 – 005
14	61	67	
15	270	337	
16	491	828	
17	153	981	
18	15	996	
19	4	1000	
	1000		

As the cumulative frequency of the second class is 67 while that of the first class was 6, therefore we allocate sampling numbers 006 to 066 to the 61 students who belong to this class.

AGE X	No. of Students f	cf	Sampling Numbers
13	6	6	000 – 005
14	61	67	006 – 066
15	270	337	
16	491	828	
17	153	981	
18	15	996	
19	4	1000	
	1000		

As the cumulative frequency of the third class is 337 while that of the second class was 67, therefore we allocate sampling numbers 067 to 336 to the 270 students who belong to this class.

AGE X	No. of Students f	cf	Sampling Numbers
13	6	6	000 – 005
14	61	67	006 – 066
15	270	337	067 – 336
16	491	828	
17	153	981	
18	15	996	
19	4	1000	
	1000		

Proceeding in this manner, we obtain the column of sampling numbers.

AGE X	**No. of Students f**	**cf**	**Sampling Numbers**
13	**6**	**6**	**000 – 005**
14	**61**	**67**	**006 – 066**
15	**270**	**337**	**067 – 336**
16	**491**	**828**	**337 – 827**
17	**153**	**981**	**828 – 980**
18	**15**	**996**	**981 – 995**
19	**4**	**1000**	**996 - 999**
	1000		

The column implies that the first student of the first class has been allocated the sampling number 000, the second student has been allocated the sampling 001, and, proceeding in this fashion, the last student i.e. the 1000th student has been allocated the sampling number 999.

The question is: Why did we not allot the number 0001 to the first student and the number 1000 to the 1000th student? The answer is that we could do that but that would have meant that every student would have been allocated a four-digit number, whereas by shifting the number backward by 1, we are able to allocate to every student a three-digit number --- which is obviously simpler.
The next step is to SELECT 10 RANDOM NUMBERS from the random number table. This is accomplished by closing one's eyes and letting one's finger land anywhere on the random number table. In this example, since all our sampling numbers are three-digit numbers, hence we will read three digits that are adjacent to each other at that position where our finger landed. Suppose that we adopt this procedure and our random numbers come out to be 041, 103, 374, 171, 508, 652, 880, 066, 715, 471
Selected Random Numbers:
041, 103, 374, 171, 508, 652, 880, 066, 715, 471
Thus the corresponding ages are:
14, 15, 16, 15, 16, 16, 17, 15, 16, 16

EXPLANATION

Our first selected random number is 041 which mean that we have to pick up the 42nd student. The cumulative frequency of the first class is 6 whereas the cumulative frequency of the second class is 67. This means that definitely the 42nd student does not belong to the first class but does belong to the second class.

AGE X	No. of Students f	cf
13	6	6
14	61	67
15	270	337
16	491	828
17	153	981
18	15	996
19	4	1000
	1000	

The age of each student in this class is 14 years; hence, obviously, the age of the 42nd student is also 14 years. This is how we are able to ascertain the ages of all the students who have been selected in our sampling. You will recall that in this example, our aim was to draw a sample from the population of college students, and to compare the sample's mean age with the population mean age. The population mean age comes out to be 15.785 years.

AGE X	No. of Students f	fX
13	6	78
14	61	854
15	270	4050
16	491	7856
17	153	2601
18	15	270
19	4	76
	1000	15785

The population mean age is :

$$\mu = \frac{\sum fx}{\sum f} = \frac{15785}{1000}$$
$$= 15.785 \ years$$

The above formula is a slightly modified form of the basic formula that you have done ever-since school-days i.e. the mean is equal to the sum of all the observations divided by the total number of observations.
Next, we compute the sample mean age.
Adding the 10 values and dividing by 10, we obtain:
Ages of students selected in the sample (in years):
14, 15, 16, 15, 16, 16, 17, 15, 16, 16
Hence the sample mean age is: 15.6, comparing the sample mean age of 15.6 years with the population mean age of 15.785 years, we note that the difference is really quite slight, and hence the sampling error is equal to

Sampling Error

$$\bar{X} - \mu = 15.6 - 15.785$$
$$= -0.185 \ years$$

And the reason for such a small error is that we have adopted the RANDOM sampling method. The basic advantage of random sampling is that the probability is very high that the sample will be a good representative of the population from which it has been drawn, and any quantity computed from the sample will be a good estimate of the corresponding quantity computed from the population! Actually, a sample is supposed to be a MINIATURE REPLICA of the population. As stated earlier, there are various other types of random sampling.

OTHER TYPES OF RANDOM SAMPLING

- ·Stratified sampling (if the population is heterogeneous)
- Systematic sampling (practically, more convenient than simple random sampling)
- Cluster sampling (sometimes the sampling units exist in natural clusters)
- Multi-stage sampling

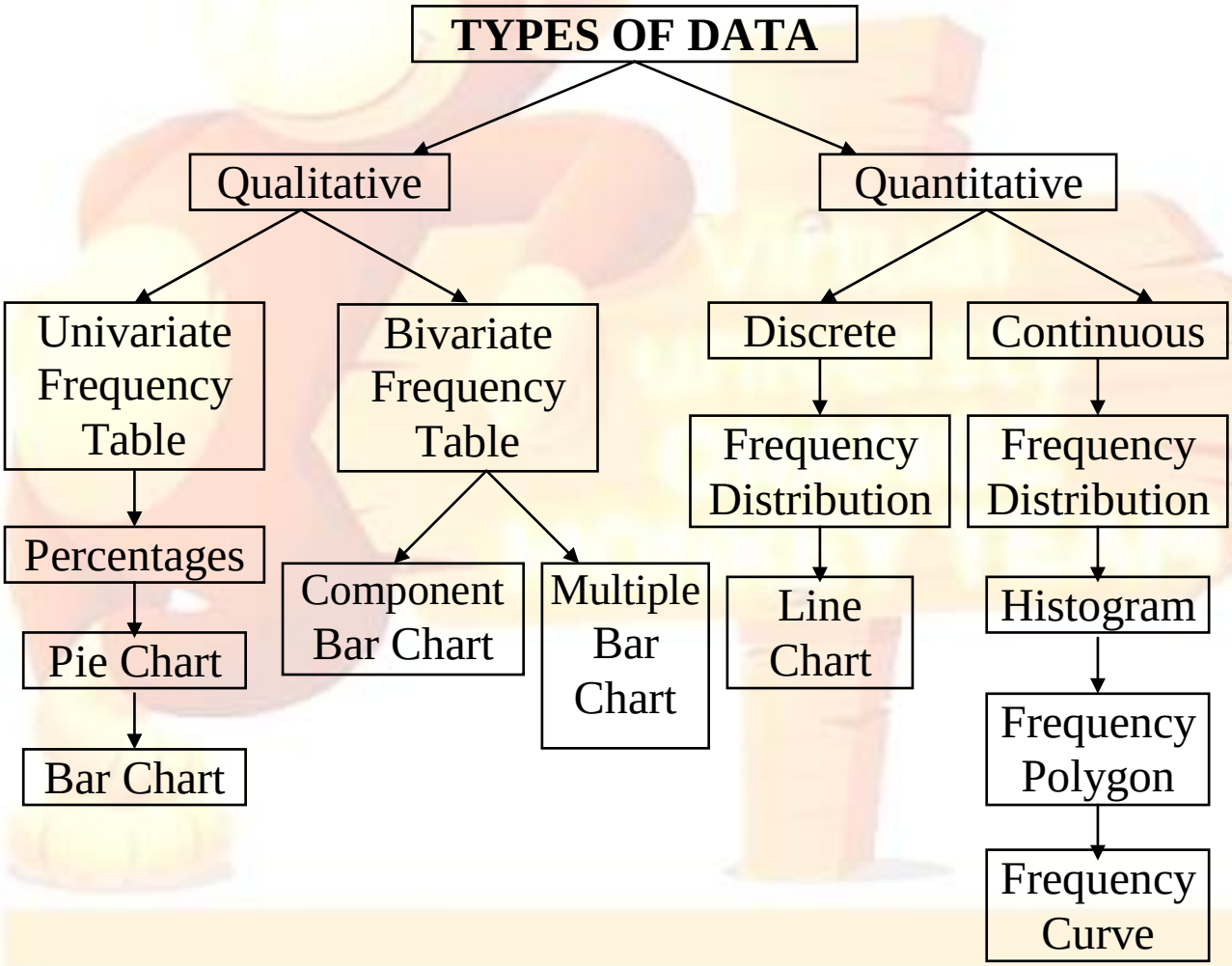
All these designs rest upon random or quasi-random sampling. They are various forms of PROBABILITY sampling --- that in which each sampling unit has a known (but not necessarily equal) probability of being selected.
Because of this knowledge, there exist methods by which the precision and the reliability of the estimates can be calculated OBJECTIVELY.
It should be realized that in practice, several sampling techniques are incorporated into each survey design, and only rarely will simple random sample be used, or a multi-stage design be employed, without stratification.
The point to remember is that whatever method be adopted, care should be exercised at every step so as to make the results as reliable as possible.



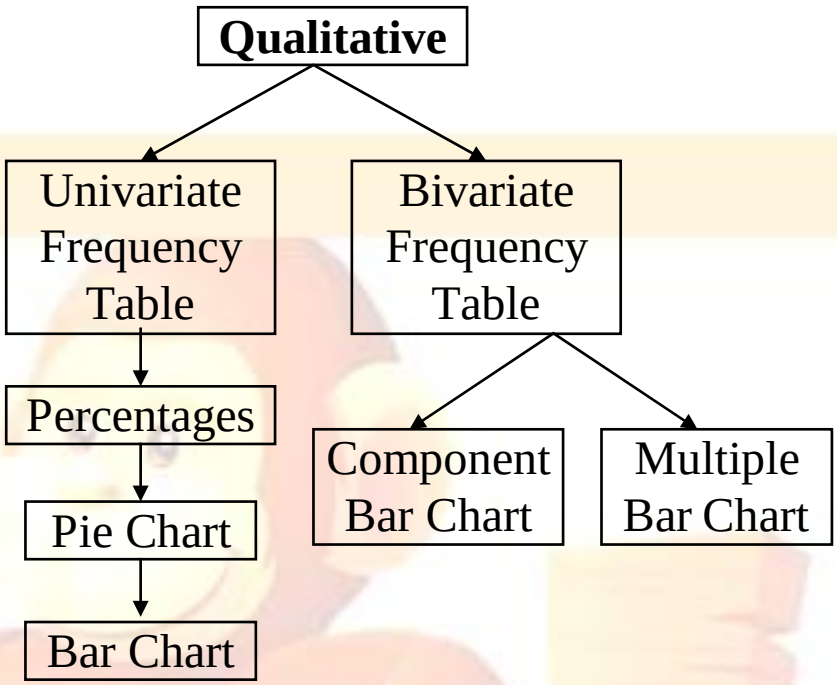
LECTURE NO. 3

- Tabulation
- Simple bar chart
- Component bar chart
- Multiple bar chart
- Pie chart

As indicated in the last lecture, there are two broad categories of data … qualitative data and quantitative data. A variety of methods exist for summarizing and describing these two types of data. The tree-diagram below presents an outline of the various techniques



In today's lecture, we will be dealing with various techniques for summarizing and describing qualitative data.



We will begin with the univariate situation, and will proceed to the bivariate situation.

EXAMPLE

Suppose that we are carrying out a survey of the students of first year studying in a co-educational college of Lahore. Suppose that in all there are 1200 students of first year in this large college. We wish to determine what proportion of these students have come from Urdu medium schools and what proportion has come from English medium schools. So we will interview the students and we will inquire from each one of them about their schooling. As a result, we will obtain a set of data as you can now see on the screen.
We will have an array of observations as follows:

U, U, E, U, E, E, E, U, ……

(U : URDU MEDIUM)
(E : ENGLISH MEDIUM)

Now, the question is what should we do with this data?
Obviously, the first thing that comes to mind is to count the number of students who said "Urdu medium" as well as the number of students who said "English medium". This will result in the following table:

Medium of Institution	No. of Students (f)
Urdu	719
English	481
	1200

The technical term for the numbers given in the second column of this table is "frequency". It means "how frequently something happens?" Out of the 1200 students, 719 stated that they had come from Urdu medium schools. So in this example, the frequency of the first category of responses is 719 whereas the frequency of the second category of responses is 481.

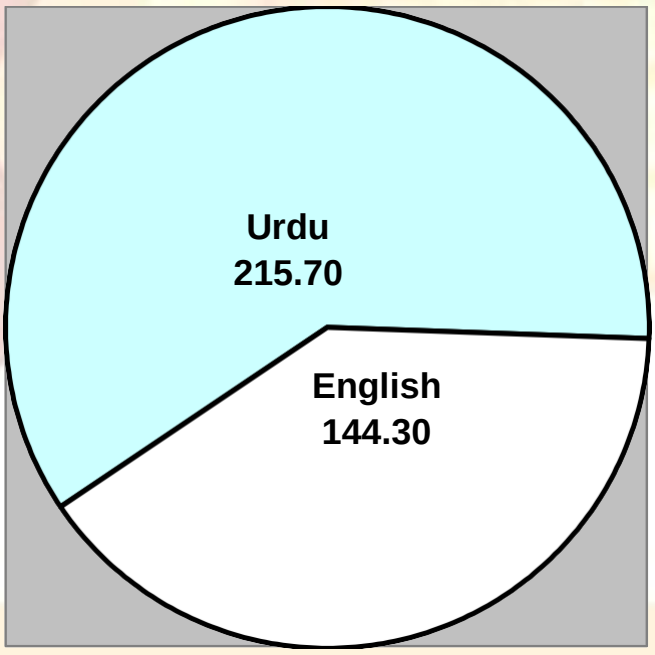
It is evident that this information is not as useful as if we compute the proportion or percentage of students falling in each category. Dividing the cell frequencies by the total frequency and multiplying by 100 we obtain the following:

Medium of Institution	**f**	**%**
Urdu	719	59.9 = 60%
English	481	40.1 = 40%
	1200	

What we have just accomplished is an example of a univariate frequency table pertaining to qualitative data.
Let us now see how we can represent this information in the form of a diagram.
One good way of representing the above information is in the form of a pie chart.
A pie chart consists of a circle which is divided into two or more parts in accordance with the number of distinct categories that we have in our data.
For the example that we have just considered, the circle is divided into two sectors, the larger sector pertaining to students coming from Urdu medium schools and the smaller sector pertaining to students coming from English medium schools.
How do we decide where to cut the circle?
The answer is very simple! All we have to do is to divide the cell frequency by the total frequency and multiply by 360.
This process will give us the exact value of the angle at which we should cut the circle.

PIE CHART

Medium of Institution	**f**	**Angle**
Urdu	719	215.7^0
English	481	144.3^0
	1200	



SIMPLE BAR CHART:

The next diagram to be considered is the simple bar chart.

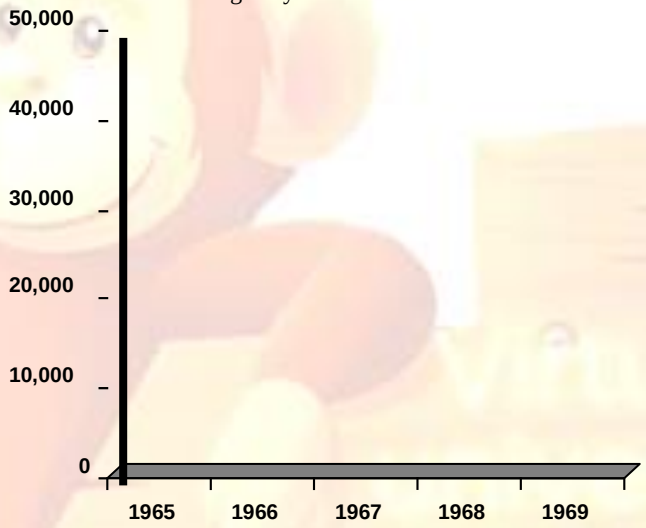
A simple bar chart consists of horizontal or vertical bars of equal width and lengths proportional to values they represent.

As the basis of comparison is one-dimensional, the widths of these bars have no mathematical significance but are taken in order to make the chart look attractive.
Let us consider an example.

Suppose we have available to us information regarding the turnover of a company for 5 years as given in the table below:

Years	1965	1966	1967	1968	1969
Turnover (Rupees)	35,000	42,000	43,500	48,000	48,500

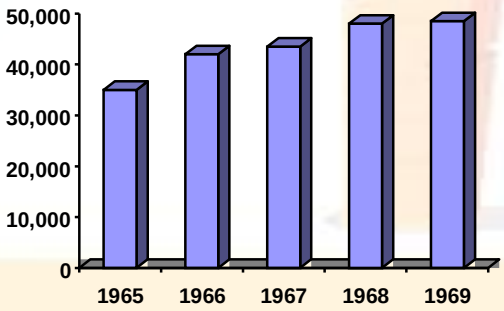
In order to represent the above information in the form of a bar chart, all we have to do is to take the year along the x-axis and construct a scale for turnover along the y-axis.



Next, against each year, we will draw vertical bars of equal width and different heights in accordance with the turn-over figures that we have in our table.

As a result we obtain a simple and attractive diagram as shown below.
When our values do not relate to time, they should be arranged in ascending or descending order before-charting.

BIVARIATE FREQUENCY TABLE



What we have just considered was the univariate situation. In each of the two examples, we were dealing with one single variable. In the example of the first year students of a college, our lone variable of interest was 'medium of schooling'. And in the second example, our one single variable of interest was turnover. Now let us expand the discussion a little, and consider the bivariate situation.

Going back to the example of the first year students, suppose that alongwith the enquiry about the Medium of Institution, you are also recording the sex of the student.
Suppose that our survey results in the following information:

Student No.	Medium	Gender
1	U	F
2	U	M
3	E	M
4	U	F
5	E	M
6	E	F
7	U	M
8	E	M
:	:	:
:	:	:

Now this is a bivariate situation; we have two variables, medium of schooling and sex of the student.
In order to summarize the above information, we will construct a table containing a box head and a stub as shown below:

Sex / Med.	MALE	Female	Total
Urdu			
English			
Total			

The top row of this kind of a table is known as the boxhead and the first column of the table is known as stub.
Next, we will count the number of students falling in each of the following four categories:

1. Male student coming from an Urdu medium school.
2. Female student coming from an Urdu medium school.
3. Male student coming from an English medium school.
4. Female student coming from an English medium school.

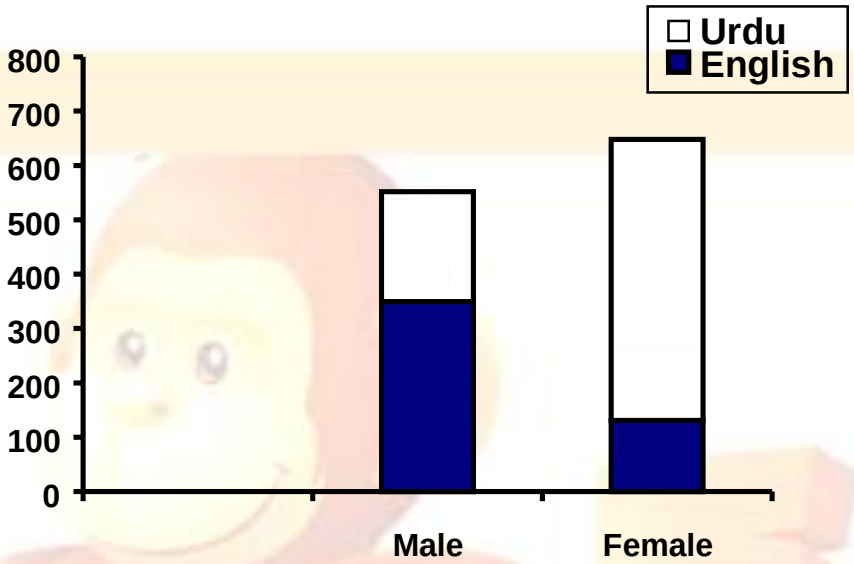
As a result, suppose we obtain the following figures:

Sex / Med.	MALE	Female	Total
Urdu	202	517	719
English	350	131	481
Total	552	648	1200

What we have just accomplished is an example of a bivariate frequency table pertaining to two qualitative variables.

COMPONENT BAR CHAR:

Let us now consider how we will depict the above information diagrammatically.
This can be accomplished by constructing the component bar chart (also known as the subdivided bar chart) as shown below:



In the above figure, each bar has been divided into two parts. The first bar represents the total number of male students whereas the second bar represents the total number of female students.

As far as the medium of schooling is concerned, the lower part of each bar represents the students coming from English medium schools. Whereas the upper part of each bar represents the students coming from the Urdu medium schools. The advantage of this kind of a diagram is that we are able to ascertain the situation of both the variables at a glance.

We can compare the number of male students in the college with the number of female students, and at the same time we can compare the number of English medium students among the males with the number of English medium students among the females.

MULTIPLE BAR CHARTS

The next diagram to be considered is the multiple bar charts. Let us consider an example.
Suppose we have information regarding the imports and exports of Pakistan for the years 1970-71 to 1974-75 as shown in the table below:

Years	Imports (Crores of Rs.)	Exports (Crores of Rs.)
1970-71	370	200
1971-72	350	337
1972-73	840	855
1973-74	1438	1016
1974-75	2092	1029

Source: State Bank of Pakistan

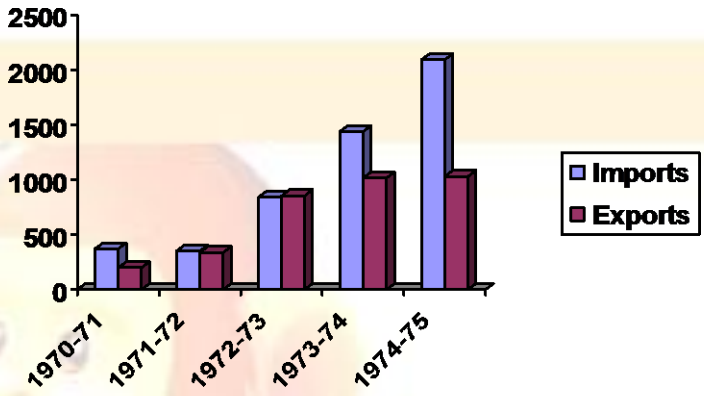
A multiple bar chart is a very useful and effective way of presenting this kind of information.

This kind of a chart consists of a set of grouped bars, the lengths of which are proportionate to the values of our variables, and each of which is shaded or colored differently in order to aid identification. With reference to the above example, we obtain the multiple bar chart shown below:

Multiple Bar Chart Showing

Imports & Exports
of Pakistan 1970-71 to 1974-75

This is a very good device for the comparison of two different kinds of information.



If, in addition to information regarding imports and exports, we also had information regarding production, we could have compared them from year to year by grouping the three bars together.
The question is, what is the basic difference between a component bar chart and a multiple bar chart?
The component bar chart should be used when we have available to us information regarding totals and their components.

For example, the total number of male students out of which some are Urdu medium and some are English medium. The number of Urdu medium male students and the number of English medium male students add up to give us the total number of male students.
On the contrary, in the example of exports and imports, the imports and exports do not add up to give us the totality of some one thing!

LECTURE NO. 4

In THIS Lecture, we will discuss the frequency distribution of a continuous variable & the graphical ways of representing data pertaining to a continuous variable i.e. histogram, frequency polygon and frequency curve.
You will recall that in Lecture No. 1, it was mentioned that a continuous variable takes values over a continuous interval (e.g. a normal Pakistani adult male's height may lie anywhere between 5.25 feet and 6.5 feet).
Hence, in such a situation, the method of constructing a frequency distribution is somewhat different from the one that was discussed in the last lecture.

EXAMPLE:

Suppose that the Environmental Protection Agency of a developed country performs extensive tests on all new car models in order to determine their mileage rating. Suppose that the following 30 measurements are obtained by conducting such tests on a particular new car model.

EPA MILEAGE RATINGS ON 30 CARS (MILES PER GALLON)		
36.3	42.1	44.9
30.1	37.5	32.9
40.5	40.0	40.2
36.2	35.6	35.9
38.5	38.8	38.6
36.3	38.4	40.5
41.0	39.0	37.0
37.0	36.7	37.1
37.1	34.8	33.9
39.9	38.1	39.8

EPA: Environmental Protection Agency

There are a few steps in the construction of a frequency distribution for this type of a variable.

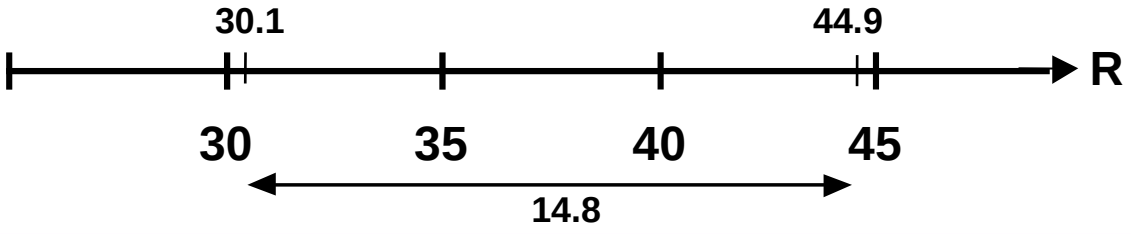
CONSTRUCTION OF A FREQUENCY DISTRIBUTION

Step-1

Identify the smallest and the largest measurements in the data set.
In our example:
Smallest value (X0) = 30.1,
Largest Value (Xm) = 44.9,

Step-2

Find the range which is defined as the difference between the largest value and the smallest value
In our example:
Range = Xm – X0
= 44.9 – 30.1
= 14.8
Let us now look at the graphical picture of what we have just computed.



(Range)

Step-3

Decide on the number of classes into which the data are to be grouped.

(By classes, we mean small sub-intervals of the total interval which, in this example, is 14.8 units long.)There are no hard and fast rules for this purpose. The decision will depend on the size of the data. When the data are sufficiently large, the number of classes is usually taken between 10 and 20.In this example, suppose that we decide to form 5 classes (as there are only 30 observations).

Step-4

Divide the range by the chosen number of classes in order to obtain the approximate value of the class interval i.e. the width of our classes.

Class interval is usually denoted by h. Hence, in this example

Class interval = h = 14.8 / 5 = 2.96

Rounding the number 2.96, we obtain 3, and hence we take h = 3.

This means that our big interval will be divided into small sub-intervals, each of which will be 3 units long.

Step-5

Decide the lower class limit of the lowest class. Where should we start from?

The answer is that we should start constructing our classes from a number equal to or slightly less than the smallest value in the data.

In this example, smallest value = 30.1

So we may choose the lower class limit of the lowest class to be 30.0.

Step-6

Determine the lower class limits of the successive classes by adding h = 3 successively. Hence, we obtain the following table:

Class Number	Lower Class Limit
1	30.0
2	30.0 + 3 = 33.0
3	33.0 + 3 = 36.0
4	36.0 + 3 = 39.0
5	39.0 + 3 = 42.0

Step-7

Determine the upper class limit of every class. The upper class limit of the highest class should cover the largest value in the data. It should be noted that the upper class limits will also have a difference of h between them. Hence, we obtain the upper class limits that are visible in the third column of the following table.

Class Number	Lower Class Limit	Upper Class Limit
1	30.0	32.9
2	30.0 + 3 = 33.0	32.9 + 3 = 35.9
3	33.0 + 3 = 36.0	35.9 + 3 = 38.9
4	36.0 + 3 = 39.0	38.9 + 3 = 41.9
5	39.0 + 3 = 42.0	41.9 + 3 = 44.9

Hence we obtain the following classes:

Classes
30.0 – 32.9
33.0 – 35.9
36.0 – 38.9
39.0 – 41.9
42.0 – 44.9

The question arises: why did we not write 33 instead of 32.9? Why did we not write 36 instead of 35.9? and so on.

The reason is that if we wrote 30 to 33 and then 33 to 36, we would have trouble when tallying our data into these classes. Where should I put the value 33? Should I put it in the first class, or should I put it in the second class? By writing 30.0 to 32.9 and 33.0 to 35.9, we avoid this problem. And the point to be noted is that the class interval is still 3, and not 2.9 as it appears to be. This point will be better understood when we discuss the concept of class boundaries … which will come a little later in today's lecture.

Step-8

After forming the classes, distribute the data into the appropriate classes and find the frequency of each class, in this example:

Class	Tally	Frequency
30.0 – 32.9	||	2
33.0 – 35.9	||||	4
36.0 – 38.9	𝍸 𝍸 ||||	14
39.0 – 41.9	𝍸 |||	8
42.0 – 44.9	||	2
	Total	30

This is a simple example of the frequency distribution of a continuous or, in other words, measurable variable.

CLASS BOUNDARIES:

The true class limits of a class are known as its class boundaries. In this example:

Class Limit	Class Boundaries	Frequency
30.0 – 32.9	29.95 – 32.95	2
33.0 – 35.9	32.95 – 35.95	4
36.0 – 38.9	35.95 – 38.95	14
39.0 – 41.9	38.95 – 41.95	8
42.0 – 44.9	41.95 – 44.95	2
	Total	30

It should be noted that the difference between the upper class boundary and the lower class boundary of any class is equal to the class interval h = 3.
32.95 minus 29.95 is equal to 3, 35.95 minus 32.95 is equal to 3, and so on.
A key point in this entire discussion is that the class boundaries should be taken up to one decimal place more than the given data. In this way, the possibility of an observation falling exactly on the boundary is avoided. (The observed value will either be greater than or less than a particular boundary and hence will conveniently fall in its appropriate class).Next, we consider the concept of the relative frequency distribution and the percentage frequency distribution.
Next, we consider the concept of the relative frequency distribution and the percentage frequency distribution.
This concept has already been discussed when we considered the frequency distribution of a discrete variable.
Dividing each frequency of a frequency distribution by the total number of observations, we obtain the relative frequency distribution.
Multiplying each relative frequency by 100, we obtain the percentage of frequency distribution.
In this way, we obtain the relative frequencies and the percentage frequencies shown below

Class Limit	Frequency	Relative Frequency	%age Frequency
30.0 – 32.9	2	2/30 = 0.067	6.7
33.0 – 35.9	4	4/30 = 0.133	13.3
36.0 – 38.9	14	14/30 = 0.467	4.67
39.0 – 41.9	8	8/30 = 0.267	26.7
42.0 – 44.9	2	2/30 = 0.067	6.7
	30		

The term **'relative frequencies'** simply means that we are considering the frequencies of the various classes relative to the total number of observations.
The advantage of constructing a relative frequency distribution is that comparison is possible between two sets of data having similar classes.
For example, suppose that the Environment Protection Agency perform tests on two car models A and B, and obtains the frequency distributions shown below:

MILEAGE	FREQUENCY	
	Model A	Model B
30.0 – 32.9	2	7
33.0 – 35.9	4	10
36.0 – 38.9	14	16
39.0 – 41.9	8	9
42.0 – 44.9	2	8
	30	50

In order to be able to compare the performance of the two car models, we construct the relative frequency distributions in the percentage form:

MILEAGE	Model A	Model B
30.0-32.9	2/30 x 100 = 6.7	7/50 x 100 = 14
33.0-35.9	4/30 x 100 = 13.3	10/50 x 100 = 20
36.0-38.9	14/30 x 100 = 46.7	16/50 x 100 = 32
39.0-41.9	8/30 x 100 = 26.7	9/50 x 100 = 18
42.0-44.9	2/30 x 100 = 6.7	8/50 x 100 = 16

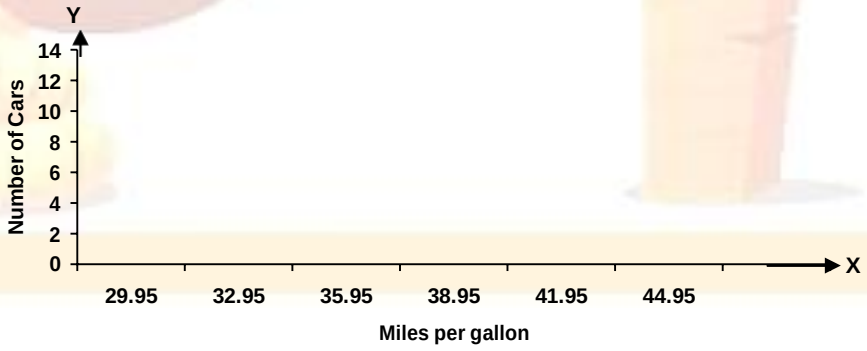
From the table it is clear that whereas 6.7% of the cars of model A fall in the mileage group 42.0 to 44.9, as many as 16% of the cars of model B fall in this group. Other comparisons can similarly be made.
Let us now turn to the visual representation of a continuous frequency distribution. In this context, we will discuss three different types of graphs i.e. the histogram, the frequency polygon, and the frequency curve.

HISTOGRAM:

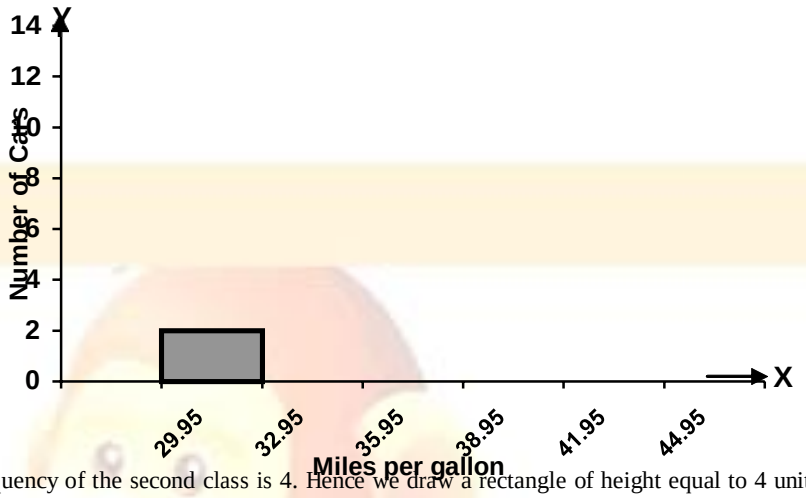
A histogram consists of a set of adjacent rectangles whose bases are marked off by class boundaries along the X-axis, and whose heights are proportional to the frequencies associated with the respective classes.
It will be recalled that, in the last lecture, we were considering the mileage ratings of the cars that had been inspected by the Environment Protection Agency. Our frequency table came out as shown below:

Class Limit	Class Boundaries	Frequency
30.0 – 32.9	29.95 – 32.95	2
33.0 – 35.9	32.95 – 35.95	4
36.0 – 38.9	35.95 – 38.95	14
39.0 – 41.9	38.95 – 41.95	8
42.0 – 44.9	41.95 – 44.95	2
	Total	30

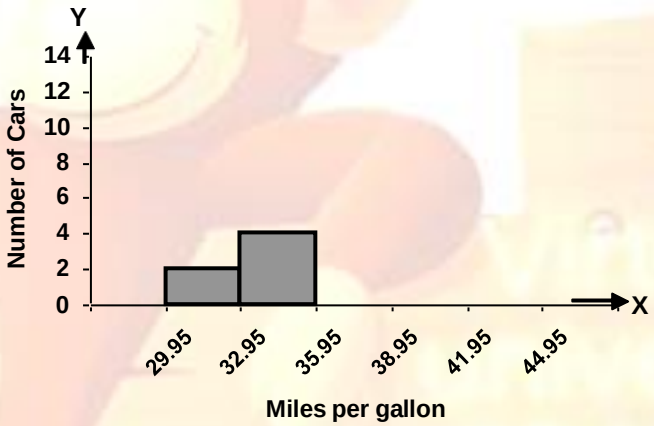
In accordance with the procedure that I just mentioned, we need to take the class boundaries along the X axis
We obtain



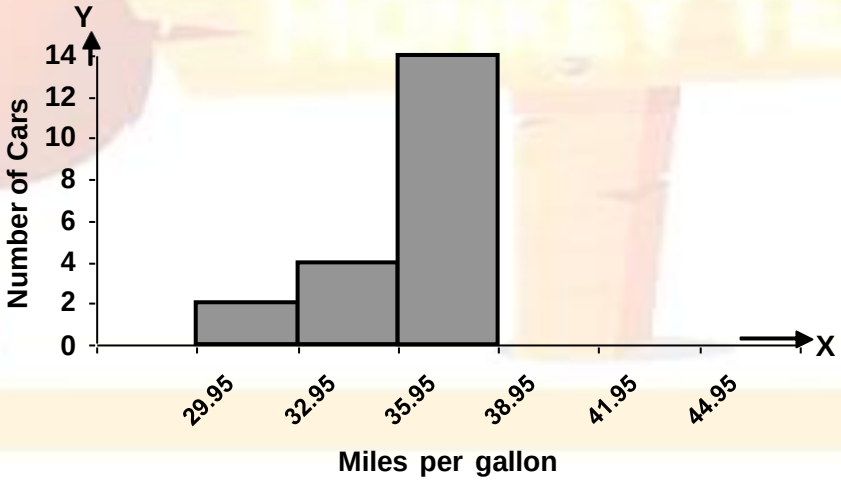
Now, as seen in the frequency table, the frequency of the first class is 2. As such, we will draw a rectangle of height equal to 2 units and obtain the following figure:



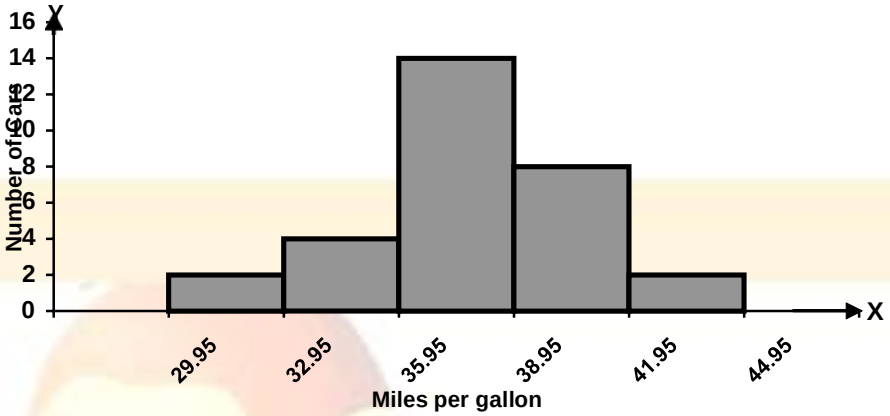
The frequency of the second class is 4. Hence we draw a rectangle of height equal to 4 units against the second class, and thus obtain the following situation:



The frequency of the third class is 14. Hence we draw a rectangle of height equal to 14 units against the third class, and thus obtain the following picture:



Continuing in this fashion, we obtain the following attractive diagram:



This diagram is known as the histogram, and it gives an indication of the overall pattern of our frequency distribution.

FREQUENCY POLYGON:

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments. In our example of the EPA mileage ratings, the classes are

Class Boundaries
29.95 – 32.95
32.95 – 35.95
35.95 – 38.95
38.95 – 41.95
41.95 – 44.95

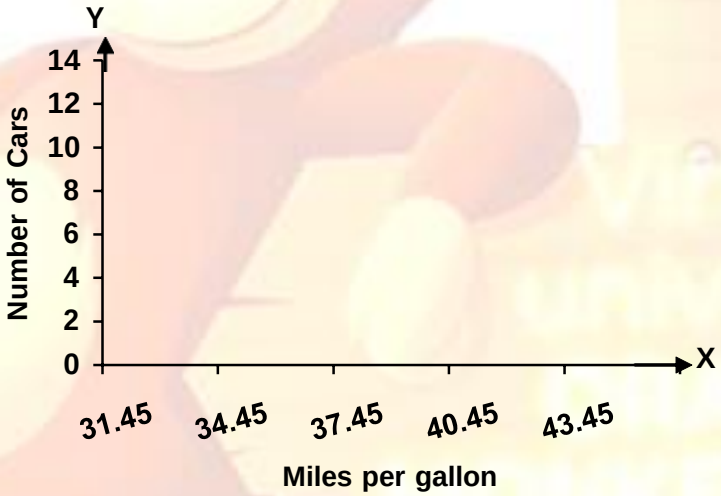
These mid-points are denoted by X.
Now let us add two classes to my frequency table, one class in the very beginning, and one class at the very end.

Class Boundaries	Mid-Point (X)	Frequency (f)
26.95 – 29.95	28.45	
29.95 – 32.95	31.45	2
32.95 – 35.95	34.45	4
35.95 – 38.95	37.45	14
38.95 – 41.95	40.45	8
41.95 – 44.95	43.45	2
44.95 – 47.95	46.45	

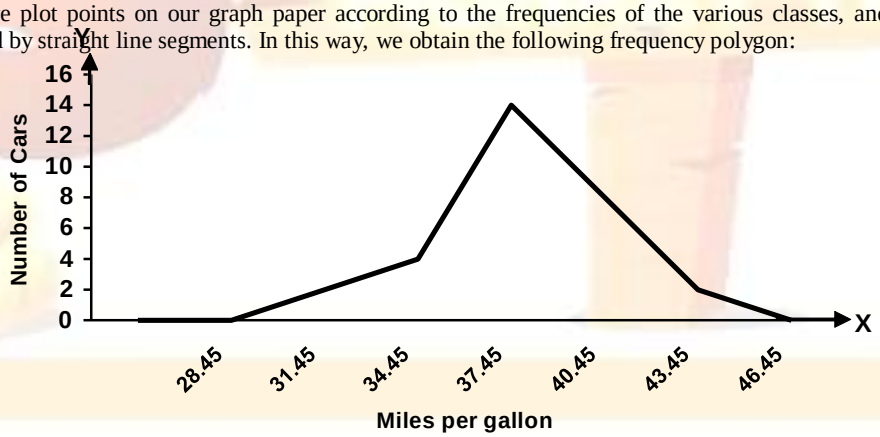
The frequency of each of these two classes is 0, as in our data set, no value falls in these classes.

Class Boundaries	Mid-Point (X)	Frequency (f)
26.95 – 29.95	28.45	0
29.95 – 32.95	31.45	2
32.95 – 35.95	34.45	4
35.95 – 38.95	37.45	14
38.95 – 41.95	40.45	8
41.95 – 44.95	43.45	2
44.95 – 47.95	46.45	0

Now, in order to construct the frequency polygon, the mid-points of the classes are taken along the X-axis and the frequencies along the Y-axis, as shown below:

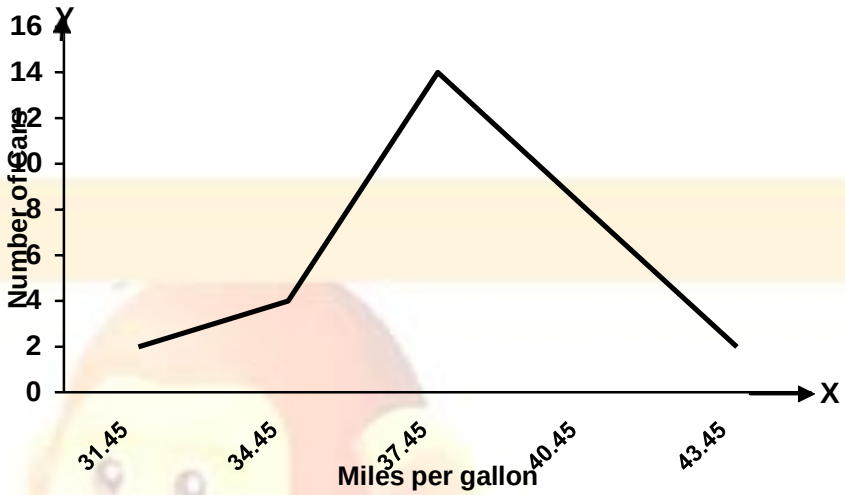


Next, we plot points on our graph paper according to the frequencies of the various classes, and join the points so obtained by straight line segments. In this way, we obtain the following frequency polygon:



It is well-known that the term 'polygon' implies a many-sided closed figure. As such, we want our frequency polygon to be a closed figure. This is exactly the reason why we added two classes to our table, each having zero frequency. Because of the frequency being zero, the line segment touches the X-axis both at the beginning and at the end, and our figure becomes a closed figure.

Had we not carried out this step, our graph would have been as follows:

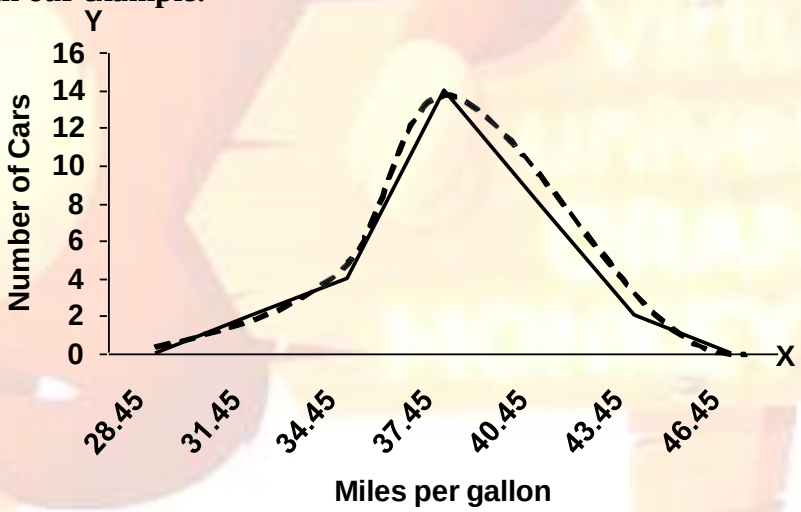


And since this graph is touching the X-axis, hence it cannot be called a frequency polygon (because it is not a closed figure)!

FREQUENCY CURVE:

When the frequency polygon is **smoothed**, we obtain what may be called the frequency curve.

In our example:



LECTURE NO. 5

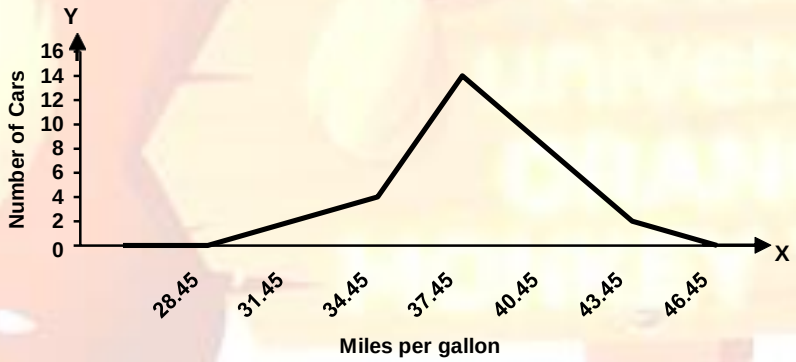
Today's lecture is in continuation with the last lecture, and today we will begin with various types of frequency curves that are encountered in practice. Also, we will discuss the cumulative frequency distribution and cumulative frequency polygon for a continuous variable.

FREQUENCY POLYGON:

A frequency polygon is obtained by plotting the class frequencies against the mid-points of the classes, and connecting the points so obtained by straight line segments. In our example of the EPA mileage ratings, the classes were:

Class Boundaries	Mid-Point (X)	Frequency (f)
26.95 – 29.95	28.45	
29.95 – 32.95	31.45	2
32.95 – 35.95	34.45	4
35.95 – 38.95	37.45	14
38.95 – 41.95	40.45	8
41.95 – 44.95	43.45	2
44.95 – 47.95	46.45	

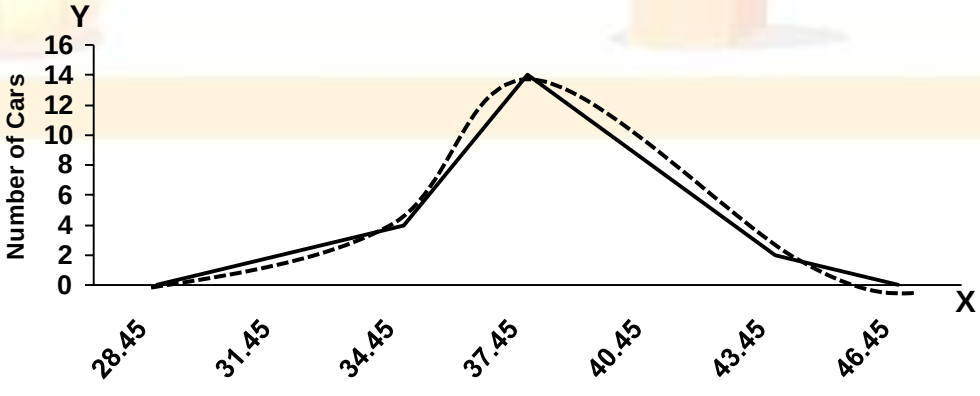
And our frequency polygon came out to be:



Also, it was mentioned that, when the frequency polygon is smoothed, we obtain what may be called the

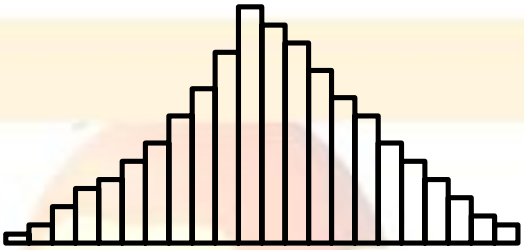
FREQUENCY CURVE

In our example:



ORANGE MONKEY TEAM

In the above figure, the dotted line represents the frequency curve. It should be noted that it is not necessary that our frequency curve must touch all the points. The purpose of the frequency curve is simply to display the overall pattern of the distribution. Hence we draw the curve by the free-hand method, and hence it does not have to touch all the plotted points. It should be realized that the frequency curve is actually a theoretical concept.
If the class interval of a histogram is made very small, and the number of classes is very large, the rectangles of the histogram will be narrow as shown below:



The smaller the class interval and the larger the number of classes, the narrower the rectangles will be. In this way, the histogram approaches a smooth curve as shown below:



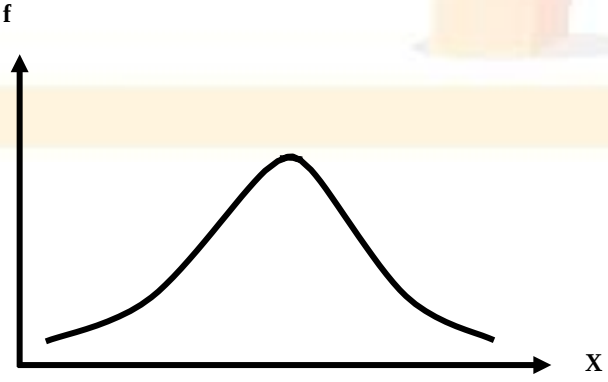
In spite of the fact that the frequency curve is a theoretical concept, it is useful in analyzing real-world problems. The reason is that very close approximations to theoretical curves are often generated in the real world so close that it is quite valid to utilize the properties of various types of mathematical curves in order to aid analysis of the real-world problem at hand.

VARIOUS TYPES OF FREQUENCY CURVES

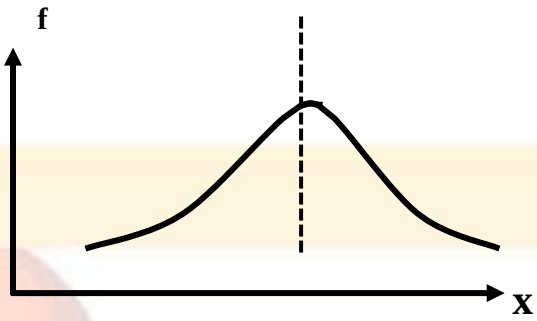
- the symmetrical frequency curve
- the moderately skewed frequency curve
- the extremely skewed frequency curve
- the U-shaped frequency curve

Let us discuss them one by one. First of all, the symmetrical frequency curve is of the following shape:

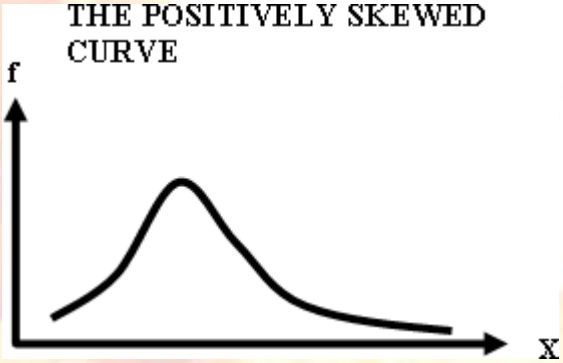
THE SYMMETRIC CURVE



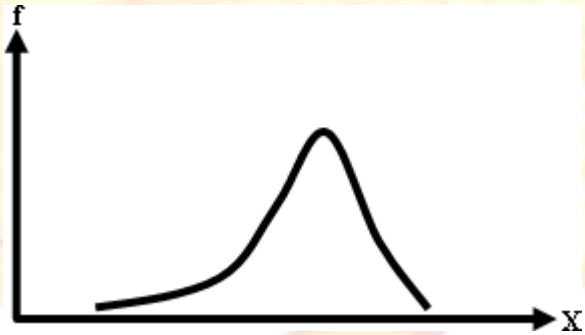
If we place a vertical mirror in the centre of this graph, the left hand side will be the mirror image of the right hand side.



Next, we consider the moderately skewed frequency curve. We have the positively skewed curve and the negatively skewed curve. The positively skewed curve is that one whose right tail is longer than its left tail, as shown below

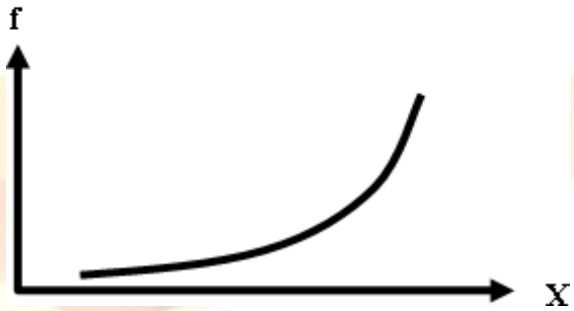


On the other hand, the negatively skewed frequency curve is the one for which the left tail is longer than the right tail.



Both of these that we have just considered are moderately positively and negatively skewed.
Sometimes, we have the extreme case when we obtain the EXTREMELY skewed frequency curve. An extremely negatively skewed curve is of the type shown below:

THE EXTREMELY NEGATIVELY SKEWED (J-SHAPED) CURVE



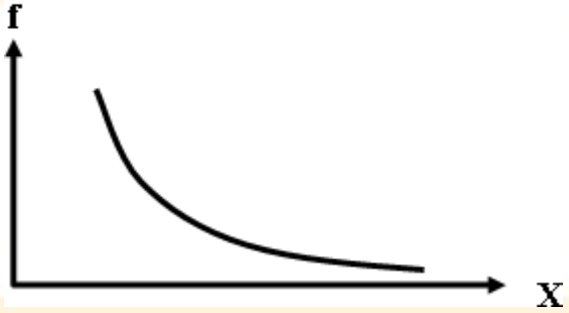
This is the case when the maximum frequency occurs at the end of the frequency table.
For example, if we think of the death rates of adult males of various age groups starting from age 20 and going up to age 79 years, we might obtain something like this:

DEATH RATES BY AGE GROUP

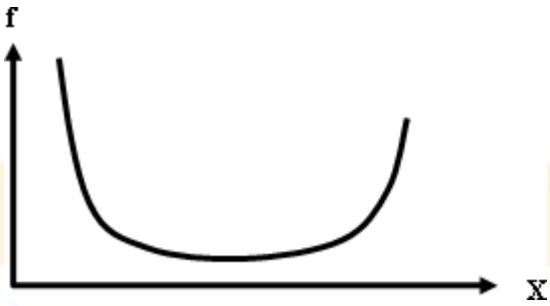
Age Group	No. of deaths per thousand
20 – 29	2.1
30 – 39	4.3
40 – 49	5.7
50 – 59	8.9
60 – 69	12.4
70 – 79	16.7

This will result in a J-shaped distribution similar to the one shown above.
Similarly, the extremely positively skewed distribution is known as the REVERSE J-shaped distribution.

THE EXTREMELY POSITIVELY SKEWED (REVERSE J-SHAPED) CURVE



A relatively LESS frequently encountered frequency distribution is the U-shaped distribution.



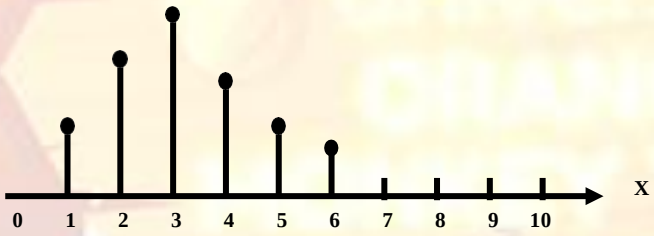
If we consider the example of the death rates not for only the adult population but for the population of ALL the age groups, we will obtain the **U-shaped distribution.**

Out of all these curves, the MOST frequently encountered frequency distribution is the moderately skewed frequency distribution. There are thousands of natural and social phenomena which yield the moderately skewed frequency distribution. Suppose that we walk into a school and collect data of the weights, heights, marks, shoulder-lengths, finger-lengths or any other such variable pertaining to the children of any one class.

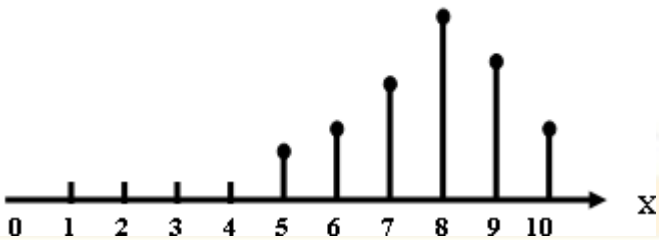
If we construct a frequency distribution of this data, and draw its histogram and its frequency curve, we will find that our data will generate a moderately skewed distribution. Until now, we have discussed the various possible shapes of the frequency distribution of a continuous variable. Similar shapes are possible for the frequency distribution of a discrete variable.

VARIOUS TYPES OF DISCRETE FREQUENCY DISTRIBUTION

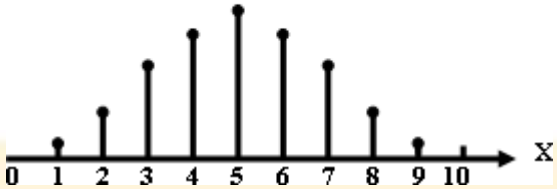
I. Positively Skewed Distribution



II. Negatively Skewed Distribution



III. Symmetric Distribution



Let us now consider another aspect of the frequency distribution i.e.

CUMULATIVE FREQUENCY DISTRIBUTION

As in the case of the frequency distribution of a discrete variable, if we start adding the frequencies of our frequency table column-wise, we obtain the column of cumulative frequencies.

In our example, we obtain the cumulative frequencies shown below:

CUMULATIVE FREQUENCY DISTRIBUTION

Class Boundaries	Frequency	Cumulative Frequency
29.95 – 32.95	2	2
32.95 – 35.95	4	2+4 = 6
35.95 – 38.95	14	6+14 = 20
38.95 – 41.95	8	20+8 = 28
41.95 – 44.95	2	28+2 = 30
	30	

In the above table, 2+4 gives 6, 6+14 gives 20, and so on.

The question arises: "What is the purpose of making this column?"

You will recall that, when we were discussing the frequency distribution of a discrete variable, any particular cumulative frequency meant that we were counting the number of observations starting from the very first value of X and going up to THAT particular value of X against which that particular cumulative frequency was falling.

In case of a the distribution of a continuous variable, each of these cumulative frequencies represents the total frequency of a frequency distribution from the lower class boundary of the lowest class to the UPPER class boundary of THAT class whose cumulative frequency we are considering.

In the above table, the total number of cars showing mileage less than 35.95 miles per gallon is 6, the total number of car showing mileage less than 41.95 miles per gallon is 28, etc.

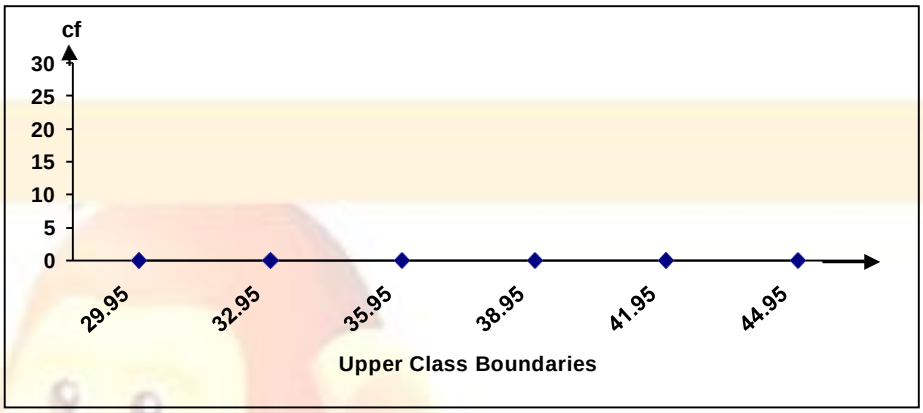
CUMULATIVE FREQUENCY DISTRIBUTION

Class Boundaries	Frequency	Cumulative Frequency
29.95 – 32.95	2	2
32.95 – 35.95	4	2+4 = 6
35.95 – 38.95	14	6+14 = 20
38.95 – 41.95	8	20+8 = 28
41.95 – 44.95	2	28+2 = 30
	30	

Such a cumulative frequency distribution is called a "**less than**" type of a cumulative frequency distribution. The graph of a cumulative frequency distribution is called a

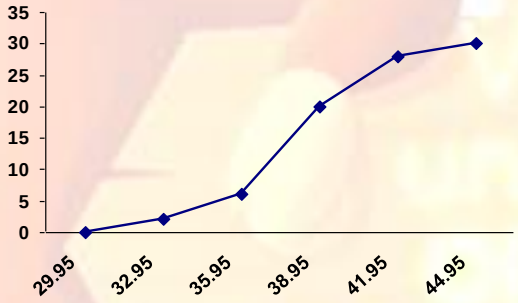
CUMULATIVE FREQUENCY POLYGON or OGIVE

A "less than" type ogive is obtained by marking off the upper class boundaries of the various classes along the X-axis and the cumulative frequencies along the y-axis, as shown below:



The cumulative frequencies are plotted on the graph paper against the upper class boundaries, and the points so obtained are joined by means of straight line segments.
Hence we obtain the cumulative frequency polygon shown below:

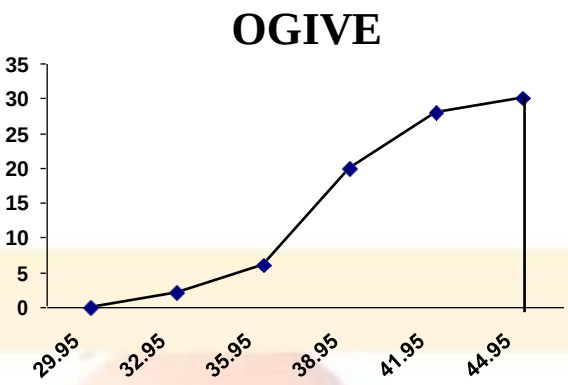
Cumulative Frequency Polygon or OGIVE



It should be noted that this graph is touching the X-Axis on the left-hand side. This is achieved by ADDING a class having zero frequency in the beginning of our frequency distribution, as shown below:

Class Boundaries	Frequency	Cumulative Frequency
26.95 – 29.95	0	0
29.95 – 32.95	2	0+2 = 2
32.95 – 35.95	4	2+4 = 6
35.95 – 38.95	14	6+14 = 20
38.95 – 41.95	8	20+8 = 28
41.95 – 44.95	2	28+2 = 30
	30	

Since the frequency of the first class is zero, hence the cumulative frequency of the first class will also be zero, and hence, automatically, the cumulative frequency polygon will touch the X-Axis from the left hand side. If we want our cumulative frequency polygon to be closed from the right-hand side also, we can achieve this by connecting the last point on our graph paper with the X-axis by means of a vertical line, as shown below:



In the example of EPA mileage ratings, all the data-values were correct to one decimal place. Let us now consider another example:

EXAMPLE:

For a sample of 40 pizza products, the following data represent cost of a slice in dollars (S Cost).

PRODUCT	S cost
Pizza Hut Hand Tossed	1.51
Domino's Deep Dish	1.53
Pizza Hut Pan Pizza	1.51
Domino's Hand Tossed	1.90
Little Caesars Pan! Pizza!	1.23

PRODUCT	S Cost
Boboli crust with Boboli sauce	1.00
Jack's Super Cheese	0.69
Pappalo's Three Cheese	0.75
Tombstone Original Extra Cheese	0.81
Master Choice Gourmet Four Cheese	0.90
Celeste Pizza For One	0.92
Totino's Party	0.64
The New Weight Watchers Extra Cheese	1.54
Jeno's Crisp'N Tasty	0.72
Stouffer's French Bread 2-Cheese	1.15

PRODUCT	S Cost
Ellio's 9-slice	0.52
Kroger	0.72
Healthy Choice French Bread	1.50
Lean Cuisine French Bread	1.49
DiGiorno Rising Crust	0.87
Tombstone Special Order	0.81
Pappalo's	0.73
Jack's New More Cheese!	0.64
Tombstone Original	0.77
Red Baron Premium	0.80

PRODUCT	Scost
Tony's Italian Style Pastry Cruse	0.83
Red Baron Deep Dish Singles	1.13
Totino's Party	0.62
The New Weight Watchers	1.52
Jeno's Crisp'N Tasty	0.71
Stouffer's French Bread	1.14
Celeste Pizza For One	1.11
Tombstone For One French Bread	1.11
Healthy Choice French Bread	1.46
Lean Cuisine French Bread	1.71

PRODUCT	Scost
Little Caesars Pizza! Pizza!	1.28
Pizza Hut Stuffed Crust	1.23
DiGiorno Rising Crust Four Cheese	0.90
Tombstone Speical Order Four Cheese	0.85
Red Baron Premium 4-Cheese	0.80

Source: "Pizza," Copyright 1997 by Consumers Union of United States, Inc., Yonkers, N.Y. 10703.

Example taken from "Business Statistics – A First Course" by Mark L. Berenson & David M. Levine (International Edition), Prentice-Hall International, Inc., Copyright © 1998.

In order to construct the frequency distribution of the above data, the first thing to note is that, in this example, all our data values are correct to two decimal places. As such, we should construct the class limits correct to TWO decimal places, and the class boundaries correct to three decimal places.

As in the last example, first of all, let us find the maximum and the minimum values in our data, and compute the RANGE.

Minimum value X0 = 0.52
Maximum value Xm = 1.90
Hence:

Range = 1.90 - 0.52 = 1.38

Desired number of classes = 8
Hence:

Class interval h = RANGE/No. of classes

= 1.38 / 8 = 0.1725 ~ 0.20

Lower limit of the first class = 0.51

Hence, our successive class limits come out to be:

Class Limits
0.51 – 0.70
0.71 – 0.90
0.91 – 1.10
1.11 – 1.30
1.31 – 1.50
1.51 – 1.70
1.71 – 1.90

Stretching the class limits to the left and to the right, we obtain class boundaries as shown below:

Class Limits	Class Boundaries
0.51 – 0.70	0.505 – 0.705
0.71 – 0.90	0.705 – 0.905
0.91 – 1.10	0.905 – 1.105
1.11 – 1.30	1.105 – 1.305
1.31 – 1.50	1.305 – 1.505
1.51 – 1.70	1.505 – 1.705
1.71 – 1.90	1.705 – 1.905

By tallying the data-values in the appropriate classes, we will obtain a frequency distribution similar to the one that we obtained in the examples of the EPA mileage ratings.
By constructing the histogram of this data-set, we will be able to decide whether our distribution is symmetric, positively skewed or negatively skewed. This may please be attempted as an exercise.

LECTURE NO. 6

This plot was introduced by the famous statistician *John Tukey in 1977*. A frequency table has the disadvantage that the identity of individual observations is lost in grouping process. To overcome this drawback, John Tukey (1977) introduced this particular technique (**known as the Stem-and-Leaf Display).**

This technique offers a quick and novel way for simultaneously sorting and displaying data sets where each number in the data set is divided into two parts, a Stem and a Leaf.

A stem is the leading digit(s) of each number and is used in sorting, while a leaf is the rest of the number or the trailing digit(s) and shown in display. A vertical line separates the leaf (or leaves) from the stem.

For example, the number 243 could be split in two ways:

Leading Digit	Trailing Digits
2	43
Stem	Leaf

OR

Leading Digit	Trailing Digit
24	3
Stem	Leaf

How do we construct a stem and leaf display when we have a whole set of values? This is explained by way of the following example:

EXAMPLE:

The ages of 30 patients admitted to a certain hospital during a particular week were as follows:

48, 31, 54, 37, 18, 64, 61, 43, 40, 71, 51, 12, 52, 65, 53, 42, 39, 62, 74, 48, 29, 67, 30, 49, 68, 35, 57, 26, 27, 58.

Construct a stem-and-leaf display from the data and list the data in an array.

A scan of the data indicates that the observations range (in age) from 12 to 74. We use the first (or leading) digit as the stem and the second (or trailing) digit as the leaf. The first observation is 48, which has a stem of 4 and a leaf of 8, the second a stem of 3 and a leaf of 1, etc. Placing the leaves in the order in which they APPEAR in the data, we get the stem-and-leaf display as shown below:

Stem (Leading Digit)	Leaf (Trailing Digit)
1	8 2
2	9 6 7
3	1 7 9 0 5
4	8 3 0 2 8 9
5	4 1 2 3 7 8
6	4 1 5 2 7 8
7	1 4

But it is a common practice to ARRANGE the trailing digits in each row from smallest to highest. In this example, in order to obtain an array, we associate the leaves in order of size with the stems as shown below:

DATA IN THE FORM OF AN ARRAY (in ascending order):

12, 18, 26, 27, 29, 30, 31, 35, 37, 39, 40, 42, 43, 48, 48, 49, 51, 52, 53, 54, 57, 58, 61, 62, 64, 65, 67, 68, 71, 74.

Hence we obtain the stem and leaf plot shown below:

STEM AND LEAF DISPLAY

Stem (Leading Digit)	Leaf (Trailing Digit)
1	2 8
2	6 7 9
3	0 1 5 7 9
4	0 2 3 8 8 9
5	1 2 3 4 7 8
6	1 2 4 5 7 8
7	1 4

The stem-and-leaf table provides a useful description of the data set and, if we so desire, can easily be converted to a frequency table. In this example, the frequency of the class 10-19 is 2, the frequency of the class 20-29 is 3, and the frequency of the class 30-39 is 5, and so on.

STEM AND LEAF DISPLAY

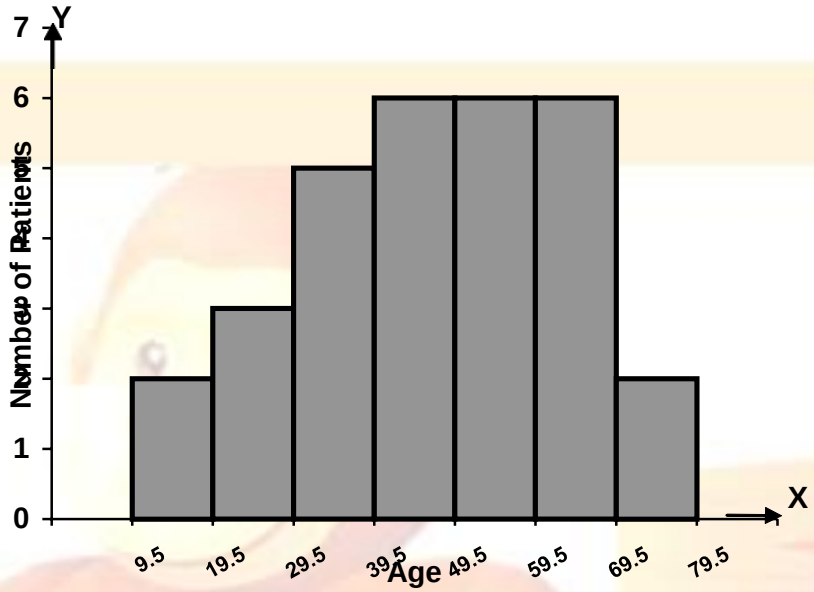
Stem (Leading Digit)	Leaf (Trailing Digit)
1	2 8
2	6 7 9
3	0 1 5 7 9
4	0 2 3 8 8 9
5	1 2 3 4 7 8
6	1 2 4 5 7 8
7	1 4

Hence, this stem and leaf plot conveniently converts into the frequency distribution shown below:

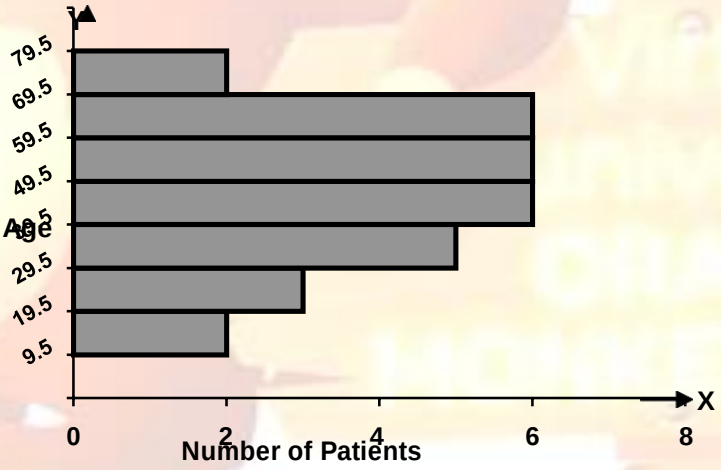
FREQUENCY DISTRIBUTION

Class Limits	Class Boundaries	Tally Marks	Frequency
10 – 19	9.5 – 19.5	//	2
20 – 29	19.5 – 29.5	///	3
30 – 39	29.5 – 39.5	~~///~~	5
40 – 49	39.5 – 49.5	~~///~~ /	6
50 – 59	49.5 – 59.5	~~///~~ /	6
60 – 69	59.5 – 69.5	~~///~~ /	6
70 - 79	69.5 – 79.5	//	2

Converting this frequency **distribution into a histogram,** we obtain:



If we rotate this histogram by 90 degrees, we will obtain:



Let us re-consider the stem and leaf plot that we obtained a short while ago.

STEM AND LEAF DISPLAY

Stem (Leading Digit)	Leaf (Trailing Digit)
7	1 4
6	1 2 4 5 7 8
5	1 2 3 4 7 8
4	0 2 3 8 8 9
3	0 1 5 7 9
2	6 7 9
1	2 8

It is noteworthy that the shape of the stem and leaf display is exactly like the shape of our histogram. Let us now consider another example.

EXAMPLE

Construct a stem-and-leaf display for the data of mean annual death rates per thousand at ages 20-65 given below:
7.5, 8.2, 7.2, 8.9, 7.8, 5.4, 9.4, 9.9, 10.9, 10.8, 7.4, 9.7, 11.6, 12.6, 5.0, 10.2, 9.2, 12.0, 9.9, 7.3, 7.3, 8.4, 10.3, 10.1, 10.0, 11.1, 6.5, 12.5, 7.8, 6.5, 8.7, 9.3, 12.4, 10.6, 9.1, 9.7, 9.3, 6.2, 10.3, 6.6, 7.4, 8.6, 7.7, 9.4, 7.7, 12.8, 8.7, 5.5, 8.6, 9.6, 11.9, 10.4, 7.8, 7.6, 12.1, 4.6, 14.0, 8.1, 11.4, 10.6, 11.6, 10.4, 8.1, 4.6, 6.6, 12.8, 6.8, 7.1, 6.6, 8.8, 8.8, 10.7, 10.8, 6.0, 7.9, 7.3, 9.3, 9.3, 8.9, 10.1, 3.9, 6.0, 6.9, 9.0, 8.8, 9.4, 11.4, 10.9

Using the decimal part in each number as the leaf and the rest of the digits as the stem, we get the ordered stem-and-leaf display shown below:

STEM AND LEAF DISPLAY

Stem	Leaf
3	9
4	6 6
5	0 4 5
6	0 0 2 2 5 5 6 6 6 8 9
7	1 3 3 3 4 4 5 6 7 7 8 8 8 9
8	1 1 2 4 6 6 7 7 8 8 8 9 9
9	0 1 2 3 3 3 3 4 4 4 6 7 7 9 9
10	0 1 1 2 3 3 4 4 6 6 7 8 8 9 9
11	1 4 4 6 6 9
12	0 1 4 5 6 8 8
14	0

EXERCISE

- The above data may be converted into a stem and leaf plot (so as to verify that the one shown above is correct).
- Various variations of the stem and leaf display may be studied on your own.

The next concept that we are going to consider is the concept of the central tendency of a data-set. In this context, the first thing to note is that in any data-based study, our data is always going to be variable, and hence, first of all, we will need to describe the data that is available to us.

DESCRIPTION OF VARIABLE DATA:

Regarding any statistical enquiry, primarily we need some means of describing the situation with which we are confronted. A concise numerical description is often preferable to a lengthy tabulation, and if this form of description also enables us to form a mental image of the data and interpret its significance, so much the better.

MEASURES OF CENTRAL TENDENCY AND MEASURES OF DISPERSION

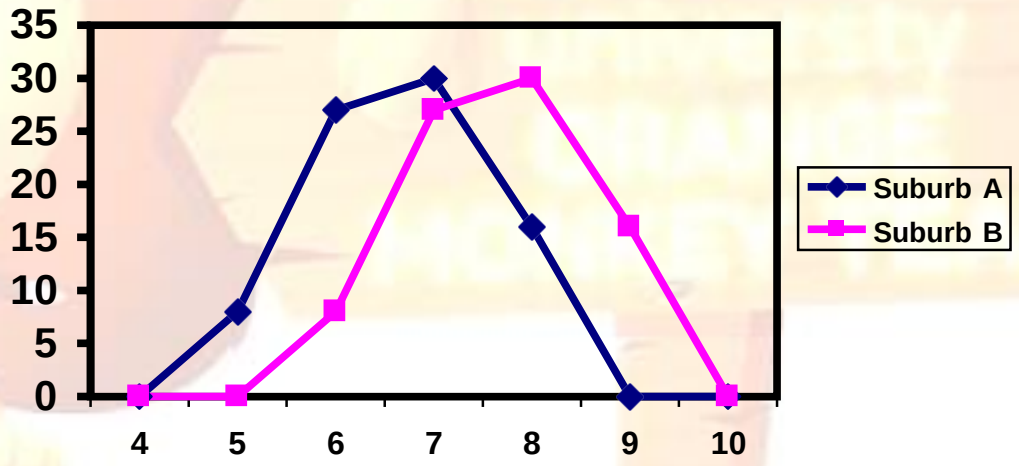
- Averages enable us to measure the central tendency of variable data
- Measures of dispersion enable us to measure its variability.

AVERAGES (I.E. MEASURES OF CENTRAL TENDENCY)

An average is a single value which is intended to represent a set of data or a distribution as a whole. It is more or less CENTRAL value ROUND which the observations in the set of data or distribution usually tend to cluster.
As a measure of central tendency (i.e. an average) indicates the location or general position of the distribution on the X-axis, it is also known as a measure of location or position.
Let us consider an example: Suppose that we have the following two frequency distributions:

EXAMPLE:

Looking at these two frequency distributions, we should ask ourselves what exactly is the distinguishing feature?
If we draw the frequency polygon of the two frequency distributions, we obtain



Inspection of these frequency polygons shows that they have exactly the same shape. It is their position relative to the horizontal axis (X-axis) which distinguishes them.
If we compute the mean number of rooms per house for each of the two suburbs, we will find that the average number of rooms per house in A is 6.67 while in B it is 7.67.
This difference of 1 is equivalent to the difference in position of the two frequency polygons.

Our interpretation of the above situation would be that there are LARGER houses in suburb B than in suburb A, to the extent that there are on the *average.*

VARIOUS TYPES OF AVERAGES:

There are several types of averages each of which has a use in specifically defined circumstances.
The most common types of averages are:

- The arithmetic mean,
- The geometric mean,

- The harmonic mean
- The median, and
- The mode

The ***Arithmetic, Geometric and Harmonic means*** are averages that are mathematical in character, and give an indication of the magnitude of the observed values.
The ***Median*** indicates the middle position while the mode provides information about the most frequent value in the distribution or the set of data. THE MODE:
The ***Mode*** is defined as that value which occurs most frequently in a set of data i.e. it indicates the most common result.

EXAMPLE:

Suppose that the marks of eight students in a particular test are as follows:

2, 7, 9, 5, 8, 9, 10, 9

Obviously, the most common mark is 9. In other words, Mode = 9.

MODE IN CASE OF RAW DATA PERTAINING TO A CONTINUOUS VARIABLE

In case of a set of values (pertaining to a continuous variable) that have not been grouped into a frequency distribution (i.e. in case of raw data pertaining to a continuous variable), the mode is obtained by counting the number of times each value occurs.

EXAMPLE:

Suppose that the government of a country collected data regarding the percentages of revenues spent on Research and Development by 49 different companies, and obtained the following figures:

Percentage of Revenues Spent on Research and Development

Company	Percentage	Company	Percentage
1	13.5	14	9.5
2	8.4	15	8.1
3	10.5	16	13.5
4	9.0	17	9.9
5	9.2	18	6.9
6	9.7	19	7.5
7	6.6	20	11.1
8	10.6	21	8.2
9	10.1	22	8.0
10	7.1	23	7.7
11	8.0	24	7.4
12	7.9	25	6.5
13	6.8	26	9.5

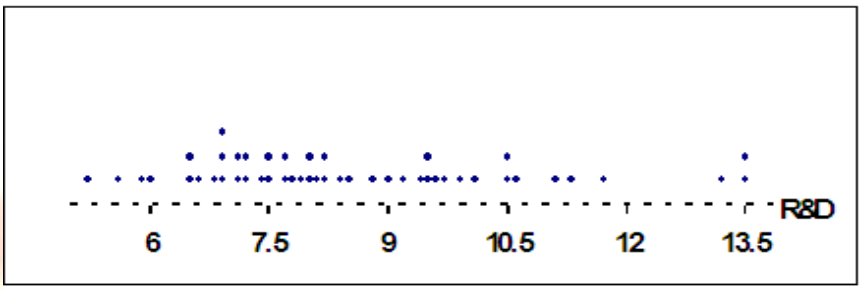
Company	Percentage	Company	Percentage
27	8.2	39	6.5
28	6.9	40	7.5
29	7.2	41	7.1
30	8.2	42	13.2
31	9.6	43	7.7
32	7.2	44	5.9
33	8.8	45	5.2
34	11.3	46	5.6
35	8.5	47	11.7
36	9.4	48	6.0
37	10.5	49	7.8
38	6.9		

We can represent this data by means of a plot that is called dot plot.

DOT PLOT:

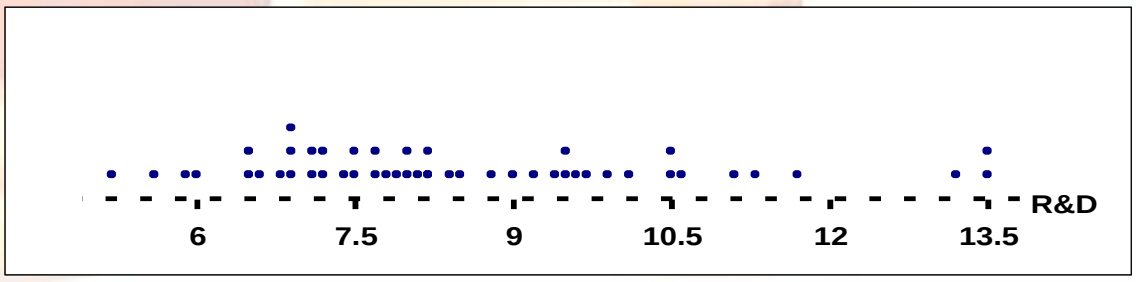
The horizontal axis of a dot plot contains a scale for the quantitative variable that we want to represent. The numerical value of each measurement in the data set is located on the horizontal scale by a dot. When data values repeat, the dots are placed above one another, forming a pile at that particular numerical location.
In this example

Dot Plot

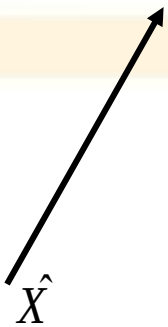


The above material has been taken from "Statistics for Business and Economics" by James T. McClave, P. George Benson and Terry Sincich (Seventh Edition), © 1998 Prentice-Hal International, **Inc.**

As is obvious from the above diagram, the value 6.9 occurs 3 times whereas all the other values are occurring either once or twice.
Hence the modal value is 6.9.



Dot Plot



= 6.9

Also, this dot plot shows that

- almost all of the R&D percentages are falling between 6% and 12%,
- most of the percentages are falling between 7% and 9%.

THE MODE IN CASE OF A DISCRETE FREQUENCY DISTRIBUTION:

In case of a discrete frequency distribution, identification of the mode is immediate; one simply finds that value which has the highest frequency.

EXAMPLE:

An airline found the following numbers of passengers in fifty flights of a forty-seated plane

No. of Passengers X	No. of Flights f
28	1
33	1
34	2
35	3
36	5
37	7
38	10
39	13
40	8
Total	**50**

Highest Frequency fm = 13
Occurs against the X value 39
Hence: Mode = x= 39
The mode is obviously 39 passengers and the company should be quite satisfied that a 40 seater is the correct-size aircraft for this particular route.

THE MODE IN CASE OF THE *FREQUENCY DISTRIBUTION* OF A CONTINUOUS VARIABLE

In case of grouped data, the modal group is easily recognizable (the one that has the highest frequency).
At what point within the modal group does the mode lie?
The answer is contained in the following formula:

Mode:

$$\hat{X} = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$$

Where

l = lower class boundary of the modal class,

f_m = frequency of the modal class,

f_1 = frequency of the class preceding the modal class

f_2 = frequency of the class following modal

class

h = length of class interval of the modal class

Going back to the example of EPA mileage ratings, we have:

EPA MILEAGE RATINGS

Mileage Rating	Class Boundaries	No. of Cars
30.0 – 32.9	29.95 – 32.95	2
33.0 – 35.9	32.95 – 35.95	4 = f_1
36.0 – 38.9	35.95 – 38.95	14 = f_m
39.0 – 41.9	38.95 – 41.95	8 = f_2
42.0 – 44.9	41.95 – 44.95	2

It is evident that the third class is the modal class. The mode lies somewhere between 35.95 and 38.95.
In order to apply the formula for the mode, we note that fm = 14, f1 = 4 and f2 = 8.
Hence we obtain:

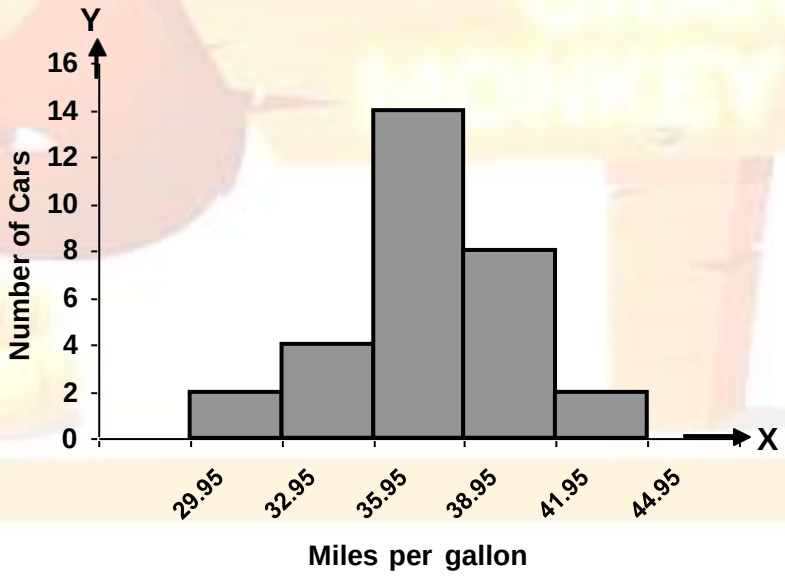
$$\hat{X} = 35.95 + \frac{14-4}{(14-4)+(14-8)} \times 3$$

$$= 35.95 + \frac{10}{10+6} \times 3$$

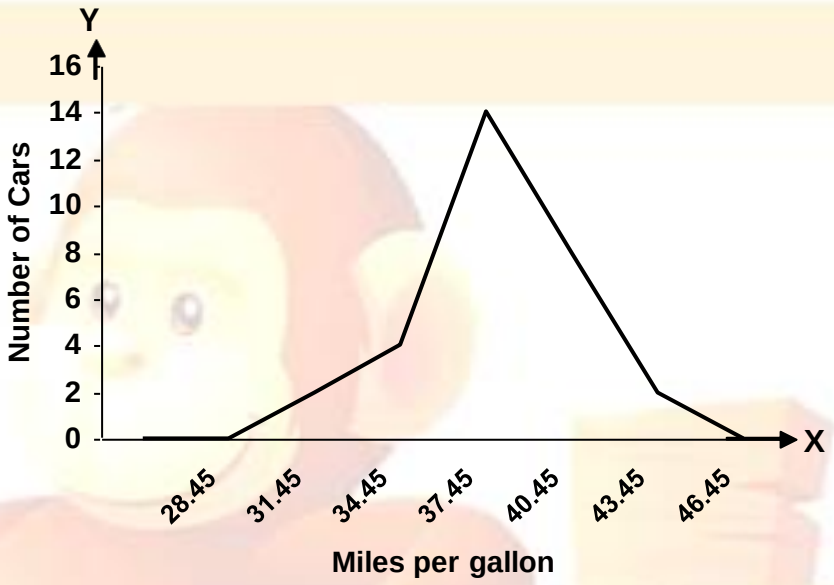
$$= 35.95 + 1.875$$

$$= 37.825$$

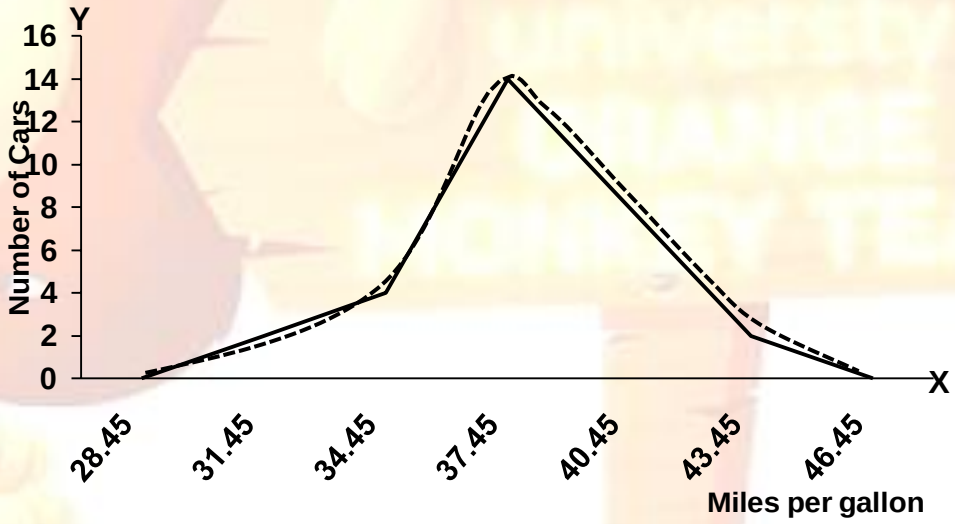
Let us now perceive the mode by considering the graphical representation of our frequency distribution. You will recall that, for the example of EPA Mileage Ratings, the histogram was as shown below:



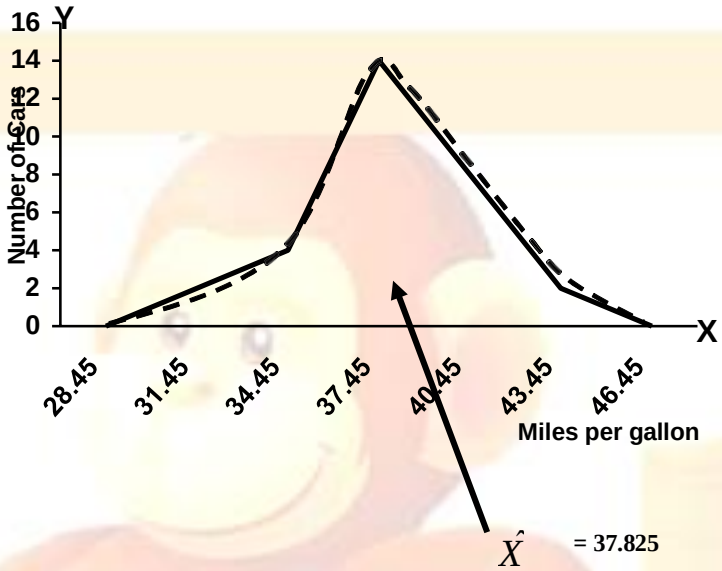
The frequency polygon of the same distribution was:



And the frequency curve was as indicated by the dotted line in the following figure:



In this example, the mode is 37.825, and if we locate this value on the X-axis, we obtain the following picture:



Since, in most of the situations the mode exists somewhere in the middle of our data-values, hence it is thought of as a measure of central tendency.

LECTURE NO. 7

In general, it was noted that, for most of the frequency distributions, the mode lies somewhere in the middle of our frequency distribution, and hence is eligible to be called a measure of central tendency.
The mode has some very desirable properties.

DESIRABLE PROPERTIES OF THE MODE:

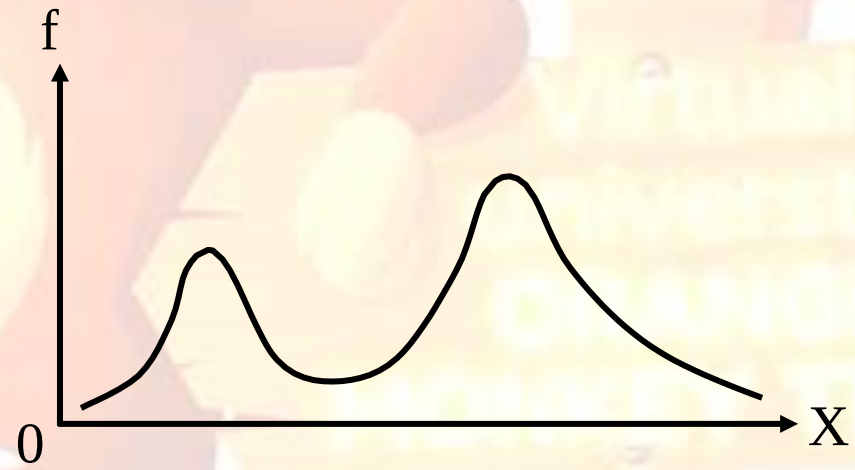
- The mode is easily understood and easily ascertained in case of a discrete frequency distribution.
- It is not affected by a few very high or low values.

The question arises, "When should we use the mode?" The answer to this question is that the mode is a valuable concept in certain situations such as the one described below:

Suppose the manager of a men's clothing store is asked about the average size of hats sold. He will probably think not of the arithmetic or geometric mean size, or indeed the median size. Instead, he will in all likelihood quote that particular size which is sold ***most often.*** This average is of far more use to him as a businessman than the arithmetic mean, geometric mean or the median. The modal size of all clothing is the size which the businessman must stock in the greatest quantity and variety in comparison with other sizes. Indeed, in most inventory (stock level) problems, one needs the mode more often than any other measure of central tendency. It should be noted that in some situations there may be no mode in a simple series where no value occurs more than once.

On the other hand, sometimes a frequency distribution contains two modes in which case it is called a bi-modal distribution as shown below:

THE BI-MODAL FREQUENCY DISTRIBUTION



The next measure of central tendency to be discussed is the arithmetic mean.

THE ARITHMETIC MEAN

The arithmetic mean is the statistician's term for what the layman knows as the average. It can be thought of as that value of the variable series which is *numerically* MOST representative of the whole series. Certainly, this is the most widely used average in statistics. Easiest In addition, it is probably the to calculate.
Its formal definition is:
"The arithmetic mean or simply the mean is a value obtained by dividing the sum of all the observations by their number."

$$\overline{X} = \frac{\text{Sum of all the observations}}{\text{Number of the observations}}$$

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Where n represents the number of observations in the sample that has been the ith observation in the sample (i = 1, 2, 3, …, n), and represents the mean of the sample.
For simplicity, the above formula can be written as

$$\bar{X} = \frac{\sum X}{n}$$

In other words, it is not necessary to insert the subscript 'i'.)

EXAMPLE:

Information regarding the receipts of a news agent for seven days of a particular week are given below

Day	Receipt of News Agent
Monday	£ 9.90
Tuesday	£ 7.75
Wednesday	£ 19.50
Thursday	£ 32.75
Friday	£ 63.75
Saturday	£ 75.50
Sunday	£ 50.70
Week Total	£ 259.85

Mean sales per day in this week:

= £ 259.85/7 = £ 37.12 (To the nearest penny).

INTERPRETATION:

The mean, £ 37.12, represents the amount (in pounds sterling) that would have been obtained on each day if the same amount were to be obtained on each day. The above example pertained to the computation of the arithmetic mean in case of ungrouped data i.e. raw data.

Let us now consider the case of data that has been grouped into a frequency distribution. When data pertaining to a continuous variable has been grouped into a frequency distribution, the frequency distribution is used to calculate the approximate values of descriptive measures --- as the identity of the observations is lost.
To calculate the approximate value of the mean, the observations in each class are assumed to be identical with the class midpoint Xi.
The mid-point of every class is known as its class-mark. In other words, the midpoint of a class 'marks' that class. As was just mentioned, the observations in each class are assumed to be identical with the midpoint i.e. the class-mark. (This is based on the assumption that the observations in the group are evenly scattered between the two extremes of the class interval).

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FREQUENCY DISTRIBUTION

Mid Point X	Frequency f
X_1	f_1
X_2	f_2
X_3	f_3
:	:
:	:
:	:
X_k	f_k

In case of a frequency distribution, the arithmetic mean is defined as:
ARITHMETIC MEAN

$$\bar{X} = \frac{\sum_{i=1}^{k} f_i X_i}{\sum_{i=1}^{k} f_i} = \frac{\sum_{i=1}^{k} f_i X_i}{n}$$

For simplicity, the above formula can be written as

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{\sum fX}{n}$$ (The subscript 'i' can be dropped.)

Let us understand this point with the help of an example:
Going back to the example of EPA mileage ratings, that we dealt with when discussing the formation of a frequency distribution. The frequency distribution that we obtained was:

EPA MILEAGE RATINGS OF 30 CARS OF A CERTAIN MODEL

Class (Mileage Rating)	Frequency (No. of Cars)
30.0 – 32.9	2
33.0 – 35.9	4
36.0 – 38.9	14
39.0 – 41.9	8
42.0 – 44.9	2
Total	30

The first step is to compute the mid-point of every class.
(You will recall that the concept of the mid-point has already been discussed in an earlier lecture.)

CLASS-MARK (MID-POINT):

The mid-point of each class is obtained by adding the sum of the two limits of the class and dividing by 2.
Hence, in this example, our mid-points are computed in this manner:

30.0 plus 32.9 divided by 2 is equal to 31.45,
33.0 plus 35.9 divided by 2 is equal to 34.45,

And so on.

Class (Mileage Rating)	Class-mark (Midpoint) X
30.0 – 32.9	31.45
33.0 – 35.9	34.45
36.0 – 38.9	37.45
39.0 – 41.9	40.45
42.0 – 44.9	43.45

In order to compute the arithmetic mean, we first need to construct the column of fX, as shown below:

Class-mark (Midpoint) X	Frequency f	fX
31.45	2	62.9
34.45	4	137.8
37.45	14	524.3
40.45	8	323.6
43.45	2	86.9
	30	1135.5

Applying the formula

$$\bar{X} = \frac{\sum fX}{\sum f},$$

We obtain

$$\bar{X} = \frac{1135.5}{30} = 37.85$$

INTERPRETATION:

The average mileage rating of the 30 cars tested by the Environmental Protection Agency is 37.85 – on the average, these cars run 37.85 miles per gallon. An important concept to be discussed at this point is the concept of grouping error.

GROUPING ERROR:

"Grouping error" refers to the error that is introduced by the assumption that all the values falling in a class are equal to the mid-point of the class interval. In reality, it is highly improbable to have a class for which all the values lying in that class are equal to the mid-point of that class. This is why the mean that we calculate from a frequency distribution does not give exactly the same answer as what we would get by computing the mean of our raw data.

As indicated earlier, a frequency distribution is used to calculate the approximate values of various descriptive measures.(The word 'approximate' is being used because of the grouping error that was just discussed.) This grouping error arises in the computation of many descriptive measures such as the geometric mean, harmonic mean, mean deviation and standard deviation. But, experience has shown that in the calculation of the arithmetic mean, this error is usually small and never serious. Only a slight difference occurs between the true answer that we would get from the raw data, and the answer that we get from the data that has been grouped in the form of a frequency distribution.

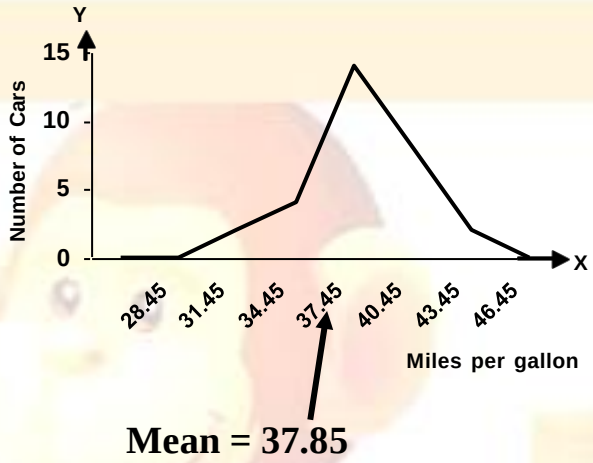
In this example, if we calculate the arithmetic mean directly from the 30 EPA mileage ratings, we obtain:

Arithmetic mean computed from raw data of the EPA mileage ratings:

$$\bar{X} = \frac{36.3 + 30.1 + \ldots\ldots + 33.9 + 39.8}{30}$$

$$= \frac{1134.7}{30} = 37.82$$

The difference between the true value of i.e. 37.82 and the value obtained from the frequency distribution i.e. 37.85 is indeed very slight. The arithmetic mean is predominantly used as a measure of central tendency.
The question is, "Why is it that the arithmetic mean is known as a measure of central tendency?"
The answer to this question is that we have just obtained i.e. 37.85 falls more or less in the centre of our frequency distribution.



As indicated earlier, the arithmetic mean is predominantly used as a measure of central tendency.
It has many desirable properties:

DESIRABLE PROPERTIES OF THE ARITHMETIC MEAN

- Best understood average in statistics.
- Relatively easy to calculate
- Takes into account every value in the series.

But there is one limitation to the use of the arithmetic mean:
As we are aware, every value in a data-set is included in the calculation of the mean, whether the value be high or low. Where there are a few very high or very low values in the series, their effect can be to *drag* the arithmetic mean towards them. This may make the mean unrepresentative.

EXAMPLE:

Example of the Case Where the Arithmetic Mean Is Not a Proper Representative of the Data:
Suppose one walks down the main street of a large city centre and counts the number of floors in each building.
Suppose, the following answers are obtained:
5, 4, 3, 4, 5, 4, 3, 4, 5, 20, 5, 6, 32, 8, 27
The mean number of floors is 9 even though 12 out of 15 of the buildings have 6 floors or less.
The three skyscraper blocks are having a disproportionate effect on the arithmetic mean (Some other average in this case would be more representative). The concept that we just considered was the concept of the simple arithmetic mean.
Let us now discuss the concept of the weighted arithmetic mean.
Consider the following example:

EXAMPLE:

Suppose that in a particular high school, there are:-

100	–	freshmen
80	–	sophomores
70	–	juniors
50	–	seniors

And suppose that on a given day, 15% of freshmen, 5% of sophomores, 10% of juniors, 2% of seniors are *absent*.
The problem is that: What percentage of students is absent for the school as a whole on that particular day?
Now a student is likely to attempt to find the answer by adding the percentages and dividing by 4
i.e.

$$\frac{15+5+10+2}{4}=\frac{32}{4}=8$$

But the fact of the matter is that the above calculation gives a wrong answer. In order to figure out why this is a wrong calculation, consider the following: As we have already noted, 15% of the freshmen are absent on this particular day. Since, in all, there are 100 freshmen in the school, hence the total number of freshmen who are absent is also 15.

But as far as the sophomores are concerned, the total number of them in the school is 80, and if 5% of them are absent on this particular day, this means that the total number of sophomores who are absent is only 4. Proceeding in this manner, we obtain the following table.

Category of Student	Number of Students in the school	Number of Students who are absent
Freshman	100	15
Sophomore	80	4
Junior	70	7
Senior	50	1
TOTAL	300	27

Dividing the total number of students who are absent by the total number of students enrolled in the school, and multiplying by 100, we obtain:

$$\frac{27}{300} \times 100 = 9$$

Thus its very clear that previous result was not correct. This situation leads us to a very important observation, i.e. here our figures pertaining to absenteeism in various categories of students cannot be regarded as having equal weightage.

When we have such a situation, the concept of "weighing" applies i.e. every data value in the data set is assigned a certain weight according to a suitable criterion. In this way, we will have a weighted series of data instead of an un-weighted one. In this example, the number of students enrolled in each category acts as the weight for the number of absences pertaining to that category i.e.

Category of Student	Percentage of Students who are absent X_i	Number of students enrolled in the school (Weights) W_i	W_iX_i (Weighted X_i)
Freshman	15	100	$100 \times 15 = 1500$
Sophomore	5	80	$80 \times 5 = 400$
Junior	10	70	$70 \times 10 = 700$
Senior	2	50	$50 \times 2 = 100$
	Total	$\Sigma W_i = 300$	$\Sigma W_iX_i = 2700$

The formula for the weighted arithmetic mean is:

WEIGHTED MEAN

$$\overline{X}_W = \frac{\sum W_i X_i}{\sum W_i}$$

And, in this example, the weighted mean is equal to:

$$\bar{X}_w = \frac{\sum W_i X_i}{\sum W_i}$$

$$= \frac{2700}{300}$$

$$= 9$$

Thus we note that, in this example, the weighted mean yields exactly the same as the answer that we obtained earlier.

As obvious, the weighing process leads us to a correct answer under the situation where we have data that cannot be regarded as being such that each value should be given equal weightage.

An important point to note here is the *criterion* for assigning weights. Weights can be assigned in a number of ways depending on the situation and the problem domain.

The next measure of central tendency that we will discuss is the median.

Let us understand this concept with the help of an example.

Let us return to the problem of the 'average' number of floors in the buildings at the centre of a city. We saw that the arithmetic mean was distorted towards the few extremely high values in this series and became unrepresentative.

We could more appropriately and easily employ the median as the 'average' in these circumstances.

MEDIAN

The median is the middle value of the series when the variable values are placed in order of magnitude.

The median is defined as a "value which divides a set of data into two halves, one half comprising of observations greater than and the other half smaller than it. More precisely, the median is a value at or below which 50% of the data lie."

The median value can be ascertained by inspection in many series. For instance, in this very example, the data that we obtained was:

EXAMPLE-1

The average number of floors in the buildings at the centre of a city:

5, 4, 3, 4, 5, 4, 3, 4, 5, 20, 5, 6, 32, 8, 27

Arranging these values in ascending order, we obtain

3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 6, 8, 20, 27, 32

Picking up the middle value, we obtain the median equal to 5.

INTERPRETATION

The median number of floors is 5. Out of those 15 buildings, 7 have unto 5 floors and 7 have 5 floors or more. We noticed earlier that the arithmetic mean was distorted toward the few extremely high values in the series and hence became unrepresentative. The median = 5 is much more representative of this series.

Height of buildings (number of floors)	
3	
3	
4	
4	7 lower
4	
5	
5	
5 = median height	
5	
5	
6	
8	7 higher
20	
27	
32	

EXAMPLE 2

Retail price of motor-car (£) (several makes and sizes)	
415	
480	4 above
525	
608	
719	= median price
1,090	
2,059	4 above
4,000	
6,000	

A slight complication arises when there are even numbers of observations in the series, for now there are two middle values.
The expedient of taking the arithmetic mean of the two is adopted as explained below:

EXAMPLE-3

Number of passengers travelling on a bus at six Different times during the day	
4	
9	
14	= median value
18	
23	
47	
Median = $\frac{14+18}{2}$ = 16 passengers	

EXAMPLE -4:

The number of passengers traveling on a bus at six different times during a day is as follows:
5, 14, 47, 34, 18, 23
Find the median.

Solution:
Arranging the values in ascending order, we obtain
5, 14, 18, 23, 34, 47
As before, a slight complication has arisen because of the fact that there are even numbers of observations in the series and, as such, there are two middle values. As before, we take the arithmetic mean of the two middle values.
Hence we obtain:

Median:

$$\tilde{X} = \frac{18+23}{2} = 20$$

A very important point to be noted here is that we must arrange the data in ascending order before searching for the two middle values. All the above examples pertained to raw data. Let us now consider the case of grouped data.
We begin by discussing the case of discrete data grouped into a frequency table.
As stated earlier, a discrete frequency distribution is no more than a concise representation of a simple series pertaining to a discrete variable, so that the same approach as the one discussed just now would seem relevant.

EXAMPLE OF A DISCRETE FREQUENCY DISTRIBUTION

Comprehensive School

Number of pupils per class	Number of Classes
23	1
24	0
25	1
26	3
27	6
28	9
29	8
30	10
31	7
	45

In order to locate the middle value, the best thing is to first of all construct a column of cumulative frequencies:

Comprehensive School

Number of pupils per class X	Number of Classes f	Cumulative Frequency cf
23	1	1
24	0	1
25	1	2
26	3	5
27	6	11
28	9	20
29	8	28
30	10	38
31	7	45
	45	

In this school, there are 45 classes in all, so that we require as the median that class-size below which there are 22 classes and above which also there are 22 classes.
In other words, we must find the 23rd class in an ordered list. We could simply count down noticing that there is 1 class of 23 children, 2 classes with up to 25 children, 5 classes with up to 26 children. Proceeding in this manner, we find that 20 classes contain up to 28 children whereas 28 classes contain up to 29 children. This means that the 23rd class --- the one that we are looking for --- is the one which contains exactly 29 children.

Comprehensive School

Number of pupils per class X	Number of Classes f	Cumulative Frequency cf
23	1	1
24	0	1
25	1	2
26	3	5
27	6	11
28	9	20
29	8	28
30	10	38
31	7	45
	45	

Median number of pupils per class:

$$\tilde{X} = 29$$

This means that 29 is the middle size of the class. In other words, 22 classes are such which contain 29 or less than 29 children, and 22 classes are such which contain 29 or more than 29 children.

LECTURE NO. 8

- Median in case of a frequency distribution of a continuous variable
- Median in case of an open-ended frequency distribution
- Empirical relation between the mean, median and the mode
- Quantiles (quartiles, deciles & percentiles)
- Graphic location of quantiles.

MEDIAN IN CASE OF A FREQUENCY DISTRIBUTION OF A CONTINUOUS VARIABLE:

In case of a frequency distribution, the median is given by the formula

$$\tilde{X} = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$

Where
l =lower class boundary of the median class (i.e. that class for which the cumulative frequency is just in excess of n/2).
h=class interval size of the median class
f =frequency of the median class
n=Σf (the total number of observations)
c =cumulative frequency of the class preceding the median class

Note:
This formula is based on the assumption that the observations in each class are evenly distributed between the two class limits.

EXAMPLE:

Going back to the example of the EPA mileage ratings, we have

Mileage Rating	No. of Cars	Class Boundaries	Cumulative Frequency
30.0 – 32.9	2	29.95 – 32.95	2
33.0 – 35.9	4	32.95 – 35.95	6
36.0 – 38.9	14	35.95 – 38.95	20
39.0 – 41.9	8	38.95 – 41.95	28
42.0 – 44.9	2	41.95 – 44.95	30

In this example, n = 30 and n/2 = 15.
Thus the third class is the median class. The median lies somewhere between 35.95 and 38.95. Applying the above formula, we obtain

$$\tilde{X} = 35.95 + \frac{3}{14}(15 - 6)$$
$$= 35.95 + 1.93$$
$$= 37.88$$
$$\simeq 37.9$$

INTERPRETATION

This result implies that half of the cars have mileage less than or up to 37.88 miles per gallon whereas the other half of the cars has mileage greater than 37.88 miles per gallon. As discussed earlier, the median is preferable to the arithmetic

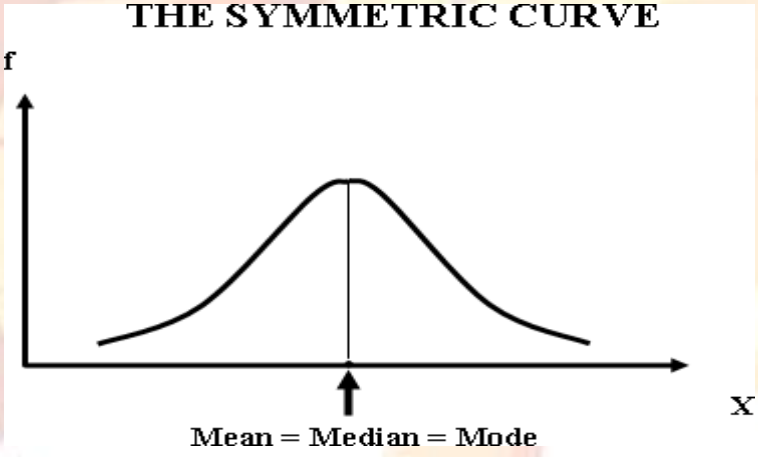
mean when there are a few very high or low figures in a series. It is also exceedingly valuable when one encounters a frequency distribution having open-ended class intervals.
The concept of open-ended frequency distribution can be understood with the help of the following example.

Example:

WAGES OF WORKERS IN A FACTORY	
Monthly Income (in Rupees)	No. of Workers
Less than 2000/-	100
2000/- to 2999/-	300
3000/- to 3999/-	500
4000/- to 4999/-	250
5000/- and above	50
Total	1200

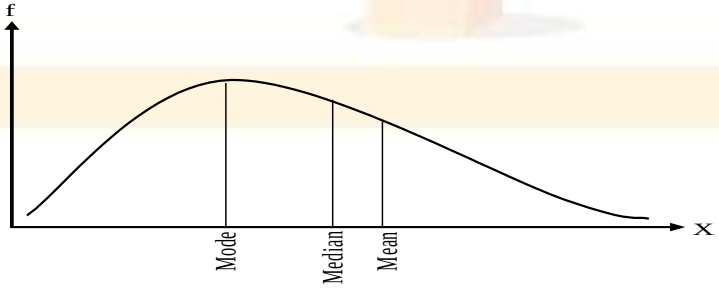
In this example, both the first class and the last class are open-ended classes. This is so because of the fact that we do not have exact figures to begin the first class or to end the last class. The advantage of computing the median in the case of an open-ended frequency distribution is that, except in the unlikely event of the median falling within an open-ended group occurring in the beginning of our frequency distribution, there is no need to estimate the upper or lower boundary. This is so because of the fact that, if the median is falling in an intermediate class, then, obviously, the first class is not being involved in its computation. The next concept that we will discuss is the empirical relation between the mean, median and the mode. This is a concept which is not based on a rigid mathematical formula; rather, it is based on observation. In fact, the word 'empirical' implies 'based on observation'.

This concept relates to the relative positions of the mean, median and the mode in case of a hump-shaped distribution. In a single-peaked frequency distribution, the values of the mean, median and mode coincide if the frequency distribution is absolutely symmetrical.



But in the case of a skewed distribution, the mean, median and mode do not all lie on the same point. They are pulled apart from each other, and the empirical relation explains the way in which this happens. Experience tells us that in a unimodal curve of moderate skewness, the median is usually sandwiched between the mean and the mode.

The second point is that, in the case of many real-life data-sets, it has been observed that the distance between the mode and the median is approximately double of the distance between the median and the mean, as shown below:



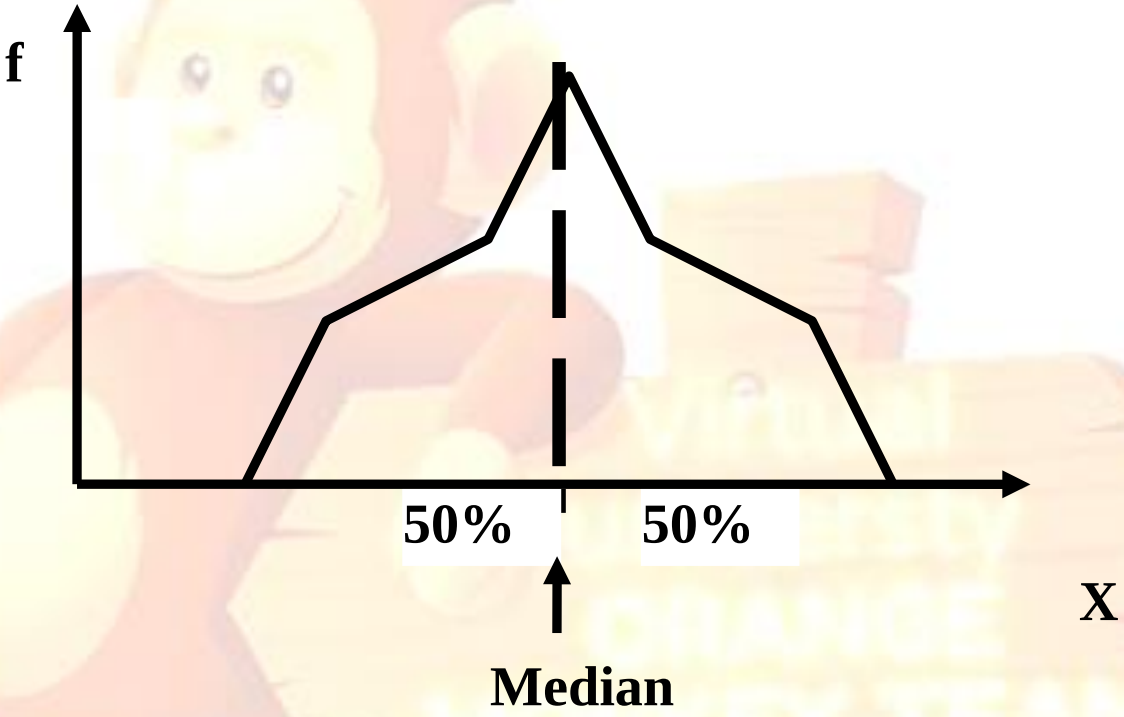
This diagrammatic picture is equivalent to the following algebraic expression:
Median - Mode 2 (Mean - Median) ---- (1)
The above-mentioned point can also be expressed in the following way:

Mean – Mode = 3 (Mean – Median) -----(2)
Equation (1) as well as equation (2) yields the approximate relation given below:

EMPIRICAL RELATION BETWEEN THE MEAN, MEDIAN AND THE MODE

Mode = 3 Median – 2 Mean
An exactly similar situation holds in case of a moderately negatively skewed distribution.
An important point to note is that this empirical relation does not hold in case of a J-shaped or an extremely skewed distribution.

Let us now extend the concept of partitioning of the frequency distribution by taking up the concept of quantiles (i.e. quartiles, deciles and percentiles).
We have already seen that the median divides the area under the frequency polygon into two equal halves:



A further split to produce quarters, tenths or hundredths of the total area under the frequency polygon is equally possible, and may be extremely useful for analysis. (We are often interested in the highest 10% of some group of values or the middle 50% another.

QUARTILES

The quartiles, together with the median, achieve the division of the total area into four equal parts.
The first, second and third quartiles are given by the formulae:

1. FIRST QUARTILE

$$Q_1 = l + \frac{h}{f}\left(\frac{n}{4} - c\right)$$

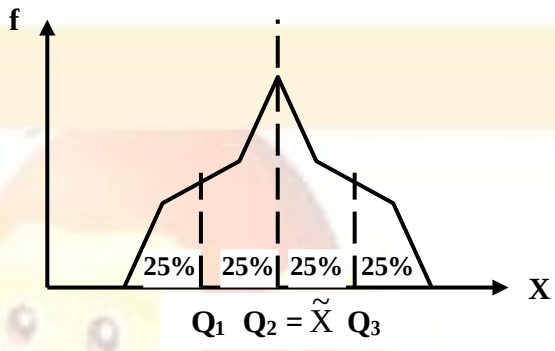
2. SECOND QUARTILE (I.E. MEDIAN)

$$Q_2 = l + \frac{h}{f}\left(\frac{2n}{4} - c\right) = l + \frac{h}{f}\left(\frac{n}{2} - c\right)$$

3. THIRD QUARTILE

$$Q_3 = l + \frac{h}{f}\left(\frac{3n}{4} - c\right)$$

It is clear from the formula of the second quartile that the second quartile is the same as the median.



DECILES & PERCENTILES

The deciles and the percentiles give the division of the total area into 10 and 100 equal parts respectively.
The formula for the first decile is

$$D_1 = l + \frac{h}{f}\left(\frac{n}{10} - c\right)$$

The formulae for the subsequent deciles are

$$D_2 = l + \frac{h}{f}\left(\frac{2n}{10} - c\right)$$

$$D_3 = l + \frac{h}{f}\left(\frac{3n}{10} - c\right)$$

and so on.

It is easily seen that the 5th decile is the same quantity as the median.
The formula for the first percentile is

$$P_1 = l + \frac{h}{f}\left(\frac{n}{100} - c\right)$$

The formulae for the subsequent percentiles are

$$P_2 = l + \frac{h}{f}\left(\frac{2n}{100} - c\right)$$

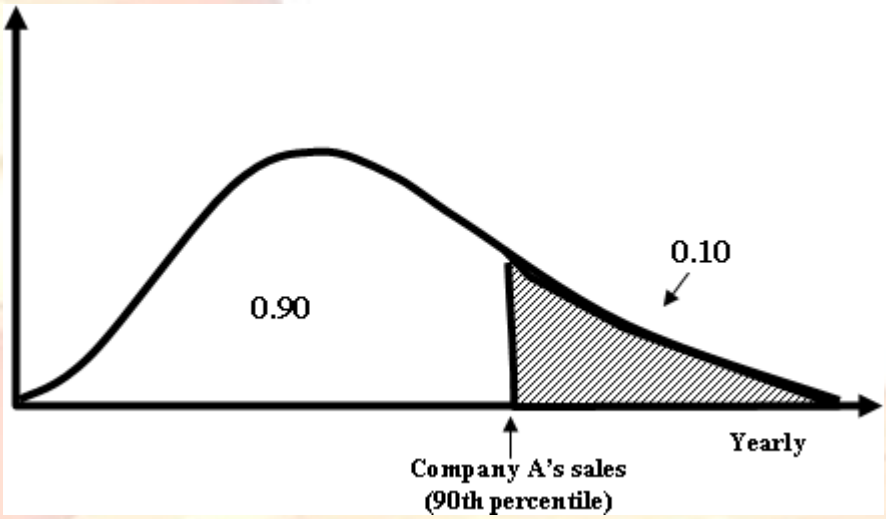
$$P_3 = l + \frac{h}{f}\left(\frac{3n}{100} - c\right)$$

and so on.
Again, it is easily seen that the 50th percentile is the same as the median, the 25th percentile is the same as the 1st quartile, the 75th percentile is the same as the 3rd quartile, the 40th percentile is the same as the 4th decile, and so on.

All these measures i.e. the median, quartiles, deciles and percentiles are collectively called quantiles. The question is, "What is the significance of this concept of partitioning? *Why* is it that we wish to divide our frequency distribution into two, four, ten or hundred parts?"
The answer to the above questions is: In certain situations, we may be interested in describing the *relative* quantitative location of a particular measurement within a data set. Quantiles provide us with an easy way of achieving this. Out of these various quantiles, one of the most frequently used is percentile ranking.
Let us understand this point with the help of an example.

EXAMPLE

If oil company 'A' reports that its yearly sales are at the 90th percentile of all companies in the industry, the implication is that 90% of all oil companies have yearly sales *less* than company A's, and only 10% have yearly sales exceeding company A's,this is demonstrated in the following figure:



It is evident from the above example that the concept of percentile ranking is quite a useful concept, but it should be kept in mind that percentile rankings are of practical value only for large data sets.
It is evident from the above example that the concept of percentile ranking is quite a useful concept, but it should be kept in mind that percentile rankings are of practical value only for large data sets. The next concept that we will discuss is the graphic location of quantiles.
Let us go back to the example of the EPA mileage ratings of 30 cars that was discussed in an earlier lecture.

EXAMPLE

Suppose that the Environmental Protection Agency of a developed country performs extensive tests on all new car models in order to determine their mileage rating. Suppose that the following 30 measurements are obtained by conducting such tests on a particular new car model.

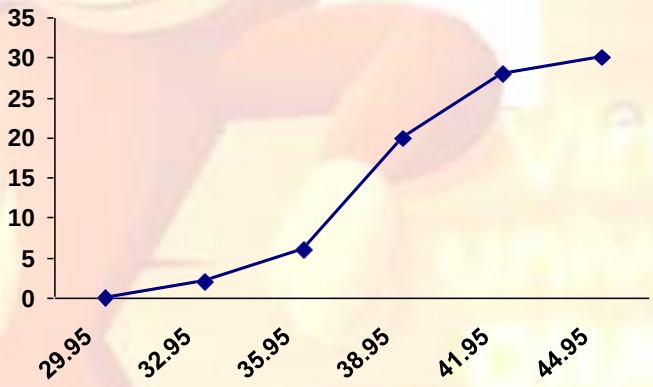
EPA MILEAGE RATINGS ON 30 CARS (MILES PER GALLON)		
36.3	42.1	44.9
30.1	37.5	32.9
40.5	40.0	40.2
36.2	35.6	35.9
38.5	38.8	38.6
36.3	38.4	40.5
41.0	39.0	37.0
37.0	36.7	37.1
37.1	34.8	33.9
39.9	38.1	39.8

When the above data was converted to a frequency distribution, we obtained:

Class Limit	Frequency
30.0 – 32.9	2
33.0 – 35.9	4
36.0 – 38.9	14
39.0 – 41.9	8
42.0 – 44.9	2
	30

Also, we considered the graphical representation of this distribution.
The cumulative frequency polygon of this distribution came out to be as shown in the following figure:

Cumulative Frequency Polygon or OGIVE

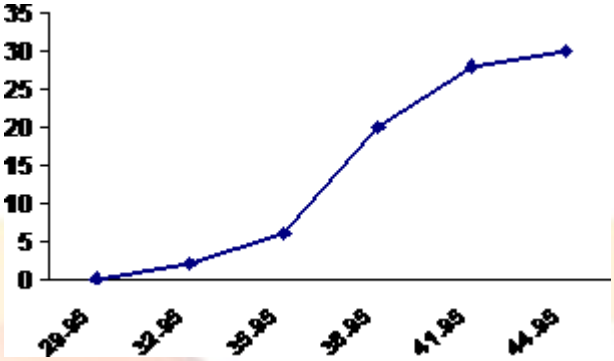


This ogive enables us to find the median and any other quantile that we may be interested in very conveniently. And this process is known as the graphic location of quantiles.
Let us begin with the graphical location of the median:

Because of the fact that the median is that value before which half of the data lies, the first step is to divide the total number of observations n by 2.
In this example:

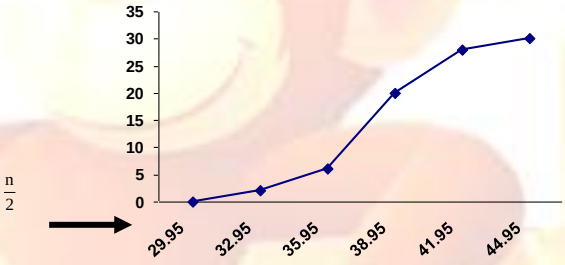
$$\frac{n}{2} = \frac{30}{2} = 15$$

The next step is to locate this number 15 on the y-axis of the cumulative frequency polygon.

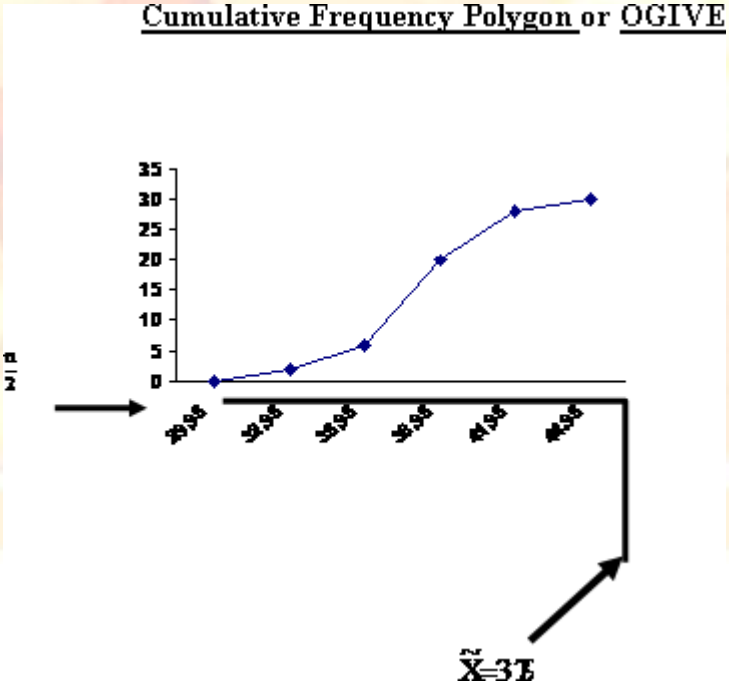


Lastly, we drop a vertical line from the cumulative frequency polygon down to the x-axis.

Cumulative Frequency Polygon or OGIVE

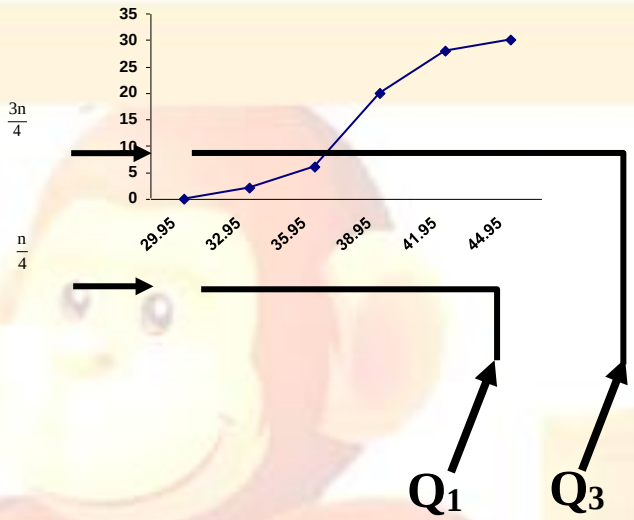


Now, if we read the x-value where our perpendicular touches the x-axis, students, we find that this value is approximately the same as what we obtained from our formula.



It is evident from the above example that the cumulative frequency polygon is a very useful device to find the value of the median very quickly. In a similar way, we can locate the quartiles, deciles and percentiles. To obtain the first quartile, the horizontal line will be drawn against the value n/4, and for the third quartile, the horizontal line will be drawn against the value 3n/4.

Cumulative Frequency Polygon or OGIVE



For the deciles, the horizontal lines will be against the values n/10, 2n/10, 3n/10, and so on. And for the percentiles, the horizontal lines will be against the values n/100, 2n/100, 3n/100, and so on.

The graphic location of the quartiles as well as of a few deciles and percentiles for the data-set of the EPA mileage ratings may be taken up as an exercise:

This brings us to the end of our discussion regarding quantiles which are sometimes also known as fractiles --- this terminology because of the fact that they divide the frequency distribution into various parts or fractions.

LECTURE NO. 9

- Geometric mean
- Harmonic mean
- Relation between the arithmetic, geometric and harmonic means
- Some other measures of central tendency

GEOMETRIC MEAN

The geometric mean, G, of a set of n positive values X1, X2,…,Xn is defined as the positive nth root of their product.

$$G = \sqrt[n]{X_1 X_2 ... X_n}$$

(Where $X_i > 0$)
When n is large, the computation of the geometric mean becomes laborious as we have to extract the nth root of the product of all the values.
The arithmetic is simplified by the use of logarithms.

Taking logarithms to the base 10, we get

$$\log G = \frac{1}{n}\left[\log X_1 + \log X_2 + ... + \log X_n\right]$$

Hence

$$= \frac{\sum \log X}{n}$$

$$G = \text{antilog}\left[\frac{\sum \log X}{n}\right]$$

EXAMPLE

Find the geometric mean of numbers:

45, 32, 37, 46, 39, 36, 41, 48, 36

Solution:
We need to compute the numerical value of

$$= \sqrt[9]{45 \times 32 \times 37 \times 46 \times 39 \times 36 \times 41 \times 48 \times 36}$$

But, obviously, it is a bit cumbersome to find the ninth root of a quantity. So we make use of logarithms, as shown below:

X	log X
45	1.6532
32	1.5052
37	1.5682
46	1.6628
39	1.5911
36	1.5563
41	1.6128
48	1.6812
36	1.5563
	14.3870

$$\log G = \frac{\sum \log X}{n}$$

$$= \frac{14.3870}{9} = 1.5986$$

$$\text{Hence } G = \text{antilog } 1.5986$$

$$= 39.68$$

The above example pertained to the computation of the geometric mean in case of raw data.
Next, we consider the computation of the geometric mean in the case of grouped data.

GEOMETRICMEAN FOR GROUPED DATA

In case of a frequency distribution having k classes with midpoints X1, X2, …,Xk and the corresponding frequencies f1, f2, …, fk (such that ∑fi = n), the geometric mean is given by

$$G = \sqrt[n]{X_1^{f} X_2^{f} \cdots X_k^{f}}$$

Each value of X thus has to be multiplied by itself f times, and the whole procedure becomes quite a formidable task! In terms of logarithms, the formula becomes

$$\log G = \frac{1}{n}\left[f_1 \log X_1 + f_2 \log X_2 + \ldots + f_k \log X_k\right]$$

Hence

$$G = anti\log\left[\frac{\sum f \log X}{n}\right]$$

Obviously, the above formula is much easier to handle. Let us now apply it to an example.
Going back to the example of the EPA mileage ratings, we have:

Mileage Rating	No. of Cars	Class-mark (midpoint) X	log X	f log X
30.0 - 32.9	2	31.45	1.4976	2.9952
33.0 - 35.9	4	34.45	1.5372	6.1488
36.0 - 38.9	14	37.45	1.5735	22.0290
39.0 - 41.9	8	40.45	1.6069	12.8552
42.0 - 44.9	2	43.45	1.6380	3.2760
	30			47.3042

G = antilog $\frac{47.3042}{30}$

= antilog 1.5768 = 37.74

This means that, if we use the geometric mean to measures the central tendency of this data set, then the central value of the mileage of those 30 cars comes out to be 37.74 miles per gallon.
The question is, "When should we use the geometric mean?"
The answer to this question is that when relative changes in some variable quantity are averaged, we prefer the geometric mean.

EXAMPLE

Suppose it is discovered that a firm's turnover has increased during 4 years by the following amounts:

Year	Turnover	Percentage Compared With Year Earlier
1958	£ 2,000	–
1959	£ 2,500	125
1960	£ 5,000	200
1961	£ 7,500	150
1962	£ 10,500	140

The yearly increase is shown in a percentage form in the right-hand column i.e. the turnover of 1959 is 125 percent of the turnover of 1958, the turnover of 1960 is 200 percent of the turnover of 1959, and so on. The firm's owner may be interested in knowing his average rate of turnover growth.
If the arithmetic mean is adopted he finds his answer to be:

Arithmetic Mean:

$$\frac{125+200+150+140}{4}$$

$$=153.75$$

i.e. we are concluding that the turnover for any year is 153.75% of the turnover for the previous year. In other words, the turnover in each of the years considered appears to be 53.75 per cent higher than in the previous year.
If *this* percentage is used to calculate the turnover from 1958 to 1962 inclusive, we obtain:

153.75% of £ 2,000 = £ 3,075
153.75% of £ 3,075 = £ 4,728
153.75% of £ 4,728 = £ 7,269
153.75% of £ 7,269 = £ 11,176

Whereas the actual turnover figures were

Year	Turnover
1958	£ 2,000
1959	£ 2,500
1960	£ 5,000
1961	£ 7,500
1962	£ 10,500

It seems that both the individual figures and, more important, the total at the end of the period, are incorrect. Using the arithmetic mean has exaggerated the 'average' annual rate of increase in the turnover of this firm. Obviously, we would like to rectify this false impression. The geometric mean enables us to do so:
Geometric mean of the turnover figures:

$$\sqrt[4]{(125\times200\times150\times140)}$$

$$=\sqrt[4]{525000000}$$

$$=151.37\%$$

Now, if we utilize this particular value to obtain the individual turnover figures, we find that:
151.37% of £2,000 = £3,027
151.37% of £3,027 = £4,583
151.37% of £4,583 = £6,937
151.37% of £6,937 = £10,500
So that the turnover figure of 1962 is exactly the same as what we had in the original data.

INTERPRETATION

If the turnover of this company were to increase annually at a constant rate, then the annual increase would have been 51.37 percent.(On the average, each year's turnover is 51.37% higher than that in the previous year.) The above example clearly indicates the significance of the geometric mean in a situation when relative changes in a variable quantity are to be averaged.

But we should bear in mind that such situations are not encountered too often, and that the occasion to calculate the geometric mean arises less frequently than the arithmetic mean.(The most frequently used measure of central tendency is the arithmetic mean.)
The next measure of central tendency that we will discuss is the harmonic mean.

HARMONIC MEAN

The harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the values.
In case of raw data:

$$H.M.=\frac{n}{\sum\left(\frac{1}{x}\right)}$$

In case of grouped data (data grouped into a frequency distribution):

$$H.M. = \frac{n}{\sum f\left(\frac{1}{X}\right)}$$

(Where X represents the midpoints of the various classes)

EXAMPLE

Suppose a car travels 100 miles with 10 stops, each stop after an interval of 10 miles. Suppose that the speeds at which the car travels these 10 intervals are 30, 35, 40, 40, 45, 40, 50, 55, 55 and 30 miles per hours respectively.
What is the average speed with which the car traveled the total distance of 100 miles?
If we find the arithmetic mean of the 10 speeds, we obtain:
Arithmetic mean of the 10 speeds:

$$\frac{30+35+....+30}{10} = \frac{420}{10} = 42 \text{ miles per hour.}$$

But, if we study the problem carefully, we find that the above answer is incorrect.
By definition, the average speed is the speed with which the car would have traveled the 100 mile distance if it had maintained a constant speed throughout the 10 intervals of 10 miles each.

$$\textbf{Average speed} = \frac{\textbf{Total distance travelled}}{\textbf{Total time taken}}$$

Now, *total distance traveled* = 100 miles. *Total time taken* will be computed as shown below:

Interval	Distance	Speed = Distance / Time	Time = Distance / Speed
1	10 miles	30 mph	10/30 = 0.3333 hrs
2	10 miles	35 mph	10/35 = 0.2857 hrs
3	10 miles	40 mph	10/40 = 0.2500 hrs
4	10 miles	40 mph	10/40 = 0.2500 hrs
5	10 miles	45 mph	10/45 = 0.2222 hrs
6	10 miles	40 mph	10/40 = 0.2500 hrs
7	10 miles	50 mph	10/50 = 0.2000 hrs
8	10 miles	55 mph	10/55 = 0.1818 hrs
9	10 miles	55 mph	10/55 = 0.1818 hrs
10	10 miles	30 mph	10/30 = 0.333 hrs
Total = 100 miles		Total Time = 2.4881 hrs	

Hence
Average speed = $\frac{100}{2.4881} = 40.2 \ mph$

which is not the same as 42 miles per hour.
Let us now try the harmonic mean to find the average speed of the car.

$$H.M. = \frac{n}{\sum\left(\frac{1}{X}\right)}$$

where n is the no. of terms)
We have:

X	1/X
30	1/30 = 0.0333
35	1/35 = 0.0286
40	1/40 = 0.0250
40	1/40 = 0.0250
45	1/45 = 0.0222
40	1/40 = 0.0250
50	1/50 = 0.0200
55	1/55 = 0.0182
55	1/55 = 0.0182
30	1/30 = 0.0333
	$\sum \frac{1}{X} = 0.2488$

$$H.M. = \frac{n}{\sum \frac{1}{X}}$$

$$= \frac{10}{0.2488}$$

$$= 40.2 \text{ mph}$$

Hence it is clear that the harm gives the totally correct resul

The key question is, "When should we compute the harmonic mean of a data set?" The answer to this question will be easy to understand if we consider the following rules:

RULES

- When values are given as x per y where x is constant and y is variable, the Harmonic Mean is the appropriate average to use.
- When values are given as x per y where y is constant and x is variable, the Arithmetic Mean is the appropriate average to use.
- When relative changes in some variable quantity are to be averaged, the geometric mean is the appropriate average to use.

We have already discussed the geometric and the harmonic means. Let us now try to understand Rule No. 1 with the help of an example:

EXAMPLE

If 10 students have obtained the following marks (in a test) out of 20:
13, 11, 9, 9, 6, 5, 19, 17, 12, 9
Then the average marks (by the formula of the arithmetic mean) are:

$$\frac{13+11+9+9+6+5+19+17+12+9}{10}$$

$$= \frac{110}{10} = 11$$

This is equivalent to

$$\frac{\frac{13}{20}+\frac{11}{20}+\frac{9}{20}+\frac{9}{20}+\frac{6}{20}+\frac{5}{20}+\frac{19}{20}+\frac{17}{20}+\frac{12}{20}+\frac{9}{20}}{10}$$

$$= \frac{\frac{110}{20}}{10} = \frac{110}{10 \times 20} = \frac{11}{20}$$

(i.e. the average marks of this group of students are 11 out of 20).In the above example, the point to be noted was that all the marks were expressible as x per y where the denominator y was constant i.e. equal to 20, and hence, it was appropriate to compute the arithmetic mean.
Let us now consider a mathematical relationship exists between these three measures of central tendency.

RELATION BETWEEN ARITHMETIC, GEOMETRIC AND HARMONIC MEANS

Arithmetic Mean ≥ Geometric Mean ≥Harmonic Mean
We have considered the five most well-known measures of central tendency i.e. arithmetic mean, median, mode, geometric mean and harmonic mean. It is interesting to note that there are some other measures of central tendency as well. Two of these are the mid range, and the mid quartile range.
Let us consider these one by one:

MID-RANGE

If there are n observations with x0 and xm as their smallest and largest observations respectively, then their mid-range is defined as

$$\text{mid} - \text{range} = \frac{x_0 + x_m}{2}$$

It is obvious that if we add the smallest value with the largest, and divide by 2, we will get a value which is more or less in the middle of the data-set.

MID-QUARTILE RANGE

If x1, x2… xn are n observations with Q1 and Q3 as their first and third quartiles respectively, then their mid-quartile range is defined as

$$\text{mid} - \text{quartile range} = \frac{Q_1 + Q_3}{2}$$

Similar to the case of the mid-range, if we take the arithmetic mean of the upper and lower quartiles, we will obtain a value that is somewhere in the middle of the data-set. The mid-quartile range is also known as the mid-hinge.
Let us now revise briefly the core concept of central tendency: Masses of data are usually expressed in the form of frequency tables so that it becomes easy to comprehend the data. Usually, a statistician would like to go a step ahead and to compute a number that will represent the data in some definite way.
Any such single number that represents a whole set of data is called **'Average'**.
Technically speaking, there are many kinds of averages (i.e. there are several ways to compute them). These quantities that represent the data-set are called "measures of central tendency".

LECTURE NO. 10

- Concept of dispersion
- Absolute and relative measures of dispersion
- Range
- Coefficient of dispersion
- Quartile deviation
- Coefficient of quartile deviation

Let us begin the concept of DISPERSION.
Just as variable series differ with respect to their location on the horizontal axis (having different 'average' values); similarly, they differ in terms of the amount of variability which they exhibit. Let us understand this point with the help of an example:

EXAMPLE

In a technical college, it may well be the case that the ages of a group of first-year students are quite consistent, e.g. 17, 18, 18, 19, 18, 19, 19, 18, 17, 18 and 18 years.
A class of evening students undertaking a course of study in their spare time may show just the opposite situation, e.g. 35, 23, 19, 48, 32, 24, 29, 37, 58, 18, 21 and 30.
It is very clear from this example that the variation that exists between the various values of a data-set is of substantial importance. We obviously need to be aware of the amount of variability present in a data-set if we are to come to useful conclusions about the situation under review. This is perhaps best seen from studying the two frequency distributions given below.

EXAMPLE

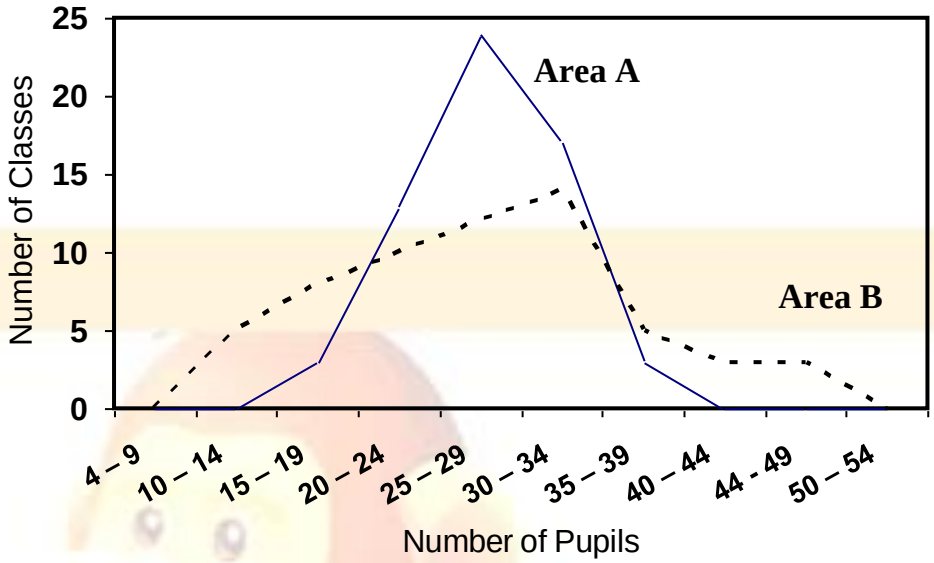
The sizes of the classes in two comprehensive schools in different areas are as follows:

Number of Pupils	Number of Classes	
	Area A	Area B
10 – 14	0	5
15 – 19	3	8
20 – 24	13	10
25 – 29	24	12
30 – 34	17	14
35 – 39	3	5
40 – 44	0	3
45 - 49	0	3
	60	60

If the arithmetic mean size of class is calculated, we discover that the answer is identical: 27.33 pupils in both areas.
Average class-size of each school

$$\overline{X} = 27.33$$

Even though these two distributions share a common average, it can readily be seen that they are entirely DIFFERENT.
And the graphs of the two distributions (given below) clearly indicate this fact.



The question which must be posed and answered is 'In what way can these two situations be distinguished?' We need a measure of variability or DISPERSION to accompany the relevant measure of position or 'average' used.
The word 'relevant' is important here for we shall find one measure of dispersion which expresses the scatter of values round the arithmetic mean, another the scatter of values round the median, and so forth. Each measure of dispersion is associated with a particular 'average'.

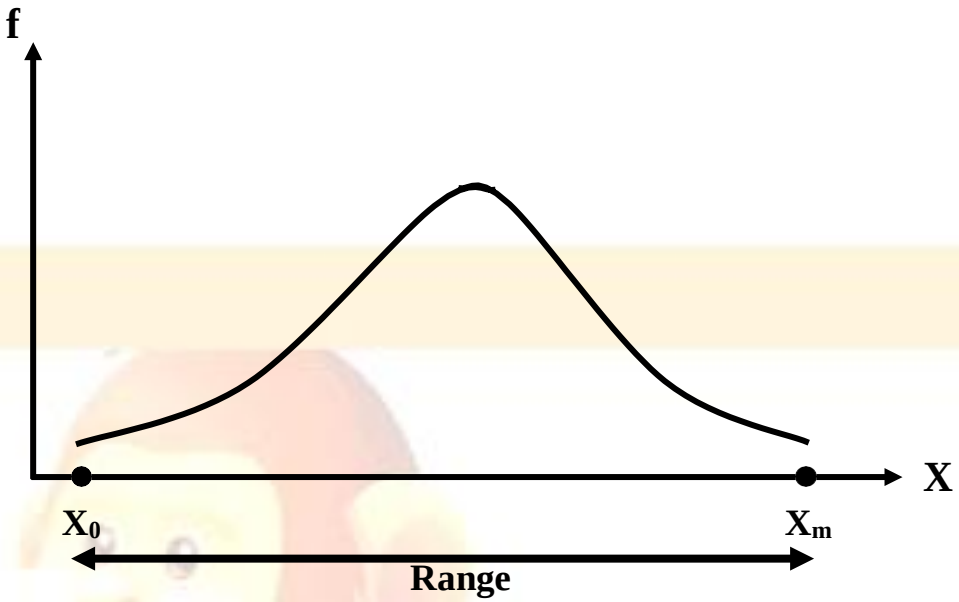
ABSOLUTE VERSUS RELATIVE MEASURES OF DISPERSION

There are two types of measurements of dispersion: absolute and relative.
An absolute measure of dispersion is one that measures the dispersion in terms of the same units or in the square of units, as the units of the data.
For example, if the units of the data are rupees, meters, kilograms, etc., the units of the measures of dispersion will also be rupees, meters, kilograms, etc.
On the other hand, relative measure of dispersion is one that is expressed in the form of a ratio, co-efficient of percentage and is independent of the units of measurement.
A *relative* measure of dispersion is useful for comparison of data of different nature. A measure of central tendency together with a measure of dispersion gives an adequate description of data. We will be discussing FOUR measures of dispersion i.e. the range, the quartile deviation, the mean deviation, and the standard deviation.

RANGE

The range is defined as the difference between the two extreme values of a data-set, i.e. R = Xm – X0 where Xm represents the highest value and X0 the lowest.
Evidently, the calculation of the range is a simple question of MENTAL arithmetic.

The simplicity of the concept does not necessarily invalidate it, but in general it gives no idea of the DISTRIBUTION of the observations between the two ends of the series. For this reason it is used principally as a supplementary aid in the description of variable data, in conjunction with other measures of dispersion. When the data are grouped into a frequency distribution, the range is estimated by finding the difference between the upper boundary of the highest class and the lower boundary of the lowest class.
We now consider the graphical representation of the range:



Obviously, the greater the difference between the largest and the smallest values, the greater will be the range. As stated earlier, the range is a simple concept and is easy to compute. However, because of the fact that it is computed from only the two extreme values in a data-set, it has two serious disadvantages.

- It *ignores* all the INFORMATION available from the intermediate observations.
- It might give a MISLEADING picture of the spread in the data.

From THIS point of view, it is an unsatisfactory measure of dispersion. However, it is APPROPRIATELY used in statistical quality control charts of manufactured products, daily temperatures, stock prices, etc. It is interesting to note that the range can also be viewed in the following way.
It is twice of the arithmetic mean of the deviations of the smallest and largest values round the mid-range i.e.

$$\frac{(\text{Midrange} - X_0) + (X_m - \text{Midrange})}{2}$$

$$= \frac{\text{Midrange} - X_0 + X_m - \text{Midr}}{2}$$

$$= \frac{X_m - X_0}{2}$$

Because of what has been just explained, the range can be regarded as that measure of dispersion which is associated with the mid-range. As such, the range may be employed to indicate dispersion when the mid-range has been adopted as the most appropriate average.
The range is an *absolute* measure of dispersion. Its *relative* measure is known as the CO-EFFICIENT OF DISPERSION, and is defined by the relation given below:
\

COEFFICIENT OF DISPERSION

$$= \frac{\frac{1}{2}(\text{Range})}{\text{Mid} - \text{Range}}$$

$$= \frac{\frac{X_m - X_0}{2}}{\frac{X_m + X_0}{2}} = \frac{X_m - X_0}{X_m + X_0}$$

This is a pure (i.e. dimensionless) number and is used for the purposes of COMPARISON. (This is so because a pure number can be compared with another pure number.)

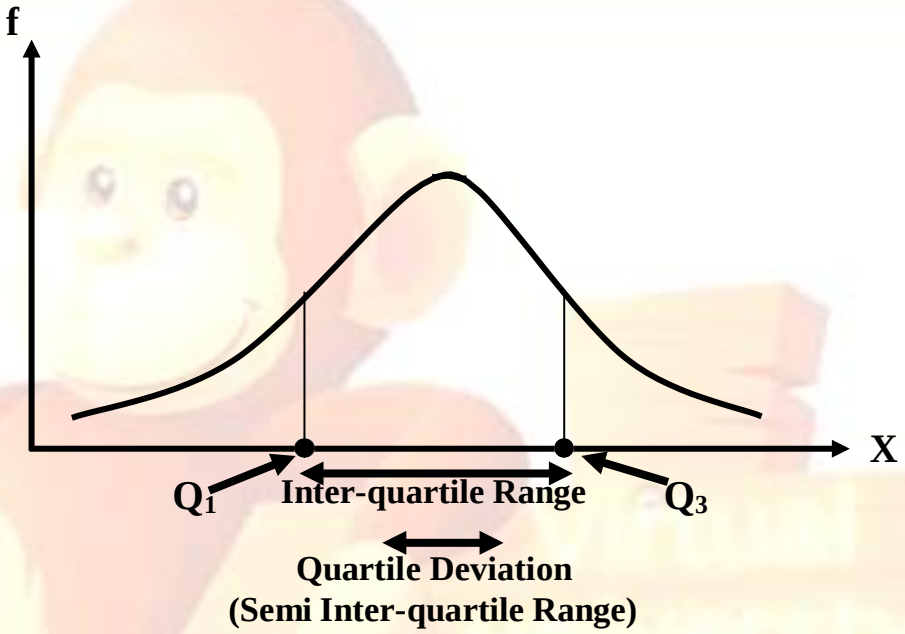
For example, if the coefficient of dispersion for one data-set comes out to be 0.6 whereas the coefficient of dispersion for another data-set comes out to be 0.4, then it is obvious that there is greater amount of dispersion in the first data-set as compared with the second.

QUARTILE DEVIATION

The quartile deviation is defined as half of the difference between the third and first quartiles i.e.

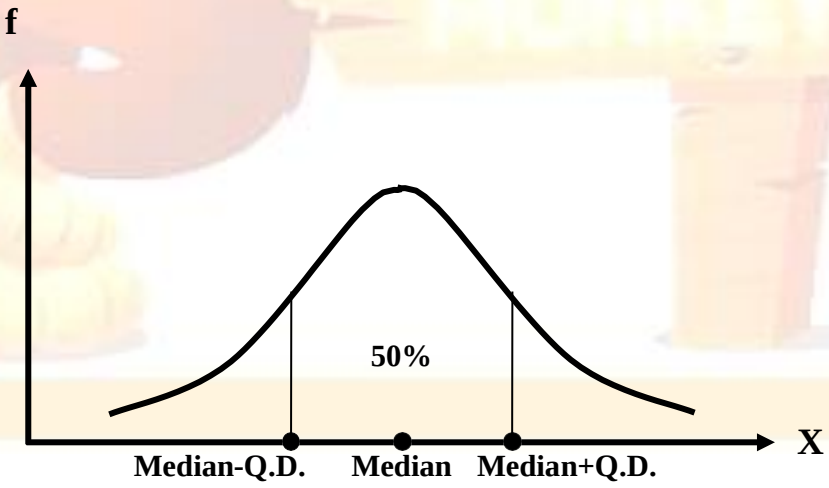
$$Q.D. = \frac{Q_3 - Q_1}{2}$$

It is also known as **semi-interquartile range.** Let us now consider the graphical representation of the quartile deviation:



Although simple to compute, it is NOT an extremely satisfactory measure of dispersion because it takes into account the spread of only two values of the variable round the median, and this gives no idea of the rest of the dispersion within the distribution.

The quartile deviation has an attractive feature that the range "Median + Q.D." contains approximately 50% of the data. This is illustrated in the figure given below:



Let us now apply the concept of quartile deviation to the following example:

EXAMPLE

The shareholding structure of two companies is given below:

	Company X	Company Y
1st quartile	60 shares	165 shares
Median	185 shares	185 shares
3rd quartile	270 shares	210 shares

The quartile deviation for company X is

$$\frac{270-60}{2}=105 \quad \text{Shares}$$

For company Y, it is

$$\frac{210-165}{2}=22 \text{Shares}$$

A comparison of the above two results indicate that there is a considerable concentration of shareholders about the MEDIAN number of shares in company Y, whereas in company X, there does not exist this kind of a concentration around the median. (In company X, there is approximately the SAME numbers of small, medium and large shareholders.)

From the above example, it is obvious that the larger the quartile deviation, the greater is the scatter of values within the series. The quartile deviation is superior to range as it is not affected by extremely large or small observations. It is simple to understand and easy to calculate.

The mean deviation can also be viewed in another way: It is the arithmetic mean of the deviations of the first and third quartiles round the median i.e.

$$\frac{(M-Q_1)+(Q_3-M)}{2}$$

$$=\frac{M-Q_1+Q_3-M}{2}$$

$$=\frac{Q_3-Q_1}{2}$$

Because of what has been just explained, the quartile deviation is regarded as that measure of dispersion which is associated with the median. As such, the quartile deviation should always be employed to indicate dispersion when the median has been adopted as the most appropriate average.

The quartile deviation is also an *absolute* measure of dispersion. Its *relative* measure called the CO-EFFICIENT OF QUARTILE DEVIATION or of Semi-Inter-quartile Range, is defined by the relation:

COEFFICIENT OF QUARTILE DEVIATION

$$=\frac{Quartile\ Deviation}{Mid-Quartile\ Range}$$

$$=\frac{\frac{Q_3-Q_1}{2}}{\frac{Q_3+Q_1}{2}}=\frac{Q_3-Q_1}{Q_3+Q_1},$$

The Coefficient of Quartile Deviation is a pure number and is used for *COMPARING* the variation in two or more sets of data.

The next two measures of dispersion to be discussed are the Mean Deviation and the Standard Deviation. In this regard, the first thing to note is that, whereas the range as well as the quartile deviation are two such measures of dispersion which are NOT based on all the values, the mean deviation and the standard deviation are two such measures of dispersion that involve each and every data-value in their computation.

The range measures the dispersion of the data-set around the *mid-range*, whereas the quartile deviation measures the dispersion of the data-set around the *median*.

How are we to decide upon the amount of dispersion round the *arithmetic mean*?

It would seem reasonable to compute the *DISTANCE* of each observed value in the series from the arithmetic mean of the series.

But the problem is that the sum of the deviations of the values from the mean is ZERO! (No matter what the amount of dispersion in a data-set is, this quantity will *always* be zero, and hence it cannot be used to measure the dispersion in the data-set.)

Then, the question arises, 'HOW will we be able to measure the dispersion present in our data-set?' In an attempt to answer this question, we might look at the numerical differences between the mean and the data values WITHOUT considering whether these are positive or negative. By ignoring the sign of the deviations we will achieve a NON-ZERO sum, and averaging these absolute differences, again, we obtain a non-zero quantity which can be used as a measure of dispersion. (The larger this quantity, the greater is the dispersion in the data-set).

This quantity is known as the **MEAN DEVIATION.**

Let us denote these absolute differences

by 'modulus of d'

or 'mod d'. Then, the mean deviation is given by

MEAN DEVIATION

$$M.D. = \frac{\sum |d|}{n}$$

As the absolute deviations of the observations from their mean are being averaged, therefore the complete name of this measure is Mean Absolute Deviation --- but generally, it is simply called "Mean Deviation". In the next lecture, this concept will be discussed in detail. (The case of raw data as well as the case of grouped data will be considered.)Next, we will discuss the most important and the most widely used measure of dispersion i.e. the Standard Deviation.

LECTURE NO. 11

- Mean Deviation
- Standard Deviation and Variance
- Coefficient of variation

First, we will discuss it for the case of raw data, and then we will go on to the case of a frequency distribution. The first thing to note is that, whereas the range as well as the quartile deviation are two such measures of dispersion which are NOT based on all the values, the mean deviation and the standard deviation are two such measures of dispersion that involve each and every data-value in their computation.

You must have noted that the range was measuring the dispersion of the data-set around the mid-range, whereas the quartile deviation was measuring the dispersion of the data-set around the median.

How are we to decide upon the amount of dispersion round the arithmetic mean? It would seem reasonable to compute the DISTANCE of each observed value in the series from the arithmetic mean of the series.

Let us do this for a simple data-set shown below:

THE NUMBER OF FATALITIES IN MOTORWAY ACCIDENTS IN ONE WEEK

Day	Number of fatalities X
Sunday	4
Monday	6
Tuesday	2
Wednesday	0
Thursday	3
Friday	5
Saturday	8
Total	28

Let us do this for a simple data-set shown below:

THE NUMBER OF FATALITIES IN MOTORWAY ACCIDENTS IN ONE WEEK

Day	Number of fatalities X
Sunday	4
Monday	6
Tuesday	2
Wednesday	0
Thursday	3
Friday	5
Saturday	8
Total	28

The arithmetic mean number of fatalities per day is

$$\bar{X} = \frac{\sum X}{n} = \frac{28}{7} = 4$$

In order to determine the distances of the data-values from the mean, we subtract our value of the arithmetic mean from each daily figure, and this gives us the deviations that occur in the third column of the table below

Day	Number of fatalities X	$X - \overline{X}$
Sunday	4	0
Monday	6	+ 2
Tuesday	2	– 2
Wednesday	0	– 4
Thursday	3	– 1
Friday	5	+ 1
Saturday	8	+ 4
TOTAL	28	0

The deviations are negative when the daily figure is less than the mean (4 accidents) and positive when the figure is higher than the mean. It does seem, however, that our efforts for computing the dispersion of this data set have been in vain, for we find that the total amount of dispersion obtained by summing the $(x - \bar{x})$ column comes out to be zero! In fact, this should be no surprise, for it is a basic property of the arithmetic means that: The sum of the deviations of the values from the mean is zero. The question arises:

How will we measure the dispersion that is actually present in our data-set?

Our problem might at first sight seem irresolvable, for by this criterion it appears that no series has any dispersion. Yet we know that this is absolutely incorrect, and we must think of some other way of handling this situation. Surely, we might look at the numerical difference between the mean and the daily fatality figures without considering whether these are positive or negative. Let us denote these absolute differences by 'modulus of d' or 'mod d'.

This is evident from the third column of the table below

X	$X - \overline{X} = d$	$\lvert d \rvert$
4	0	0
6	2	2
2	–2	2
0	–4	4
3	–1	1
5	1	1
8	4	4
	Total	14

By ignoring the sign of the deviations we have achieved a non-zero sum in our second column. Averaging these absolute differences, we obtain a measure of dispersion known as the mean deviation.

In other words, the mean deviation is given by the formula:

MEAN DEVIATION

$$M.D. = \frac{\sum |d_i|}{n}$$

As we are averaging the absolute deviations of the observations from their mean, therefore the complete name of this measure is mean absolute deviation --- but generally we just say "mean deviation". Applying this formula in our example, we find that, the mean deviation of the number of fatalities is

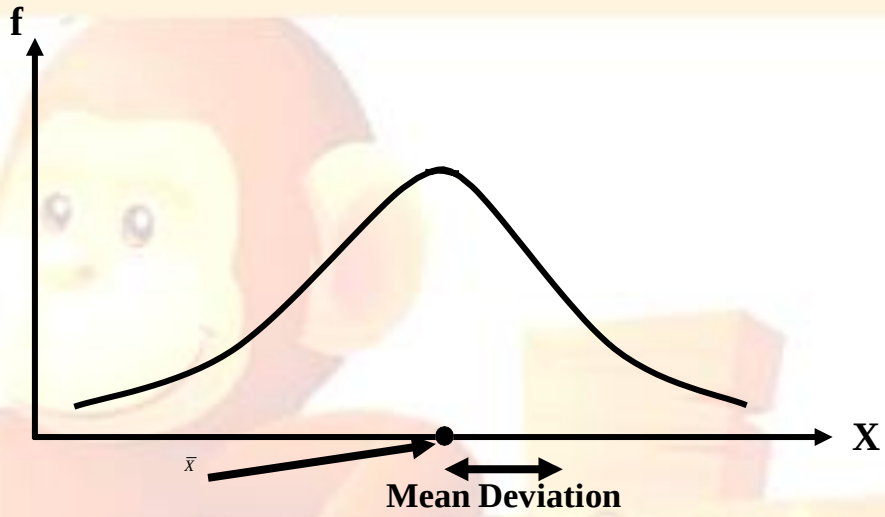
$$M.D. = \frac{14}{7} = 2.$$

The formula that we have just considered is valid in the case of raw data. In case of grouped data i.e. a frequency distribution, the formula becomes

MEAN DEVIATION FOR GROUPED DATA

$$M.D. = \frac{\sum f_i \left| x_i - \bar{x} \right|}{n} = \frac{\sum f_i \left| d_i \right|}{n}$$

As far as the graphical representation of the mean deviation is concerned, it can be depicted by a horizontal line segment drawn below the X-axis on the graph of the frequency distribution, as shown below



The approach which we have adopted in the concept of the mean deviation is both quick and simple. But the problem is that we introduce a kind of artificiality in its calculation by ignoring the algebraic signs of the deviations.
In problems involving descriptions and comparisons alone, the mean deviation can validly be applied; but because the negative signs have been discarded, further theoretical development or application of the concept is impossible.

Mean deviation is an absolute measure of dispersion. Its relative measure, known as the co-efficient of mean deviation, is obtained by dividing the mean deviation by the average used in the calculation of deviations i.e. the arithmetic mean. Thus

CO-EFFICIENT OF M.D

Sometimes, the mean deviation is computed by averaging the absolute deviations of the data-values from the median i.e.

$$\text{Mean deviation} = \frac{\sum \left| x - \tilde{x} \right|}{n}$$

And when will we have a situation when we will be using the median instead of the mean? As discussed earlier, the median will be more appropriate than the mean in those cases where our data-set contains a few very high or very low values. In such a situation, the coefficient of mean deviation is given by:

Co-efficient of M.D:

$$= \frac{M.D.}{\text{Median}} \qquad = \frac{M.D.}{\text{Mean}}$$

Let us now consider the ***standard deviation*** --- that statistic which is the most important and the most widely used measure of dispersion.
The point that made earlier that from the mathematical point of view, it is not very preferable to take the absolute values of the deviations, *This problem is overcome by computing the standard deviation.*
In order to compute the standard deviation, rather than taking the absolute values of the deviations, we square the deviations.

Averaging these squared deviations, we obtain a statistic that is known as the variance.

VARIANCE

$$= \frac{\sum(x - \bar{x})^2}{n}$$

Let us compute this quantity for the data of the above example.
Our X-values were:

X
4
6
2
0
3
5
8

Taking the deviations of the X-values from their mean, and then squaring these deviations, we obtain:

X	$(x - \bar{x})$	$(x - \bar{x})^2$
4	0	0
6	+ 2	4
2	– 2	4
0	– 4	16
3	– 1	1
5	+ 1	1
8	+ 4	16
		42

Obviously, both (– 2)2 and (2)2 equal 4, both (– 4)2 and (4)2 equal 16, and both (– 1)2 and (1)2 = 1
Hence $\sum(x - \bar{x})^2 = 42$ is now positive, and this positive value has been achieved without 'bending' the rules of mathematics. Averaging these squared deviations, the variance is given by:

Variance:

$$= \frac{\sum(x - \bar{x})^2}{n}$$

$$= \frac{42}{7} = 6$$

The variance is frequently employed in statistical work, but it should be noted that the figure achieved is in 'squared' units of measurement.
In the example that we have just considered, the variance has come out to be "6 squared fatalities", which does not seem to make much sense! In order to obtain an answer which is in the original unit of measurement, we take the positive square root of the variance. The result is known as the standard deviation.

STANDARD DEVIATION

$$S = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

Hence, in this example, our standard deviation has come out to be 2.45 fatalities.

In computing the standard deviation (or variance) it can be tedious to first ascertain the arithmetic mean of a series, then subtract it from each value of the variable in the series, and finally to square each deviation and then sum. It is very much more straight-forward to use the short cut formula given below:

SHORT CUT FORMULA FOR THE STANDARD DEVIATION

$$S = \sqrt{\left\{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2\right\}}$$

In order to apply the short cut formula, we require only the aggregate of the series ($\sum x$) and the aggregate of the squares of the individual values in the series ($\sum x2$).

In other words, only two columns of figures are called for. The number of individual calculations is also considerably reduced, as seen below:

	X	X^2
	4	16
	6	36
	2	4
	0	0
	3	9
	5	25
	8	64
Total	28	154

Therefore

$$S = \sqrt{\left\{\frac{154}{7} - \left(\frac{28}{7}\right)^2\right\}} = \sqrt{(22 - 16)}$$

$$= \sqrt{6} = 2.45 \text{ fatalities}$$

The formulae that we have just discussed are valid in case of raw data. In case of grouped data i.e. a frequency distribution, each squared deviation round the mean must be multiplied by the appropriate frequency figure i.e.

STANDARD DEVIATION IN CASE OF GROUPED DATA

$$S = \sqrt{\frac{\sum f(x - \bar{x})^2}{n}}$$

And the short cut formula in case of a frequency distribution is:

SHORT CUT FORMULA OF THE STANDARD DEVIATION IN CASE OF GROUPED DATA

$$S = \sqrt{\left\{\frac{\sum fx^2}{n} - \left(\frac{\sum fx}{n}\right)^2\right\}}$$

Which is again preferred from the computational standpoint
For example, the standard deviation life of a batch of electric light bulbs would be calculated as follows:

EXAMPLE

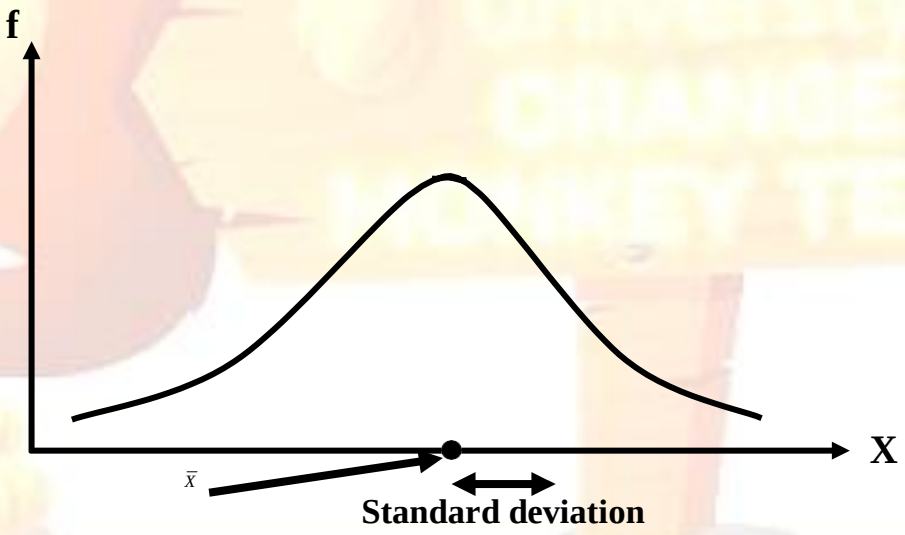
Life (in Hundreds of Hours)	No. of Bulbs f	Mid-point x	fx	fx^2
0 – 5	4	2.5	10.0	25.0
5 – 10	9	7.5	67.5	506.25
10 – 20	38	15.0	570.0	8550.0
20 – 40	33	30.0	990.0	29700.0
40 and over	16	50.0	800.0	40000.0
	100		2437.5	78781.25

Therefore, standard deviation:

$$S = \sqrt{\left\{\frac{78781.25}{100} - \left(\frac{2437.5}{100}\right)^2\right\}}$$

=13.9hundredhours
= 1390 hours

As far as the graphical representation of the standard deviation is concerned, a horizontal line segment is drawn below the X-axis on the graph of the frequency distribution --- just as in the case of the mean deviation.



The standard deviation is an absolute measure of dispersion. Its relative measure called coefficient of standard deviation is defined as:

COEFFICIENT OF S.D

$$= \frac{\text{Standard Deviation}}{\text{Mean}}$$

And, multiplying this quantity by 100, we obtain a very important and well-known measure called the coefficient of variation.

COEFFICIENT OF VARIATION

$$C.V. = \frac{S}{\overline{X}} \times 100$$

As mentioned earlier, the standard deviation is expressed in absolute terms and is given in the same unit of measurement as the variable itself.

There are occasions, however, when this absolute measure of dispersion is inadequate and a relative form becomes preferable. For example, if a comparison between the variability of distributions with different variables is required, or when we need to compare the dispersion of distributions with the same variable but with very different arithmetic means. To illustrate the usefulness of the coefficient of variation, let us consider the following two examples.

EXAMPLE-1

Suppose that, in a particular year, the mean weekly earnings of skilled factory workers in one particular country were $ 19.50 with a standard deviation of $ 4, while for its neighboring country the figures were Rs. 75 and Rs. 28 respectively.

From these figures, it is not immediately apparent which country has the GREATER VARIABILITY in earnings. The coefficient of variation quickly provides the answer:

For country No. 1:

$$\frac{4}{19.5} \times 100 = 20.5 \text{ per cent,}$$

And for country No. 2:

$$\frac{28}{75} \times 100 = 37.3 \text{ per cent.}$$

From these calculations, it is immediately obvious that the spread of earnings in country No. 2 is greater than that in country No. 1, and the reasons for this could then be sought.

EXAMPLE-2:

The crop yield from 20 acre plots of wheat-land cultivated by ordinary methods averages 35 bushels with a standard deviation of 10 bushels. The yield from similar land treated with a new fertilizer averages 58 bushels, also with a standard deviation of 10 bushels. At first glance, the yield variability may seem to be the same, but in fact it has improved (i.e. decreased) in view of the higher average to which it relates.

Again, the coefficient of variation shows this very clearly:

Untreated land:

$$\frac{10}{35} \times 100 = 28.57 \text{ per cent}$$

Treated land:

$$\frac{10}{58} \times 100 = 17.24 \text{ per cent}$$

The coefficient of variation for the untreated land has come out to be 28.57 percent, whereas the coefficient of variation for the treated land is only 17.24 percent.

LECTURE NO. 12

- Chebychev's Inequality
- The Empirical Rule
- The Five-Number Summary

In the last lecture, we discussed the concept of standard deviation in quite a lot of detail.
It is an extremely important concept, and it is very important that we appreciate and understand its role in statistical analysis. We've seen that if we are comparing the variability of two samples selected from a population, the sample with the larger standard deviation is the more variable of the two.
Thus, we know how to interpret the standard deviation on a relative or comparative basis, but we haven't considered how it provides a measure of variability for a single sample.
To understand how the standard deviation provides a measure of variability of a data set, consider a specific data set and answer the following questions:

Question-1

'How many measurements are within 1 standard deviation of the mean?'

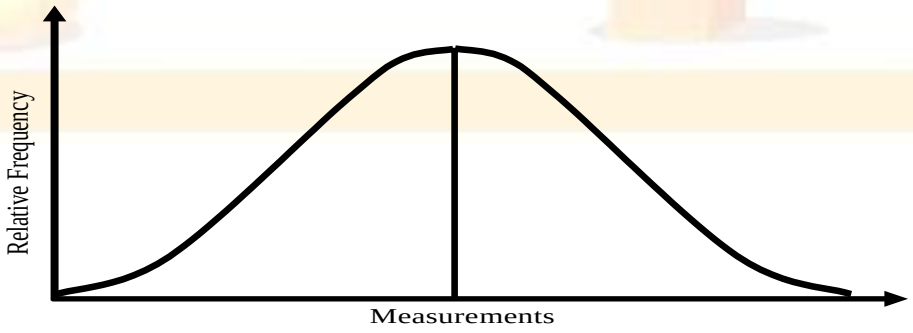
Question-2

'How many measurements are within 2 standard deviations?'
and so on.
For any specific data set, we can answer these questions by counting the number of measurements in each of the intervals. However, if we are interested in obtaining a general answer to these questions the problem is a bit more difficult. We will discuss to you two sets of answers to the questions of how many measurements fall within 1, 2, and 3 standard deviations of the mean. General answer to these questions the problem is a bit more difficult. The first, which applies to any set of data, is derived from a theorem proved by Russian mathematician, P.L. Chebychev (1821-1894).The second, which applies to mound-shaped, symmetric distributions of data, is based upon empirical evidence that has accumulated over the years. And this set of answers is valid and applicable even if our distribution is slightly skewed, Let us begin with the Chebychev's theorem.
Chebychev's Rule applies to any data set, regardless of the shape of the frequency distribution of the data.

CHEBYCHEV'S THEOREM

For any number k greater than 1, at least 1 – 1/k2 of the data-values fall within k standard deviations of the mean, i.e., within the interval ($\bar{X}$ – kS, $\bar{X}$ + kS)
This means that:
a) At least 1-1/22 = 3/4 will fall within 2 standard deviations of the mean, i.e. within the interval ($\bar{X}$ – 2S, $\bar{X}$ + 2S).
b) At least 1-1/32=8/9 of the data-values will fall within 3 standard deviations of the mean, i.e. within the interval ($\bar{X}$ – 3S, $\bar{X}$ + 3S)
Because of the fact that Chebychev's theorem requires k to be greater than 1, therefore no useful information is provided by this theorem on the fraction of measurements that fall within 1 standard deviation of the mean, i.e. within the interval (X–S, $\bar{X}$+S).

Next, let us consider the ***Empirical Rule mentioned above.***
This is a rule of thumb that applies to data sets with frequency distributions that are mound-shaped and symmetric, as follows:



According to this empirical rule:

- Approximately 68% of the measurements will fall within 1 standard deviation of the mean, i.e. within the interval ($\bar{X} - S$, $\bar{X} + S$)
- Approximately 95% of the measurements will fall within 2 standard deviations of the mean, i.e. within the interval ($\bar{X} - 2S$, $\bar{X} + 2S$).
- Approximately 100% (practically all) of the measurements will fall within 3 standard deviations of the mean, i.e. within the interval ($\bar{X} - 3S$, $\bar{X} + 3S$).

Let us understand this point with the help of an example:

EXAMPLE

The 50 companies' percentages of revenues spent on R&D (i.e. Research and Development) are:

13.5	9.5	8.2	6.5	8.4	8.1	6.9	7.5	10.5	13.5
7.2	7.1	9.0	9.9	8.2	13.2	9.2	6.9	9.6	7.7
9.7	7.5	7.2	5.9	6.6	11.1	8.8	5.2	10.6	8.2
11.3	5.6	10.1	8.0	8.5	11.7	7.1	7.7	9.4	6.0
8.0	7.4	10.5	7.8	7.9	6.5	6.9	6.5	6.8	9.5

Calculate the proportions of these measurements that lie within the intervals $\bar{X} \pm S$, $\bar{X} \pm 2S$, and $\bar{X} \pm 3S$, and compare the results with the theoretical values. The mean and standard deviation of these data come out to be 8.49 and 1.98, respectively.

Mean:

$\bar{X} = 8.49$

Standard deviation:

$S = 1.98$

Hence

$(\bar{X} - S, \bar{X} + S)$

$= (8.49 - 1.98, 8.49 + 1.98)$

$= (6.51, 10.47)$

A check of the measurement reveals that 34 of the 50 measurements, or 68%, fall between 6.51and 10.47.

Similarly, the interval

$(\bar{X} - 2S, \bar{X} + 2S)$

$= (8.49 - 3.96, 8.49 + 3.96)$

$= (4.53, 12.45)$

Contains 47 of the 50 measurements, i.e. 94% of the data-values

Finally, the 3-standard deviation interval around $\bar{X}$, i.e. $(\bar{X} - 3S, \bar{X} + 3S)$

$= (8.49 - 5.94, 8.49 + 5.94)$

$= (2.55, 14.43)$ contains all, or 100%, of the measurements.

In spite of the fact that the distribution of these data is skewed to the right, the percentages of data-values falling within 1, 2, and 3 standard deviations of the mean are remarkably close to the theoretical values (68%, 95%, and 100%) given by the Empirical Rule.

The fact of the matter is that, unless the distribution is extremely skewed, the mound-shaped approximations will be reasonably accurate. Of course, no matter what the shape of the distribution, Chebychev's Rule, assures that at least 75% and at least 89% (8/9) of the measurements will lie within 2 and 3 standard deviations of the mean, respectively.

In this example, 94% of the values are lying inside the interval $\bar{X} + 2S$, and this percentage IS greater than 75%.

Similarly,100% of the values are lying inside the interval $\bar{X} + 3S$, and this percentage IS greater than 89%.

But, before we discuss all these new concepts, let us revise the concept of the Chebychev's Inequality. In the last lecture, we noted that when all the values in a set of data are located near their mean, they exhibit a small amount of variation or dispersion.

And those sets of data in which some values are located far from their mean have a large amount of dispersion. Expressing these relationships in terms of the standard deviation, which measures dispersion, we can say that when the values of a set of data are concentrated near their mean, the standard deviation is small. And when the values of a set of data are scattered widely about the mean, the standard deviation is large. In exactly the same way, if the standard deviation computed from a set of data is large, the values from which it is computed are dispersed widely about their mean. A useful rule that illustrates the relationship between dispersion and standard deviation is given by *Chebychev's theorem*, named after the Russian mathematician P.L. Chebychev (1821-1894). This theorem enables us to calculate for any set of data the minimum proportion of values that can be expected to lie within a specified number of standard deviations of the mean.

The theorem tells us that at least 75% of the values in a set of data can be expected to fall within two standard deviations of the mean, at least 89% (8/9) within three standard deviations of the mean, and at least 94% (15/16) within four standard deviations of the mean.
In general, Chebychev's theorem may be stated as follows:

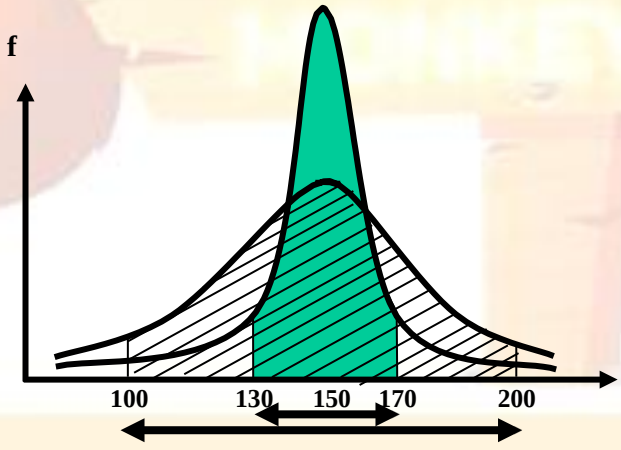
CHEBYCHEV'S THEOREM

Given a set of n observations x1, x2, x3… xn on the variable X, the probability is at least (1 – 1/k2) that X will take on a value within k standard deviations of the mean of the set of observations (where k > 1).
Chebychev's theorem is applicable to any set of observations, so we can use it for either samples or populations. Let us now see how we can suppose that a set of data has a mean of 150 and a standard deviation of 25. Putting k = 2 in the Chebychev's theorem, at least 1 – 1/ (2)2 = 75% of the data-values will take on a value within two standard deviations of the mean.
Apply it in practice.
Since the standard deviation is 25, hence 2(25) = 50, and at least 75% of the data-values will take on a value between 150 – 50 = 100 and 150 + 50 = 200. Consequently, we can say that we can expect at least 75% of the values to be between 100 and 200.By similar calculations we find that we can expect at least 89% to be between 75 and 225, and at least 96% to be between 25 and 275.
(The last statement has been made by putting k = 5 in the formula 1 - 1/k2)
Suppose that another set of data has the same mean as before, i.e. 150, but a standard deviation of 10. Applying Chebychev's theorem, for this set of data we can expect at least 75% of the values to be between 130 and 170, at least 89% to be between 120 and 180, and at least 96% to be between 100 and 200.
The above results are summarized in the following table:

PERCENTAGE OF DATA	FOR DATA-SET NO. 1	FOR DATA-SET NO. 2
At least 75 %	Lies Between 100 & 200	Lies Between 130 & 170
At least 89 %	Lies Between 75 & 225	Lies Between 120 & 180
At least 96 %	Lies Between 25 & 275	Lies Between 100 & 200

Thus the intervals computed for the latter set of data are all narrower than those for the former.
For two symmetric, hump-shaped distributions having the same mean, this point is depicted in the following diagram:

THE SYMMETRIC CURVE



Therefore, we see that for a set of data with a small standard deviation, a larger proportion of the values will be concentrated near the mean than for a set of data with a large standard deviation.
A limitation of the Chebychev's theorem is that it gives no information at all about the probability of observing a value within one standard deviation of the mean, since 1 – 1/k2 = 0 when k = 1. Also, it should be noted that the Chebychev's theorem provides weak information for our variable of interest. For many random variables, the probability of observing a value within 2 standard deviations of the mean is far greater than 1 – 1/22 = 0.75.
In this way, the Chebychev's theorem and the Empirical Rule play an important role in understanding the nature and importance of the standard deviation as a measure of dispersion.

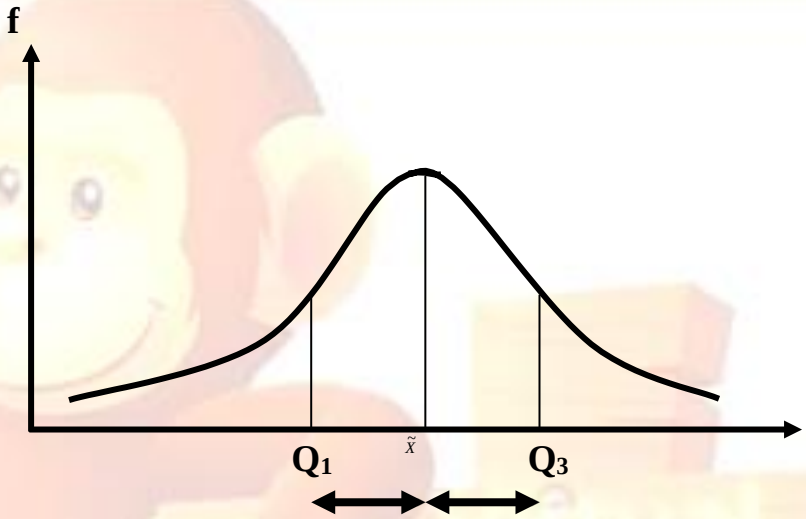
The next topic of today's lecture is the five-number summary. (Now that we have studied the three major properties of numerical data (i.e. central tendency, variation, and shape), it is important that we identify and describe the major features of the data in a summarized format.)
One approach to this "exploratory data analysis" is to develop a five-number summary.

FIVE-NUMBER SUMMARY

A five-number summary consists of X0,Q1, Median, Q3, and Xm ; It provides us quite a good idea about the shape of the distribution. If the data were perfectly symmetrical, the following would be true:

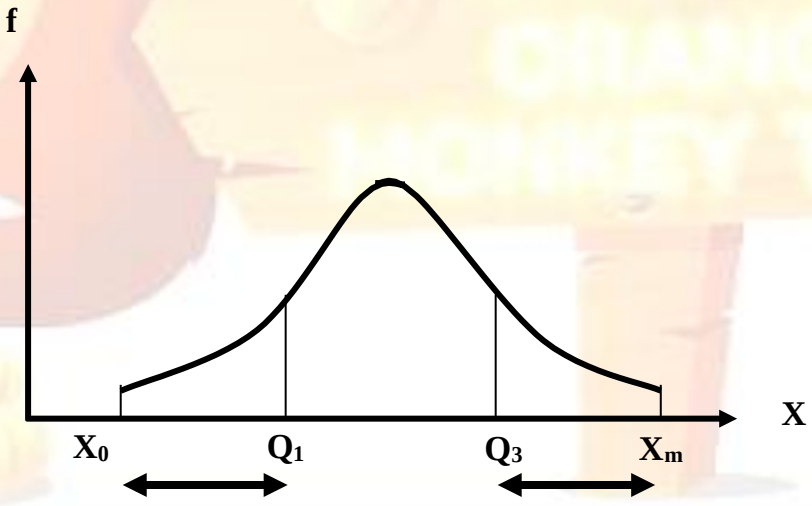
1. The distance from Q1 to the median would be equal to the distance from the median to Q3:

THE SYMMETRIC CURVE



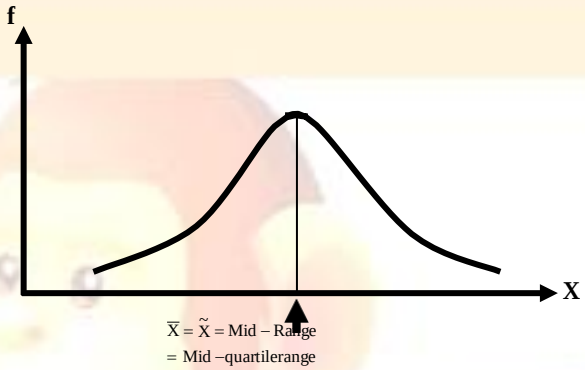
3. The distance from X_0 to Q_1 would be equal to the distance from Q_3 to X_m.

THE SYMMETRIC CURVE



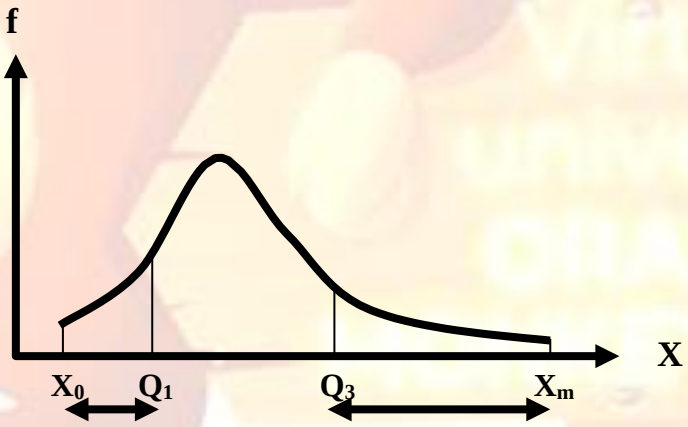
3. The median, the mid-quartile range, and the midrange would all be equal. All these measures would also be equal to the arithmetic mean of the data:

THE SYMMETRIC CURVE



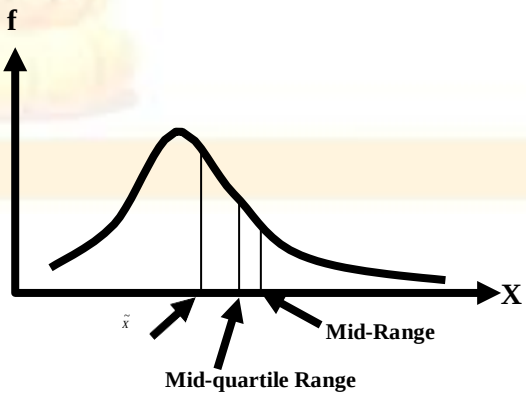
On the other hand, for non-symmetrical distributions, the following would be true:
1. In right-skewed distributions the distance from Q_3 to X_m greatly exceeds the distance from X_0 to Q_1.

THE POSITIVELY SKEWED CURVE



2. in right-skewed distributions,
median < mid-quartile range < midrange:

THE POSITIVELY SKEWED CURVE



Similarly, in left-skewed distributions, the distance from X0 to Q1 greatly exceeds the distance from Q3 to Xm.Also, in left-skewed distributions, midrange < mid-quartile range < median.
Let us try to understand this concept with the help of an example

EXAMPLE

Suppose that a study is being conducted regarding the annual costs incurred by students attending public versus private colleges and universities in the United States of America. In particular, suppose, for exploratory purposes, our sample consists of 10 Universities whose athletic programs are members of the 'Big Ten' Conference. The annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference are given as follows:

Name of University	Annual Costs (in $000)
Indiana University	15.6
Michigan State University	17.0
Ohio State University	15.2
Pennsylvania State University	16.4
Purdue University	15.2
University of Illinois	15.4
University of Iowa	13.0
University of Michigan	23.1
University of Minnesota	14.3
University of Wisconsin	14.9

If we wish to state the five-number summary for these data, the first step will be to arrange our data-set in ascending order:
Ordered Array:

$X_0 = 13.0$	14.3	14.9	15.2	15.2	15.4	15.6	16.4	17.0	$X_m = 23.1$

And if we carry out the relevant computations, we find that:

- The median for this data comes out to be 15.30 thousand dollars.
- The first quartile comes out to be 14.90 thousand dollars, and
- The third quartile comes out to be 16.40 thousand dollars.

Therefore, the five-number summary for this data-set is:
The Five-Number Summary:

X_0	Q_1	$\tilde{X}$	Q_3	X_m
13.0	14.9	15.3	16.4	23.1

If we apply the rules that I am conveyed to you a short while ago, it is clear that the annual cost data for our sample are right-skewed. We come to this conclusion because of two reasons:

- The distance from Q3 to Xm (i.e., 6.7) greatly exceeds the distance from X0 to Q1 (i.e., 1.9).
- If we compare the median (which is 15.3), the mid-quartile range (which is 15.65), and the midrange (which is 18.05), we observe that the median < the mid-quartile range < the midrange.

Both these points clearly indicate that our distribution is positively skewed.
The gist of the above discussion is that the five-number summary is a simple yet effective way of determining the shape of our frequency distribution --- without actually drawing the graph of the frequency distribution.

LECTURE NO. 13

- Box and Whisker Plot
- Pearson's Coefficient of Skewness

Prior to discussing the THE BOX-AND-WHISKER PLOT, let us review the concept of THE FIVE-NUMBER SUMMARY. As indicated in the last lecture, once we have studied the three major properties of numerical data (i.e. central tendency, variation, and shape), it is important that we identify and describe the major features of the data in a SUMMARIZED format. One way of doing this is to develop a five-number summary.

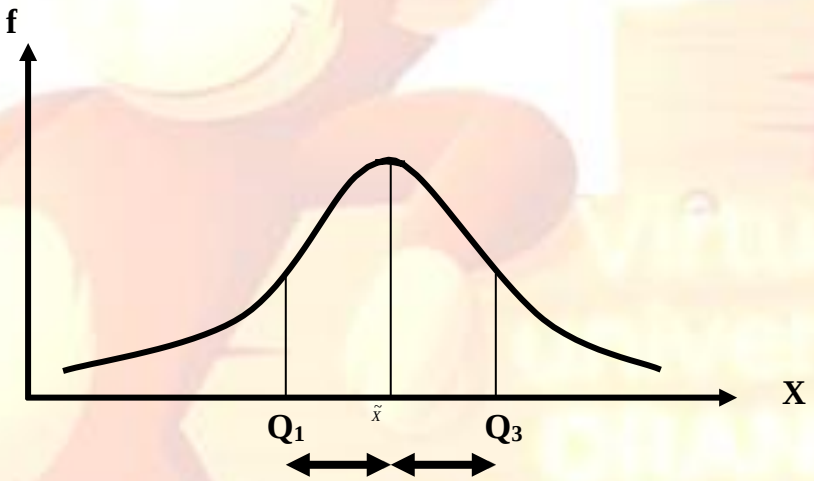
FIVE-NUMBER SUMMARY

A five-number summary consists of X_0,Q_1, Median, Q_3; Xm.It provides us a better idea as to the SHAPE of the distribution, as explained below:

If the data were perfectly symmetrical, the following would be true:

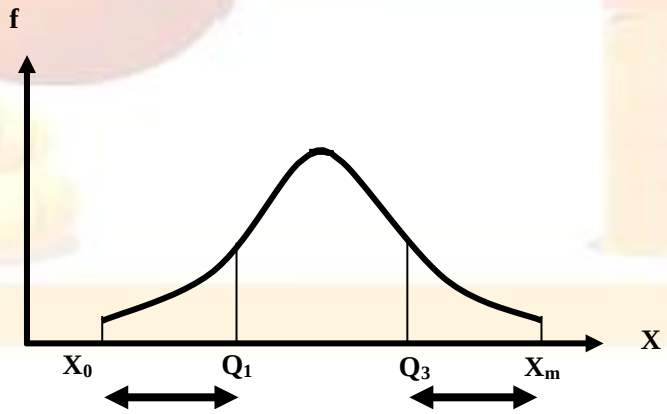
1. The distance from Q1 to the median would be equal to the distance from the median to Q3, as shown below:

THE SYMMETRIC CURVE



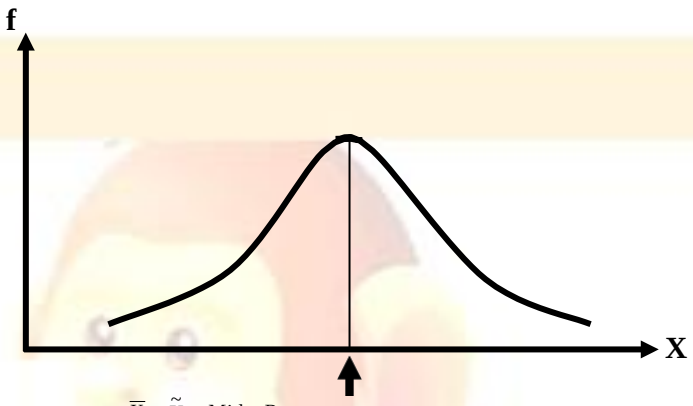
2. The distance from X0 to Q1 would be equal to the distance from Q3 to Xm, as shown below:

THE SYMMETRIC CURVE



3. The median, the mid-quartile range, and the midrange would ALL be equal.
These measures would also be equal to the arithmetic mean of the data, as shown below:

THE SYMMETRIC CURVE

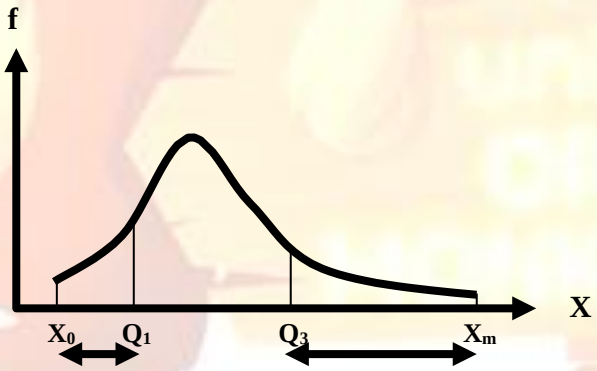


$\overline{X} = \tilde{X} = \text{Mid} - \text{Range}$
$= \text{Mid} - \text{quartilerange}$

On the other hand, for non-symmetrical distributions, the following would be true:

1. In right-skewed (positively-skewed) distributions the distance from Q3 to Xm greatly EXCEEDS the distance from X0 to Q1, as shown below:

THE POSITIVELY SKEWED CURVE

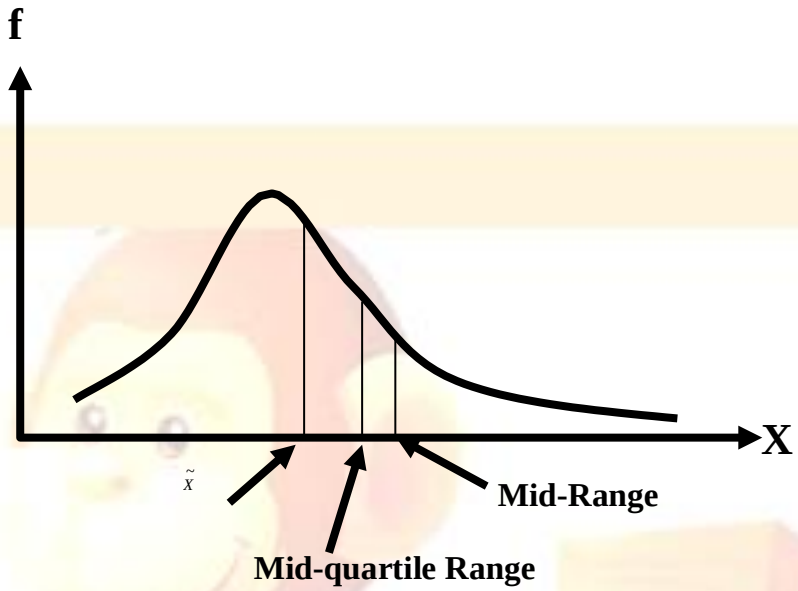


2. In right-skewed distributions,

median < mid-quartile range < midrange

This is indicated in the following figure:

THE POSITIVELY SKEWED CURVE



Similarly, in left-skewed distributions, the distance from X0 to Q1 greatly exceeds the distance from Q3 to Xm.Also, in left-skewed distributions, midrange < mid-quartile range < median.
Let us try to understand this concept with the help of an example:

EXAMPLE

Suppose that a study is being conducted regarding the annual costs incurred by students attending public versus private colleges and universities in the United States of America. In particular, suppose, for exploratory purposes, our sample consists of 10 Universities whose athletic programs are members of the 'Big Ten' Conference? The annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference are given in the following table; state the five-number summary for these data.

Annual Costs Incurred on Tuition Fees, etc.

Name of University	Annual Costs (in $000)
Indiana University	15.6
Michigan State University	17.0
Ohio State University	15.2
Pennsylvania State University	16.4
Purdue University	15.2
University of Illinois	15.4
University of Iowa	13.0
University of Michigan	23.1
University of Minnesota	14.3
University of Wisconsin	14.9

SOLUTION:
For our sample, the ordered array is

X_0 = 13.0	14.3	14.9	15.2	15.2	15.4	15.6	16.4	17.0	X_m = 23.1

The median for this data comes out to be 15.30 thousand dollars. The first quartile comes out to be 14.90 thousand dollars, and the third quartile comes out to be 16.40 thousand dollars.
Therefore, the five-number summary is:

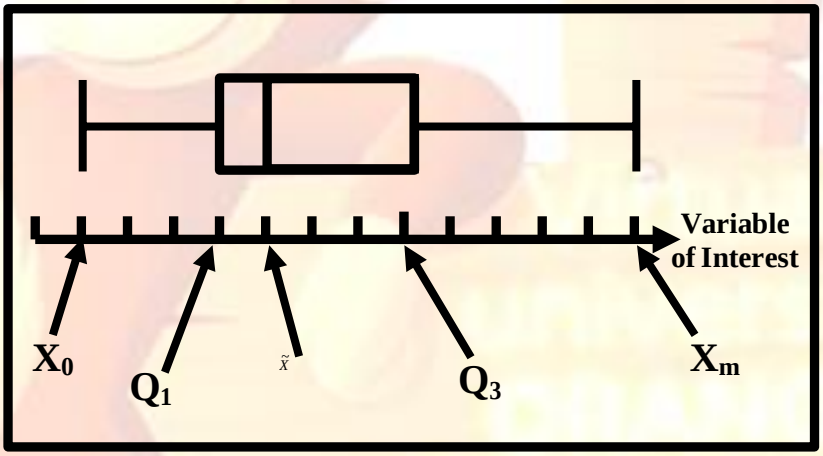
X_0	Q_1	$\tilde{X}$	Q_3	X_m
13.0	14.9	15.3	16.4	23.1

We may now use the five-number summary to study the *shape* of this distribution:
We notice that
1. The distance from Q3 to Xm (i.e., 6.7) greatly exceeds the distance from X0 to Q1 (i.e., 1.9).
2. If we compare the median (which is 15.3), the mid-quartile range (which is 15.65), and the midrange (which is 18.05), we observe that the median < the mid-quartile range < the midrange.
Hence, from the preceding rules, it is clear that the annual cost data for our sample are *right-skewed*. The gist of the above discussion is that the five-number summary is a SIMPLE yet effective way of determining the shape of our frequency distribution --- WITHOUT actually drawing the graph of the frequency distribution. The concept of the five number summary is directly linked with the concept of the box and whisker plot:

BOX AND WHISKER PLOT

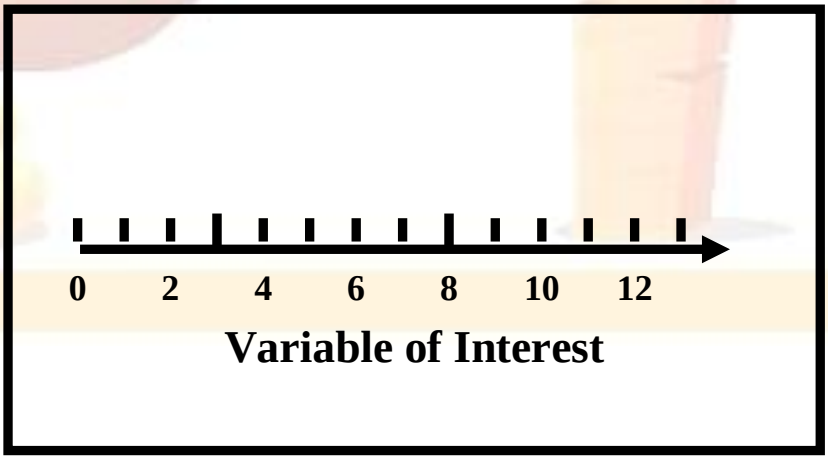
In its simplest form, a box-and-whisker plot provides a graphical representation of the data through its five-number summary.

Box and Whisker Plot

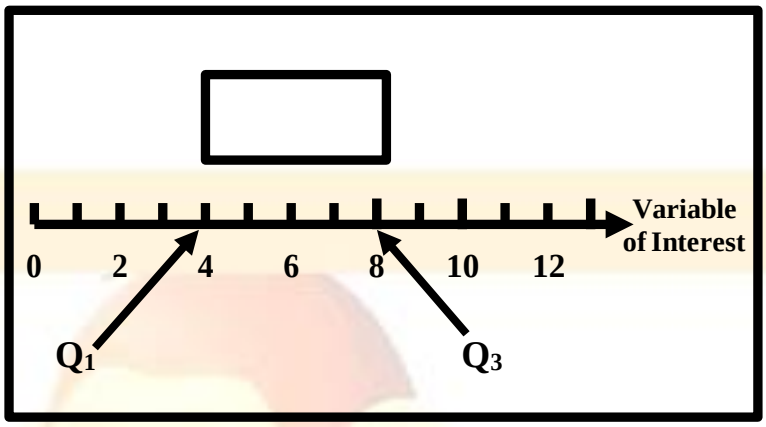


To construct a box-and-whisker plot, we proceed as follows:
Steps involved in the construction of the Box and Whisker Plot:

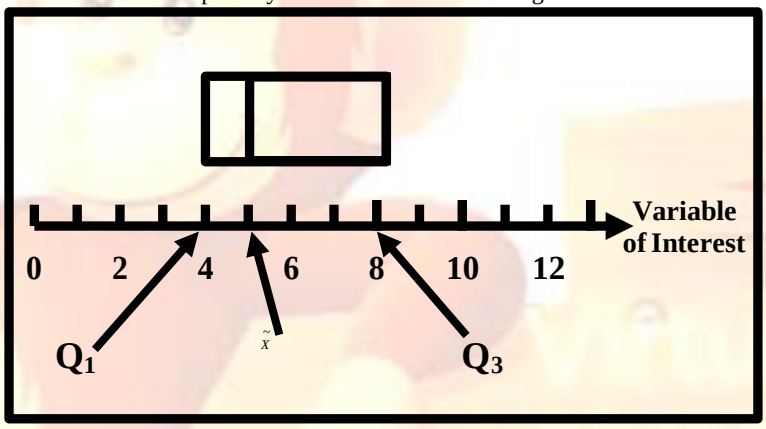
1. The variable of interest in represented on the horizontal axis.



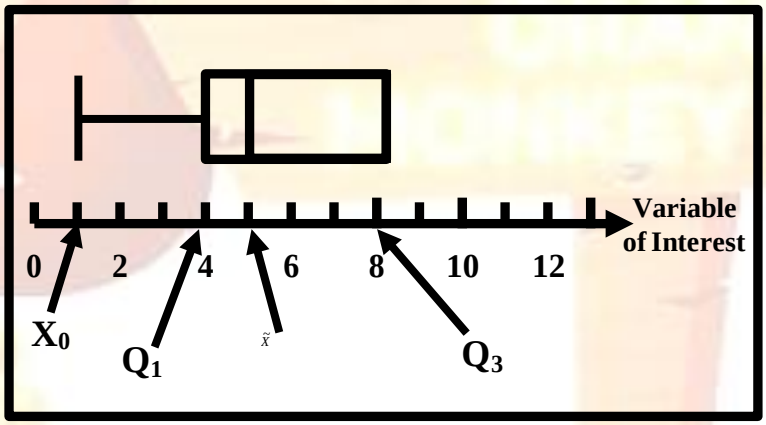
2. A BOX is drawn in the space above the horizontal axis in such a way that the left end of the box aligns with the first quartile Q1 and the right end of the box is aligned with the third quartile Q3.



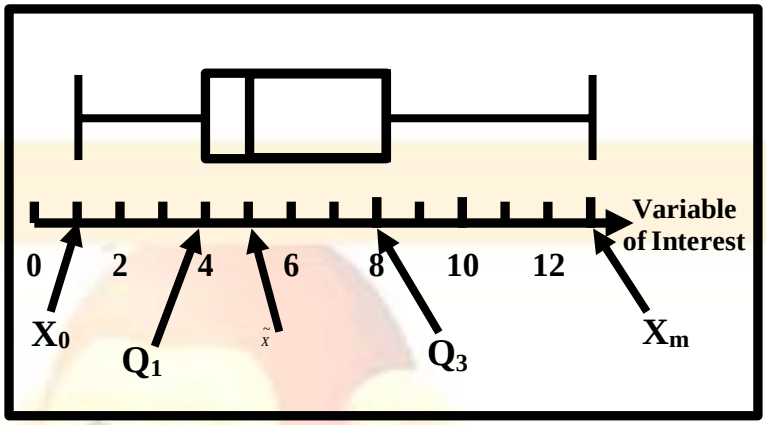
3. The box is divided into two parts by a VERTICAL line that aligns with the MEDIAN.



4. A line, called a whisker, is extended from the LEFT end of the box to a point that aligns with X0, the smallest measurement in the data set.



5. Another line, or whisker, is extended from the RIGHT end of the box to a point that aligns with the LARGEST measurement in the data set.



Let us understand the construction of the box-and-whisker plot with reference to an example:

EXAMPLE

The following table shows the downtime, in hours, recorded for 30 machines owned by a large manufacturing company. The period of time covered was the same for all machines.

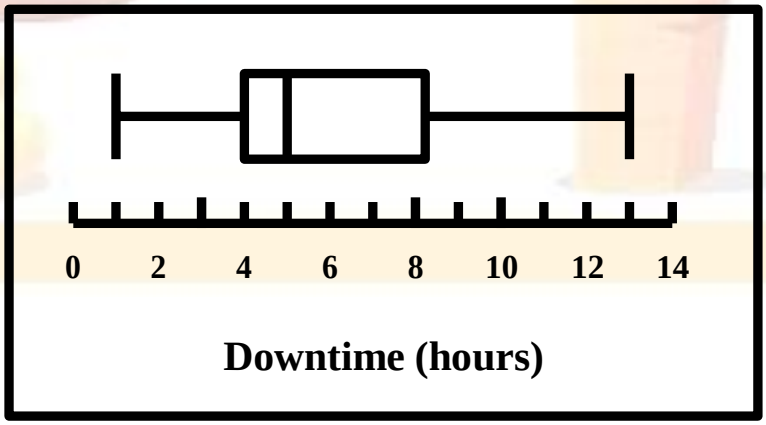
DOWNTIME IN HOURS OF 30 MACHINES

4	4	1	4	1	4
6	10	5	5	8	2
1	6	10	1	13	5
8	4	3	9	4	9
1	4	4	11	8	9

In order to construct a box-and-whisker plot for these data, we proceed as follows:
First of all, we determine the two extreme values in our data-set:
The smallest and largest values are X0 = 1 and Xm = 13, respectively.
As far as the computation of the quartiles is concerned, we note that, in this example, we are dealing with raw data.

The first quartile is the (30 + 1)/4 = 7.75th ordered measurement and is equal to 4.
The median is the (30 + 1)/2 = 15.5th measurement, or 5,and
The third quartile is the 3(30 + 1)/4 = 23.25th ordered measurement, which is 8.25.
As a result, we obtain the following box and whisker plot:

Box and Whisker Plot



INTERPRETATION OF THE BOX AND WHISKER PLOT

With regard to the *interpretation* of the Box and Whisker Plot, it should be noted that, by looking at a box-and-whisker plot, one can quickly form an impression regarding the amount of SPREAD, location of CONCENTRATION, and SYMMETRY of our data set.
A glance at the box and whisker plot of the example that we just considered reveals that:

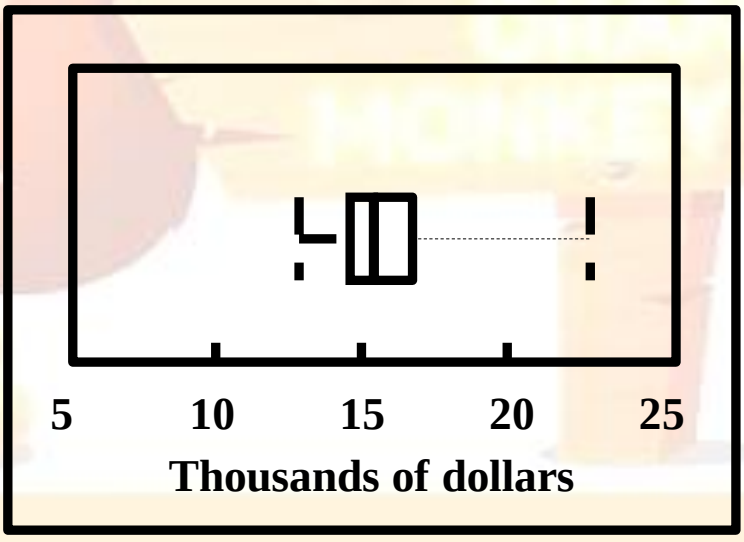
- 50% of the measurements are between 4 and 8.25.
- The median is 5, and the range is 12. and, most importantly:
- Since the median line is closer to the *left* end of the box, hence the data are SKEWED to the RIGHT.(The fundamental point is that in a perfectly symmetrical data set, the median line will be EXACTLY HALFWAY between the two ends of the box, and in a data set that is skewed to the LEFT, the median line will be CLOSER TO THE RIGHT END of the box.)

Let us consolidate all the above ideas by going back to the example of the Big Ten Universities in which the annual costs incurred for tuition fees, room, and board at 10 schools belonging to Big Ten Conference were given as follows:

Name of University	Annual Costs (in $000)
Indiana University	15.6
Michigan State University	17.0
Ohio State University	15.2
Pennsylvania State University	16.4
Purdue University	15.2
University of Illinois	15.4
University of Iowa	13.0
University of Michigan	23.1
University of Minnesota	14.3
University of Wisconsin	14.9

As stated earlier, the Five-Number Summary of this data-set is :
For this data, the Box and Whisker Plot is of the form given below:

Box and Whisker Plot



As indicated earlier, the vertical line drawn within the box represents the location of the median value in the data; the vertical line at the LEFT side of the box represents the location of Q1, and the vertical line at the RIGHT side of the box represents the location of Q3. Therefore, the BOX contains the middle 50% of the observations in the distribution. The lower 25% of the data are represented by the whisker that connects the *left* side of the box to the location of the *smallest* value, X0, and the upper 25% of the data are represented by the whisker connecting the *right* side of the box to Xm.

INTERPRETATION OF THE BOX AND WHISKER PLOT

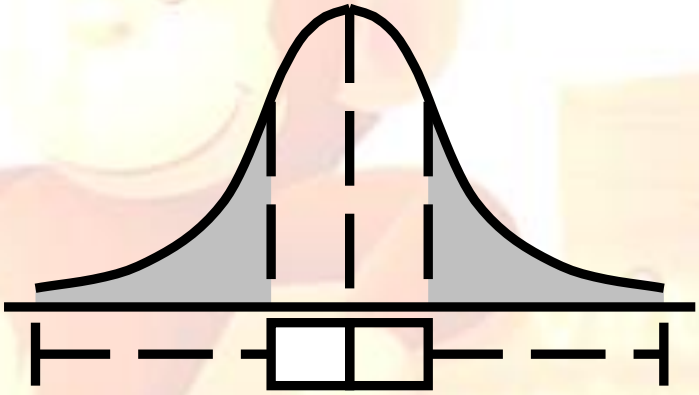
We note that (1) the vertical median line is CLOSER to the left side of the box, and (2) the left side whisker length is clearly SMALLER than the right side whisker length .Because of these observations, we The gist of the above discussion is that if the median line is at a greater distance from the left side of the box as compared with its distance from the right side of the box, our distribution will be skewed to the left.

In this situation, the whisker appearing on the left side of the box and whisker plot will be longer than the whisker of the right side. Conclude that the data-set of the annual costs is RIGHT-skewed.

The gist of the above discussion is that if the median line is at a greater distance from the left side of the box as compared with its distance from the right side of the box, our distribution will be skewed to the left. In this situation, the whisker appearing on the left side of the box and whisker plot will be longer than the whisker of the right side. The Box and Whisker Plot comes under the realm of "exploratory data analysis" (EDA) which is a relatively new area of statistics. The following figures provide a comparison between the Box and Whisker Plot and the traditional procedures such as the frequency polygon and the frequency curve with reference to the SKEWNESS present in the data-set.

Four different types of hypothetical distributions are depicted through their box-and-whisker plots and corresponding frequency curves.

1) When a data set is perfectly symmetrical, as is the case in the following two figures, the mean, median, midrange, and mid-quartile range will be the SAME:



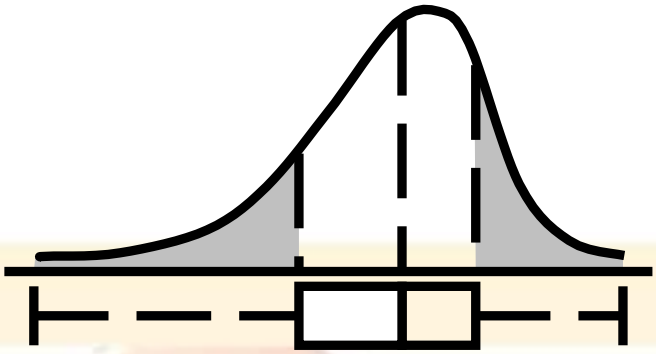
(a) Bell-shaped distribution



(b) Rectangular distribution

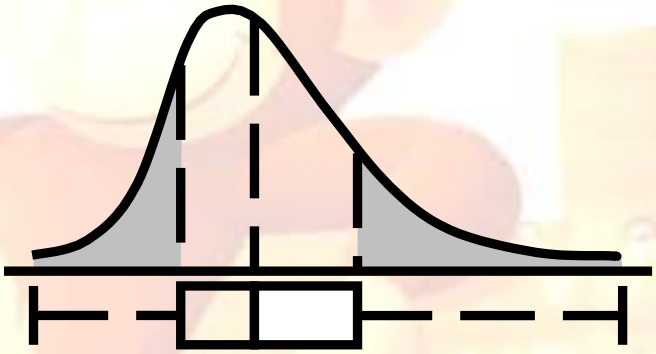
In ADDITION, the length of the left whisker will be equal to the length of the right whisker, and the median line will divide the box in HALF. (In *practice*, it is unlikely that we will observe a data set that is perfectly symmetrical. However, we should be able to state that our data set is *approximately* symmetrical *if* the lengths of the two whiskers are *almost* equal and the *median* line *almost* divides the box in HALF.)

2) When our data set is LEFT-skewed as in the following figure, the few small observations pull the midrange and mean toward the LEFT tail:



Left-skewed distribution

For this LEFT-skewed distribution, we observe that the skewed nature of the data set indicates that there is a HEAVY CLUSTERING of observations at the HIGH END of the scale (i.e., the RIGHT side).
75% of all data values are found between the left edge of the box (Q1) and the end of the right whisker (Xm). Therefore, the LONG left whisker contains the distribution of only the smallest 25% of the observations, demonstrating the distortion from symmetry in this data set.
3) If the data set is RIGHT-skewed as shown in the following figure, the few large observations PULL the midrange and mean toward the right tail.



Right-skewed distribution

For the right-skewed data set, the concentration of data points is on the LOW end of the scale (i.e., the left side of the box-and-whisker plot). Here, 75% of all data values are found between the beginning of the left whisker (X0) and the RIGHT edge of the box (Q3), and the remaining 25% of the observations are DISPERSED ALONG the LONG right whisker at the upper end of the scale. This brings us to the end of the discussion of the five number summary and the box and whisker plot.
Next, we discuss *another* way of determining the skewness of the data-set and that is the **PEARSON'S**

COEFFICIENT OF SKEWNESS

In this connection, the first thing to note is that, by providing information about the location of a series and the dispersion within that series it might appear that we have achieved a PERFECTLY adequate overall description of the data. But, the fact of the matter is that, it is quite possible that two series are decidedly dissimilar and yet have exactly the same arithmetic mean AND standard deviation: Let us understand this point with the help of an example:

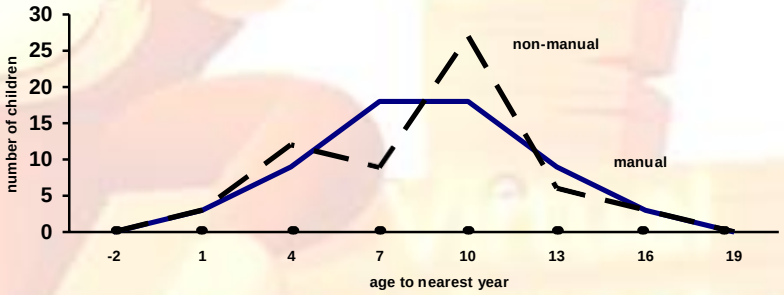
EXAMPLE:

Age of Onset of Nervous Asthma in Children (to Nearest Year)	Children of Manual Workers	Children of Non-Manual Workers
0 – 2	3	3
3 – 5	9	12
6 – 8	18	9
9 – 11	18	27
12 – 14	9	6
15 – 17	3	3
	60	60

In order to compute the mean and standard deviation for each distribution, we carry out the following calculations:

Age of Onset of Nervous Asthma in Children (to Nearest Year)		Children of Manual Workers			Children of Non-Manual Workers		
Age Group	X	f_1	f_1X	f_1X^2	f_2	f_2X	f_2X^2
0 – 2	1	3	3	3	3	3	3
3 – 5	4	9	36	144	12	48	192
6 – 8	7	18	126	882	9	63	441
9 – 11	10	18	180	1800	27	270	2700
12 – 14	13	9	117	1521	6	78	1014
15 – 17	16	3	48	768	3	48	768
	51	60	510	5118	60	510	5118

We find that, for each of the two distributions, the mean is 8.5 years and the standard deviation is 3.61 years.
The frequency polygons of the two distributions are as follows:



By inspecting these, it can be seen that one distribution is symmetrical while the other is quite different. The distinguishing feature here is the degree of asymmetry or *SKEWNESS* in the two polygons. In order to *measure* the skewness in our distribution, we compute the **PEARSON's COEFFICIENT OF SKEWNESS** which is defined as:
Pearson's Coefficient of Skewness:

$$\frac{mean - mode}{standard\ deviation}$$

Applying the *empirical relation* between the mean, median and the mode, the Pearson's Coefficient of Skewness is given by:

Pearson's Coefficient of Skewness

$$= \frac{3\,(\text{mean} - \text{median})}{\text{standard deviation}}$$

For a *symmetrical* distribution the coefficient will always be ZERO, for a distribution skewed to the RIGHT the answer will always be positive, and for one skewed to the LEFT the answer will always be negative.
Let us now calculate this coefficient for the example of the children of the manual and non-manual workers.
Sample statistics pertaining to the ages of these children are as follows:

	Children of Manual Workers	Children of Non-Manual Workers
Mean	8.50 years	8.50 years
Standard deviation	3.61 years	3.61 years
Median	8.50 years	9.16 years
Q_1	6.00 years	5.50 years
Q_3	11.00 years	10.83 years
Quartile deviation	2.50 years	2.66 years

The Pearson's Coefficient of Skewness is calculated for each of the two categories of children, as shown below: Pearson's Coefficient of Skewness (Modified):

Ages of Children of Manual Workers

$$\frac{3(8.50-8.50)}{3.61}$$

$= 0$

Ages of Children of Non-Manual Workers

$$\frac{3(8.50-9.16)}{3.61}$$

$= -0.55$

For the data pertaining to children of manual workers, the coefficient is zero, whereas, for the children of non-manual workers, the coefficient has turned out to be a negative number. This indicates that the distribution of the ages of the children of the manual workers is symmetric whereas the distribution of the ages of the children of the non-manual workers is negatively skewed.

The students are encouraged to draw the frequency polygon and the frequency curve for each of the two distributions, and to compare the results that have just been obtained with the *shapes* of the two distributions.

LECTURE NO. 14

- Bowley's coefficient of skewness
- The Concept of Kurtosis
- Percentile Coefficient of Kurtosis
- Moments & Moment Ratios
- Sheppard's Corrections
- The Role of Moments in Describing Frequency Distributions

You will recall that the Pearson's coefficient of skewness is defined as (mean - mode)/standard deviation, and if we apply the empirical relation between the mean, median and the mode, then the coefficient is given by:

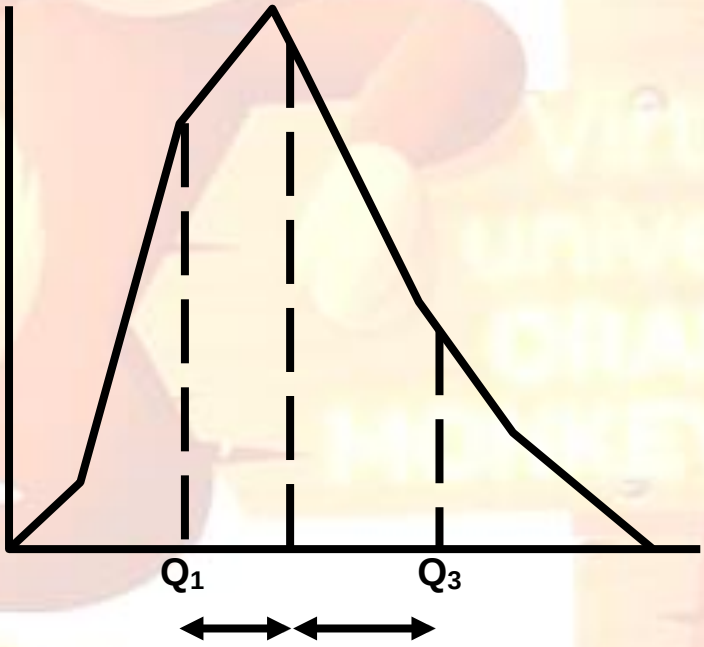
PEARSON'S COEFFICIENT OF SKEWNESS:

$$= \frac{3\,(\text{mean} - \text{median})}{\text{standard deviation}}$$

As you can see, this coefficient involves the calculation of the mean as well as the standard deviation. Actually, the numerator is divided by the standard deviation in order to obtain a pure number. If the analysis of a data-set is being undertaken using the median and quartiles alone, then we use a measure called Bowley's coefficient of skewness.

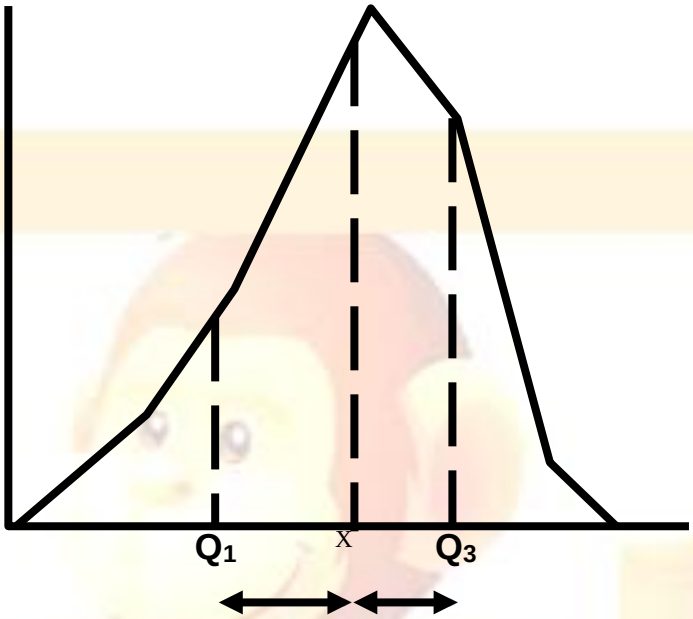
The advantage of this particular formula is that it requires NO KNOWLEDGE of the MEAN or STANDARD DEVIATION. In an asymmetrical distribution, the quartiles will NOT be equidistant from the median, and the AMOUNT by which each one deviates will give an indication of skewness. Where the distribution is positively skewed, Q1 will be closer to the median than Q3.In other words, the distance between Q3 and the median will be greater than the distance between the median and Q1.

POSITIVE SKEWNESS



And hence, if we subtract the distance median - Q1 from the distance Q3 - median, we will obtain a positive answer.
In case of a positively skewed distribution:
(Q3 - median) - (Median - Q1) > 0
i.e. Q1 + Q3 - 2 median > 0
The opposite is true for skewness to the left

NEGATIVE SKEWNESS



In this case:

(Q3 - median) - (Median - Q1) < 0 i.e.

Q1 + Q3 - 2 median < 0

The gist of the above discussion is that in case of a positively skewed distribution, the quantity

$$\mathbf{Q1 + Q3 -}\ 2\tilde{X}$$

will be positive, whereas in case of a negatively distribution, this quantity will be negative.

A RELATIVE measure of skewness is obtained by dividing

$$\mathbf{Q1 + Q3 -}\ 2\tilde{X}$$

by the inter-quartile range i.e. Q3 - Q1, so that Bowley's coefficient of skewness is given by:

Bowley's coefficient of Skewness

$$= \frac{\left(Q_1 + Q_3 - 2\tilde{X}\right)}{Q_3 - Q_1}$$

It is a pure (unit less) number, and its value lies between 0 and ± 1.

For a positively skewed distribution, this coefficient will turn out to be positive, and for a negatively skewed distribution this coefficient will come out to be negative. Let us apply this concept to the example regarding the ages of children of the manual and non-manual workers that we considered in the last lecture.

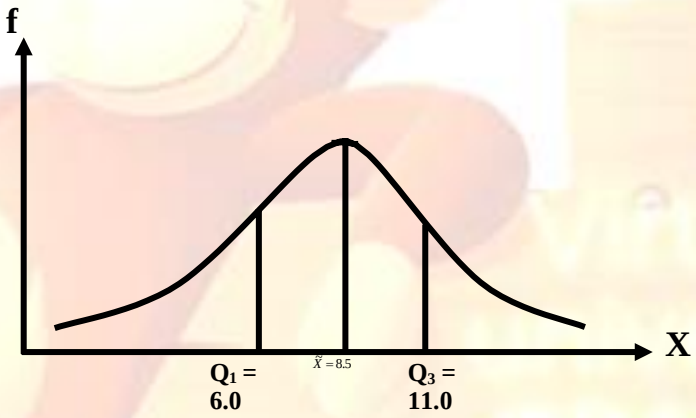
Age of Onset of Nervous Asthma in Children (to Nearest Year)	Children of Manual Workers	Children of Non-Manual Workers
0 – 2	3	3
3 – 5	9	12
6 – 8	18	9
9 – 11	18	27
12 – 14	9	6
15 – 17	3	3
	60	60

EXAMPLE:
Sample statistics pertaining to ages of children of manual and non-manual workers:

	Children of Manual Workers	Children of Non-Manual Workers
Mean	8.50 years	8.50 years
Standard deviation	3.61 years	3.61 years
Median	8.50 years	9.16 years
Q_1	6.00 years	5.50 years
Q_3	11.00 years	10.83 years
Quartile deviation	2.50 years	2.66 years

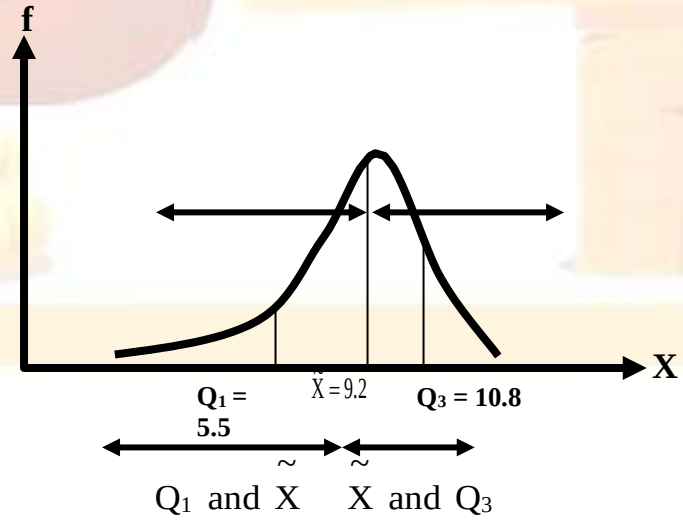
The statistics pertaining to children of manual workers yield the following PICTURE:

Ages of Children of Manual Workers



On the other hand, the statistics pertaining to children of non-manual workers yield the following PICTURE:

Ages of Children of Non-Manual Workers



The diagram pertaining to children of non-manual workers clearly shows that the distance between

is much greater than the distance between
which happens whenever we are dealing with a negatively skewed distribution. If we compute the Bowley's coefficient of skewness for each of these two data-sets, we obtain:

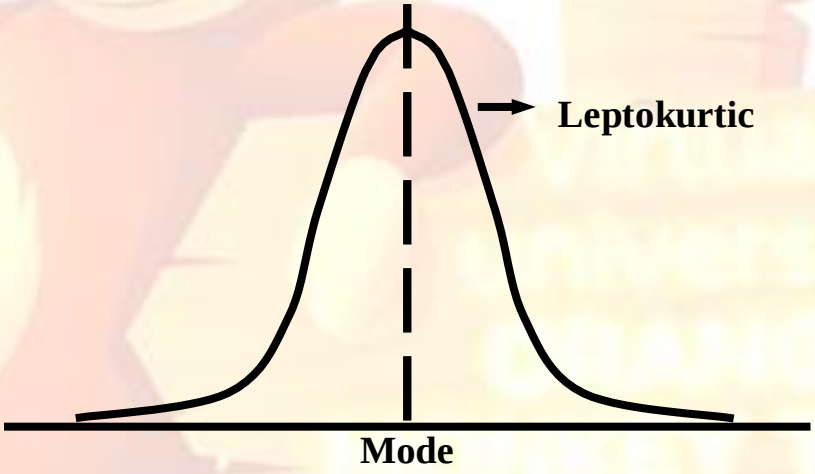
Bowley's Coefficient of Skewness

Ages of Children of Manual Workers	**Ages of Children of Non-Manual Workers**
$= \frac{11.00 + 6.00 - 2\times 8.50}{2.50}$	$\frac{10.83 + 5.50 - 2\times 9.16}{10.83 - 5.50}$
= 0	**= – 0.37**

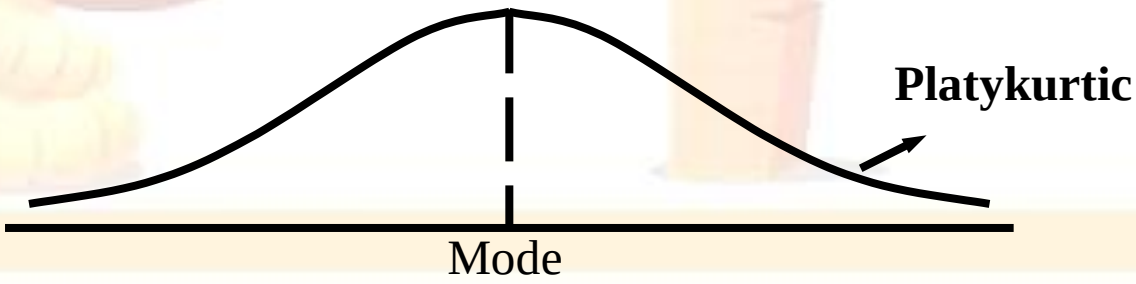
As you have noticed, for the children of the manual workers, the Bowley's coefficient has come out to be zero, whereas for the children of the non-manual workers, the coefficient has come out to be negative. This indicates that the distribution of the ages of the children of manual workers is symmetrical whereas the distribution of the ages of the children of the non-manual workers IS negatively skewed --- EXACTLY the same conclusion that we obtained when we computed the Pearson's coefficient of skewness.

KURTOSIS

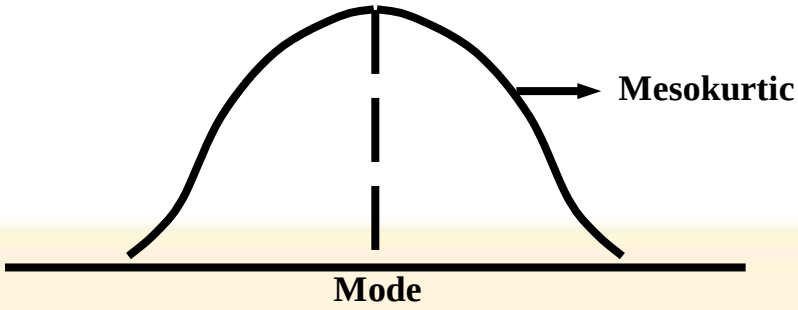
The term kurtosis was introduced by Karl Pearson. This word literally means 'the amount of hump', and is used to represent the degree of PEAKEDNESS or flatness of a unimodal frequency curve.
When the values of a variable are closely BUNCHED round the mode in such a way that the peak of the curve becomes relatively high, we say that the curve is LEPTOKURTIC.



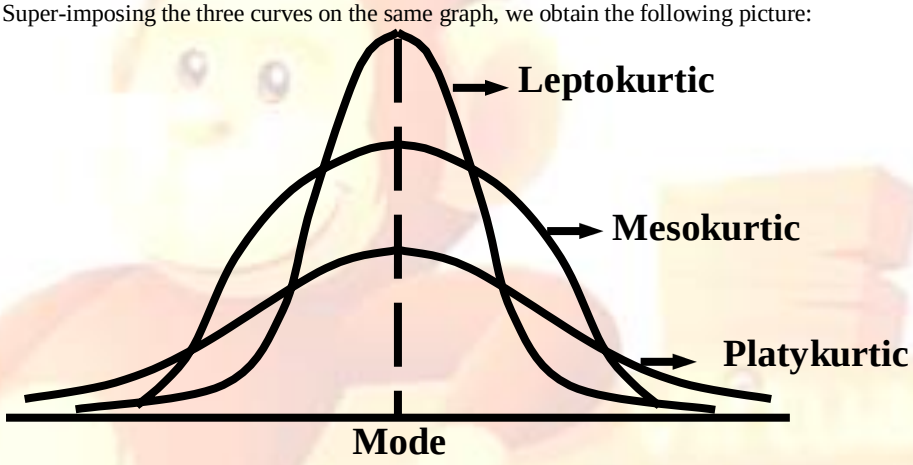
On the other hand, if the curve is flat-topped, we say that the curve is **PLATYKURTIC:**



The NORMAL curve is a curve which is neither very peaked nor very flat, and hence it is taken as A BASIS FOR COMPARISON. The normal curve itself is called **MESOKURTIC.**
I will discuss with you the normal in detail when we discuss continuous probability distributions.
At the moment, just think of the symmetric hump shaped curve shown below:



Super-imposing the three curves on the same graph, we obtain the following picture:



The tallest one is called leptokurtic, the intermediate one is called mesokurtic, and the flat one is called platykurtic.
The question arises, "How will we MEASURE the degree of peakedness or kurtosis of a data-set?"
A MEASURE of kurtosis based on quartiles and percentiles is

$$K = \frac{Q.D.}{P_{90} - P_{10}},$$

This is known as the *PERCENTILE COEFFICIENT OF KURTOSIS.*
It has been shown that K for a normal distribution is 0.263 and that it lies between 0 and 0.50.
In case of a leptokurtic distribution, the percentile coefficient of kurtosis comes out to be LESS THAN 0.263, and in the case of a platykurtic distribution, the percentile coefficient of kurtosis comes out to be GREATER THAN 0.263.The next concept that I am going to discuss with you is the concept of moments --- a MATHEMATICAL concept, and a very important concept in statistics.

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MOMENTS

A *moment* designates the power to which deviations are raised before averaging them.
For example, the quantity

$$\frac{1}{n}\sum(x_i - \bar{x})^1 = \frac{1}{n}\sum(x_i - \bar{x})$$

is called the first sample moment about the mean, and is denoted by m1.
Similarly, the quantity

$$\frac{1}{n}\sum(x_i - \bar{x})^2$$

is called the second sample moment about the mean, and is denoted by m2.In general, the rth moment about the mean is: the arithmetic mean of the *rth* power of the deviations of the observations from the mean. In symbols, this means that

$$m_r = \frac{1}{n}\sum(x_i - \bar{x})^r \qquad \text{for sample data.}$$

Moments about the mean are also called the central moments or the mean moments.
In a similar way, moments about an arbitrary origin, say α, are defined by the relation

$$m'_r = \frac{1}{n}\sum(x_i - \alpha)^r \qquad \text{for sample data}$$

For r = 1, we have

$$m_1 = \frac{1}{n}\sum(x_i - \bar{x}) = \frac{\sum x_i}{n} - \bar{x} = \bar{x} - \bar{x} = 0,$$

and

$$m'_1 = \frac{1}{n}\sum(x_i - \alpha) = \frac{\sum x_i}{n} - \alpha = \bar{x} - \alpha.$$

Putting r = 2 in the relation for mean moments, we see that

$$m_2 = \frac{1}{n}\sum(x_i - \bar{x})^2$$

which is exactly the same as the sample variance.
If we take the positive square root of this quantity, we obtain the standard deviation.
In the formula,

$$m'_r = \frac{1}{n}\sum(x_i - \alpha)^r$$

if we put α = 0, we obtain

$$m'_r = \frac{1}{n}\sum x_i^r$$

and this is called the rth moment about zero, or the rth moment about the origin.
Let us now consolidate the idea of moments by considering an example.

EXAMPLE

Calculate the first four moments about the mean for the following set of examination marks: 45, 32, 37, 46, 39, 36, 41, 48 & 36.
For convenience, the observed values are written in an increasing sequence. The necessary calculations appear in the table below:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})^3$	$(x_i - \bar{x})^4$
32	– 8	64	– 512	4096
36	– 4	16	– 64	256
36	– 4	16	– 64	256
37	– 3	9	– 27	81
39	– 1	1	– 1	1
41	1	1	1	1
45	5	25	125	625
46	6	36	216	1296
48	8	64	512	4096
360	0	232	186	10708

Now $\bar{x} = \frac{\sum x_i}{n} = \frac{360}{9} = 40$ marks.

Therefore

$$m_1 = \frac{\sum(x_i - \bar{x})}{n} = 0$$

$$m_2 = \frac{\sum(x_i - \bar{x})^2}{n} = \frac{232}{9} = 25.78 \ (\text{marks})^2$$

$$m_3 = \frac{\sum(x_i - \bar{x})^3}{n} = \frac{186}{9} = 20.67 \ (\text{marks})^3$$

$$m_4 = \frac{\sum(x_i - \bar{x})^4}{n} = \frac{10708}{9} = 1189.78 \ (\text{marks})^4$$

All the formulae that I have discussed until now pertain to the case of raw data.
How will we compute the various moments in the case of grouped data?

MOMENTS IN THE CASE OF GROUPED DATA

When the sample data are grouped into a frequency distribution having k classes with midpoints x1, x2, …, xk and the corresponding frequencies f1, f2, …,fk , (∑fi = n), the rth sample moments are given by

$$m_r = \frac{1}{n}\sum f_i (x_i - \bar{x})^r \ , \text{ and}$$

$$m'_r = \frac{1}{n}\sum f_i (x_i - \alpha)^r .$$

In the calculation of moments from a grouped frequency distribution, an error is introduced by the assumption that the frequencies associated with a class are located at the MIDPOINT of the class interval. You remember the concept of grouping error that I discussed with you in an earlier lecture? Our moments therefore need corrections.
These corrections were introduced by W.F. Sheppard, and hence they are known as **SHEPPARD'S CORRECTIONS:**
Sheppard's Corrections for Grouping Error:
It has been shown by W.F. Sheppard that, if the frequency distribution (i) is continuous and (ii) tails off to zero at each end, the corrected moments are as given below:

$$m_2 \text{ (corrected)} = m_2 \text{ (uncorrected)} - \frac{h^2}{12};$$

$$m_3 \text{ (corrected)} = m_3 \text{ (uncorrected)};$$

$$m_4 \text{ (corrected)} = m_4 \text{ (uncorrected)} - \frac{h^2}{2} . \ m_2 \text{ (uncorrected)} + \frac{7}{240} . \ h^4;$$

where **h** denotes the uniform class-interval.
The important point to note here is that these corrections are NOT applicable to highly skewed distributions and distributions having unequal class-intervals. I am now going to discuss with you certain mathematical RELATIONSHIPS that exist between the moments about the mean and the moments about an arbitrary origin.
The reason for doing so is that, in many situations, it is easier to calculate the moments in the first instance, about an arbitrary origin. They are then transformed to the mean-moments using the relationships that I am now going to convey to you.
The equations are:

$$m_1 = 0$$

$$m_2 = m'_2 - (m'_1)^2;$$

$$m_3 = m'_3 - 3\, m'_2\, m'_1 + 2(m'_1)^3, \text{ and}$$

$$m_4 = m'_4 - 4\, m'_3\, m'_1 + 6 m'_2 (m'_1)^2 - 3(m'_1)^4$$

In this course, I will not be discussing the mathematical derivation of these relationships. You are welcome to study the mathematics behind these formulae if you are interested. (The derivation is available in your own text book.)But I would like to give you two tips for remembering these formulae:

- In each of these relations, the sum of the coefficients of various terms on the right hand side equals zero and
- Each term on the right is of the same dimension as the term on the left.

Let us now apply these concepts to an example:

EXAMPLE

Compute the first four moments for the following distribution of marks after applying Sheppard's corrections:

Marks out of 20	5	6	7	8	9	10	11	12	13	14	15
No. of Students	1	2	5	10	20	51	22	11	5	3	1

If we wish to compute the first four moments about the mean by the direct method, first of all, we will have to compute mean itself. The mean of this particular data-set comes out to be 10.06.
But, 10.06 is not a very convenient number to work with!

This is so because when we construct the columns of $X - \overline{X}$, $(X - \overline{X})^2$ etc.,

we will have a lot many decimals. An alternative way of computing the moments is to take a convenient number as the arbitrary origin and to compute the moments about this number. Later, we utilize the relationships between the moments about the mean and the moments about the arbitrary origin in order to find the moments about the mean.
In this example, we may select 10 as the arbitrary origin, which is the X-value corresponding to the highest frequency 51, and construct the column of D which is the same as X-10. Next, we compute the columns of fD, fD2, fD3, and so on.

Earnings in Rs.(x_i)	No. of Men f_i	D_i $(x_i - 10)$	f_iD_i	$f_iD_i^2$	$f_iD_i^3$	$f_iD_i^4$
5	1	– 5	– 5	25	– 125	625
6	2	– 4	– 8	32	– 128	512
7	5	– 3	– 15	45	– 135	405
8	10	– 2	– 20	40	– 80	160
9	20	– 1	– 20	20	– 20	20
10	51	0	0	0	0	0
11	22	1	22	22	22	22
12	11	2	22	44	88	176
13	5	3	15	45	135	405
14	3	4	12	48	192	768
15	1	5	5	25	125	625
Sum	131	..	8	346	74	3718
Sum ÷ n	1	..	0.06 $=m'_1$	2.64 $=m'_2$	0.56 $=m'_3$	28.38 $=m'_4$

Moments about the mean are:

m1 = 0
m2 = m´2 – (m ´1)2 = 2.64 – (0.06)2 = 2.64

m3 = m ´3 – 3m ´2m´1 + 2 (m ´1)3
= 0.56 – 3(2.64) (0.06) + 2(0.06)3
= 0.08

m4 = m ´4 – 4m ´3m ´1 + 6m ´2 (m ´1)2 – 3(m ´1)4
= 28.38 – 4.(0.56) (0.06) + 6(2.64) (0.06)2 – 3(0.06)4

= 28.30

Applying Sheppard's corrections, we have

m2 (corrected) = m2 (uncorrected) – = 2.64 – 0.08 = 2.56,

m3 (corrected) = m3 (uncorrected) = 0.08,

m4 (corrected) = m4 (uncorrected)
– . m2 (uncorrected) +

= 28.30 – 1.32 + 0.03 = 27.01

I have discussed with you in quite a lot of detail the concept of moments.
The question arises, "Why is it that we are going through all these lengthy calculations? What is the significance of computing moments? "You will obtain the answer to this question when I discuss with you the concept of moment ratios. There are certain ratios in which both the numerators and the denominators are moments. The most common of these moment-ratios are denoted by b1 and b2, and defined by the relations:

MOMENT RATIOS:

$$b_1 = \frac{(m_3)^2}{(m_2)^3} \text{ and } b_2 = \frac{m_4}{(m_2)^2}$$

(in the case of sample data)
They are independent of origin and units of measurement, i.e. they are pure numbers.
b1 is used to measure the skewness of our distribution, and b2 is used to measure the kurtosis of the distribution.

INTERPRETATION OF b1

For symmetrical distributions, b1 is equal to zero. Hence, for any data-set, b1 comes out to be zero, we can conclude that our distribution is symmetric. It should be noted that the measure which will indicate the direction of skewness is the third moment round the mean.

If our distribution is positively skewed, m3 will be positive, and if our distribution is negatively skewed, m3 will be negative.b1 will turn out to be positive in both situations because it is given by

$$b_1 = \frac{(m_3)^2}{(m_2)^3}$$

(Since m3 is being squared, b1 will be positive regardless of the sign of m3.)

INTERPRETATION OF b2

For the normal distribution, b2 = 3.
For a leptokurtic distribution, b2 > 3, and for a platykurtic distribution, b2 < 3
You have noted that the third and fourth moments about the mean provide information about the skewness and the kurtosis of our data-set. This is so because m3 occurs in the numerator of b1 and m4 occurs in the numerator of b2.
What about the dispersion and the centre of our data-set? Do you not remember that the second moment about the mean is exactly the same thing as the variance, the positive square root of which is the standard deviation --- the most important measure of dispersion? What about the centre of the distribution? You will be interested to note that the first moment about zero is NONE OTHER than the arithmetic mean!
This is so because $\frac{1}{n}\sum(x_i - 0)^1$ is equal to $\frac{1}{n}\sum x_i$

--- none other than the arithmetic mean! In this way, the first four moments play a KEY role in describing frequency distributions.

LECTURE NO. 15

On numerous occasions, our interest lies not in just one single variable but in two, three, four or more variables. For example, if we talk about the yield of a crop, we realize that the yield of any crop depends on a variety of factors --- the fertility of the soil, the type of fertilizer used, the amount of rainfall, and so on.

- Simple Linear Regression
- Standard Error of Estimate
- Correlation

Let me begin the discussion of the bivariate situation by picking up an example.

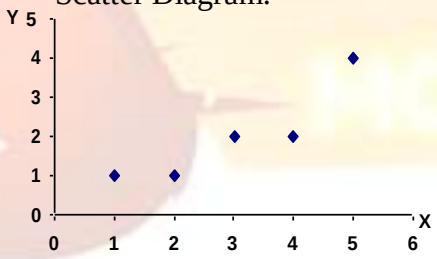
EXAMPLE:

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus.
Suppose the company administers this drug on 5 subjects and obtains the following information:

Subject	Percentage of drug	Reaction Time (milli-seconds)
	X	Y
A	1	1
B	2	1
C	3	2
D	4	2
E	5	4

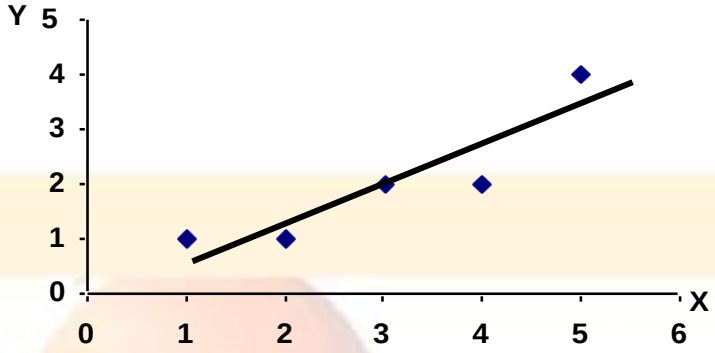
In this example, the reaction time to the stimulus will DEPEND on the amount of drug in the blood-stream. As you must know, the dependent variable is denoted by Y, and the independent variable is denoted by X.In this example, the reaction time will be denoted by Y, and the percentage of drug in the blood stream by X. Going back to the example that we were just considering, it is obvious that we are interested in determining the nature of the relationship between the amount of drug in the blood stream and the time it takes to react to a stimulus.
In order to ascertain the nature of the relationship between these two variables, the first step is to draw a **SCATTER DIAGRAM** --- which is a simple graph of the X-values against the Y-values depicted on the graph paper in the form of points.
In this example, the scatter diagram is as follows:

Scatter Diagram:



As you can see, there is an upward trend in the scatter diagram i.e. it is clear that as X increases, Y also increases. Of course, the points are not all falling on a straight line, but if we look carefully, we find an overall linear pattern as shown below:

Scatter Diagram:



It will be very RARE in the field of behavioral or social sciences to find two sets of data which are related perfectly by a straight line: it is more likely that only a general linear pattern or tendency will be apparent.
WHY is it that we will not get an exact linear relationship?
Let me explain this to you with the help of an example:
Suppose one is studying the relationship between the research and development expenditure and the profit margin on products of a number of firms. While it may be generally true to state that the two will increase together, it is INEVITABLE that some firms' profit margin will be higher than others with the SAME *R and D* expenditure, and vice versa. The reasons for this may be that the conditions under which the various firms are operating may be very different. The goods being produced, the firm's share of the market, the efficiency of the firm etc. will ALL play a part in determining the individual results.

A linear relationship between two variables is a SURPRISINGLY common occurrence, and even where a refined non-linear curve might prove slightly superior, the SIMPLER form will often be quite adequate in the context of the problem under consideration. Having plotted the n pairs of values in the form of a scatter diagram, IF an overall linear pattern emerges, then the object of regression is to superimpose on this pattern the general relationship between y and x in the linear form which will REMOVE the effect of outside factors. I am sure that you are aware of the equation of a straight line.
Do you not remember the equation **Y = mX + c,** where Y represents the slope of the line, and c represent the Y-intercept?
This equation can also be stated as Y = c + mX, and if we rename c and m as 'a' and 'b', the equation becomes **Y = a + bX.**

EQUATION OF A STRAIGHT LINE

$$Y = a + bX$$

Where

- Y represents the dependent variable
- X represents the independent variable
- a represents the Y-intercept

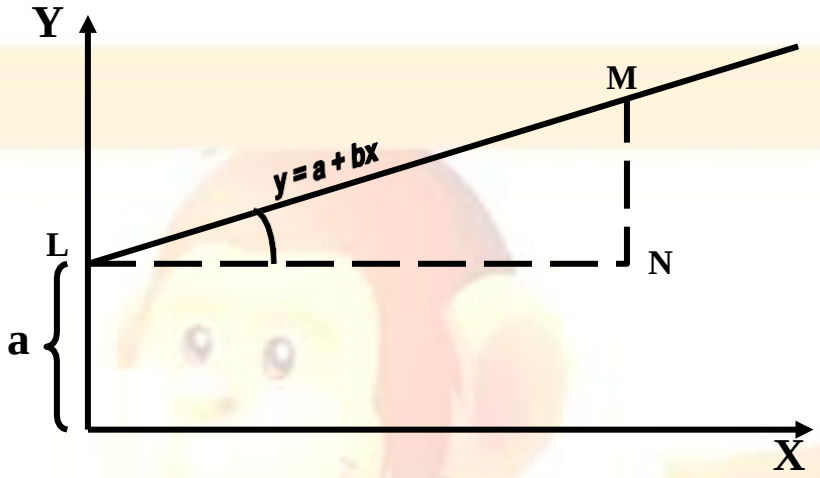
(i.e. the value of Y when X is equal to zero)

- b represents the slope of the line

(i.e. the value of the tan θ, where θ represents the angle between the line and the horizontal axis)

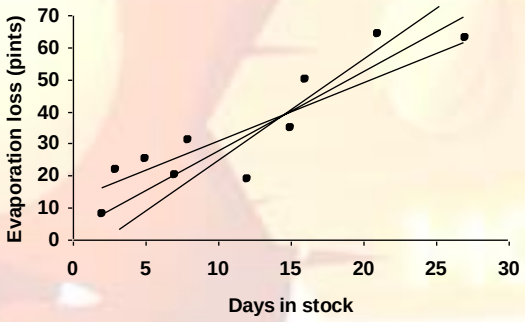
Interpretation of 'a' and 'b':

$$b = \tan\theta = \frac{MN}{NL}$$



A very important point to note is that MANY lines can be drawn through the same scatter diagram.

THE LINEAR PATTERN:



Even with the greatest care and skill, a line drawn between the points with a ruler will be highly SUBJECTIVE, and different individuals will arrive at different lines.

The *real* objective is to find the line of BEST fit. For this, we use a method known as THE METHOD OF LEAST SQUARES. The line of best fit obtained by the method of least squares is called the REGRESSION LINE of Y on X.

And, this whole process is known as simple linear regression. A very important point to note here is that, from the MATHEMATICAL standpoint, simple linear regression requires that X is a NON-RANDOM variable, whereas Y is a RANDOM variable. For example, consider the case of agricultural experiments. If we conduct an experiment to determine the optimal amount of a particular fertilizer to obtain the maximum yield of a certain crop, then the amount of fertilizer is a non-random variable whereas the yield is a random variable. This is so because the amount of fertilizer is in our OWN control. But, the yield is a random variable because it is NOT in our control. In connection with determining the line of BEST fit, the first point is that.

If we use the 'FREE-hand' method of curve-fitting in order to represent the relationship between X and Y as portrayed by the scatter diagram, one tends, consciously or subconsciously, to draw the straight line such that there is EQUAL numbers of points located on either side of the line.

What is more, the eye will automatically try to judge and EQUATE the total distances between the points above and below the line.

This is a recognition of the fact that the line of best fit must be an 'AVERAGE' line in the true sense.

You will recall that the sum of the deviations round the ARITHMETIC MEAN of a data-set is always equal to zero i.e. the positive and negative deviations CANCEL each other.

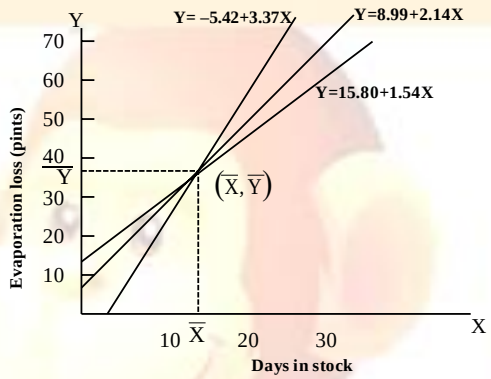
Similarly, POSITIVE and NEGATIVE deviations round a line of BEST fit must CANCEL out.

This is the first of the conditions or requirements for an optimal line.
But, the point to understand is that there are an INFINITELY large number of straight lines which will satisfy this condition.

Any line that passes through the point $(\bar{X}, \bar{Y})$ will satisfy this condition, and as shown in the following figure, numerous lines can pass through the point $(\bar{X}, \bar{Y})$

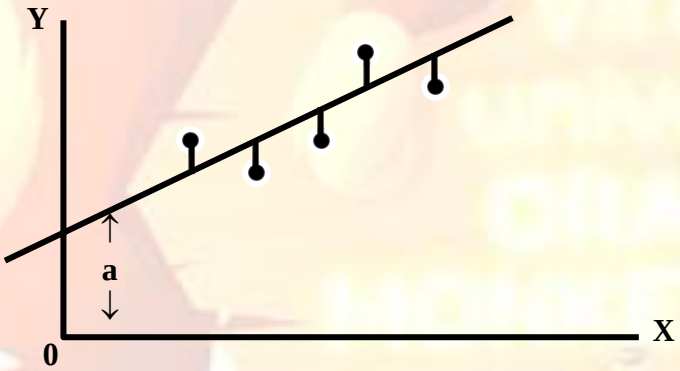
Three equations where

$\sum(Y - \hat{Y}) = 0$:



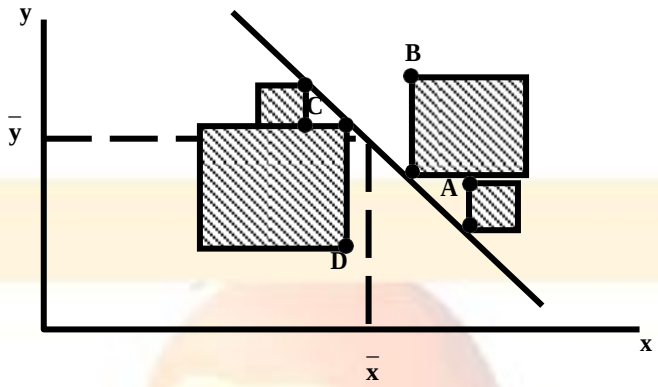
.

For each of the three lines that you see, the SUM of the VERTICAL deviations between the data-points and the line is ZERO. These deviations are depicted by the following diagram:

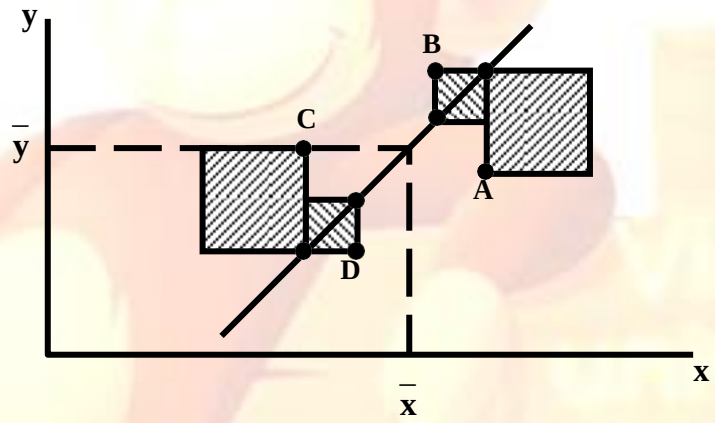


How will we calculate these vertical deviations?
The values of Y obtained from the line are denoted by .
And the deviations of the actual Y-values from the corresponding Y-values obtained from the line are obtained by subtracting from Y.
In all the cases --- as long as our line passes through the point $(\bar{x}, \bar{y})$ --- , we find that the sum of the deviations of the *actual* Y-values from the corresponding Y-values obtained from the *line* is zero.
Hence, it appears that we need some SECOND criterion for establishing a *unique* position for the BEST-fitting line.
Our interest is NOT simply in achieving a non-zero sum: it is the MAGNITUDE of the sum which is our main concern.
In the two figures that follow, the DIFFERENCES in the sums of squared deviations for two different lines passing through the SAME scatter diagram are clearly portrayed. In position 1, the shaded areas are relatively large, but AS the line is rotated around the point $(\bar{X}, \bar{Y})$ in a clockwise direction to position 2, the areas become smaller.

Position 1



Position 2



There would seem to be some UNIQUE position at which the sum of the square deviations is at a MINIMUM. *This* is the position of *least* squares.
If we can ascertain the location of THIS particular straight line in terms of the constants a and b of the linear equation Y = a + bX, then we have found the line of BEST fit.
The rationale is that the SMALLER the sum of the squared deviations round the mean, the LESS dispersed are the data points around the fitted line.

THE PRINCIPAL OF LEAST SQUARES

According to the principal of least squares, the best-fitting line to a set of points is the one for which the sum of the squares of the vertical distances between the points and the line is minimum.
The line Y = a + bX is the one that best fits the given set of points according to the principal of least squares.
And, this best fitting line is obtained by solving simultaneously two equations which are known as the normal equations.

NORMAL EQUATIONS

$$\left.\begin{aligned} \sum Y &= na + b\sum X \\ \sum XY &= a\sum X + b\sum X^2 \end{aligned}\right\}$$

In connection with these two equations, two points should be noted:
1) I will not be discussing the mathematical derivation of these equations.
2) The word "normal" here has nothing to do with the well-known normal distribution.
For any bivariate data-set, obviously we will have available to us two columns, a column of X and a column of Y.
Hence, obviously, we will be in a position to compute sums like ∑X, ∑Y, ∑XY, and so on.

Hence, the only unknown quantities in the two normal equations are a and b, as shown below:

NORMAL EQUATIONS

$$\left.\begin{aligned} \sum Y &= na + b\sum X \\ \sum XY &= a\sum X + b\sum X^2 \end{aligned}\right\}$$

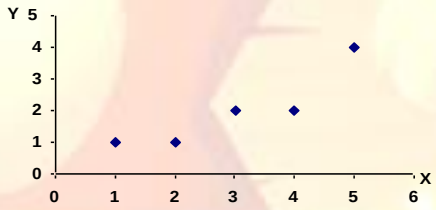
Hence, when we solve the two normal equations simultaneously, we will obtain the values of a and b, and these are EXACTLY the two quantities that we need in order to obtain the BEST-fitting line.
Let me explain this whole concept to you with the help of the same example that I picked up in the beginning of today's lecture:

EXAMPLE

An important concern for any pharmaceutical company producing drugs is to determine how a particular drug will affect one's perception or general awareness. Suppose one such company wants to establish a relationship between the PERCENTAGE of a drug in the blood-stream and the LENGTH OF TIME it takes to respond to a stimulus. Suppose the company administers this drug on 5 subjects and obtains the following information:

Subject	Percentage of drug	Reaction Time (milli-seconds)
	X	Y
A	1	1
B	2	1
C	3	2
D	4	2
E	5	4

Scatter Diagram:



In order to find a and b, we need to solve the two normal equations, and for this purpose, we will carry out computations as shown below:

X	Y	X^2	XY
1	1	1	1
2	1	4	2
3	2	9	6
4	2	16	8
5	4	25	20
15	10	55	37

$$10 = 5a + 15\,b \quad \text{— 1}$$
$$37 = 15a + 55b \quad \text{— 2}$$
$$-7 = -10b$$
$$b = \frac{-7}{-10} = 0.7$$
$$10 = 5a + 15b$$
$$\therefore \quad 10 - 15b = 5a$$
$$\therefore \quad 10 - 15(0.7) = 5a$$
$$\therefore \quad 10 - 10.5 = 5a$$
$$\therefore \quad -0.5 = 5a \qquad \therefore \quad a = \frac{-0.5}{5} = -0.1$$

Hence our straight line is given by

$$Y = -0.1 + 0.7\,X$$

A hat is placed on top of the Y so as to differentiate the Y values obtained from the line from the ones that pertain to the actual data-points.

The question is, "What is the advantage of fitting this line?"

The answer to this question is that this line can be used to ESTIMATE the value of the dependent variable corresponding to some particular value of the independent variable. In this example, suppose that we are interested in finding out what will be the reaction time of a person who has 4.33% of the drug in his blood stream? The answer will be obtained by putting X=4.33 in the equation that we just obtained.

Our regression line is

$$= -0.1 + 0.7\,X$$

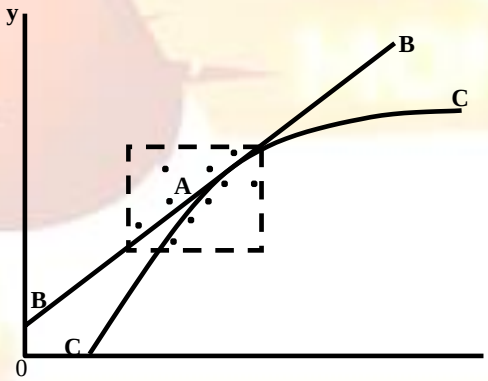
Putting X = 4.33, we obtain

$$= -0.1 + 0.7\,(4.33)$$
$$= -0.1 + 3.031 = 2.931$$

Hence we conclude that it can be expected that a person having 4.33% of the drug in his blood stream will take 2.9 milli-seconds to react to the stimulus. A point to be noted here is that this procedure of estimating the value of the dependent variable should not be used for extrapolation. Extrapolation means the making of estimates or predictions outside the range of the experimental data, and in some situations, this can be very unwise.

Let me explain this point with the help of the following diagram:

The extrapolation trap



A = region of interpolation
B = regions of extrapolation
C = true relationship in regions of extrapolation

While a set of observations may show a good linear relationship between the variables, there is NEVER any *guarantee* that the SAME linear form is present over THOSE ranges of the variable NOT under consideration. I would now like to convey to you another point: All the discussion that I have done until now assumes that Y is the dependent variable and X is the independent variable, and therefore, we are regressing Y on X. But, in some situations, we may be interested in just the OPPOSITE --- i.e. we may wish to regress X on Y. In this situation, all we have to do is to interchange the roles of X and Y. I would like to encourage you to work on this on your own, and to establish the normal equations that will

be required in this situation. You may be thinking, "Why should we go through this hassle? Can't we use the equation that we have just fitted i.e. Y = a + bX to estimate X from Y?"
It is important to note that this is not the case. If we are confronted with a situation where we require to predict Y from X AND X from Y, then *two* DISTINCT equations need to be found. The regressions of Y on X and X on Y for the same bivariate data are NOT identical.
The next concept that I am going to discuss with you is the STANDARD ERROR OF ESTIMATE.

STANDARD ERROR OF ESTIMATE

The observed values of (X, Y) do not all fall on the regression line but they scatter away from it. The degree of scatter of the observed values about the regression line is measured by what is called the standard deviation of regression or the standard error of estimate of Y on X.
For sample data, the standard error of estimate is obtained from the formula

$$s_{y.x} = \sqrt{\frac{\sum\left(Y - \hat{Y}\right)^2}{n-2}}$$

where Y denotes an observed values, denotes the corresponding values obtained from the least-squares line and n denotes the sample size.
The formula that I just conveyed to you is a bit cumbersome to apply because, in order to apply it, we first need to compute corresponding to all our X-values.
Alternative formula for Sy.x
The standard error of estimate can be more conveniently computed from the alternative formula

$$s_{y.x} = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$$

INTERPRETATION OF Sy.x

The range within which sy.x lies is given by 0 < sy.x < sy (where sy denotes the standard deviation of the y values). sy.x will be zero when all the observed points fall on the regression line (denoting perfect relationship between the two variables). sy.x will be equal to sy when there is no relationship between the two variables. Hence the closer sy.x is to zero (the further away it is from sy), the closer the points are to the line, and the more RELIABLE is our line for purposes of prediction.
Let us apply this concept to the example of the amount of drug in the blood stream and the time taken to react to a stimulus:

X	Y	X^2	Y^2	XY
1	1	1	1	1
2	1	4	1	2
3	2	9	4	6
4	2	16	4	8
5	4	25	16	20
15	10	55	26	37

a = -0.1
b = 0.7

$$Sy.x = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}$$

$$= \sqrt{\frac{26 - (-0.1)(10) - (0.7)(37)}{5-2}}$$

$$= \sqrt{\frac{26 + 1 - 25.9}{3}}$$

$$= \sqrt{\frac{1.1}{3}} = \sqrt{0.3667} = 0.61$$

$$\text{Also, } s_y = \sqrt{\frac{\sum Y^2}{n} - \left(\frac{\sum Y}{n}\right)^2}$$

$$= \sqrt{\frac{26}{5} - \left(\frac{10}{5}\right)^2} = \sqrt{5.2 - 4}$$

$$= \sqrt{1.2} \qquad = 1.10$$

INTERPRETATION

Sy.x is NOT very small compared with sy, hence our least-squares line.

$$\hat{Y} = -0.1 + 0.7X$$

is probably NOT very reliable for purposes of prediction. As it explained a short while ago, the smaller our Standard Error of Estimate, the closer the data-points will be to the line --- i.e. the more REPRESENTATIVE our line will be of the data-points --- and, hence, the more RELIABLE our line will be for estimation purposes.
The next concept that I am going to discuss with you is the concept of CORRELATION.
It is a concept that is very closely linked with the concept of linear regression.

CORRELATION

is a measure of the strength or the degree of relationship between two RANDOM variables.

A numerical measure of the strength of the linear relationship between two random variables X and Y is known as Pearson's Product-Moment Coefficient of Correlation.

PEARSON'S COEFFICIENT OF CORRELATION

where, covariance of X and Y is defined as $r = \dfrac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}}$

$$Cov(X,Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{n}$$

This formula is a bit cumbersome to apply. Therefore, we may use the following short cut formula:

SHORT CUT FORMULA FOR THE PEARSON'S COEFFICIENT OF CORRELATION

$$r = \frac{\sum XY - \left(\sum X\right)\left(\sum Y\right)/n}{\sqrt{\left[\sum X^2 - \left(\sum X\right)^2/n\right]\left[\sum Y^2 - \left(\sum Y\right)^2/n\right]}}$$

It should be noted that
r is a pure number that lies between –1 and 1 i.e.

$-1 < r < 1$

Actually, the mathematical expressions that you have just seen is a combination of three different mathematical expressions:

Case 1:

Positive correlation: $0 < r < 1$

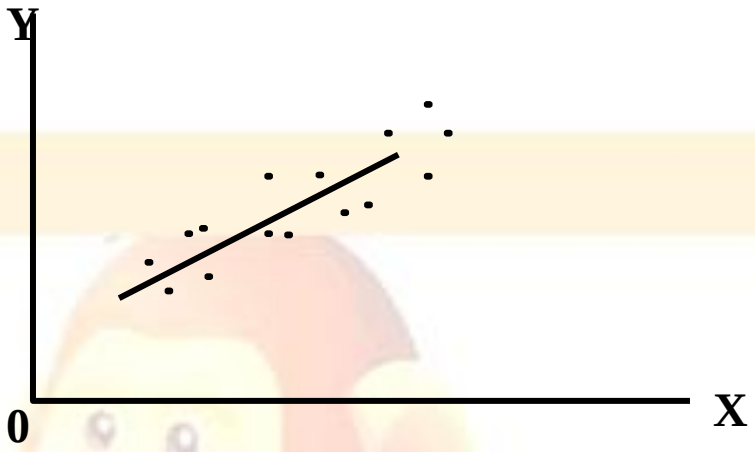
Case 2:

No correlation: $r = 0$

Case 3:

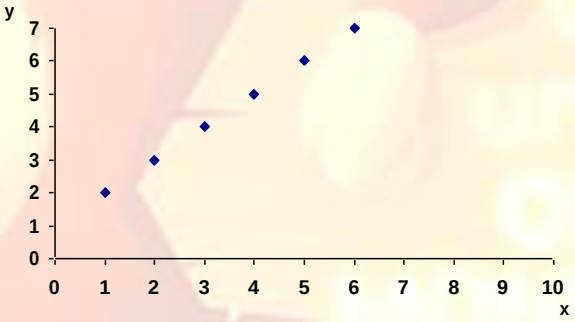
Negative correlation: $-1 < r < 0$

Case 1:
In case of a positive linear relationship, r lies between 0 and 1.



In this case, the closer the points are to the UPWARD-going line, the STRONGER is the positive linear relationship, and the closer r is to 1.

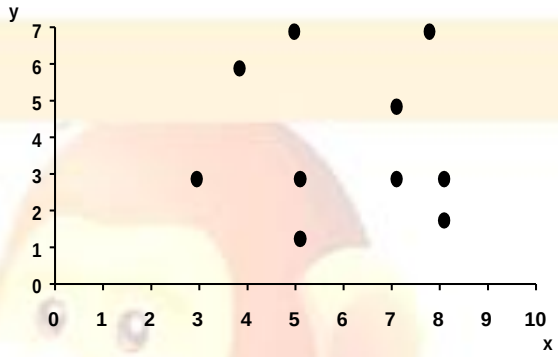
***Perfect* Positive Linear Correlation (r = 1):**



In this case, the closer the points are to the DOWNWARD-going line, the stronger is the linear relationship, and the closer r is to –1.

Case 3:

In a situation where neither an upward linear trend nor a downward linear trend can be visualized, r ≃ 0



Here, the bivariate data seem to be *completely* random.

The *extreme* of dissociation (*zero* correlation (r = 0)):



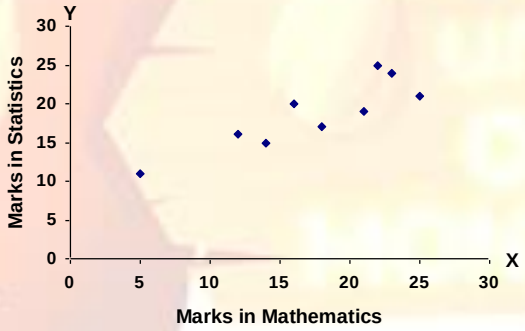
In such a situation, X and Y are said to be *uncorrelated*.

EXAMPLE

Suppose that the principal of a college wants to know if there exists any correlation between grades in Mathematics and grades in Statistics. Suppose that he selects a random sample of 9 students out of all those who take this combination of subjects. The following information is obtained:

Student	Marks in Mathematics (Total Marks: 25)	Marks in Statistics (Total Marks: 25)
	X	Y
A	5	11
B	12	16
C	14	15
D	16	20
E	18	17
F	21	19
G	22	25
H	23	24
I	25	21

SCATTER DIAGRAM



In order to compute the correlation coefficient, we carry out the following computations,

X	Y	X^2	Y^2	XY
5	11	25	121	55
12	16	144	256	192
14	15	196	225	210
16	20	256	400	320
18	17	324	289	306
21	19	441	361	399
22	25	484	625	550
23	24	529	576	552
25	21	625	441	525
156	168	3024	3294	3109

$$r = \frac{\sum XY - (\sum X)(\sum Y)/n}{\sqrt{\left[\sum X^2 - (\sum X)^2/n\right]\left[\sum Y^2 - (\sum Y)^2/n\right]}}$$

$$= \frac{3109 - (156)(168)/9}{\sqrt{\left[3024 - (156)^2/9\right]\left[3294 - (168)^2/9\right]}}$$

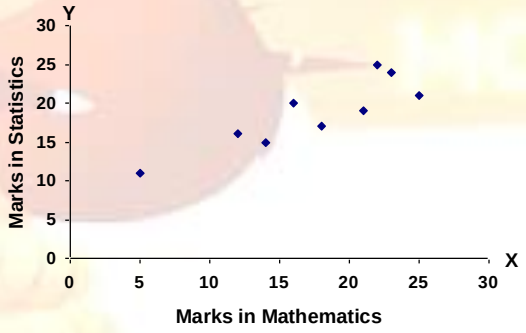
$$= \frac{3109 - 2912}{\sqrt{\left[3024 - 2704\right]\left[3294 - 3136\right]}}$$

$$= \frac{197}{\sqrt{320 \times 158}} = \frac{197}{224.86} = 0.88$$

INTERPRETATION

There exists a strong *positive* linear correlation between marks in Mathematics and marks in Statistics for these 9 students who have been taken into consideration.
The conclusion that we have just drawn i.e. strong positive linear correlation --- this conclusion is supported by the scatter diagram.

SCATTER DIAGRAM



As you can see in the scatter diagram, the data-points appear to follow a linear pattern quite strongly.
In today's lecture, I have discussed with you the concept of regression and correlation. Although I have conveyed to you a number of interesting concepts, believe me, this is only the BEGINNING of a very vast and important area of Statistics. You can study this concept further, and, if possible, to study a little bit about MULTIPLE regression and correlation as well --- the situation when we try to study the relationship between three or more variables.
This brings us to the end of the FIRST part of this course i.e. Descriptive Statistics.
This brings us to the end of the FIRST part of this course i.e. Descriptive Statistics.

LECTURE NO. 16

- Set Theory
- Counting Rules:
- The Rule of Multiplication

"SET"

A set is any well-defined collection or list of distinct objects, e.g. a group of students, the books in a library, the integers between 1 and 100, all human beings on the earth, etc. The term well-defined here means that any object must be classified as either belonging or not belonging to the set under consideration, and the term distinct implies that each object must appear only once. The objects that are in a set, are called members or elements of that set. Sets are usually denoted by capital letters such as A, B, C, Y, etc., while their elements are represented by small letters such as, a, b, c, y, etc.

Elements are enclosed by parentheses to represent a set.

For example:

EXAMPLES OF SETS:

A = {a, b, c, d} or
B = {1, 2, 3, 7}

The Number of a set A, written as n(A), is defined as the number of elements in A.

If x is an element of a set A, we write $x \in A$ which is read as "x belongs to A" or x is in A. If x does not belong to A, i.e. x is not an element of A, we write $x \notin A$.

A set that has no elements is called an empty or a null set and is denoted by the symbol ϕ. (It must be noted that {0} is not an empty set as it contains an element 0.)

If a set contains only one element, it is called a unit set or a singleton set.

It is also important to note the difference between an element "x" and a unit set {x}.

A set may be specified in two ways:

1. We may give a list of all the elements of a set (the "Roster" method),

e.g.

A = {1, 3, 5, 7, 9, 11} ;
B = {a book, a city, a clock, a teacher};

2. We may state a rule that enables us to determine whether or not a given object is a member of the set(the "Rule" method or the "Set Builder" method),

e.g.

A = {x | x is an odd number and x < 12} meaning that A is a set of all elements x such that x is an odd number and x is less than 12. (The vertical line is read as "such that".). *An important point to note is that:* The repetition or the order in which the elements of a set occur, does not change the nature of the set. The size of a set is given by the number of elements present in it. This number may be finite or infinite. Thus a set is finite when it contains a finite number of elements; otherwise it is an infinite set.

The Empty set is regarded as a Finite set.

EXAMPLES OF FINITE SETS

i) A = {1, 2, 3……., 99, 100};

ii) B = {x | x is a month of the year};

iii) C = {x | x is a printing mistake in a book};

iv) D = {x | x is a living citizen of Pakistan};

Examples of infinite sets:

i) A = {x | x is an even integer};

ii) B = {x | x is a real number between 0 and 1 inclusive},
i.e. B = (x | x 0 < x < 1}

iii) C = {x | x is a point on a line};

iv) D = {x | x is a sentence in a English language}; etc

SUBSETS

A set that consists of some elements of another set, is called a subset of that set.
For example, if B is a subset of A, then every member of set B is also a member of set A.
If B is a subset of A, we write:
$B \subset A$ or equivalently:
$A \supset B$
'B is a subset of A' is also read as 'B is contained in A',
or 'A contains B'.

EXAMPLE

If A = {1, 2, 3, 4, 5, 10}
and B {1, 3, 5}
then $B \subset A$,
i.e. B is contained in A.
It should be noted that any set is always regarded a subset of itself.
and an empty set ϕ is considered to be a subset of every set.
Two sets A and B are Equal or Identical, if and only if they contain exactly the same elements.
In other words, A = B if and only if $A \subset B$ and $B \subset A$.

PROPER SUBSET

If a set B contains some but not all of the elements of another set A, while A contains each element of B, i.e. if
$B \subset A$ and $B \neq A$
then the set B is defined to be a proper subset of A.

Universal Set:

The original set of which all the sets we talk about, are subsets, is called the universal set (or the space) and is generally denoted by S or Ω.
The universal set thus contains all possible elements under consideration.
A set S with n elements will produce 2^n subsets, including S and ϕ.

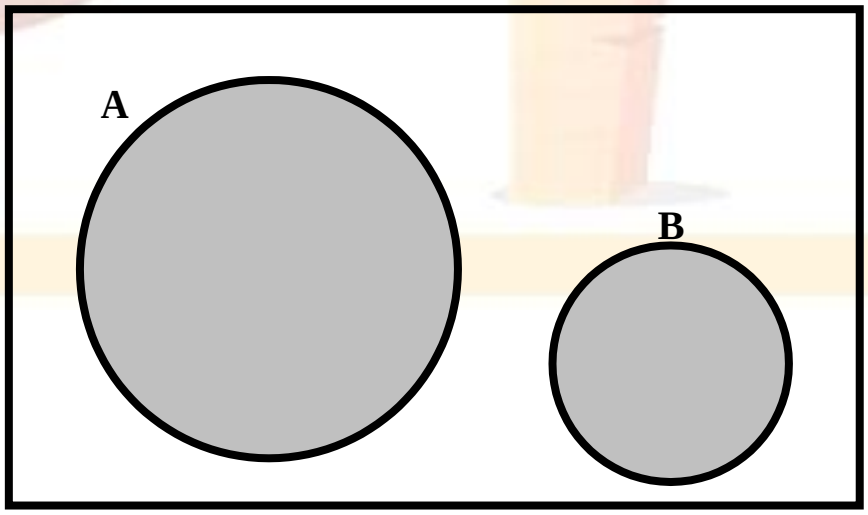
EXAMPLE;

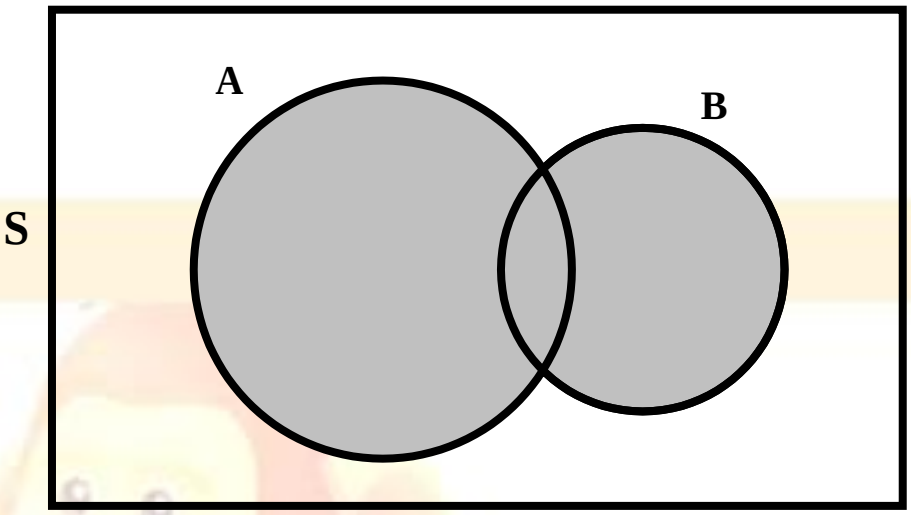
Consider the set A = {1, 2, 3}.
All possible subsets of this set are:
ϕ, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3} and {1, 2, 3}
Hence, there are $2^3 = 8$ subsets of the set A.

VENN DIAGRAM

A diagram that is understood to represent sets by circular regions, parts of circular regions or their complements with respect to a rectangle representing the space S is called a Venn diagram, named after the English logician John Venn (1834-1923).
The Venn diagrams are used to represent sets and subsets in a pictorial way and to verify the relationship among sets and subsets.
A Simple Venn diagram:

Disjoint Sets



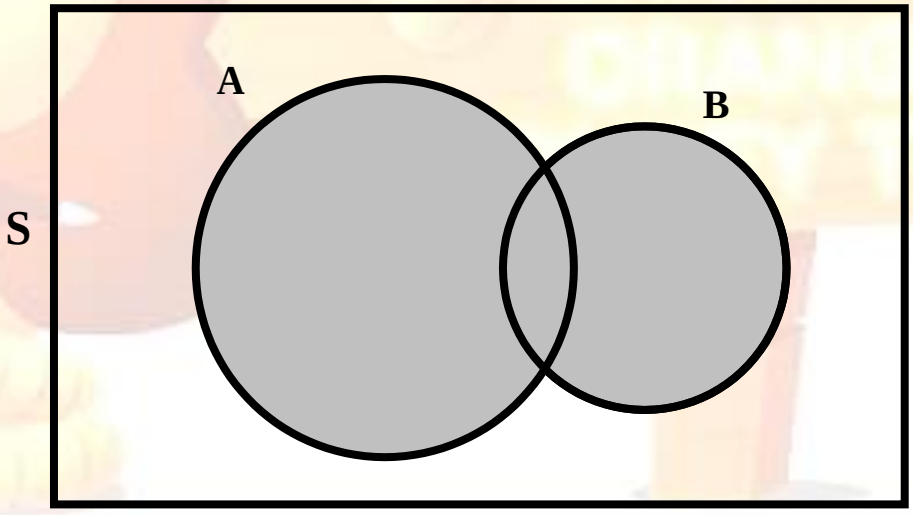


OPERATIONS ON SETS

Let the sets A and B be the subsets of some universal set S. Then these sets may be combined and operated on in various ways to form new sets which are also subsets of S.
The basic operations are union, intersection, difference and complementation.

UNION OF SETS

The union or sum of two sets A and B, denoted by $A \cup B$, and read as "A union B", means the set of all elements that belong to at least one of the sets A and B, that is
$A \cup B = \{ x \mid x \in A \text{ or } x \in B\}$
By means of a Venn Diagram, $A \cup B$ is shown by the shaded area as below:



$A \cup B$ is shaded

EXAMPLE

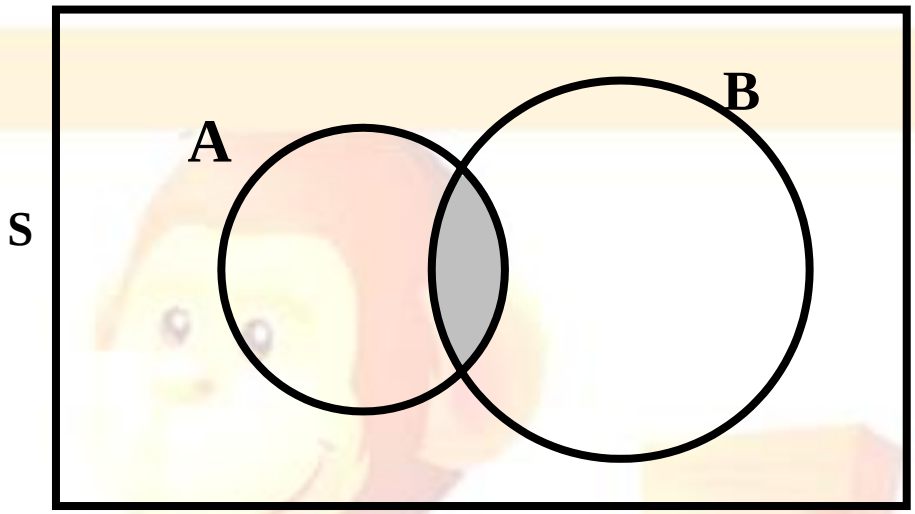
Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$
Then $A \cup B = \{1, 2, 3, 4, 5, 6\}$

INTERSECTION OF SETS

The intersection of two sets A and B, denoted by $A \cap B$, and read as "A intersection B", means that the set of all elements that belong to both A and B; that is

$A \cap B = \{x \mid x \in \text{and } x \in B\}$.

Diagrammatically, $A \cap B$ is shown by the shaded area as below:



$A \cap B$ is shaded

EXAMPLE

Let A = {1, 2, 3, 4} and B = {3, 4, 5, 6}

Then $A \cap B = \{3, 4\}$

The operations of union and intersection that have been defined for two sets may conveniently be extended to any finite number of sets.

DISJOINT SETS

Two sets A and B are defined to be disjoint or mutually exclusive or non-overlapping when they have no elements in common, i.e. when their intersection is an empty set

i.e. $A \cap B = \phi$.

On the other hand, two sets A and B are said to be conjoint when the have at least one element in common.

SET DIFFERENCE

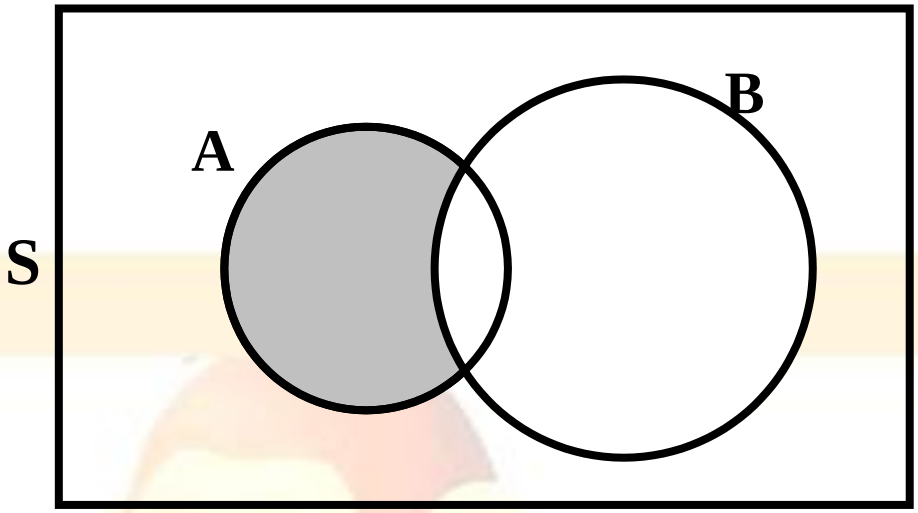
The difference of two sets A and B, denoted by A – B or by $A - (A \cap B)$, is the set of all elements of A which do not belong to B.

Symbolically,

$A - B = \{x \mid x \in A \text{ and } x \notin B\}$

It is to be pointed out that in general $A - B \neq B - A$.

The shaded area of the following Venn diagram shows the difference A – B:



Difference A – B is shaded

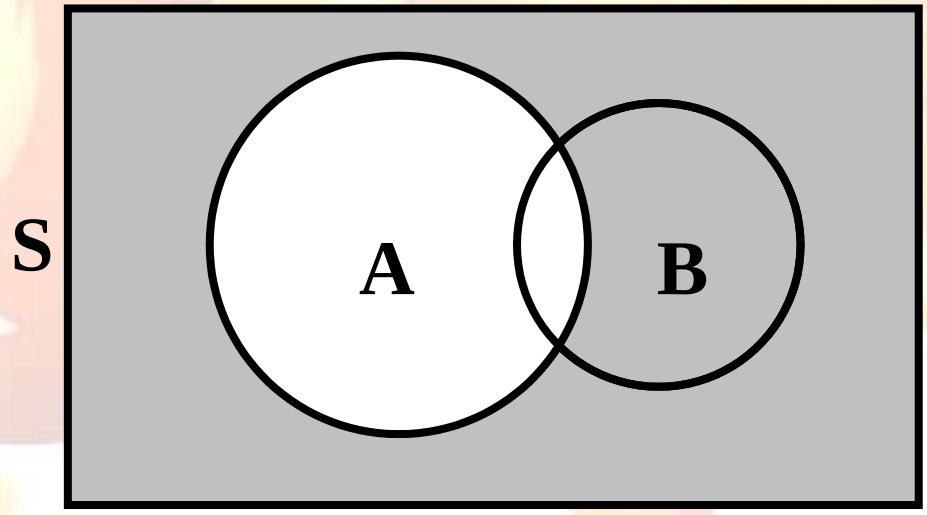
It is to be noted that A – B and B are disjoint sets. If A and B are disjoint, then the difference A – B coincides with the set A.

COMPLEMENTATION

The particular difference S – A, that is, the set of all those elements of S which do not belong to A, is called the complement of A and is denoted by $\bar{A}$ or by Ac.
In symbols:

$$\bar{A} = \{x \mid x \in S \text{ and } s \notin A\}$$

The complement of S is the empty set ϕ.
The complement of A is shown by the shaded portion in the following Venn diagram.



$\bar{A}$ is shaded

It should be noted that A – B and A $\cap$ $\bar{B}$, where $\bar{B}$ is the complement of set B, are the same set. Next, we consider the Algebra of Sets. The algebra of sets provides us with laws which can be used to solve many problems in probability calculations.
Let A, B and C be any subsets of the universal set S. Then, we have:

1. Commutative laws:

$A \cup B = B \cup A$
$A \cap B = B \cap A$

2. Associative laws:

$(A \cup B) \cup C = A \cup (B \cup C)$

$(A \cap B) \cap C = A \cap (B \cap C)$

3. Distributive laws

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

4. Idempotent laws

$A \cup A = A$
$A \cap A = A$

5. Identity laws

$A \cup S = S$,
$A \cap S = A$,
$A \cup \phi = A$, and
$A \cap \phi = \phi$.

6. Complementation laws

$A \cup \bar{A} = S$,
$A \cap \bar{A} = \phi$,

$(\bar{A}) = A$,

$\bar{S} = \phi$, and
$\bar{\phi} = S$

7. De Morgan's laws:

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B},$$

and $$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$$

PARTITION OF SETS

A partition of a set S is a sub-division of the set into non-empty subsets that are disjoint and exhaustive, i.e. their union is the set S itself.
This implies that:

- i)$A_i \cap A_j = \phi$, where $i \neq j$;
- ii)$A_1 \cap A_2 \cup \ldots \cup A_n = S$.

The subsets in a partition are called cells.

EXAMPLE

Let us consider a set S = {a, b, c, d, e}.
Then {a, b}, and {c, d, e} is a partition of S as each element of S belongs to exactly one cell.

CLASS OF SETS

A set of sets is called a class. For example, in a set of lines, each line is a set of points.

POWER SET

The class of ALL subsets of a set A is called the Power Set of A and is denoted by P(A).
For example, if A = {H, T}, then P(A) = {ϕ, {H}, {T}, {H, T}}.

CARTESIAN PRODUCT OF SETS

The Cartesian product of sets A and B, denoted by $A \times B$, (read as "A cross B"), is a set that contains all ordered pairs (x, y), where x belongs to A and y belongs to B.
Symbolically, we write
$A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$
This set is also called the Cartesian set of A and B set of A and B, named after the French mathematician Rene' Descartes (1596-1605).
The product of a set A by itself is denoted by A^2.
This concept of product may be extended to any finite number of sets.

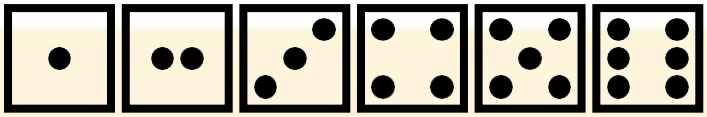
EXAMPLE

Let A = {H, T} and B = {1, 2, 3, 4, 5, 6}. Then the Cartesian product set is the collection of the following twelve (2 × 6) ordered pairs:

A×B = {(H, 1); (H, 2);(H, 3); (H, 4);(H, 6); (H, 6);(T, 1); (T, 2); (T, 3); (T, 4); (T, 5); (T, 6) }

Clearly, these twelve elements together make up the universal set S when a COIN and a DIE are tossed together.

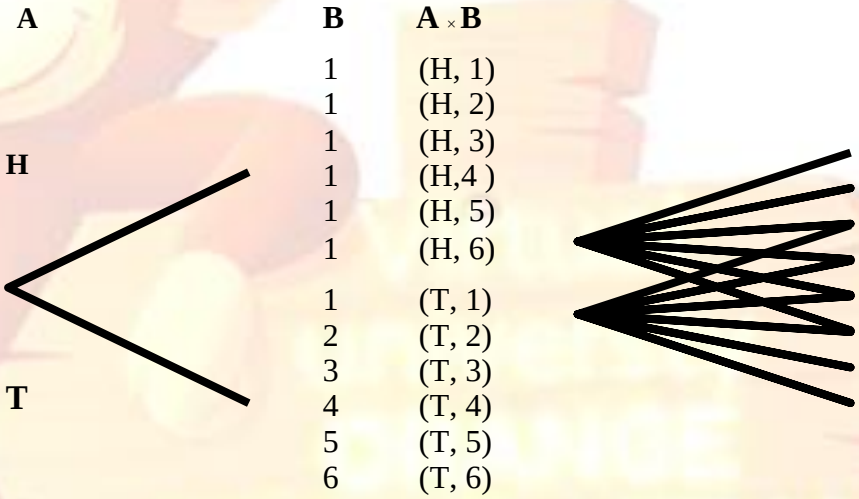
A die is a cube of wood or ivory whose six faces are marked with dots are shown below:



The plural of the word 'die' is 'dice'.

The product A × B may conveniently be found by means of the so-called tree diagram shown below:

Tree Diagram



TREE DIAGRAM

The "tree" is constructed from the left to the right. A "tree diagram" is a useful device for enumerating all the possible outcomes of two or more sequential events.

The possible outcomes are represented by the individual paths or branches of the tree.

It is relevant to note that, in general

$$A \times B \neq B \times A.$$

Having reviewed the basics of set theory, let us now review the COUNTING RULES that facilitate the computation of probabilities in a number of problems.

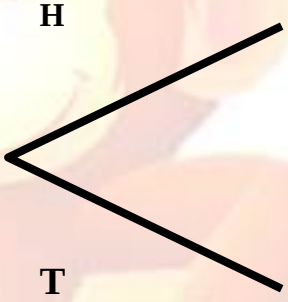
RULE OF MULTIPLICATION

If a compound experiment consists of two experiments which that the first experiment has exactly m distinct outcomes and, if corresponding to each outcome of the first experiment there can be n distinct outcomes of the second experiment, then the compound experiment has exactly mn outcomes.

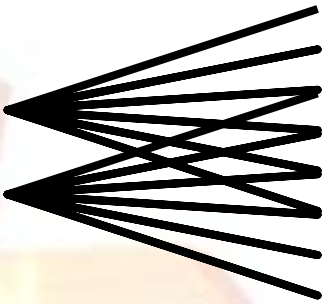
EXAMPLE

The compound experiment of tossing a coin and throwing a die together consists of two experiments. The coin-tossing experiment consists of two distinct outcomes (H, T), and the die-throwing experiment consists of six distinct outcomes (1, 2, 3, 4, 5, 6).
The total number of possible distinct outcomes of the compound experiment is therefore $2 \times 6 = 12$ as each of the two outcomes of the coin-tossing experiment can occur with each of the six outcomes of die-throwing experiment. As stated earlier, if A = {H, T} and B = {1, 2, 3, 4, 5, 6}, then the Cartesian product set is the collection of the following twelve (2×6) ordered pairs:
A×B = { (H, 1); (H, 2);(H, 3); (H, 4);
(H, 6); (H, 6);(T, 1); (T, 2);
(T, 3); (T, 4); (T, 5); (T, 6) }

Tree Diagram



B	A × B
1	(H, 1)
1	(H, 2)
1	(H, 3)
1	(H,4)
1	(H, 5)
1	(H, 6)
1	(T, 1)
2	(T, 2)
3	(T, 3)
4	(T, 4)
5	(T, 5)
6	(T, 6)



The rule of multiplication can be readily extended to compound experiments consisting of any number of experiments performed in a given sequence.

This rule can also be called the Multiple Choice Rule, as illustrated by the following example:

EXAMPLE

Suppose that a restaurant offers three types of soups, four types of sandwiches, and two types of desserts. Then, a customer can order any one out of $3 \times 4 \times 2 = 24$ different meals.

EXAMPLE

Suppose that we have a combination lock on which there are eight rings.
In how many ways can the lock be adjusted?
Solution:
The logical way to look at this problem is to see that there are eight rings on the lock, each of which can have any of the 10 figures 0 to 9:

$$\bar{A}\ \bar{B}\ \bar{C}\ \bar{D}\ \bar{E}\ \bar{F}\ \bar{G}\ \bar{H}$$

ring A can have any of the digits 0 to 9 and ring B can have any of the digits 0 to 9 and ring C can have any of the digits 0 to 9 and
.
.
.
ring H can have any of the digits 0 to 9
Hence the total No. of ways in which these 8 rings can be filled is 8
$10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10$
$= 10$
100,000,000 — one hundred million.

LECTURE NO. 17

- Permutations
- Combinations
- Random Experiment
- Sample Space
- Events
  - Mutually Exclusive Events
  - Exhaustive Events
  - Equally Likely Events

COUNTING RULES

As discussed in the last lecture, there are certain rules that facilitate the calculations of probabilities in certain situations. They are known as counting rules and include concepts of;

- Multiple Choice
- Permutations
- Combinations

We have already discussed the rule of multiplication in the last lecture.
Let us now consider the rule of permutations.

RULE OF PERMUTATION

A permutation is any ordered subset from a set of n distinct objects.
For example, if we have the set {a, b}, then one permutation is ab, and the other permutation is ba. The number of permutations of r objects, selected in a definite order from n distinct objects is denoted by the symbol ${}^nP_r$ and is given by

${}^nP_r = n(n-1)(n-2)\ldots(n-r+1)$

$$= \frac{n!}{(n-r)!}.$$

FACTORIALS

$7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$
$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$
.
.
.
$1! = 1$

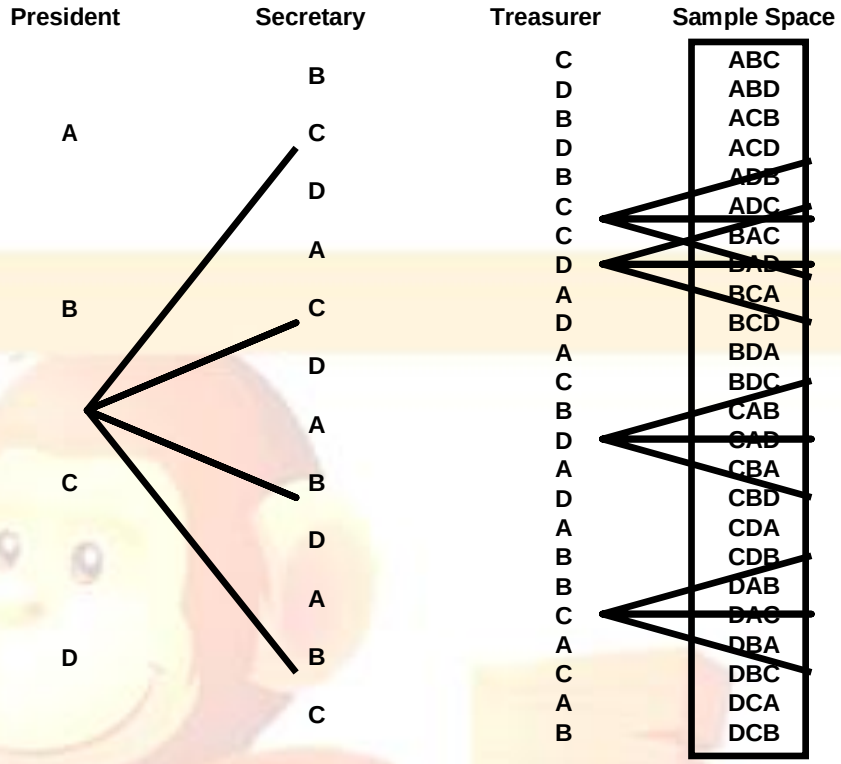
Also, we define $0! = 1$.

EXAMPLE

A club consists of four members. How many ways are there of selecting three officers: president, secretary and treasurer? It is evident that the order, in which 3 officers are to be chosen, is of significance. Thus there are 4 choices for the first office, 3 choices for the second office, and 2 choices for the third office. Hence the total number of ways in which the three offices can be filled is $4 \times 3 \times 2 = 24$.
The same result is obtained by applying the rule of permutations:

$$
\begin{aligned}
{}^4P_3 &= \frac{4!}{(4-3)!} \\
&= 4 \times 3 \times 2 \\
&= 24
\end{aligned}
$$

Let the four members be, A, B, C and D. Then a tree diagram which provides an organized way of listing the possible arrangements, for this example, is given below:



PERMUTATIONS

In the formula of ${}^{n}P_{r}$, if we put r = n, we obtain:

${}^{n}P_{n}$ = n (n – 1) (n – 2) … 3 × 2 × 1

= n!

I.e. the total number of permutations of n distinct objects, taking all n at a time, is equal to n!

EXAMPLE

Suppose that there are three persons A, B & D, and that they wish to have a photograph taken. The total number of ways in which they can be seated on three chairs (placed side by side) is

3P3 = 3! = 6

These are:
ABD,
ADB,
BAD,
BDA,
DAB,
DBA

The above discussion pertained to the case when all the objects under consideration are distinct objects. If some of the objects are not distinct, the formula of permutations modifies as given below:

The number of permutations of n objects, selected all at a time, when n objects consist of n1 of one kind, n2 of a second kind, …, nk of a kth kind,

$$\text{is } P = \frac{n!}{n_1!\, n_2!\, ...\, n_k!}.$$

$$(\text{where } \sum n_i = n)$$

EXAMPLE

How many different (meaningless) words can be formed from the word 'committee'?
In this example:
n = 9 (because the total number of letters in this word is 9)
n1 = 1 (because there is one c)
n2 = 1 (because there is one o)
n3 = 2 (because there are two m's)
n4 = 1 (because there is one i)
n5 = 2 (because there are two t's)
and
n6 = 2 (because there are two e's)
Hence, the total number of (meaningless) words
(permutations) is:

$$P = \frac{n!}{n_1!\, n_2!\, ...n_k!}.$$

$$= \frac{9!}{1!\ 1!\ 2!\ 1!\ 2!\ 2!}$$

$$= \frac{9\times 8\times 7\times 6\times 5\times 4\times 3\times 2\times 1}{1\times 1\times 2\times 1\times 1\times 2\times 1\times 2\times 1}$$

$$= 45360$$

Next, let us consider the rule of combinations.

RULE OF COMBINATION

A combination is any subset of r objects, selected without regard to their order, from a set of n distinct objects. The total number of such combinations is denoted by the symbol

and is given by $^{n}C_{r}$ or $\binom{n}{r}$,

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

where r < n.

It should be noted that

$$^{n}P_{r} = r!\binom{n}{r}$$

In other words, every combination of r objects (out of n objects) generates r! Permutations

EXAMPLE

Suppose we have a group of three persons, A, B, & C. If we wish to select a group of two persons out of these three, the three possible groups are {A, B}, {A, C} and {B, C}.In other words, the total number of combinations of size two out of this set of size three is 3.

Now, suppose that our interest lies in forming a committee of two persons, one of whom is to be the president and the other the secretary of a club.
The six possible committees are:

(A, B), (B, A),
(A, C), (C, A),
(B, C) & (C, B)

In other words, the total number of permutations of two persons out of three is 6.
And the point to note is that each of three combinations mentioned earlier generates 2 = 2! Permutations, I.e. the combination {A, B} generates the permutations (A, B) and (B, A) and the combination {A, C} generates the permutations (A, C) and (C, A); and the combination {B, C} generates the permutations (B, C) and (C, B).

The quantity

$$\binom{n}{r}$$

or ${}^{n}C_{r}$ is also called a binomial co-efficient because of its appearance in the binomial expansion of

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r}\ b^r .$$

The binomial co-efficient has two important properties.

i) $\binom{n}{r} = \binom{n}{n-r}$, and

ii) $\binom{n}{n-r} + \binom{n}{r} = \binom{n+1}{r}$

Also, it should be noted that

$$\binom{n}{0} = 1 = \binom{n}{n}$$

and

$$\binom{n}{1} = n = \binom{n}{n-1}$$

EXAMPLE

A three-person committee is to be formed out of a group of ten persons. In how many ways can this be done?
Since the order in which the three persons of the committee are chosen, is unimportant, it is therefore an example of a problem involving combinations. Thus the desired number of combinations is

$$\binom{n}{r} = \binom{10}{3} = \frac{10!}{3!\,(10-3)!} = \frac{10!}{3!\,7!}$$

$$= \frac{10\times 9\times 8\times 7\times 6\times 5\times 4\times 3\times 2\times 1}{3\times 2\times 1\times 7\times 6\times 5\times 4\times 3\times 2\times 1}$$

$$= 120$$

In other words, there are one hundred and twenty different ways of forming a three-person committee out of a group of only ten persons!

EXAMPLE

In how many ways can a person draw a hand of 5 cards from a well-shuffled ordinary deck of 52 cards?
The total number of ways of doing so is given by

$$\binom{n}{r} = \binom{52}{5} = \frac{52\times 51\times 50\times 49\times 48}{5\times 4\times 3\times 2\times 1} = 2{,}598{,}960$$

Having reviewed the counting rules that facilitate calculations of probabilities in a number of problems, let us now begin the discussion of concepts that lead to the formal definitions of probability. The first concept in this regard is the concept of Random Experiment. The term experiment means a planned activity or process whose results yield a set of data. A single performance of an experiment is called a trial. The result obtained from an experiment or a trial is called an outcome.

RANDOM EXPERIMENT

An experiment which produces different results even though it is repeated a large number of times under essentially similar conditions is called a Random Experiment.

The tossing of a fair coin, the throwing of a balanced die, drawing of a card from a well-shuffled deck of 52 playing cards, selecting a sample, etc. are examples of random experiments.

PROPERTIES OF A RANDOM EXPERIMENT

A random experiment has three properties:

- The experiment can be repeated, practically or theoretically, any number of times.
- The experiment always has two or more possible outcomes. An experiment that has only one possible outcome is not a random experiment.
- The outcome of each repetition is unpredictable, i.e. it has some degree of uncertainty.

Considering a more realistic example, interviewing a person to find out whether or not he or she is a smoker is an example of a random experiment. This is so because this example fulfils all the three properties that have just been discussed:

- This process of interviewing can be repeated a large number of times.
- To each interview, there are at least two possible replies: 'I am a smoker' and 'I am not a smoker'.
- For any interview, the answer is not known in advance i.e. there is an element of uncertainty regarding the person's reply.

A concept that is closely related with the concept of a random experiment is the concept of the Sample Space.

SAMPLE SPACE

A set consisting of all possible outcomes that can result from a random experiment (real or conceptual), can be defined as the sample space for the experiment and is denoted by the letter S. Each possible outcome is a member of the sample space, and is called a sample point in that space. Let us consider a few examples:

EXAMPLE-1

The experiment of tossing a coin results in either of the two possible outcomes: a head (H) or a tail (T). (We assume that it is not possible for the coin to land on its edge or to roll away). The sample space for this experiment may be expressed in set notation as S = {H, T}. 'H' and 'T' are the two sample points.

EXAMPLE-2

The sample space for tossing two coins once (or tossing a coin twice) will contain four possible outcomes denoted by

S = {HH, HT, TH, TT}.

In this example, clearly, S is the Cartesian product A × A, where A = {H, T}.

EXAMPLE-3

The sample space S for the random experiment of throwing two six-sided dice can be described by the Cartesian product A × A, where A = {1, 2, 3, 4, 5,6}. In other words, S = A × A = {(x, y) | x ∈ A and y ∈ A}, Where x denotes the number of dots on the upper face of the first die, and y denotes the number of dots on the upper face of the second die. Hence, S contains 36 outcomes or sample points, as shown below:

S = {(1, 1), (1, 2), (1, 3), (1, 5), (1, 6),
(2, 1), (2, 2), (2, 3), (2, 5), (2, 6),
(3, 1), (3, 2), (3, 3), (3, 5), (3, 6),
(4, 1), (4, 2), (4, 3), (4, 5), (4, 6),
(5, 1), (5, 2), (5, 3), (5, 5), (5, 6),
(6, 1), (6, 2), (6, 3), (6, 5), (6, 6)}

The next concept is that of events.

EVENTS

Any subset of a sample space S of a random experiment, is called an event. In other words, an event is an individual outcome or any number of outcomes (sample points) of a random experiment.

SIMPLE & COMPOUND EVENTS

An event that contains exactly one sample point is defined as a simple event. A compound event contains more than one sample point, and is produced by the union of simple events.

EXAMPLE

The occurrence of a 6 when a die is thrown, is a simple event, while the occurrence of a sum of 10 with a pair of dice, is a compound event, as it can be decomposed into three simple events (4, 6), (5, 5) and (6, 4).

OCCURRENCE OF AN EVENT

An event A is said to occur if and only if the outcome of the experiment corresponds to some element of A.

EXAMPLE

Suppose we toss a die, and we are interested in the occurrence of an even number.
If ANY of the three numbers '2', '4' or '6' occurs, we say that the event of our interest has occurred.
In this example, the event A is represented by the set {2, 4, 6}, and if the outcome '2' occurs, then, since this outcome is corresponding to the first element of the set A, therefore, we say that A has occurred.

COMPLEMENTARY EVENT

The event "not-A" is denoted by $\bar{A}$ or Ac and called the negation (or complementary event) of A.

EXAMPLE

If we toss a coin once, then the complement of "heads" is "tails". If we toss a coin four times, then the complement of "at least one head" is "no heads". A sample space consisting of n sample points can produce 2n different subsets (or simple and compound events).

EXAMPLE

Consider a sample space S containing 3 sample points, i.e. S = {a, b, c}.
Then the 23 = 8 possible subsets are

ϕ, {a}, {b}, {c}, {a, b},
{a, c}, {b, c}, {a, b, c}

Each of these subsets is an event. The subset {a, b, c} is the sample space itself and is also an event. It always occurs and is known as the certain or sure event. The empty set ϕ is also an event, sometimes known as impossible event, because it can never occur.

MUTUALLY EXCLUSIVE EVENTS

Two events A and B of a single experiment are said to be mutually exclusive or disjoint if and only if they cannot both occur at the same time i.e. they have no points in common.

EXAMPLE-1

When we toss a coin, we get *either* a head *or* a tail, but *not* both at the same time. The two events head and tail are therefore mutually exclusive.

EXAMPLE-2

When a die is rolled, the events 'even number' and 'odd number' are mutually exclusive as we can get either an even number or an odd number in one throw, not both at the same time. Similarly, a student *either* qualifies or fails, a single birth must be *either* a boy or a girl, it cannot be both, etc., etc. Three or more events originating from the same experiment are mutually exclusive if pair wise they are mutually exclusive. If the two events *can* occur at the same time, they are not mutually exclusive, e.g., if we draw a card from an ordinary deck of 52 playing cars, it *can* be both a king and a diamond.
Therefore, kings and diamonds are not mutually exclusive. Similarly, inflation and recession are not mutually exclusive events. Speaking of playing cards, it is to be remembered that an ordinary deck of playing cards contains 52 cards arranged in 4 suits of 13 each. The four suits are called diamonds, hearts, clubs and spades; the first two are red and the last two are black. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, …, 10, jack, queen and king. The *face cards* are king, queen and jack. These cards are used for various games such as whist, bridge, poker, etc. We have discussed the concepts of mutually exclusive events. Another important concept is that of exhaustive events.

EXHAUSTIVE EVENTS

Events are said to be collectively exhaustive, when the union of mutually exclusive events is equal to the entire sample space S.

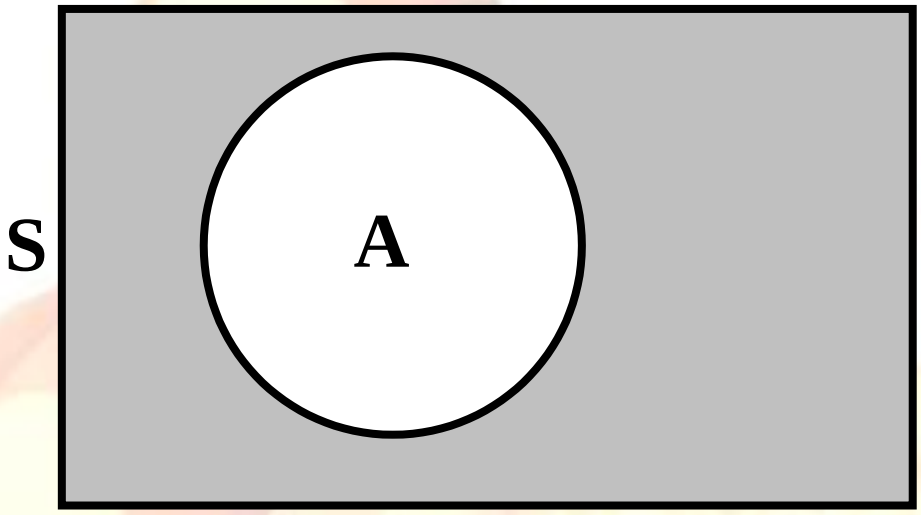
EXAMPLES

- In the coin-tossing experiment, 'head' and 'tail' are collectively exhaustive events.
- In the die-tossing experiment, 'even number' and 'odd number' are collectively exhaustive events.

In conformity with what was discussed in the last lecture:

PARTITION OF THE SAMPLE SPACE

A group of mutually exclusive and exhaustive events belonging to a sample space is called a partition of the sample space. With reference to any sample space S, events A and $\bar{A}$ form a partition as they are mutually exclusive and their union is the entire sample space. The Venn Diagram below clearly indicates this point.



$\bar{A}$ is shaded

Next, we consider the concept of equally likely events:

EQUALLY LIKELY EVENTS

Two events A and B are said to be equally likely, when one event is as likely to occur as the other. In other words, each event should occur in equal number in repeated trials.

EXAMPLE

When a fair coin is tossed, the head is as likely to appear as the tail, and the proportion of times each side is expected to appear is 1/2.

EXAMPLE

If a card is drawn out of a deck of well-shuffled cards, each card is equally likely to be drawn, and the probability that any card will be drawn is 1/52.

LECTURE NO. 18

DEFINITIONS OF PROBABILITY

- Subjective Approach to Probability
- Objective Approach:
- Classical Definition of Probability

RELATIVE FREQUENCY DEFINITION OF PROBABILITY

Before we begin the various definitions of probability, let us revise the concepts of:

- Mutually Exclusive Events
- Exhaustive Events
- Equally Likely Events

MUTUALLY EXCLUSIVE EVENTS

Two events A and B of a single experiment are said to be mutually exclusive or disjoint if and only if they cannot both occur at the same time i.e. they have no points in common.

EXAMPLE-1

When we toss a coin, we get *either* a head *or* a tail, but *not* both at the same time. The two events head and tail are therefore mutually exclusive.

EXAMPLE-2

When a die is rolled, the events 'even number' and 'odd number' are mutually exclusive as we can get either an even number or an odd number in one throw, not both at the same time. Similarly, a student *either* qualifies *or* fails, a person is either a teenager or not a teenager, etc., etc.

Three or more events originating from the same experiment are mutually exclusive if pair wise they are mutually exclusive. If the two events *can* occur at the same time, they are not mutually exclusive, e.g., if we draw a card from an ordinary deck of 52 playing cars, it *can* be both a king and a diamond.

Therefore, kings and diamonds are not mutually exclusive. Speaking of playing cards, it is to be remembered that an ordinary deck of playing cards contains 52 cards arranged in 4 suits of 13 each. The four suits are called diamonds, hearts, clubs and spades; the first two are red and the last two are black. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, …, 10, jack, queen and king. The face values called denominations, of the 13 cards in each suit are ace, 2, 3, …, 10, jack, queen and king. We have discussed the concepts of mutually exclusive events.

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EXAMPLES

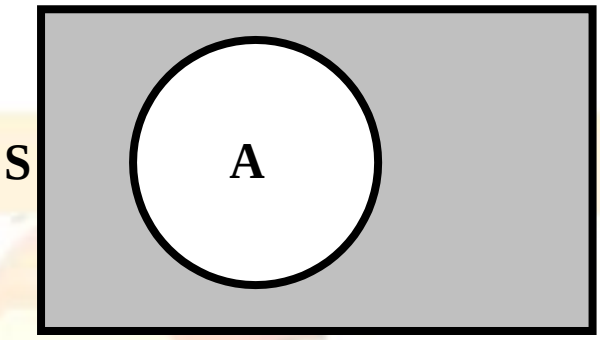
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In conformity with what was discussed in the last lecture:

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A group of mutually exclusive and exhaustive events belonging to a sample space is called a partition of the sample space. With reference to any sample space S, events A and $\bar{A}$ form a partition as they are mutually exclusive and their union is the entire sample space. The Venn Diagram below clearly indicates this point.

Venn Diagram



$\bar{A}$ is shaded

EQUALLY LIKELY EVENTS

Two events A and B are said to be equally likely, when one event is as likely to occur as the other. In other words, each event should occur in equal number in repeated trials.

EXAMPLE

When a fair coin is tossed, the head is as likely to appear as the tail, and the proportion of times each side is expected to appear is 1/2.

EXAMPLE

If a card is drawn out of a deck of well-shuffled cards, each card is equally likely to be drawn, and the proportion of times each card can be expected to be drawn in a *very* large number of draws is 1/52.Having discussed basic concepts related to probability theory, we now begin the discussion of THE CONCEPT AND DEFINITIONS OF PROBABILITY. Probability can be discussed from two points of view: the subjective approach, and the objective approach.

SUBJECTIVE OR PERSONALISTIC PROBABILITY

As its name suggests, the subjective or personality probability is a measure of the strength of a person's belief regarding the occurrence of an event A. Probability in this sense is purely subjective, and is based on whatever evidence is available to the individual. It has a disadvantage that two or more persons faced with the same evidence may arrive at different probabilities.
For example, suppose that a panel of three judges is hearing a trial. It is possible that, based on the evidence that is presented, two of them arrive at the conclusion that the accused is guilty while one of them decides that the evidence is NOT strong enough to draw this conclusion. On the other hand, objective probability relates to those situations where everyone will arrive at the same conclusion.
It can be classified into two broad categories, each of which is briefly described as follows:

1. THE CLASSICAL OR 'A PRIORI' DEFINITION OF PROBABILITY

f a random experiment can produce n mutually exclusive and equally likely outcomes, and if m out to these outcomes are considered favorable to the occurrence of a certain event A, then the probability of the event A, denoted by P(A), is defined as the ratio m/n.
Symbolically, we write

$$P(A) = \frac{m}{n}$$

$$= \frac{\text{Number of favourable outcomes}}{\text{Total numberof possibleoutcomes}}$$

This definition was formulated by the French mathematician P.S. Laplace (1949-1827) and can be very conveniently used in experiments where the total number of possible outcomes and the number of outcomes favorable to an event can be DETERMINED.

Let us now consider a few examples to illustrate the classical definition of probability:

EXAMPLE-1

If a card is drawn from an ordinary deck of 52 playing cards, find the probability that i) the card is a red card, ii) the card is a 10.

SOLUTION:
The total number of possible outcomes is 13+13+13+13 = 52, and we assume that all possible outcomes are equally likely.(It is well-known that an ordinary deck of cards contains 13 cards of diamonds, 13 cards of hearts, 13 cards of clubs, and 13 cards of spades.)
(i) Let A represent the event that the card drawn is a red card.
Then the number of outcomes favorable to the event A is 26 (since the 13 cards of diamonds and the 13 cards of hearts are *red*).

Hence

$$P(A) = \frac{m}{n}$$

$$= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{26}{52} = \frac{1}{2}$$

Thus $P(B) = \frac{4}{52} = \frac{1}{13}$.

EXAMPLE-2

A fair coin is tossed three times. What is the probability that at least one head appears?

SOLUTION

The sample space for this experiment is

S = {HHH, HHT, HTH, THH,
HTT, THT, TTH, TTT}

and thus the total number of sample points is 8 i.e. n(S) = 8.Let A denote the event that at least one head appears. Then

A = {HHH, HHT, HTH, THH, HTT, THT, TTH}
Therefore n(A) = 7.

Hence

$$P(A) = \frac{n(A)}{n(S)} = \frac{7}{8}.$$

EXAMPLE-3

Four items are taken at random from a box of 12 items and inspected. The box is *rejected* if more than 1 item is found to be faulty. If there are 3 faulty items in the box, find the probability that the box is *accepted.*

SOLUTION

The sample space S contains $\binom{12}{4} = 495$ Sample points

(Because there are $\binom{12}{4}$ ways of selecting four items out of twelve)

The box contains 3 faulty and 9 good items. The box is accepted if there is (i) no faulty items, or (ii) one faulty item in the sample of 4 items *selected*.
Let A denote the event the number of faulty items chosen is 0 or 1.
Then

$$n(A) = \binom{3}{0}\binom{9}{4} + \binom{3}{1}\binom{9}{3}$$

$$= 126 + 252 = 378 \quad sample \quad point s.$$

$$\therefore \quad P(A) = \frac{m}{n} = \frac{378}{495} = 0.76$$

Hence the probability that the box is accepted is 76% , (in spite of the fact that the box *contains* 3 faulty items).

THE CLASSICAL DEFINITION HAS THE FOLLOWING SHORTCOMINGS

- This definition is said to involve circular reasoning as the term equally likely really means equally probable.
- Thus probability is defined by introducing concepts that presume a *prior* knowledge of the *meaning* of probability.
- This definition becomes vague when the possible outcomes are INFINITE in number, or uncountable.
- This definition is NOT applicable when the assumption of equally likely does *not* hold. And the fact of the matter is that there are NUMEROUS situations where the assumption of equally likely cannot hold.

And these are the situations where we have to look for another definition of probability!

THE **RELATIVE FREQUENCY DEFINITION OF PROBABILITY**

The essence of this definition is that if an experiment is repeated a large number of times under (more or less) identical conditions, and if the event of our interest occurs a certain number of times, then the *proportion* in which this event occurs is regarded as the probability of that event.
For example, we know that a large number of students sit for the matric examination every year. Also, we know that a certain proportion of these students will obtain the first division, a certain proportion will obtain the second division, --- and a certain proportion of the students will fail.
Since the total number of students appearing for the matric exam is very large, hence:

- The proportion of students who obtain the first division --- this proportion can be regarded as the *probability* of obtaining the first division,
- The proportion of students who obtain the second division --- this proportion can be regarded as the *probability* of obtaining the second division, and so on.

ORANGE MONKEY TEAM

LECTURE NO. 19

- Relative Frequency Definition of Probability
- Axiomatic Definition of Probability
- Laws of Probability
  - Rule of Complementation
  - Addition Theorem

THE RELATIVE FREQUENCY DEFINITION OF PROBABILITY ('A POSTERIORI' DEFINITION OF PROBABILITY)

If a random experiment is repeated a large number of times, say n times, under identical conditions and if an event A is observed to occur m times, then the probability of the event A is defined as the LIMIT of the relative frequency m/n as n tends to infinitely.
Symbolically, we write

$$P(A) = \lim_{n\to\infty} \frac{m}{n}$$

The definition assumes that as n increases indefinitely, the ratio m/n tends to become stable at the numerical value P(A).
The relationship between relative frequency and probability can also be represented as follows:

Relative Frequency $\rightarrow$ Probability
as $n \rightarrow \infty$

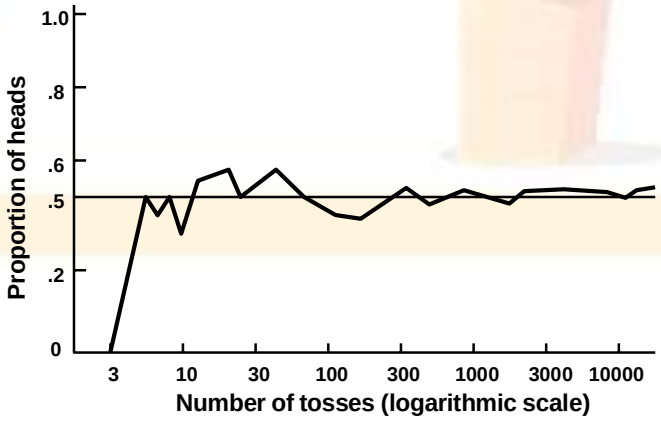
As its name suggests, the relative frequency definition relates to the relative frequency with which are event occurs in the *long run*. In situations where we can say that an experiment has been repeated a very large number of times, the *relative frequency definition* can be applied.
As such, this definition is very useful in those practical situations where we are interested in computing a probability in numerical form but where the classical definition cannot be applied.(Numerous real-life situations are such where various possible outcomes of an experiment are NOT equally likely). This type of probability is also called empirical probability as it is based on *EMPIRICAL* evidence i.e. on *OBSERVATIONAL* data.
It can also be called ***STATISTICAL PROBABILITY*** for it is this very probability that forms the basis of mathematical statistics.
Let us try to understand this concept by means of two examples:

- from a coin-tossing experiment and
- From data on the numbers of boys and girls born.

EXAMPLE-1

Coin-Tossing:
No one can tell which way a coin will fall but we expect the proportion of leads and tails after a large no. of tosses to be nearly equal. An experiment to demonstrate this point was performed by Kerrich in Denmark in 1946. He tossed a coin 10,000 times, and obtained altogether 5067 heads and 4933 tails.
The behavior of the proportion of heads throughout the experiment is shown as in the following figure:

The proportion; of heads in a sequence of tosses of a coin (Kerrich, 1946):



As you can see, the curve fluctuates *widely* at first, but begins to settle down to a more or less *stable* value as the number of spins increases. It seems reasonable to suppose that the fluctuations would continue to diminish if the experiment were continued *indefinitely*, and the proportion of heads would cluster more and more *closely* about a *limiting* value which would be *very near*, if not exactly, one-half. This hypothetical *limiting* value is the (statistical) probability of heads.

Let us now take an example closely related to our *daily* lives --- that relating to the sex ratio:-

In this context, the first point to note is that it has been known since the eighteenth century that in *reliable* birth statistics based on sufficiently *large* numbers (in at least some parts of the world), there is always a slight *excess* of boys, Laplace records that, among the 215,599 births in thirty districts of France in the years 1800 to 1802, there were 110,312 boys and 105,287 girls. The proportions of boys and girls were thus 0.512 and 0.488 respectively (indicating a slight *excess* of boys over girls).In a *smaller* number of births one would, however, expect considerable *deviations* from these proportions. This point can be illustrated with the help of the following example:

EXAMPLE-2

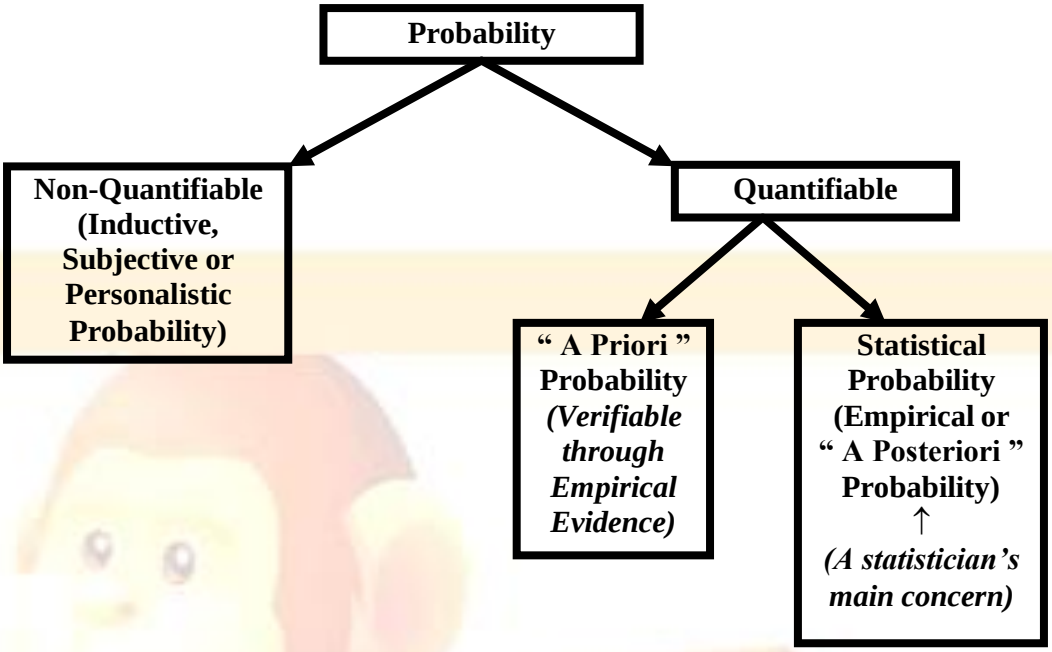
The following table shows the proportions of male births that have been worked out for the major regions of England as well as the rural districts of Dorset (for the year 1956).

Proportions of Male Births in various Regions and Rural Districts of England in 1956

Region of England	Proportion of Male Births	Rural Districts of Dorset	Proportion of Male Births
Northern	.514	Beaminster	.38
E. & W. Riding	.513	Blandford	.47
North Western	.512	Bridport	.53
North Midland	.517	Dorchester	.50
Midland	.514	Shaftesbury	.59
Eastern	.516	Sherborne	.44
London and S. Eastern	.514	Sturminster	.54
Southern	.514	Wareham and Purbeck	.53
South Western	.513	Wimborne & Cranborne	.54
Whole country	.514	All Rural District's of Dorset	.512

(Source: Annual Statistical Review)

As you can see, the figures for the rural districts of Dorset, based on about 200 births each, fluctuate between 0.38 and 0.59. While those for the major regions of England, which are each based on about 100,000 births, do not fluctuate much, rather, they range between 0.512 and 0.517 only. The larger sample size is clearly the reason for the greater constancy of the latter. We can imagine that if the sample were increased indefinitely, the proportion of boys would tend to a *limiting* value which is unlikely to differ much from 0.514, the proportion of male births for the *whole* country. This hypothetical *limiting* value is the (statistical) *probability* of a male birth. The overall discussion regarding the various ways in which probability can be defined is presented in the following diagram:



As far as *quantifiable* probability is concerned, in those situations where the various possible outcomes of our experiment are equally likely, we can compute the probability *prior* to actually conducting the experiment --- otherwise, as is generally the case, we can compute a probability only *after* the experiment has been conducted (and this is why it is also called 'a posteriori' probability*). Non-quantifiable probability is the one that is called Inductive Probability.*
It refers to the degree of belief which it is reasonable to place in a proposition on *given* evidence.
An important point to be noted is that it is difficult to express inductive probabilities numerically — to construct a numerical scale of inductive probabilities, with 0 standing for impossibility and for logical certainty. An important point to be noted is that it is difficult to express inductive probabilities numerically — to construct a numerical scale of inductive probabilities, with 0 standing for impossibility and for logical certainty. Most statisticians have arrived at the conclusion that inductive probability cannot, in general, he measured and, therefore cannot be use in the mathematical theory of statistics.
This conclusion is not, perhaps, very surprising since there seems no reason why rational degree of belief should be measurable any more than, say, degrees of beauty. Some paintings are very beautiful, some are quite beautiful, and some are ugly, but it would be observed to try to construct a numerical scale of beauty, on which Mona Lisa had a beauty value of 0.96.Similarly some propositions are highly probable, some are quite probable and some are improbable, but it does not seem possible to construct a numerical scale of such (inductive) probabilities .Because of the fact that inductive probabilities are not quantifiable and cannot be employed in a mathematical argument, this is the reason why the usual methods of statistical inference such as tests of significance and confidence interval are based entirely on the concept of statistical probability. Although we have discussed three different ways of defining probability, the most formal definition is yet to come.
This is The Axiomatic Definition of Probability.

THE AXIOMATIC DEFINITION OF PROBABILITY

This definition, introduced in 1933 by the Russian mathematician Andrei N. Kolmogrov, is based on a set of AXIOMS. Let S be a sample space with the sample points E1, E2, … Ei, …En. To each sample point, we assign a real number, denoted by the symbol P(Ei), and called the probability of Ei, that must satisfy the following basic axioms:

Axiom 1:

For any event Ei,
$0 < P(Ei) < 1$.

Axiom 2:

$P(S) = 1$
for the sure event S.

Axiom 3:
If A and B are mutually exclusive events (subsets of S), then

$P(A \cup B) = P(A) + P(B)$.

It is to be emphasized that According to the axiomatic theory of probability:

SOME probability defined as a non-negative real number is to be ATTACHED to each sample point Ei *such that the sum of all such numbers must equal ONE*. The ASSIGNMENT of probabilities may be based on past evidence or on some other underlying conditions. (If this assignment of probabilities is based on past evidence, we are talking about EMPIRICAL probability, and if this assignment is based on underlying conditions that ensure that the various possible outcomes of a random experiment are EQUALLY LIKELY, then we are talking about the CLASSICAL definition of probability. Let us consider another example:

EXAMPLE

Table given below shows the numbers of births in England and Wales in 1956 classified by (a) sex and (b) whether live born or stillborn.

Table-1

Number of births in England and Wales in 1956 by sex and whether live- or still born

(Source Annual Statistical Review)

	Liveborn	Stillborn	Total
Male	359,881 (A)	8,609 (B)	368,490
Female	340,454 (B)	7,796 (D)	348,250
Total	700,335	16,405	716,740

There are four possible events in this double classification:

- Male livebirth (denoted by A),
- Male stillbirth (denoted by B),
- Female livebirth (denoted by C)
- Female stillbirth (denoted by D),

The relative frequencies corresponding to the figures of Table-1 are given in Table-2:

Table-2

Proportion of births in England and Wales in 1956 by sex and whether live- or stillborn

(Source *Annual Statistical Review*)

	Liveborn	Stillborn	Total
Male	.5021	.0120	.5141
Female	.4750	.0109	.4859
Total	.9771	.0229	1.0000

The total number of births is large enough for these relative frequencies to be treated for all practical purposes as *PROBABILITIES*.

Let us denote the compound events 'Male birth' and 'Stillbirth' by the letters M and S. Now a male birth occurs whenever either a male live birth or a male stillbirth occurs, and so the proportion of male birth, regardless of whether they are live-or stillborn, is equal to the sum of the proportions of these two types of birth; that is to say,

$$p(M) = p(A \text{ or } B) = p(A) + p(B)$$
$$= .5021 + .0120 = .5141$$

Similarly, a stillbirth occurs whenever either a male stillbirth or a female stillbirth occurs and so the proportion of stillbirths, regardless of sex, is equal to the sum of the proportions of these two events:

$$p(S) = p(B \text{ or } D) = p(B) + p(D)$$
$$= .0120 + .0109 = .0229$$

Let us now consider some basic LAWS of probability. These laws have important applications in solving probability problems.

LAW OF COMPLEMENTATION

If $\bar{A}$ is the complement of an event A relative to the sample space S, then

$$P(\bar{A}) = 1 - P(A).$$

Hence the probability of the complement of an event is equal to one minus the probability of the event. Complementary probabilities are very useful when we are wanting to solve questions of the type 'What is the probability that, in tossing two fair dice, at least one even number will appear?'

EXAMPLE

A coin is tossed 4 times in succession. What is the probability that at least one head occurs?

- The sample space S for this experiment consists of 24 = 16 sample points (as each toss can result in 2 outcomes),and
- We assume that each outcome is equally likely.

If we let A represent the event that at least one head occurs, then A will consist of MANY sample points, and the process of computing the probability of this event will become somewhat cumbersome! So, instead of denoting this particular event by A, let us denote its complement i.e. "No head" by A.
Thus the event A consists of the SINGLE sample point {TTTT}.
Therefore P(A) = 1/16.
Hence by the law of complementation, we have

$$P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{16} = \frac{15}{16}.$$

The next law that we will consider is the Addition Law or the General Addition Theorem of Probability:

ADDITION LAW

If A and B are any two events defined in a sample space S, then
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
In words, this law may be stated as follows:
"If two events A and B are not mutually exclusive, then the probability that at least one of them occurs, is given by the sum of the separate probabilities of events A and B minus the probability of the joint event $A \cap B$.**"**

LECTURE NO. 20

- Application of Addition Theorem
- Conditional Probability
- Multiplication Theorem

First of all, let us consider in some detail the Addition Law or the General Addition Theorem of Probability:

ADDITION LAW

If A and B are any two events defined in a sample space S, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
In words, this law may be stated as follows:

"If two events A and B are not mutually exclusive, then the probability that at least one of them occurs, is given by the sum of the separate probabilities of events A and B minus the probability of the joint event $A \cap B$."

EXAMPLE

If one card is selected at random from a deck of 52 playing cards, what is the probability that the card is a club or a face card or both?
Let A represent the event that the card selected is a club, B, the event that the card selected is a face card, and $A \cap B$, the event that the card selected is both a club and a face card. Then we need $P(A \cup B)$
Now P(A) = 13/52, as there are 13 clubs, P(B) = 12/52, as there are 12 faces cards,
$P(A \cap B) = 3/52$, since 3 of clubs are also face cards.
Therefore the desired probability is

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$= 13/52 + 12/52 - 3/52$
$= 22/52.$

COROLLARY-1

If A and B are mutually exclusive events, then
$P(A \cup B) = P(A) + P(B)$ (Since $A \cap B$ is an impossible event, hence $P(A \cap B) = 0$)

EXAMPLE

Suppose that we toss a pair of dice, and we are interested in the event that we get a total of 5 or a total of 11.What is the probability of this event?

SOLUTION

In this context, the first thing to note is that 'getting a total of 5' and 'getting a total of 11' are mutually exclusive events. Hence, we should apply the special case of the addition theorem. If we denote 'getting a total of 5' by A, and 'getting a total of 11' by B, then P (A) = 4/36 (since there are four outcomes favorable to the occurrence of a total of 5), and P(B) = 2/36 (since there are two outcomes favorable to the occurrence of a total of 11).
Hence the probability that we get a total of 5 or a total of 11 is given by

$P(A \cup B) = P(A) + P(B)$
$= 4/36 + 2/36 = 6/36 = 16.67\%.$

COROLLARY-2

If $A_1, A_2 \ldots A_k$ are k mutually exclusive events, then the probability that one of them occurs, is the sum of the probabilities of the separate events, i.e.
$P(A_1, \cup A_2 \cup \ldots \cup A_k) = P(A_1) + P(A_2) + \ldots + P(A_k)$.
Let us now consider an interesting example to illustrate the way in which probability problems can be solved:

EXAMPLE

Three horses A, B and C are in a race; A is twice as likely to win as B and B is twice as likely to win as C. What is the probability that A or B wins?
Evidently, the events mentioned in this problem are not equally likely.
Let P(C) = p
Then P (B) = 2p as B is twice as likely to win as C.
Similarly
P (A) = 2P (B) = 2(2p) = 4p
In this problem, we assume that no two of the horses A, B and C cannot win the race together (i.e. the race cannot end in a draw).
Hence, the events A, B and C are mutually exclusive.

Since A, B and C are mutually exclusive and collectively *exhaustive*, therefore the sum of their probabilities must be equal to 1.

Thus
p + 2p + 4p = 1
or p = 1/7

∴ P(C) = 1/7,
P(B) = 2(1/7) = 2/7,
and P(A) = 4(1/7) = 4/7.
Hence
P(A∪B) = P(A) + P(B)
= 4/7+ 2/7
= 6/7.

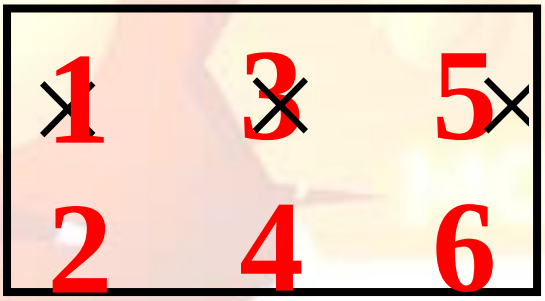
Having discussed the addition theorem in some detail, we would now like to discuss the Multiplication Theorem. But, before we are in a position to take up the multiplication theorem, we need to consider the concept of conditional probability.

CONDITIONAL PROBABILITY

The sample space for an experiment must often be changed when some additional information pertaining to the outcome of the experiment is received.
The effect of such information is to REDUCE the sample space by excluding some outcomes as being impossible which BEFORE receiving the information were believed possible. The probabilities associated with such a reduced sample space are called conditional probabilities.
The following example illustrates the concept of conditional probability

EXAMPLE

Suppose that we toss a fair die. Then the sample space of this experiment is S = {1, 2, 3, 4, 5, 6}. Suppose we wish to know the probability of the outcome that the die shows 6 (say event A).Also, suppose that, before seeing the outcome, we are told that the die shows an EVEN number of dots (say event B).
Then the information that the die shows an even number excludes the outcomes 1, 3 and 5, and thereby reduces the original sample space to a sample space that consists of three outcomes 2, 4 and 6,
i.e. the reduced sample space is
B = {2, 4, 6}.



(The sample space is reduced.)

Then, the desired probability in the reduced sample space B is 1/3.
(since each outcome in the reduced sample space is EQUALLY LIKELY). This probability 1/3 is called the conditional probability of the event A because it is computed under the CONDITION that the die has shown an even number of dots. In other words,
P(die shows 6/die shows even numbers)
= 1/3,
(Where the vertical line is read as given that and the information following the vertical line describes the conditioning event).
Sometimes, it is not very convenient to compute a conditional probability by first determining the number of sample points that belong to the reduced sample space.
In such a situation, we can utilize the following *alternative method of computing a conditional probability*

CONDITIONAL PROBABILITY

If A and B are two events in a sample space S and if P(B) is not equal to zero, then the conditional probability of the event A given that event B has occurred, written as P(A/B), is defined by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Where P (B) > 0
(If P (B) = 0, the conditional probability P(A/B) remains undefined.)
Similarly

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

where P(A) > 0.
It should be noted that P(A/B) SATISFIES all the basic axioms of probability, namely:

- 0 < P(A/B) < 1.
- **ii)** P(S/B) = 1
- P(A1∪A2/B) = P(A1/B) + P(A2/B) (provided that the events A1 and A2 are mutually exclusive).

Let us now apply this concept to a real-world example

EXAMPLE-2

At a certain elementary school in a Western country, the school-record of the past ten years shows that 75% of the students come from a two-parent home and that 20% of the students are low-achievers and belong to two-parent homes. What is the probability that such a randomly selected student will be a low achiever GIVEN THAT he or she comes from a two-parent home?

SOLUTION

Let A denote a low achiever and B a student from a two-parent home. Applying the *relative frequency definition* of probability, we have

P(B) = 0.75 and P(A ∩ B) = 0.20.

Thus, we obtain

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.20}{0.75} = 0.27$$

MULTIPLICATION THEOREM OF PROBABILITY

It is interesting to note that the multiplication theorem is obtained very conveniently from the formula of conditional probability:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

As discussed earlier, the conditional probability of A given that B has occurred has already been defined as:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \text{ Where P (B) > 0}$$

Multiplying both sides by P (B), we get

P (A ∩ B) = P (B) . P (A/B).

And if we interchange the roles of A and B, we obtain:
P (A ∩ B) = P (A) P (B/A),
Provided P(A) > 0.

MULTIPLICATION LAW

If A and B are any two events defined in a sample space S, then
P (A ∩ B)

= P (A) P (B/A), provided P (A) > 0,
= P (B) P (A/B) provided P (B) > 0.

(The second form is easily obtained by interchanging A and B. This is called the GENERAL rule of multiplication of probabilities. It can be stated as follows:

MULTIPLICATION LAW

"The probability that two events A and B will both occur is equal to the probability that one of the events will occur multiplied by the conditional probability that the other event will occur given that the first event has already occurred."
Let us apply the concept of multiplication theorem to an example

EXAMPLE

A box contains 15 items, 4 of which are defective and 11 is good. Two items are selected. What is the probability that the first is good and the second defective?
Let A represent the event that the first item selected is good, and B, the event that the second items is defective.
Then we need to calculate the probability of the JOINT event $A \cap B$ by the rule
$P(A \cap B) = P(A)P(B/A)$.
We have:

Type of Item	No. of Items
Defective	4
Good	11
Total	15

Since all the items are equally likely to be chosen, hence
P (A) = 11/15.
Given the event A has occurred, there remain 14 items of which 4 are defective. Therefore the probability of selecting a defective item after a good item has been selected is 4/14 i.e.

$$P(B/A) = 4/14.$$

Hence

$$P(A \cap B) = P(A)\,P(B/A)$$
$$= 11/15 \times 4/14$$
$$= 44/210$$
$$= 0.16.$$

In this lecture, the concepts of the Addition Theorem and the Multiplication Theorem of probability have been discussed in some detail.
In order to **differentiate** between the situation where the addition theorem is applicable and the situation where the multiplication theorem is applicable, the main point to keep in mind is that whenever we wish to compute the probability that *either* A occurs *or* B occurs, we should think of the Addition Theorem, where as, whenever we wish to compute the probability that *both* A *and* B occur, we should think of the Multiplication Theorem.

LECTURE NO. 21

- Independent and Dependent Events
- Multiplication Theorem of Probability for Independent Events
- Marginal Probability

Before we proceed the concept of independent versus dependent events, let us review the Addition and Multiplication Theorems of Probability that were discussed in the last lecture.

To this end, let us consider an interesting example that illustrates the application of both of these theorems in one problem:

EXAMPLE

A bag contains 10 white and 3 black balls. Another bag contains 3 white and 5 black balls. Two balls are transferred from first bag and placed in the second, and then one ball is taken from the latter.

What is the probability that it is a white ball?

In the beginning of the experiment, we have:

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10	3
Black	3	5
Total	13	8

Let A represent the event that 2 balls are drawn from the first bag and transferred to the second bag. Then A can occur in the following three mutually exclusive ways:

A1 = 2 white balls are transferred to the second bag.

A2 = 1 white ball and 1 black ball are transferred to the second bag.

A3 = 2 black balls are transferred to the second bag.

Then, the total number of ways in which 2 balls can be drawn out of a total of 13 balls is $\binom{13}{2}$.

And, the total number of ways in which 2 white balls can be drawn out of 10 white balls is $\binom{10}{2}$.

Thus, the probability that two white balls are selected from the first bag containing 13 balls (in order to *transfer* to the second bag) is

$$P(A_1) = \binom{10}{2} \div \binom{13}{2} = \frac{45}{78},$$

Similarly, the probability that one white ball and one black ball are selected from the first bag containing 13 balls (in order to *transfer* to the second bag) is

$$P(A_2) = \binom{10}{1}\binom{3}{1} \div \binom{13}{2} = \frac{30}{78},$$

And, the probability that two black balls are selected from the first bag containing 13 balls (in order to *transfer* to the second bag) is

$$P(A_3) = \binom{3}{2} \div \binom{13}{2} = \frac{3}{78}.$$

AFTER having transferred 2 balls from the first bag, the second bag contains

i) 5 white and 5 black balls (if 2 white balls are transferred)

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10 – 2 = 8	3 + 2 = 5
Black	3	5
Total	13 – 2 = 11	8 + 2 = 10

Hence: P(W/A1) = 5/10

ii) 4 white and 6 black balls (if 1 white and 1 black ball are transferred)

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10 – 1 = 9	3 + 1 = 4
Black	3 – 1 = 2	5 + 1 = 6
Total	13 – 2 = 11	8 + 2 = 10

Hence: P(W/A2) = 6/10

iii) 3 white and 7 black balls (if 2 black balls are transferred)

Colour of Ball	No. of Balls in Bag A	No. of Balls in Bag B
White	10	3
Black	3 – 2 = 1	5 + 2 = 7
Total	13 – 2 = 11	8 + 2 = 10

Hence: P(W/A3) = 3/10

Let W represent the event that the WHITE ball is drawn from the second bag after having transferred 2 balls from the first bag.

Then $P(W) = P(A1 \cap W) + P(A2 \cap W) + P(A3 \cap W)$

Now $P(A1 \cap W) = P(A1)\,P(W/A1)$

$= 45/78 \times 5/10$

$= 15/52$

$P(A2 \cap W) = P(A2)\,P(W/A2)$

$= 30/78 \times 4/10$

$= 2/13,$

And

$P(A3 \cap W) = P(A3)\,P(W/A3)$

$= 3/78 \times 3/10$

$= 3/260.$

Hence the required probability is

$P(W)$

$= P(A1 \cap W) + P(A2 \cap W) + P(A3 \cap W)$

$= 15/52 + 2/13 + 3/260$

$= 59/130$

$= 0.45$

INDEPENDENT EVENTS

Two events A and B in the same sample space S, are defined to be independent (or statistically independent) if the probability that one event occurs, is not affected by whether the other event has or has not occurred, that is

$P(A/B) = P(A)$ and $P(B/A) = P(B)$. It then follows that two events A and B are independent if and only if

$P(A \cap B) = P(A)\,P(B)$

and this is known as the *special case* of the Multiplication Theorem of Probability.

RATIONALE

According to the multiplication theorem of probability, we have:

$$P(A \cap B) = P(A).\,P(B/A)$$

Putting $P(B/A) = P(B)$, we obtain

$$P(A \cap B) = P(A)\,P(B)$$

The events A and B are defined to be *DEPENDENT* if $P(A \cap B) \neq P(A) \times P(B)$.

This means that the occurrence of one of the events in some way affects the probability of the occurrence of the other event. Speaking of independent events, it is to be emphasized that two events that are independent, can NEVER be mutually exclusive.

EXAMPLE

Two fair dice, one red and one green, are thrown. Let A denote the event that the red die shows an even number and let B denote the event that the green die shows a 5 or a 6. Show that the events A and B are independent.

The sample space S is represented by the following 36 outcomes:

S = {(1, 1), (1, 2), (1, 3), (1, 5), (1, 6);
(2, 1), (2, 2), (2, 3), (2, 5), (2, 6);
(3, 1), (3, 2), (3, 3), (3, 5), (3, 6);
(4, 1), (4, 2), (4, 3), (4, 5), (4, 6);
(5, 1), (5, 2), (5, 3), (5, 5), (5, 6);
(6, 1), (6, 2), (6, 3), (6, 5), (6, 6) }

Since

A represents the event that red die shows an even number, and B represents the event that green die shows a 5 or a 6,

Therefore $A \cap B$ represents the event that red die shows an even number and green die shows a 5 or a 6.

Since A represents the event that red die shows an even number, hence P(A) = 3/6. Similarly, since B represents the event that green die shows a 5 or a 6, hence P(B) = 2/6.

Now, in order to compute the probability of the joint event $A \cap B$, the first point to note is that, in all, there are 36 possible outcomes when we throw the two dice together, i.e.

S = (1, 1), (1, 2), (1, 3), (1, 5), (1, 6);
(2, 1), (2, 2), (2, 3), (2, 5), (2, 6);
(3, 1), (3, 2), (3, 3), (3, 5), (3, 6);
(4, 1), (4, 2), (4, 3), (4, 5), (4, 6);
(5, 1), (5, 2), (5, 3), (5, 5), (5, 6);
(6, 1), (6, 2), (6, 3), (6, 5), (6, 6)

The joint event $A \cap B$ contains only 6 outcomes out of the 36 possible outcomes.

These are (2, 5), (4, 5), (6, 5), (2, 6), (4, 6), and (6, 6).

and $P(A \cap B) = 6/36$.

Now

P (A) P (B)
$= 3/6 \times 2/6$
$= 6/36$
$= P (A \cap B)$.

Therefore the events A and B are independent. Let us now go back to the example pertaining to live births and stillbirths that we considered in the last lecture, and try to determine whether or not sex of the baby and nature of birth are independent.

EXAMPLE

Table-1 below shows the numbers of births in England and Wales in 1956 classified by (a) sex and (b) whether live born or stillborn.

Table-1

Number of births in England and Wales in 1956 by sex and whether live- or still born
(Source *Annual Statistical Review*)

	Liveborn	Stillborn	Total
Male	359,881 (A)	8,609 (B)	368,490
Female	340,454 (B)	7,796 (D)	348,250
Total	700,335	16,405	716,740

There are four possible events in this double classification:

- Male live birth,
- Male stillbirth,
- Female live birth, and
- Female stillbirth.

The corresponding relative frequencies are given in Table-2.

Table-2

Proportion of births in England and Wales in 1956 by sex and whether live- or stillborn
(Source Annual Statistical Review)

	Liveborn	Stillborn	Total
Male	.5021	.0120	.5141
Female	.4750	.0109	.4859
Total	.9771	.0229	1.0000

As discussed in the last lecture, the total number of births is large enough for these relative frequencies to be treated for all practical purposes as *PROBABILITIES*. The compound events 'Male birth' and 'Stillbirth' may be represented by the letters M and S. If M represents a male birth and S a stillbirth, we find that

$$\frac{n(M \text{ and } S)}{n(M)} = \frac{8609}{368490} = 0.0234$$

This figure is the proportion — and, since the sample size is large, it can be regarded as the *probability* — of males who are still born — in other words, the *CONDITIONAL* probability of a stillbirth *given that* it is a male birth. In other words, the probability of stillbirths *in* males. The corresponding proportion of stillbirths among females is

$$\frac{7796}{348258} = 0.0224.$$

These figures should be contrasted with the OVERALL, or *UNCONDITIONAL*, proportion of stillbirths, which is

$$\frac{16405}{716740} = 0.0229.$$

We observe that the conditional probability of stillbirths among boys is slightly *HIGHER* than the overall proportion. Where as the conditional proportion of stillbirths among girls is slightly *LOWER* than the overall proportion. It can be concluded that sex and stillbirth are statistically *DEPENDENT*, that is to say, the *SEX* of a baby yet to be born *has* an effect, (although a small effect), on its chance of being stillborn. The example, that we just considered point out the concept of *MARGINAL PROBABILITY*.

Let us have another look at the data regarding the live births and stillbirths in England and Wales:

Table-2Proportion of births in England and Wales in 1956 by sex and whether live- or stillborn (Source *Annual Statistical Review*)

	Liveborn	Stillborn	Total
Male	.5021	.0120	.5141
Female	.4750	.0109	.4859
Total	.9771	.0229	1.0000

And, the figures in Table-2 indicate that the probability of male birth is 0.5141, whereas the probability of female birth is 0.4859.Also, the probability of live birth is 0.9771, where as the probability of stillbirth is 0.0229. And since these probabilities appear in the margins of the Table, they are known as *Marginal Probabilities*. According to the above table, the probability that a new born baby is a male and is live born is 0.5021 whereas the probability that a new born baby is a male and is stillborn is 0.0120.Also, as stated earlier; the probability that a new born baby is a male is 0.5141, and, CLEARLY, 0.5141 = 0.5021 + 0.0120. Hence, it is clear that the joint probabilities occurring in any row of the table *ADD UP* to yield the corresponding *marginal* probability. If we reflect upon this situation carefully, we will realize that this equation is totally in accordance with the Addition Theorem of Probability for mutually exclusive events.

P (male birth) = P(male live-born *or* male stillborn)
= P (male live-born) + P (male stillborn)
= 0.5021 + 0.0120
= 0.5141

EXAMPLE

P (stillbirth/male birth)
P (male birth *and* stillbirth)/P(male birth)
=0.0120/0.5141
= 0.0233



LECTURE NO. 22

- Bayes' Theorem
- Discrete Random Variable
  - Discrete Probability Distribution
  - Graphical Representation of a Discrete Probability Distribution
  - Mean, Standard Deviation and Coefficient of Variation of a Discrete Probability Distribution
  - Distribution Function of a Discrete Random Variable.

First of all, let us discuss the BAYES' THEOREM. This theorem deals with conditional probabilities in an interesting way:

BAYES' THEOREM

If events A_1, A_2… A_k form a PARTITION of a sample space S (that is, the events Ai are mutually exclusive and exhaustive (i.e. their union is S)),
and if B is any other event of S such that it can occur ONLY IF ONE OF THE Ai OCCURS, then for any i,

$$P(A_i / B) = \frac{P(A_i)P(B/A_i)}{\sum_{i=1}^{k} P(A_i)P(B/A_i)},$$

for i = 1, 2, …, k.

Stated differently:
BAYES' THEOREM:
If A_1, A_2... and A_k are mutually exclusive events *of which one must occur*, then

$$P(A_i \mid B) = \frac{P(A_i).P(B \mid A_i)}{P(A_1).P(B \mid A_1) + P(A_2).P(B \mid A_2) + \ldots + P(A_k).P(B \mid A_k)}$$

If k = 2, we obtain:
Bayes' Theorem for two mutually exclusive events A1 and A2:

$$P(A_i \mid B) = \frac{P(A_i).P(B \mid A_i)}{P(A_1).P(B \mid A_1) + P(A_2).P(B \mid A_2)}$$

Where i = 1, 2.

In other words

$$P(A_1 \mid B) = \frac{P(A_1).P(B \mid A_1)}{P(A_1).P(B \mid A_1) + P(A_2).P(B \mid A_2)}$$

and

$$P(A_2 \mid B) = \frac{P(A_2).P(B \mid A_2)}{P(A_1).P(B \mid A_1) + P(A_2).P(B \mid A_2)}$$

EXAMPLE

In a developed country where cars are tested for the emission of pollutants, 25 percent of all cars emit excessive amounts of pollutants. When tested, 99 percent of all cars that emit excessive amounts of pollutants will fail, but 17 percent of the cars that do not emit excessive amounts of pollutants will also fail. What is the probability that a car that fails the test actually *emits* excessive amounts of pollutants?

SOLUTION

Let A1 denote the event that it emits EXCESSIVE amounts of pollutants, and let A2 denote the event that a car does NOT emit excessive amounts of pollutants. (In other words, A2 is the complement of A1.)
Also, let B denote the event that a car *FAILS* the test.
The first thing to note is that any car will either *emit* or *not* emit excessive amounts of pollutants. In other words, A1 and A2 are mutually exclusive and exhaustive events i.e. A1 and A2 form a PARTITION of the sample space S.
Hence, we are in a position to apply the Bayes' theorem.
We need to calculate P(A1|B), and, according to the Bayes' theorem:

$$P(A_1 \mid B) = \frac{P(A_1).P(B \mid A_1)}{P(A_1).P(B \mid A_1) + P(A_2).P(B \mid A_2)}$$

Now, according to the data given in this problem:

P (A1) = 0.25,

P (A2) = 0.75 (as A2 is simply the complement of A1),

P (B|A1) = 0.99,
and

P (B|A2) = 0.17
Substituting the above values in the Bayes' theorem, we obtain:

$$P(A_1|B) = \frac{P(A_1).P(B|A_1)}{P(A_1).P(B|A_1)+P(A_2).P(B|A_2)}$$

$$= \frac{(0.25)(0.99)}{(0.25)(0.99)+(0.75)(0.17)}$$

$$= \frac{0.2475}{0.2475+0.1275}$$

$$= \frac{0.2475}{0.3750}$$

$$= 0.66$$

This is the probability that a car which fails the test ACTUALLY emits excessive amounts of pollutants. The example that we just considered pertained to the simplest case when we have only two mutually exclusive and exhaustive events A1 and A2.
As stated earlier, the Bayes' theorem can be extended to the case of three, four, five or more mutually exclusive and exhaustive events.
Let us consider another example: In the following example, check the percentages of defective bolts from the recorded lecture.

EXAMPLE

In a bolt factory, 25% of the bolts are produced by machine A, 35% are produced by machine B, and the remaining 40% are produced by machine C. Of their outputs, 2%, 4% and 5% respectively are defective bolts. If a bolt is selected at random and found to be defective, what is the probability that it came from machine A?

In this example, we realize that "a bolt is produced by machine A", "a bolt is produced by machine B" and "a bolt is produced by machine C" represent three mutually exclusive and exhaustive events i.e. we can regard them as A1, A2 and A3. The event "defective bolt" represents the event B. Hence, in this example, we need to determine P(A1/B).
The students are encouraged to work on this problem on their own, in order to understand the application and significance of the Bayes' Theorem. This brings us to the *END* of the discussion of various *basic* concepts of probability. We now begin the discussion of a *very important* concept in mathematical statistics, i.e., the concept of *PROBABILITY DISTRIBUTIONS.*
As stated in the very beginning of this course, there are two types of quantitative variables --- the discrete variable, and the continuous variable. Accordingly, we have the discrete probability distribution as well as the continuous probability distribution.
We begin with the discussion of the discrete probability distribution.
In this regard, the first concept that we need to consider is the concept of Random variable.

RANDOM VARIABLE

Such a numerical quantity whose value is determined by the outcome of a random experiment is called a random variable.
For example, if we toss three dice together, and let X denote the number of heads, then the random variable X consists of the values 0, 1, 2, and 3. Obviously, in this example, X is a discrete random variable. Let us now discuss the concept of discrete probability distribution in detail with the help of the following example:
Example:

If a biologist is interested in the number of petals on a particular flower, this number may take the values 3, 4, 5, 6, 7, 8, 9, and each one of these numbers will have its own probability.

Suppose that upon observing a large no. of flowers, say 1000 flowers, of that particular species, the following results are obtained:

No. of Petals X	f
3	50
4	100
5	200
6	300
7	250
8	75
9	25
	1000

Since 1000 is quite a large number, hence the proportions f/∑f can be regarded as probabilities and hence we can write

No. of Petals X	**P(x)**
$x_1 = 3$	0.05
$x_2 = 4$	0.10
$x_3 = 5$	0.20
$x_4 = 6$	0.30
$x_5 = 7$	0.25
$x_6 = 8$	0.075
$x_7 = 9$	0.025
	1

PROPERTIES OF A DISCRETE PROBABILITY DISTRIBUTION

- (1) $0 \le P(X_i) \le 1$ • for each Xi (i = 1, 2, … 7)
- (2) $\sum p(X_i) = 1$

And, since the number of petals on a leaf can only be a *whole* number, hence the variable X is known as a *discrete* random variable, and the probability distribution of this variable is known as a *DISCRETE* probability distribution.
In other words, Any discrete variable that is associated with a random experiment, and attached to whose various values are various probabilities (Such that $\sum_{i=1}^{\infty} P(X_i) = 1$)

is known as a Discrete Random Variable, and its probability distribution is known as a Discrete Probability Distribution. Just as we can depict a frequency distribution graphically, we can draw the GRAPH of a probability distribution.

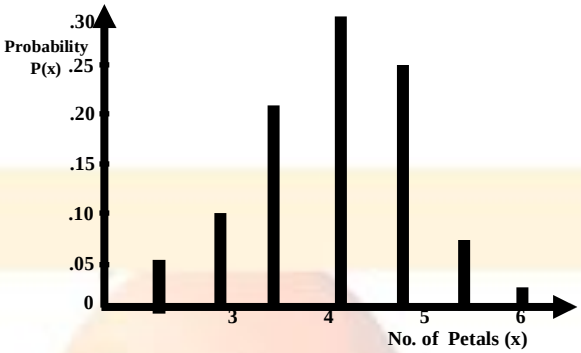
EXAMPLE

Going back to the probability distribution of the number of petals on the flowers of a particulars species, i.e.:

No. of Petals X	P(x)
$x_1 = 3$	0.05
$x_2 = 4$	0.10
$x_3 = 5$	0.20
$x_4 = 6$	0.30
$x_5 = 7$	0.25
$x_6 = 8$	0.075
$x_7 = 9$	0.025
	1

This distribution can be represented in the form of a line chart.

Line Chart Representation of the



Evidently, this particular probability distribution is approximately symmetric. In addition, this graph clearly shows that, just as in the case of a frequency distribution, every discrete probability distribution has a *CENTRAL* point and a *SPREAD*. Hence, similar to a frequency distribution, the discrete probability distribution has a *MEAN* and a *STANDARD DEVIATION*. How do we *calculate* the mean and the standard deviation of a probability distribution?
Let us first consider the computation of the MEAN:
We know that in the case of a frequency distribution such as

X	f
1	1
2	2
3	4
4	2
5	1

the mean is given by

$$\bar{X} = \frac{\sum fX}{\sum f} = \frac{\sum Xf}{\sum f}$$

In case of a discrete probability distribution, such as the one that we have been considering i.e

No. of Petals X	P(x)
$x_1 = 3$	0.05
$x_2 = 4$	0.10
$x_3 = 5$	0.20
$x_4 = 6$	0.30
$x_5 = 7$	0.25
$x_6 = 8$	0.075
$x_7 = 9$	0.025
	1

the mean is given by:

$$\mu = E(X) = \frac{\sum XP(X)}{\sum p(X)} = \frac{\sum XP(X)}{1} = \sum XP(X)$$

Hence we construct the column of XP(X), as shown below:

No. of Petals x	P(x)	xP(x)
$x_1 = 3$	0.05	0.15
$x_2 = 4$	0.10	0.40
$x_3 = 5$	0.20	1.00
$x_4 = 6$	0.30	1.80
$x_5 = 7$	0.25	1.75
$x_6 = 8$	0.075	0.60
$x_7 = 9$	0.025	0.225
Total	1	5.925

Hence $\mu = E(X) = \sum XP(X) = 5.925$ i.e. the mean of the given probability distribution is 5.925. In other words, considering a very large number of flowers of that particular species, we would expect that, on the average, a flower contains 5.925 petals --- or, *rounding* this number, 6 petals. This interpretation points to the reason why the mean of the probability distribution of a random variable X is technically called the EXPECTED VALUE of the random variable X. ("Given that the probability that the flower has 3 petals is 5%, the probability that the flower has 4 petals is 10%, and so *ON*, we *EXPECT* that on the *average* a flower contains 5.925 petals.)

COMPUTATION OF THE STANDARD DEVIATION

Just as in case of a frequency distribution, we have

$$S = \sqrt{\frac{\sum f(X-\bar{X})^2}{\sum f}}$$

$$= \sqrt{\frac{\sum fX^2}{\sum f} - \left(\frac{\sum fX}{\sum f}\right)^2} = \sqrt{\frac{\sum X^2 f}{\sum f} - \left(\frac{\sum Xf}{\sum f}\right)^2}$$

Similarly, in case of a probability distribution, we have

$$\sigma = \text{S.D.}(X) = \sqrt{\frac{\sum X^2 P(X)}{\sum P(X)} - \left[\frac{\sum XP(X)}{\sum P(X)}\right]^2}$$

$$= \sqrt{\sum X^2 P(X) - [\sum XP(X)]^2} \quad \left(\because \sum P(X) = 1\right)$$

In the above example

No. of Petals x	P(x)	xP(x)	$x^2P(x)$
$x_1 = 3$	0.05	0.15	0.45
$x_2 = 4$	0.10	0.40	1.60
$x_3 = 5$	0.20	1.00	5.00
$x_4 = 6$	0.30	1.80	10.80
$x_5 = 7$	0.25	1.75	12.25
$x_6 = 8$	0.075	0.60	4.80
$x_7 = 9$	0.025	0.225	2.025
Total	1	5.925	36.925

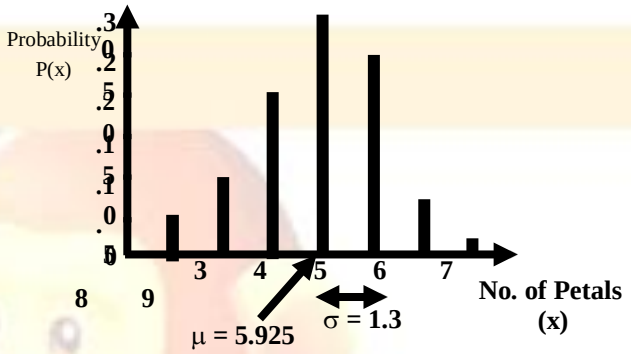
Hence

$$S.D.(X) = \sqrt{36.925 - (5.925)^2}$$

$$= \sqrt{36.925 - 35.106}$$

$$= \sqrt{1.819} = 1.3$$

Graphical Representation:



Now that we have both the mean and the standard deviation, we are in a position to compute the *coefficient of variation* of this distribution:

Coefficient of Variation

$$C.V. = \frac{\sigma}{\mu} \times 100$$

$$= \frac{1.3}{5.925} \times 100$$

$$= 21.9\ \%$$

Let us consider another example to understand the concept of discrete probability distribution.

EXAMPLE

a) Find the probability distribution of the sum of the dots when two fair dice are thrown
b) Use the probability distribution to find the probabilities of obtaining (i) a sum that is greater than 8, and (ii) a sum that is greater than 5 but less than or equal to 10.

SOLUTION

a) The sample space S is represented by the following 36 outcomes:

S = {(1, 1), (1, 2), (1, 3), (1, 5), (1, 6);
(2, 1), (2, 2), (2, 3), (2, 5), (2, 6);
(3, 1), (3, 2), (3, 3), (3, 5), (3, 6);
(4, 1), (4, 2), (4, 3), (4, 5), (4, 6);
(5, 1), (5, 2), (5, 3), (5, 5), (5, 6);
(6, 1), (6, 2), (6, 3), (6, 5), (6, 6) }

Since each of the 36 outcomes is equally likely to occur, therefore each outcome has probability 1/36.
Let X be the random variable representing the sum of dots which appear on the dice. Then the values of the r.v. are 2, 3, 4… 12.
The probabilities of these values are computed as below:

$f(2) = P(X = 2) = P[\{1,1\}] = \frac{1}{36}$, as there is only one outcome resulting in a sum of 2,

$f(3) = P(X = 3) = P[\{(1, 2), (2,1)\}] = \frac{2}{36}$,

$f(4) = P(X = 4) = P[\{(1, 3), (2, 2), (3,1)\}] = \frac{3}{36}$,

Similarly

$f(5) = \frac{4}{36}, f(6) = \frac{5}{36}, f(7) = \frac{6}{36}, f(8) = \frac{5}{36}, f(9) = \frac{4}{36}$,

$f(10) = \frac{3}{36}, f(11) = \frac{2}{36}$ and $f(12) = \frac{1}{36}$.

Therefore the desired probability distribution of the r.v X is

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The probabilities in the above table clearly indicate that if we draw the line chart of this distribution, we will obtain a triangular-shaped graph. The students are encouraged to draw the graph of this probability distribution, in order to be able to develop a visual picture in their minds.

b) Using the probability distribution, we get the required probabilities as follows:

i) P(a sum that is greater than 8)

$= P(X > 8)$

$= P(X=9) + P(X=10) + P(X=11) + P(X=12)$

$= f(9) + f(10) + f(11) + f(12)$

$= \frac{4}{36} + \frac{3}{36} + \frac{2}{36} + \frac{1}{36} = \frac{10}{36}$

ii) P(a sum that is greater than 5 but less than or equal to 10)

$= P(5 < X \leq 10)$

$= P(X = 6) + P(X = 7) + P(X = 8) + P(X = 9) + P(X = 10)$

$= f(6) + f(7) + f(8) + f(9) + f(10)$

$= \frac{5}{36} + \frac{6}{36} + \frac{5}{36} + \frac{4}{36} + \frac{3}{36} = \frac{23}{36}$.

Next, we consider the concept of the DISTRIBUTION FUNCTION of a discrete random variable:

DISTRIBUTION FUNCTION

The distribution function of a random variable X, denoted by F(x), is defined by F(x) = P(X < x).
The function F(x) gives the probability of the event that X takes a value LESS THAN OR EQUAL TO a specified value x. The distribution function is abbreviated to *d.f.* and is also called the *cumulative distribution function (cdf)* as it is the cumulative probability function of the random variable X from the smallest value upto a *specific* value x.
Let us illustrate this concept with the help of the same example that we have been considering --- that of the probability distribution of the sum of the dots when two fair dice are thrown. As explained earlier, the probability distribution of this example is:

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

The term 'distribution function' implies the cumulating of the probabilities similar to the cumulation of frequencies in the case of the frequency distribution of a discrete variable.

x_i	2	3	4	5	6	7	8	9	10	11	12
$f(x_i)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
$F(x_i)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$

If we are interested in finding the probability that we obtain a sum of five or less, the column of cumulative probabilities immediately indicates that this probability is 10/36.

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A histogram is consisting of a set of a set of adjacent rectangle whose bases are marked off by:

a. Class boundaries (Lecture no 4 Page no 32 Handouts)
b. Class limits
c. Class frequency
d. Class marks

The middle value of an ordered array of numbers is the

a. Mean
b. Median (Lecture no 7 Page no 59 Handouts)
c. Mode
d. Midpoint

If Mean =25 & S.D is 5 then C.V is:

a. 100%
b. 20% (Lecture no 11 Example Page no 88 Handouts)
c. 10%
d. 25%

Solution: C.V = (S.D/X) x 100 = (5/25) x 100 = 20% Answer

you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, you will get:

a. Ogive
b. Frequency polygon
c. Frequency curve page 38
d. Historigram

The conditional probability P (A/B) is:

a. P (A n B)/P (B) (Lecture no 20 Page no 154 Handouts)
b. P (A n B)/P (A)
c. P (A U B)/P (B)
d. P (A U B)/P (A)

Serious disadvantage of using range as a measure of dispersion is that it is based on only:

a. Minimum Values
b. Maximum Values
c. Both Minimum and Maximum values
d. None of the above

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Frequency of a variable is always in:

a. Fraction form
b. Percentage form
c. Less than form
d. Integer form

$^5C_5$ is equal to

a. 5
b. 1
c. 10
d. 24

Chebychev's inequality does not hold for k = ?

a. 3
b. 2
c. 1
d. 0 (Lecture no 12 Page no 94 Read CHEBYCHEV'S THEOREM)

If A = {1,2,3,4} and B = {3,4,5,6} then A – B will be:

a. {1,2}
b. {3,4}
c. {3,2,1}
d. {1,2,3,4,5,6}

If P (B|A) = 0.25 and P (A and B) = 0.20, then P (A) is

a. 0.05
b. 0.80
c. 0.95
d. 0.75

Solution: P (B|A) = [P (A and B)] / P (A)
P (A) = [0.20 / P (B|A)]
P (A) = [0.20 / 0.25]
P (A) = 0.80

According to Empirical rule, approximately 68% of the measurements will fall within:

a. (Mean - S.D, Mean + S.D) (Lecture no 12 Page no 90 Handouts)
b. (Mean - 2S.D, Mean + 2S.D)
c. (Mean - 3S.D, Mean + 3S.D)

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d. None of these

For Platykurtic distribution, b2 (moment ratio) will be:

a. Greater than 3
b. Less than 3 (Lecture no 14 Page no 114 Handouts)
c. Equal to 3
d. Equal to zero

There are two broad categories of data, which are:

a. Weighted and Un-weighted
b. Grouped and Un-grouped
c. Qualitative and Quantitative (Lecture no 3 Page no 16 Handouts)
d. Primary and Secondary

In a linear regression, Y= a+bX , the variable "Y" will always:

a. A random variable (Lecture no 15 Page no 117 Handouts)
b. A non-random variable
c. Qualitative variable
d. Quantitative variable

In regression line Y= a+bX , X is called:

a. Dependent variable
b. Independent variable (Lecture no 15 Page no 116 Handouts)
c. Explained variable
d. Regressand

Difference between the largest and the smallest data values is called

a. variance
b. interquartile range
c. range
d. coefficient of variation

The extremely positively skewed curve is also known as:

a. Frequency curve
b. U-shaped curve
c. J-shaped curve
d. Reverse J-shaped curve (Lecture no 5 Page no 35 Handouts)

A list of 7 pulse rates is: 70, 64, 80, 74, 92, 96, 98. What is the median for this list?

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a. 70
b. 80
c. 92
d. 98

Solution: Median is the middle value. First of all we will arrange them in ascending order 64, 70, 74, **80**, 92, 96, 98

The value of 10C9:

a. 45
b. 35
c. 35
d. 10

The most frequent value in the data is called

a. Mean
b. Median
c. Mode
d. Harmonic mean

An event that contains more than one sample points is called:

a. Mutually exclusive event
b. Not mutually exclusive event
c. Hyper event
d. Compound event (Lecture no 17 Page no 140 Handouts)

Calculate range for the following data: 22, 22, 30, 32, 37, 48, 60, 88, 90.

a. 22
b. 90
c. 37
d. 68

Solution: Largest value – Smallest value = Range
90 – 22 = **68 Answer.**

What is the median of this set of numbers: 4, 6, 7, 9, 2000000?

a. 9
b. 6
c. 7.5
d. 7

The value of the middle term in a ranked (ordered) data set is called the:

a. Mode
b. Mean

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c. Median
d. Harmonic mean

The median is _____.

a. The highest number
b. The middle point
c. The average
d. Affected by extreme scores

The Mode of 8, 5, 7, 10, 15, 21, 5, 7, 2, 5 is

a. 8
b. 5
c. 7
d. 21

Note: Mode is the most repeated value so in this case 5 is the most repeated value.

When all the values falling in a class are equal to the mid point of the class interval is called ……..?

a. Random error
b. Unbiased Error
c. Biased Error
d. Grouping Error (Lecture no 7 Page no 56 handouts)

For a Leptokurtic distribution, b2 (moment ratio) will be:

a. Greater than 3 (Lecture no 14 Page no 114 Handouts)
b. Less than 3
c. Equal to 3
d. Equal to zero

Let A = {1,2,3,4} and B = {3,4,5,6} Then A ∩ B:

a. {3,4}
b. {1,4}
c. {3,5}
d. {3,6}

The range of the scores 29, 3, 143, 27, 99 is:

a. 140
b. 143
c. 146
d. 70

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Solution: (143 – 3) = 140 Answer

In how many ways, a team of 11 players can be chosen from a total of 16 players?

a. 4368
b. 2426
c. 5400
d. 2680

Direct personal investigation is _____ when the area to be covered is vast. **(Doubtful)**

a. Costly (Lecture no 2 Page no 6 Handouts)
b. Time-consuming (Lecture no 2 Page no 6 Handouts)
c. Localized inquires
d. Inaccurate

Note: It can be both read lecture no 2 Page no 6 Handouts.

What is the mean of this set of numbers: 4, 6, 7, 9, 200000?

a. 7.5
b. 7
c. 400,005.2
d. 4

Solution: Mean = (4+6+7+9+200000)/5 = **400,005.2 Answer**

According to this empirical rule, approximately how much values will fall within (Mean – 3S.D, Mean + 3S.D)?

a. 100% (Lecture no 12 Page no 90 Handouts)
b. 95%
c. 75%
d. 68%

In a linear regression, Y= a+bX , the variable "X" will always:

a. A random variable
b. A non-random variable (Lecture no 15 Page no 117 Handouts)
c. Qualitative variable
d. Quantitative variable

Variance is expressed in _____ units as the units of data set.

a. Squared (Lesson no 11 Page no 85 Handouts)
b. Cube
c. Single
d. Same

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In a linear regression, best fitted line is obtained through:

a. Method of moment
b. Method of likelihood
c. Method of least square
d. Method of semi average

In case of frequency distribution, the median is given by the formula:

a. I+h/f (n/2-2c)
b. I+h/f (n/2-c)
c. I+f/h (n/2-c)
d. I+f/h (n/4-c)

The sum of squared deviations from mean is:

a. Maximum
b. Minimum
c. Zero
d. Undefined

In a week the prices of a bag of rice were 350,280,340,290,320,310,300.

a. 320
b. 315
c. 300
d. 420

Which scale will you use to measure the temperature?

a. Nominal scale
b. Interval scale (Lecture no 1 Page no 4 Handouts)
c. Ratio scale
d. Ordinal scale

From the given data, which one of the following chart we can draw:

Year	Turnover
1965	350000
1966	42000
1967	43500
1968	48000
1969	48500

a. Simple bar chart
b. Multiple bar charts
c. Component bar chart
d. Pie chart

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In a Venn diagram, the overlap between two circles represents:

a. The union of two sets
b. The intersection of two sets
c. The elements that are in either of two sets
d. The difference between the number of elements in two sets

Calculate range for the following data: 10, 32, 33, 34, 37, 42, 55,58, 70

a. 50
b. 60
c. 40
d. 20

Solution: Largest value – Smallest value = Range
70 – 10 = **60 Answer**

When the frequency distribution or curve departs from symmetry, is called

Skewed
Positively skewed
Negatively skewed
None of these

Measure of central tendency is used to measure:

a. Average
b. Variability
c. Location
d. Both Average and Location

Component bar charts are used when data is divided into:

a. Parts
b. Groups
c. Circles
d. None of these

In a Box and Whisker plot, right end of the box is referred as:

a. First quartile
b. Second quartile
c. Third quartile
d. Mode

Fourth moment about mean provides information about the _____ of the distribution.

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a. Centre
b. Dispersion
c. Symmetry
d. Kurtosis

Let A abd B are two dependent events such that
P(A)=1/4, P(A/B)=1/2 and P(B/A)=2/3.
Find P(A∩B).

a. 1/8
b. 1/6
c. 2/3
d. 1/4

Consider a set A = {1,2,3}. What is the number of subsets of A? **(Doubtful)**

a. 3
b. 6
c. 8
d. 9

The stem for the following data is:
22, 45,36, 15, 14, 12, 14, 14, 17, 21, 24, 24, 25, 25, 26, 26, 27, 29, 31, 34,35

a. 1,2,3,4
b. 1,2,3,4,5
c. 11, 12,13,14,15
d. 10,20,30,40,50

An event that contains more than one sample points is called: **(Doubtful)**

a. Mutually exclusive event
b. Normal event
c. Simple event
d. Compound event

Frequency distribution is considered as negatively skewed if all values of distribution moves to

a. lower tail
b. median tail
c. variance tail
d. upper tail

Which of the following is NOT a common measure of central tendency?

a. Mode

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b. Range
c. Median
d. Mean

From the table given below, how many students obtained marks between 60 and 69? **(Doubtful)**

Marks	f	Mid-Points
50-59	5	54.5
60-69	7	64.5
70-79	8	74.5
80-89	5	84.5

64.5
12
60
7

If the first and third quartiles are 22, 16 and 56,36 respectively, then the quartile deviation is:

a. 17.1
b. 30.5
c. 50.5
d. 51.3

Adding all the squared deviations taken from mean and dividing by the number of observations, we get:

a. Standard Deviation
b. Variance
c. Mean Deviation
d. None of the above

Standard deviation divided by mean is known as:

a. Co-efficient of standard deviation
b. Co-efficient of variation
c. Both
d. None

5C5 equals to:

a. 5
b. 1
c. 10
d. 15

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Which of the measure of dispersion is used to compare variation between two series?

a. C.V.
b. Q.D.
c. M.D.
d. S.D.

If Y=3X+5, then S.D of Y is equal to **(Doubtful)**

a. 9 s. d(x)
b. 3 s. d(x)
c. s. d(x)+5
d. 3 s. d(x) + 5

Which of the following technique is not used to represent the bivariate qualitative data?

a. Component Bar Chart
b. Multiple Bar Chart
c. Line Chart
d. Pie Chart

When a frequency distribution involves "open-end" classes, then which average is appropriate?

a. Mean
b. Mode
c. Median (Lecture no 8 Page no 62 Handouts)
d. None of these

Harmonic mean is extremely useful in averaging _____ types of data.

a. Ratios
b. Rates
c. Both ratios and rates
d. None of the above

According to Empirical rule, approximately how much values will fall within (Mean-3S.D, Mean+3S.D)?

a. 100% (Lecture no Page no 90 Handouts)
b. 95%
c. 75%
d. 68%

What is probability of drawing two clubs from a well shuffled pack of 52 cards?

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a. 13/51
b. 1/17
c. 1/26
d. 13/17

Solution: Total Cards = 52 cards
Number of club cards = 13 Probability of 1st card being club card = 13/52
Probability of 2nd card being club card = 12/51
∴ Total Probability
= (13/52) x (12/51)
= 1/17 Answer

Which pair of measure cannot be calculated when one of numbers in the series is zero?

a. G.M. and A.M.
b. H.M. and A.M.
c. G.M. and H.M.
d. None of these

Which of the following techniques is used to predict the value of one variable on the basis of other variables?

a. Correlation analysis
b. Coefficient of correlation
c. Covariance
d. Regression analysis

A bag contains 12 red balls and 12 blue balls. A ball is drawn at random. The probability that ball drawn is red is

a. 1/2
b. 5/11
c. 6/10
d. 1

Solution: Total Red Balls: 12
Total blue balls: 12
Total Balls: 12 + 12 = 24
Probability of ball drawn is a red ball:
P (Red) = 12/24 = **1/2 Answer**

If the GM of a set of two observations is 10 and its HM is 8, then the AM of the set of observations is

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a. 100
b. 12.5
c. 64
d. 7.5

RELATION BETWEEEN AM, GM AND HM - FORMULA

Relation between AM, GM and HM is:

AM×HM=GM^2

Solution: GM = 10, HM = 8, GM^2 = (10)^2 = 100, AM x HM = GM^2,

AM x 8 = 100, AM = 100/8 = 12.5 Answer

If A = {H, T} then which of the following is power set of A?

a. {{}, {H, T}}
b. {{H}, {T}, {H, T}}
c. {{H}, {T}, {H, T}, {T, H}}
d. {{}, {H}, {T}, {H, T}}

What do understand by the term power set?

In **set** theory, the **power set** (or **power set**) of a **Set** A is defined as the **set** of all subsets of the **Set** A including the **Set** itself and the null or empty **set**. It is denoted by P(A). Basically, this **set** is the combination of all subsets including null **set**, of a given **set**. Simple formula 2^n. n is the number of subsets. 2^2 = 4.

A= {H, T} = {{}, {H}, {T}, {H, T}} Answer

2nd Quartile = 5th Decile = 50th Percentile =?

a. Geometric Mean
b. Median
c. Mode
d. Harmonic Mean

The total number of observations, which are below a certain value are known as

A. class boundaries
B. class marks
C. cumulative frequency
D. variances

Histogram can be drawn only for:

A. Discrete frequency distribution

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B. Continuous frequency distribution

C. Continuous frequency distribution

D. Relative frequency distribution

Classification is the process of arranging data according to:

A. one characteristic

B. Two or more characteristic

C. Similar characteristic

D. None of these

Which of the following, measures the dispersion around mean?

A. Mean deviation

B. Standard deviation

C. Mean deviation and Standard deviation

D. None of these

What is mode for the following set of data: 1,1,2,2,5,5,7

a. 1

b. 1,2

c. 1,2,5

d. no mode in the data

The mode is the value that appears most frequently in a data set. A set of data may have one mode, more than one mode, or no mode at all. Other popular measures of central tendency include the mean, or the average of a set, and the median, the middle value in a set.

Important: The mode can be the same value as the mean and/or median, but this is usually not the case.

Which of the following averages give information about central value in the distribution?

a. Mean

b. Median I think

c. Mode

d. Harmonic mean

In a Pie diagram, the sector of a circle is obtained by:

a. (component part/total)*100

b. (component part/total)*360

c. (component part/total)*180

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d. (component part/total)*300

Relationship among the averages

a. HM ≥ GM ≥ AM
b. **AM ≥ GM ≥ HM**
c. GM ≤ HM ≤ AM
d. AM ≥ HM ≤ GM

Which of the scale is best to use for measuring the salary of an employee?

a. nominal
b. ordinal
c. interval
d. ratio

83. The mean of a distribution is 30, the mode is 24 and the standard deviation is 4, then the coefficient of skewness will be:

a. Less than zero
b. Equal to zero
c. **Greater than zero**
d. None of the above

Smaller standard error of estimate shows:

a. Data points are very far to the line
b. Data points are close to the line
c. There is no difference between line and points
d. Difference is additive

Data arranged in ascending or descending order of magnitude is called:

a. Ungrouped data
b. Grouped data
c. Discrete frequency distribution
d. **Arrayed data**

A circle in which sectors represents various quantities is called:

a. Histogram
b. Frequency Polygon
c**. Pie Chart**
d. Component Bar Chart

Sum of the absolute deviations of the values is least when deviations are taken from:

a. Mean
b. Median
c. Mode

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d. G.M

Chebychev's inequality is valid for the data set

a. Sample
b. Entire population
c. Both sample and entire population
d. None of the above

Which of the following terms best describes data that were originally collected at an earlier time by a different person for a different purpose?

a. Primary data
b. **Secondary data**
c. Experimental data
d. Field notes

Statistics deals with

a. Individuals
b. Isolated items
c. Observations
Aggregates of facts

. If a box contains six red, three blue and five pink ties then probability of blue ties will be equal to:

a. 1/14
b. **3/14**
c. 5/14
d. 6/14

Which of the rule is applied to any data set, regardless shape of the frequency distribution?

a. **Chebychev's rule**
b. Empirical rule
c. Combination rule
d. Permutation rule

Which average is used in the situation where the number of floors in the buildings at the center of a city?

a. Mean

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b. Median
c. Mode
d. Variance

94. Rankings of the finishes of competitors in a foot race is an example of a(n)_____.

a. ratio scale
b. ordinal scale
c. nominal scale
d. interval scale

95. Among 18 articles, six having minor Defects and three have major defects. Determine the probability that an article selected at random has major defect.

a. 1/6
b. 1/5
c. 0.25
d. 0.11

96. A series of data with exclusive classes along with the corresponding frequencies is called:

a. Discrete frequency distribution
b. Continuous frequency distribution
c. Percentage frequency distribution
d. Cumulative frequency distribution

Using the following table, calculate P(X<2)

X	0	1	2	3
f(x)	1/8	3/8	3/8	1/8

a. 1/8
b. 3/8
c. 4/8
d. 7/8

98. If a distribution has two modes then the distribution is called:

a. Uni-Modal
b. Bi-Modal
c. Tri-Modal
d. Multi-Modal

1 For any two estimators T1 and T2, if VAR(T1) < VAR(T2), then T1 is:
Consistent
Unbaised
Sufficient

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2 In a set of 10 values all the values are 5, what will be the P50?

2
5
10
20

3 What is the graphical shape of the chi-square distribution?

Positively skewed
Negatively skewed
Uniformly distributed
Norrmally distributed

4 Which of the following is a characteristic of the normal distribution:

It is a skewed distribution
It is bell-shaped
It is not asymptotic
It is leptokurtic

5 Please choose one
A five number summery consist of :

► 0 1 3 , , , , (97) m X Q Median Q andX Page
► 1 3 0 , , , , m X Q Mean Q and X
► 1 3 0 , , , , m X Q Mode Q and X
► 0 1 2 , , , , m X Q Median Q and X

6 Please choose one
in a multiplication theorem P(A∩B) equals:

►P(A) P(B)
►P(A) + P(B)
►P(A) * P(B|A) (Page 159)
►P(B\A) * P(B)

7 The probability of drawing 'White' ball from a bag containing 4 red , 8 black and 3 white ball is

►0
►3/15
►1/12
►1/2

8 If A and B are mutually exclusive events then P(A or B) equals:

►P(A) + P(B) - P(A and B)

►P(A) * P(B)
►P(A) + P(B) (Page 155)
►P(A|B) + P(B|A)

9 The simultaneous occurrence of two events is called

►Prior probability
►Subjective probability not confirmed
►Conditional probability

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►Joint probability

10 First moment about origin is always equals to:

►Mean (Page 119)
► Variance
► Standard Deviation
► Zero

11 Which one of the following measurement does not divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► Standard deviations

12 if first and third quartile are 21.16 and 56.36 respectively, then the quartile deviation is:

►17.1 (Page 84)
► 34.2
► 51.3
► 50.5

13 The height of student is 60 inch. This is an example of……?

► Continues data
► Qualitative data
► Categorical data
►Discrete data

14 Which of the statement is true regarding a sample?

►It is a part of population (Page 13)
► It must contain at least five observations
► It refers to descriptive statistics
► It produces true value

15 Which one of the following graphs is used for a time series data?

► Histogram
► Historigram
► Frequency curve
► Frequency polygon

16 Which of the following comes first to make frequency distribution

► No. of groups
► Class intervals
►Rang (Page 28)
► Tally marks

17 The average which is defines as reciprocal of arithmetic mean of the reciprocals of the values is called

► Geometric Mean
►Harmonic mean (Page 77)
► Mode

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► Median

18 If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

► Weighted mean (Not sure)
► Harmonic mean
► Arithmetic mean
► Geometric mean

19 The probability of drawing a 'jack card 'from 52 playing cards is:

► 1/52
► 4/52
► 13/52
► 26/52

20 If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

► $0 \leq r \leq 1$
► $r \geq 0$
► $r = +1$
► $r=-1$

21 If a curve has a longer tail to the right, it is called:

► Positively skewed (Page 39)
► Negatively skewed
► J-shaped
► Symmetric

22 Which one of the following is not included in measures of central tendency?

► Quartile deviation (Page 82)
► Harmonic mean
► Geometric mean
► Arithmetic mean

23 Which of the following is not based on all the observations?

► Arithmetic Mean
► Geometric Mean
► Harmonic mean
► Mode

24 What is the Standard Deviation of 7, 7, 7, 7, 7, 7, 7

► 49
► 1
► 0 (Standard deviation will always be zero if all the values in data are same)
► 7

25 10! =.....

► 362880
► 3628800

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► 362280
► 362800

26 If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:
► 1/52
► 13/52
► 4/52
► 26/52

27 Which dispersion is used to compare variation of two series?
► C.V. (Page 93)
► Q.D.
► M.D.
► S.D.

28 In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:
► ►Intercept a = 0
► Intercept a ≠ 0
► ►Slope b = 0
► ►Slope b ≠ 0

29 When two coins are tossed the probability of at least one head is:
► 1/4
► 3/4 (HH, HT, TH, TT)
► 2/4
► 4/4

30 In the model Y = mX + a, Y is also known as the:
► Predictor variable
► Independent variable
► Predicted (dependent) variable (Page 121)
► Explanatory variable

31 According to empirical rule approximately 95% of the measurements will fall under which interval?
► X ± S
► X ± 2S correct
► X ±3S
► X ± 4S

32 Which one of the following is written at the top of the table?
► Source note
► Foot note
► Prefatory note
► Title Click here for detail

33 Which one of the following is the class frequency?
► The number of observations in each class
► The difference between consecutive lower class limits
► Always contains at least 5 observations

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► Usually a multiple of the lower limit of the first class

34 If X is a discrete random variable, then the function f (x) is

► A probability function
► A probability density function
► A density function
► A distribution function (Page 173)

35 If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

► Ogive
► Frequency polygon
► Frequency curve (Page 38)
► Historigram

36 If mean of the two observations is 10.5, then median of these two observations will be

► 7.5
► 8.5
► 9.5
► 10.5 (Page 68)

37 Which one is the formula of mid range?

$=\frac{x_0+x_m}{2}$ (Page 80)

38 For the given data 2, 3, 7, 0, -8 G. M will be:

► Negative
► Positive
► Zero (Page 75)
► Undefined

39 Which one is equal to explained variation divided by total variation?

► Sum of square due to regression
► Coefficient of determinant
► Standard error estimate
► Coefficient of correlation

40 in the given series 1,2,1,1,2,2,2,3,4,5,3,2,3,1,4,2,3 mode of given is

► 4
► 3
►1
► 2

41 True for the population,

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► it must be large number of values
► It must refer to people
► It is collection of individual objects or measurement (Page 12)
► It is the small part of whole

42 Data arrangement in ascending or descending order
► Array data (Page 47)
► Group data
► Ungroup data
► Raw data

43 What is the main objective of Descriptive statistics?
► To test population properties
► To describe the data we collected
► To infer something about the population
► Making estimate

44 Which measure of central tendency?
► Variation of distribution
► Average of distribution (Page 51)
► Scattering of distribution
► Dispersion of distribution

45 If a=4 b=2 estimate line (i.e y=a+bx) and independent variable has value 3 the value of dependent variable
► 6
► 9
► 10 4+2(3)=10
► 11

46 The number of ways in which 4 books can be arranged
► 4
► 6
► 12
► 24 (Page 142)

47 If we plot paired observed (x,y)=1…….n on graph is called,
► Polygon
► Freasito diagram
► Scatter diagram
► Cumulative frequency diagram

48 The simultaneous occurrence of two events is called
► Descriptive probability
► Subjective probability
► Conditional probability
► Joint probability

49 Which one is the not measure of dispersion.
► The range
► 50th percentile

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► Inter quartile range
► Variance

50 In positively skew cure which relation is
► The mean, median and mode are equal
► Mean is greater then median
► Median is greater then mean
► Standard deviation must be greater then mean or median

51 When coin tossed we get only
►1 outcome
►2 outcomes
►3 outcomes
►4 outcomes

52 When mean is 25 and S.D is 5 then CV is
► 100%
► 25%
► 20%
► 10%

53 For a positively skewed distribution m3 will be:
► Positive (Page 119)
► Negative
► Zero
► 1

54 When data is labeled to identify an attribute of element, the measurement scale is:
► Ordinal (Page 9)
► Interval
► Nominal
► Ratio

55 Suppose the estimated equatio Yˆ = 5- 2 has been calculated for a set of data. What is slop of th line:
► 0
► 2
►-2 (Page 121)
► 5

56 If P(B|A) = 0.25 and P(A∩B) = 0.20 then P(A) is:
► 0.05
► 0.80 (Page 159)
► 0.95
► 0.75

57 Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:
► Advance

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statistics

► Probability statistics
► Descriptive statistics (Page 61)
► Inferential statistics

58 In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors."
This report is an example of :
► A sample
► A population
► Statistical inference
► Descriptive statistics

59 A set that contains all possible outcomes of a system is known as
► Finite Set
► Infinite Set
► Universal Set (Page 134)
► No of these

60 If X and Y are independent, then Var(X-Y) is equal to:
Var(X) -Var(Y)
Var(X) +Var(Y)
Var(X +Y)
Zero

61 Which of the following is the class frequency
► The number of observations in each class
► The difference between consecutive lower class limits
► Always contains at least 5 observations
► Usually a multiple of the lower limit of the first class

66 How to construct the class interval:
► Divide the class frequencies in half
► Divide the class frequency by the number of observations
► Find the difference between consecutive lower class limits
► Count the number of observations in the class

67 Data in the Population Census Report is:
► Ungrouped data (Page 11)
► Secondary data
► Primary data
► Arrayed data

68 What is the range of -2,-3,-5,-10 :
►-12 (Page 28)
► 8
► -8
► 2

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69 The algebraic sum of deviations from mean is:
- ► Maximum
- ► Minimum
- ► Zero (Page 86)
- ► Undefined

70 The sum of squares of deviations from mean is:
- ► Undefined
- ► Zero
- ► Maximum
- ► Minimum

71 Statistic is a numerical quantity, which is calculated from:
- ► Population
- ► Sample (Page 7)
- ► Data
- ► Observations

72 If any value in the data is zero, then it is not possible to have:
- ► Harmonic Mean
- ► Arithmetic Mean
- ► Median
- ► Mode

73 For a symmetrical distribution… is equidistance from median:
- ► X0 and Xm
- ► Q1 and Q3 (Page 97)
- ► X0 and Q1
- ► Xm and Q3

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75 If Mean = 25 & S.D is 5 then C.V is
- ► 100%
- ► 25%
- ► 20% (Page 93)
- ► 10%

76 A coin is tossed 3 times then, the number of sample points ii the sample space is:
- ► 3
- ► 8
- ► 6 (Page 145)
- ► 4

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77 What is the difference between a permutation and a combination:

►In a permutation order is important and in a combination it is not

► In a permutation order is not important and in a combination it is important

► A combination is based on the classical definition of probability

► A permutation is based on the classical definition of probability

78 What we consider in simple correlation analysis:

► Several independent variables

►Strength of the association between two variables (Page 128)

► Intercept with the X-axis

► Intercept with the Y-axis

79 If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':

► 0 = r = 1

► r = 0

►r = +1 (rep)

► r=-1

80 Which of the following measure of dispersion is least affected by extreme values of observations in a data:

► Range Page 82)

► Quartile deviation

► Mean absolute deviation

► Standard deviation

81 The approximate number of observations in a set of data covered by the interval, Median±Q.D are:

►50 per cent (Page 84)

► 68.5 per cent

► 95.4 per cent

► 99 per cent

82 The dispersion which is calculated from all observed values is:

►Standard deviation (Page 87)

► Quartile deviation

► Rang

► Coefficient of Rang

83 In a set of 20 values all the values are 2, what is the value of Geometric Mean:

►2 (Page 75)

►5

► 10

► 20

84 In a set of 10 values all the values are 5, what is the value of 5th Decile:

►2

►5

► 10

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► 20

85 If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the secor 30, which type of average will be used to find the average speed/hr:

► W.M
► H.M
► A.M
► G.M

86 In stem and leaf plot, data is measured on:

► Ratio Scale
► Interval Scale
► Ordinal scale
► Nominal Scale

87 In statistics, we deal with:

► Individuals
► Isolated items
► Aggregates of facts (Page 7)
► Qualitative data

88 Which of the following is a systematic arrangement of data into rows and columns:

► Component bar chart
► Classification
► Tabulation
► Bar chart

89 For any number k.....1, at least 1 – 1/k2 of the data-values fall within k standard deviations of the mean:

► Greater than 1 (Page 97)
► Less than 1
► Greater or equal to 1
► Less or equal to 1

90 When E is an impossible event,then P(E) is:

► 0 (Page 146)
► 0.5
► 1
► 2

91 The data for an ogive is found in which distribution:

► A cumulative frequency distribution (Page 43)
► A relative frequency distribution
► A frequency distribution
► A joint frequency distribution

92 Which of the following statements is true regarding a sample:

► It is a part of population (Page 13)
► It must contain at least five observations

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► It refers to descriptive statistics
► It produces True value

93 Which level of measurement is required for the median:
► Nominal
►Ordinal
► Interval
► Ratio

94 In a the regression line Y= a + bX+ € the variable which is non-random is:
►X Page 122)
►Y
► Both X and Y
► Neither X nor Y

95 The variable plotted on the horizontal or X-axis in a scatter diagram is called the:
►Scatter variable
► Independent variable
► Dependent variable
► Correlation variable

97 The descriptive measure of dispersion that is based on the concept of a deviation about the mean is:
► Range
► Inter quartile range
► Quartile deviation
►Standard deviation

98 Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:
► Advance statistics
► Probability statistics
►Descriptive statistics
► Inferential statistics

99 A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:
► Infinite population
► Finite population
► Concrete population
►Hypothetical population (Page 12)

100 When two dice are rolled, the numbers of possible sample points are:
►6
► 12
► 24

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► 36 (6*6=36)

101 The correlation of coefficient lies between :
► 0 to 1
► 0 to ∞
► - 1 to +1 (Page 128)
► +1 to ∞

102 Median can be found only when:
► Data is Discrete
► Data is Attributed Data is continuous
► Data is continuous Data is attributed
► Data is arranged (Page 52)

103 From the following observations 2,3,4,5,4,6,4, the mode is:
► 2
► 3
► 4
► 5

104 How to construct the class interval:
► Divide the class frequencies in half
► Divide the class frequency by the number of observations
► Find the difference between consecutive lower class limits
► Count the number of observations in the class

105 How many elements are in the sample space of rolling one die:
► 6
► 12
► 24
► 36

106 When two coins are tossed the probability of at most one head is:
► 1/4
► 2/4
► 3/4
► 4/4

107 In scatter diagram the variable plotted along Y-axis is:
► Independent variable
► Dependent variable
► Any one
► Undefined

108 Positive square root of variance of a distribution is:
► Rang
► Quartile deviation
► Standard deviation (Page 91)

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► only (a) &(c)

109 When more values are lying at the start of the distribution, it is a:

► Symmetrical distribution
► Positively skewed
► Negatively skewed
► U shape figure

110 What is ' m f ' in the formula of mode:

► First frequency
► Last frequency
► Maximum frequency (Page 54)
► Minimum frequency

111 If median = 7 and Mean = 5, what is the value of 2 Q :

► 1
► 3
► 5
► 7 (because median = Q2)

112 The probability of drawing a king of spade from a pack of 52 cards is:

► 1/4
► 1/13
► 1/26
► 1/52

113 When referring to a curve whose longer tail is to the left, you would call it:

► U shape
► Skewed to the left (Page 39)
► Skewed to the right
► Symmetrical

114 In statistics,we deal with:

► Individuals
► Isolated items
► Aggregates of facts (Page 7)
► Qualitative data

115 The distribution is mesokurtic if the Moment Coefficient of kurtosis b2 is:

► Equal to 0
► Equal to 3
► Less than 3
► Greater than 3

116 A histogram consists of a set of adjacent rectangles whose bases are marked off by:

► Class boundaries (Page 32)
► Class limits
► Class marks

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► Class frequency

117 Stem and leaf was introduced by:
- ► Francis Galton
- ► Friedman
- ► John Tukey (Page 47)
- ► Pearson

118 For positively skewed distribution Mean…….Median…….Mode:
- ►=
- ►<
- ►>
- ► ≠

119 For a positively skewed distribution:
- ► Mean<Mode<Median
- ► Mdian<Mode>Mean
- ► Mode>Mean>Median
- ► Mean>Median>Mode

120 $^{5}\square_{5}$ 5 C Equals :
- ►1
- ►5
- ► 10
- ► 25

121 If a curve has a longer tail to the right, it is called :
- ►Positively skewed (Page 39)
- ► negatively skewed
- ► J-shaped
- ► Symmetric

122 In measures of relative dispersion unit of measurement is:
- ► Changed
- ►Vanish (Page 82)
- ► Does not changed
- ► Dependent

123 Data used by an agency which is originally collected by them is :
- ►Primary data (Page 11)
- ► Raw data
- ► Secondary data
- ► Grouped data

124 When we toss a coin , we get only:
- ►1 outcome
- ► 2 outcomes
- ► 3 outcomes
- ► 4 outcomes

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125 Which of the following is the class frequency

►The number of observations in each class
► The difference between consecutive lower class limits
► Always contains at least 5 observations
► Usually a multiple of the lower limit of the first class

126 Standard deviation is affected by the change of__:

► Origin & scale
► Origin only
► Scale only Click here for detail
► Not origin & scale

128 A die is rolled. What is the probability that the number rolled is greater than 2 and even:

► 1/2
► 1/3
►2/3 (greater than 2 = 3,4,5,6 = 4 numbers , 4/6=2/3)
► 5/6

129 The probability of drawing any one spade card is:

► 1/52
► 4/52
►13/52
► 52/52

130 Which of the following is not the type of frequency curves?

► The symmetrical frequency curve
► The extremely skewed frequency curve
► The U-shaped frequency curve
►Frequency polygon (Page 38)

131 If the third moment about mean is zero 3 (m3 = 0) , then the distribution is:

►Symmetrical
► Negatively skewed
► Positively skewed
► Mesokurtic

1. Harmonic mean is extremely useful in averaging ---------- types of data.

a) Ratios
b) Rates
c) Both a & b

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d) None

2. Harmonic mean is the reciprocal of

a) Arithmetic mean
b) Mode
c) Geometric mean
d) Median

3. Value of Harmonic mean depends on

a) Extreme values
b) Few observations
c) Both a & b
d) All the observations

4. Harmonic mean gives more weightage to --------- values

a) Smaller values
b) Larger values
c) Both the values
d) None of the above

5. Harmonic mean cannot be calculated for the data having ----------

a) positive
b) negative
c) zero
d) all of the above

6. Harmonic mean is lower value than-------

a) Arithmetic mean

b) Geometric mean

c) **Both a & b**

d) None

7. Harmonic mean is particularly useful in computing ----------

a) Average speed
b) Average price

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c) Average rate
d) All the above

8. Harmonic mean cannot be calculated when there are--------

a) Both positive & negative items in series
b) one or more items are zero
c) both a & b
d) none

9. Relationship among the averages

A) HM ≥GM ≥AM

b) **AM≥GM≥HM**
c) GM ≤HM ≤AM
d) AM ≥HM <GM

10. If Arithmetic mean of a data is 106.6 and n=8, calculate Harmonic mean of given data

a) 0.8502

b) 0.8510

c) 0.8528

d) 0.0850

11. If reciprocal of Arithmetic mean of a data is 256.7 and N=23, calculate Harmonic mean of given data

a) 0.0892

b) 0.0893

c) 0.0894

d) 0.0895

12. Harmonic mean can be further algebraically manipulated

a) true
b) false
c) none
d) not known

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13. The pulse rate of 5 persons are given as 72, 75, 67, 73, 74 Calculate harmonic mean

a) 70.94

b) 71.94

c) 73.94

d) 72.94

14. Harmonic mean is better than other means if the data are for

a) **speed or rates**

b) heights or lenghths

c)Binary values like 0 & 1

d) ratios or proportions

15. If the AM of a set oftwo observations is 9 and its GM is 6. then the HM of the set of observations is

a) **4**

b) 36

c) 12

d) 1.5

16. For a highly variable series the most suitable mean is :

a) AM

b) GM

c**) HM**

d) None of the above

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Which of the following terms best describes data that were originally collected at an earlier time by a different person for a different purpose?
Secondary data

From the table given below, how many students obtained marks between 60 and 69? Marks f Mid-Points 50-59 5 54.5 60-69 7 64.5 70-79 8 74.5 80-89 5 84.5
7

A list of pulse rates is 70. 64. 70. 80. 74, 92. What is the mode for this list?
70

Serious disadvantage of using range as a measure of dispersion is that it is based on only
None

If the mean of two observations is 11.5. then the median of these two observations will be
11.5

A coin is tossed and a single 6-sided 11.4 is rolled. Find the probability of landing on the head side of the coin and rolling a 3 on the die_
1/12

ln measures of relative dispersion, unit of measurement remains:
Different

In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:
21/46

Suitable average for averaging the shoe sizes for children is:
Mode

The classical definition of probability is not applicable when the assumption of does not hold:
Mutually exclusive event

Quartile Deviation is based on
Not all values

A list of 5 pulse rates is: 70, 64, 80, 74. 92. What is the median for this list?
74

Which one of the following will be used to draw an OGIVE?

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Cumulative frequency distribution

What is the value of class mark for the given class 10-12?

11

It is recommended that the number of classes in a frequency distnbution be between:

5 and 20

Half of the observations are always larger than:

When data are collected irl a statistical study for only a portion or subset of all elements of interest we are using:

A sample

What type of curve you will obtain if you draw the death rates of ADULT males?

J shaped

Mean Deviation. Variance and Standard Deviation of the values 4. 4. 4, 4. 4. 4 is

0

Which of the following is not a graph of Quantitative Venable?

Piechart

For empirical rule approximately all observations fall within

X=2a

Let A and B are two dependent events such that P(B)=113. P(A/B)=1/2 and Pi B A).- 1'3 Find P(AnB).

1/6

If "6" is the leading digit in 647 then trailing digit will be:

47

The median of 7. 6. 4. 8. 2. 5. 11 is

6

When we smooth a frequency polygon, It becomes

Frequency curve

For a stem-and-leaf display

All of given option

For platykurtic distribution • b2 (moment ratio) mil be

Less than three

The Probability of a sure event is:

1

Ranking scale also include the properties of which scale,

Nominal scale

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Frequency curve is.

Asymmetric to x axis

If data is collected in rupees like income. then aboslute measure of dispersion will be in the units of

Rupees

Which method is used for obtaining the relative frequencies?

Dividing the frequency by total number of frequency

When two dice are rolled What is the probability that total is at least 12

1/36

In a lottery there are 10 prizes and 25 blanks. A lottery is drawn at random What is the probability of getting a prize?

2/7

Which of the following is not an absolute measure of dispersion----------

Standard Deviation

A case, in which the third parites or witnesses are provided the information to the investigator is called?----------

Indirect investigstion

What is the median for the following data: 1,1,2,2,5,5,7,7,7?--------

7

Relationships among the averages ----------

AM>GM>HM

Half of the observations are always larger than..-------

Median

In case of a frequency distribution, the median is given by the formula --------

MEDIAN= l+h/f[n/2-c]

1. If Q1=62 and Q3=87, then the mid-quartile range will be-------

by the 4mula: (Q3+Q2)/2= 74.5

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Range is a/an ..... measure of dispersion----------.
Absolute.

1. A circle in which sectors represents various quantities is called
a. Histogram
b. Frequency Polygon
c. Pie Chart
d. Bar Chart

2. A graph of a cumulative frequency distribution is called
a. Ogive Curve
b. Frequency Polygon
c. Pie Chart
d. Bar Chart

3. A Histogram contains a set of
a. Adjacent Rectangles
b. Non-adjacent Rectangles
c. Adjacent Squares
d. Adjacent Triangles

4. In a Pie chart one can calculate the angles for each sector by the following formula
a. (Component part / Total) X 100
b. (Component part / Total) X Pi
c. (Total/Component part) X 360
d. (Component part/Total) X 360

5. Component bar charts are used when data is divided into
a. Parts
b. Groups
c. Circles
d. None of the above
Answer: b

6. A frequency polygon is constructed by plotting frequency of the class interval and the
a. The upper limit of the class
b. The lower limit of the class
c. Mid value of the class
d. None of the above

7. A frequency polygon is a closed figure of

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a. Two sides
b. Three sides
c. Many sides
d. None of these

8. De-cumulative frequency is presented by

a. More than Ogive
b. Less than Ogive
c. Equal to Ogive
d. None of these

9. In a histogram, the area of each rectangle is proportional to

a. The class mark of the corresponding class interval
b. The class size of the corresponding class interval
c. Frequency of the corresponding class interval
d. Cumulative Frequency of the corresponding class interval

10. A frequency polygon curve touches the x-axis

a. Yes
b. No
c. Some times
d. None of the above

11. Frequency curve is

a. Asymptotic to y axis
b. Non Asymptotic to y axis
c. Asymptotic to x axis
d. None of these

12. The end of month unit values for a mutual fund are given in rupees:
9.2, 9.5, 10, 9.3, 10.2, 10.5, 10.2, 11.1, 10.5, 10.5, 10.5 and 10.
The values shown are referred to as:

a. frequency distribution.
b. histogram.
c. raw data.
d. mode.

13. The age distribution of citizens enrolled in a scheme is

Ages	Frequency
21-30	500

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31-40	200
41-50	700
51-60	600

What type of graph is needed to illustrate the data as it is organised?

a. Histogram
b. Frequency Polygon
c. Less than Cumulative Frequency diagram
d. None of the above

14. Which of the following considerations for setting up classes in a frequency distribution is correct?

a. Class widths can be different
b. Classes should not overlap
c. Open ended classes only at extremes
d. The lower limit of the first class should not be an even multiple of the class width

15. Frequency distributions can be helpful means of organising and summarising quantitative data. What type of graph is usually used to accompany such data?

a. Bar Chart
b. Histogram
c. Scatter Diagram
d. Time series

16. What are the members in the right column of a frequency distribution table called?

a. Class frequency
b. Interval frequency
c. Ordinal frequency
d. Number frequency

17. A variable is any characteristic which can assume_____ values.

a. Different
b. Similar
c. Fixed
d. Assumed

18. A _____ variable is a variable whose values can theoretically take on an infinite number of values within a given range of values.

a. Continuous

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b. Discrete
c. Random
d. Both a and b

19. The magnitude of the class is the
a. The product of lower limit and upper limit
b. The sum of lower limit and upper limit
c. The difference of upper limit and lower limit
d. None of these

20. The value lying half way between the upper limit and lower limit of the class is:
a. Class interval
b. Mid point
c. Frequency
d. None of the above

21. The classes in which the lower limit or the upper limit is not specified are known as:
a. Open end classes
b. Close end classes
c. Inclusive classes
d. Exclusive classes

22. Classes in which upper limits are excluded from the respective classes and are included in the immediate next class are:
a. Open end classes
b. Close end classes
c. Inclusive classes
d. Exclusive classes

23. The number of observations in a particular class is called:
a. Width of the class
b. Class mark
c. Frequency
d. None of the above

24. If the class mid points in a frequency distribution of age of a group of persons are 25, 32, 39, 46, 53 and 60. The size of class interval is:
a. 5
b. 7
c. 8
d. 6

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25. If the mid points of the classes are 16, 24, 32, 40, and so on, then the magnitude of the class interval is:

a. 8
b. 9
c. 7
d. 6

26. A pie diagram is also called:

a. Pictogram
b. Angular diagram
c. Line diagram
d. Bar diagram

27. The median of a frequency distribution is found graphically with the help of:

a. Histogram
b. Frequency curve
c. Frequency polygon
d. Ogive

28. The mode of a frequency distribution can be determined graphically by:

a. Histogram
b. Frequency curve
c. Frequency polygon
d. Ogive

29. The most commonly used device of presenting business and economic data is:

a. Pie diagrams
b. Pictograms
c. Bar diagrams
d. Line diagrams

30. Type of bar diagram is:

a. Pictogram
b. Sub divided diagram
c. Line diagrams
d. Pie diagram

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Dividing mean deviation by mean we get a pure number known as

Co-efficient of mean deviation*
Co-efficient
Co-efficient of quartile deviation
None of the above

Formula for Semi Interquartile Range is:

(Q3-Q1)/2*
(Q1+Q3)/2
(Q1-Q3)
(Q1+Q3)

Which scale will you use to measure the temperature?

Nominal scale
Interval scale*
Ratio scale
Ordinal scale

If Xo is the smallest value and Xm is the largest value in a data set then, range is defined by the relation:

Xo
Xm
Xm-Xo*
Xo-Xm

Half of the observations are always larger than:

Mean*
Median
Mode
Total

The difference between estimated value and the population value is called

Sampling Error
Non Random Error

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Non Sampling Error
Random Error*

The sum of squared deviations from mean is:

Maximum
Minimum
Zero*
Undefined

For the given data 20, 13, 27, 0, -8 G. M will be:
Select correct option:
Negative*
Positive
Zero
Undefined

What is the median for the following data: 1,1,2,2,5,5,7,7,7

1
2
5*
7
When the deviations (excesses and defects) from the true value tend to occur equally, it is called:

Unbiased error*
Biased error
Systematic error
Mistake

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Pie chart consists of:

CIRCLE

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The dispersion which is calculated from all observed values is:
►**Standard deviation (Page 87)**
►Quartile deviation
►Rang
►Coefficient of Rang

For any number k …….1, at least 1 – 1/k2 of the data-values fall within k standard deviations of the mean:
►**Greater than 1 (Page 97)**
►Less than 1
►Greater or equal to 1
►Less or equal to 1

If Mean = 25 & S.D is 5 then C.V is
►100%
►25%
►**20% (Page 93)rep**
►10%
When E is an impossible event,then P(E) is:
►**0 (Page 146)**
►0.5
►1
►2

When two dice are rolled, the numbers of possible sample points are:
►6
►12
►24
►**36 (6*6=36)**

The correlation of coefficient lies between :
►0 to 1
► 0 *to* ∞
►**- 1 to +1 (Page 128)**
►+1 *to* ∞

How many elements are in the sample space of rolling one die:
►**6**
►12
►24
►36

When two coins are tossed the probability of at most one head is:
►1/4
►2/4
►**3/4**
►4/4

If *A* and *B* are mutually exclusive events then *P*(*A* or *B*) equals:
►*P*(*A*) + *P*(*B*) - *P*(*A* and *B*)
►*P*(*A*) * *P*(*B*)

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►**$P(A) + P(B)$ (Page 155)**
►$P(A|B) + P(B|A)$

Positive square root of variance of a distribution is:
►Rang
►Quartile deviation
►**Standard deviation (Page 91)**

The probability of drawing a king of spade from a pack of 52 cards is:

►1/4
►1/13
►1/26
►**1/52**

$^5C_5$ Equals :
►1
►5
►10
►25

we toss a coin , we get only:
►**1 outcome**
►2 outcomes
►3 outcomes
►4 outcomes

Standard deviation is affected by the change of_____:
►Origin & scale
►Origin only
►**Scale only** Click here for detail
►Not origin & scale

A die is rolled. What is the probability that the number rolled is greater than 2 and even:
►1/2
►1/3
►**2/3 (greater than 2 = 3,4,5,6 = 4 numbers , 4/6=2/3)**
►5/6

The probability of drawing any one spade card is:
►1/52
►4/52
►**13/52**
►52/52

multiplication theorem $P(A \cap B)$ equals:
►P(A) P(B)
►P(A) + P(B)
►**P(A) * P(B|A) (Page 159)**
►P(B\A) * P(B)

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The probability of drawing 'White' ball from a bag containing 4 red , 8 black and 3 white ball is
►0
►3/15
►1/12
►1/2

If *A* and *B* are mutually exclusive events then *P*(*A* or *B*) equals:
►*P*(*A*) + *P*(*B*) - *P*(*A* and *B*)
►*P*(*A*) * *P*(*B*)
►*P*(*A*) + *P*(*B*) **(Page 155)**
►*P*(*A*|*B*) + *P*(*B*|*A*)

The simultaneous occurrence of two events is called
►Prior probability
►Subjective probability not confirmed
►Conditional probability
►Joint probability

Which one of the following measurement does ***not*** divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► Standard deviations

The height of student is 60 inch. This is an example of……?
►Continues data
►Qualitative data
►Categorical data
►Discrete data

Which of the statement is true regarding a sample?
►It is a part of population **(Page 13)**
►It must contain at least five observations
►It refers to descriptive statistics
►It produces true value

If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient '**r**':
► $0 \leq r \leq 1$
► $r \geq 0$
► r = +1
► r=-1

What is the Standard Deviation of 7, 7, 7, 7, 7, 7, 7

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►49
►1
►0 Standard deviation will always be zero if all the values in data are same
►7

10! =…….

► 362880
► 3628800
► 362280
► 362800

If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

►1/52
► 13/52
►4/52
►26/52

In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

►Intercept a ≠ 0
►Intercept a = 0 (correct)
►Slope b = 0
►Slope b ≠ 0

The probability of drawing a 'jack card 'from 52 playing cards is:
►1/52
►4/52
►13/52
►26/52

Which dispersion is used to compare variation of two series?
► C.V. (Page 93)
► Q.D.
► M.D.
► S.D.

The probability of drawing a 'jack card 'from 52 playing cards is:
►1/52
►13/52
►4/52 rep
►26/52

In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

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► **Intercept a = 0 rep**
► Intercept a ≠ 0
► Slope b = 0
► Slope b ≠ 0

Which one of the following measurement does ***not*** divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► **Standard deviations rep**

If all the values fall on the same straight line and the line has a positive slope, then what will be the value of the correlation coefficient '**r**':

► 0 ≤ r ≤ 1
► r ≥ 0
► **r = +1 (Page 129)**
► r=-1

If a=4 b=2 estimate line (i.e y=a+bx) and independent variable has value 3 the value of dependent variable

► 6
► 9
► **10 4+2(3)=10**
► 11

The number of ways in which 4 books can be arranged

► 4
► 6
► 12
► **24 (Page 142)**

The simultaneous occurrence of two events is called

► Descriptive probability
► Subjective probability
► Conditional probability
► **Joint probability**

When coin tossed we get only

►**1 outcome**
►2 outcomes
►3 outcomes
►4 outcomes

When mean is 25 and S.D is 5 then CV is

► 100%
► 25%
► **20%**
► 10%

A set that contains all possible outcomes of a system is known as

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- ► Finite Set
- ► Infinite Set
- ► **Universal Set (Page 134)**
- ► No of these

If X and Y are independent, then Var(X-Y) is equal to:

- ❖ $Var(X)-Var(Y)$
- ❖ $Var(X)+Var(Y)$
- ❖ $Var(X+Y)$
- ❖ Zero

A coin is tossed 3 times then, the number of sample points ii the sample space is:

- 3
- 8
- 6
- 4

What is the difference between a permutation and a combination.

►**In a permutation order is important and in a combination it is not**
►In a permutation order is not important and in a combination it is important
►A combination is based on the classical definition of probability
►A permutation is based on the classical definition of probability

What we consider in simple correlation analysis:
►Several independent variables
►**Strength of the association between two variables (Page 128)**
►Intercept with the X-axis
►Intercept with the *Y*-axis

If Y=bX, then variance of Y is

b*2 var(x)
var(x)
b var(x)
b square root var(x)

If f(x) is a continuous probability function, then P(X = 2) is:

1
0
1/2
2

3. Question # 3 of 10
In regression line Y=a+bX, Y is called:

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Dependent variable
Independent variable
Explanatory variable
Regressor

If A and B are mutually exclusive events with P (A) =0.25 and P (B) = 0.50, Then P (A or B) =……
Select correct option:
0.25
0.75
0.50
1

Symbolically, a conditional probability is:

P(AB)
P(A/B)
P(A)
P(AUB)

In a 52 well shuffled pack of 52 playing cards, the probability of drawing any one diamond card is
Select correct option:
1/52
4/52
13/52
52/52

Probability of a sure event is
Select correct option:
8
1
0
0.5

If Y=3X+5,then S.D of Y is equal to

9 s.d(x)
3 s.d(x)
s.d(x)+5
3s.d(x)+5

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The probability of drawing a red queen card from well-shuffled pack of 52 playing cards is

4/52
2/52
13/52
26/52

If P (B|A) = 0.25 and P (A and B) =0.20, then P (A) is

0.05
0.80
0.95
0.75

When a coin is tossed 3 times, the probability of getting 3 tails is

1/8
3/8
3/6
2/8

In how many ways can a team of 11 players be chosen from a total of 16 players?

4368(not confirmed)
2426
5400
2680

13. Question # 3 of 10

The standard deviation of c (constant) is

Select correct option:

c
c square
0
does not exist

14. Question # 4 of 10

If P (E) is the probability that an event will occur, which of the following must be false:

Select correct option:

P(E)= - 1

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P(E)=1
P(E)=1/2
P(E)=1/3

Question # 5 of 10
Let E and F be events associated with the same experiment. Suppose the E and F are independent and that P(E) = 1/4 and P(F) = 1/2 Then P(E U F) is:
Select correct option:
1/8
3/4
7/8
5/8

16. Question # 6 of 10
A student solved 25 questions from first 50 questions of a book to be solved. The probability that he will solve the remaining all questions is:
Select correct option:
0.25
0.5
1
0

17. Question # 7 of 10
If Y=bX, then variance of Y is
Select correct option:
b*2 var(x)
var(x)
b var(x)
b square root var(x)

The classical definition of probability assumes:
Select correct option:
Exhaustive events
Mutually exclusive events
Equally likely evens
Independent evens

In scatter diagram, the variable plotted along Y-axis is:
Select correct option:
Independent variable
Dependent variable
Continuous variable
Discrete variable

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Which of the following measures of dispersion are based on deviations from the mean? :

Variance

Standard deviation
Mean deviation

All of the these

What does it mean when a data set has a standard deviation equal to zero?

All values of the data appear with the same frequency.
The mean of the data is also zero.
All of the data have the same value.
There are no data to begin with.

A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.
Select correct option:

Probability distribution
The expected return
The standard deviation
Coefficient of variation

Which of the following can never be probability of an event?
Select correct option:

0
1
0.5
-0.5

24. The standard deviation of -1, -1, -1, -1 will be
Select correct option:

1
-1
0
Does not exist

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Which formula represents the probability of the complement of event A:

1 + P (A)
1 - P (A)
P (A)
P (A) -1

The Special Rule of Addition is used to combine:

Independent Events
Mutually Exclusive Events
Events that total more than 1.00
Events based on subjective probabilities

set which is the sub-set of every set is

Empty Set
Power Set
Universal Set
Super Set

E(4X + 5) =_____

12 E (X)
4 E (X) + 5
16 E (X) + 5
16 E (X)

When two dice are rolled the number of possible sample points is :

6
12
24
36

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If two events A and B are not mutually exclusive then

P (A or B) = P (A) + P (B) – P (A and B)

P (A or B) = P (A) + P (B)

P (A or B) = P (A) x P (B)

P (A or B) = P (A) + P (B)

Evaluate (10-4)!

1000
720
480
32

When E is an impossible event, then P(E) is:

0
1
2
0.5

When we toss a coin , we get only:
Select correct option:

1 outcome
2 outcome
3 outcome
4 outcome

For exhaustive events, the P(AUBUC) is equal to:
Select correct option:

P(A)
P(S)
P(A) * P(B)* P(C)
P(B)

A student solved 25 questions from first 50 questions of a book to be solved.
The probability that he will solve the remaining all questions is:
Select correct option:

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0.25
0.5
1
0

A set of possible values that a random variable can assume and their associated probabilities of occurrence are referred to as _____.
Select correct option:

Probability distribution
The expected return
The standard deviation
Coefficient of variation

If we roll a die then probability of getting a '6' will be

2/6

1/6

4/6

1

If P(A) = 0.45, P(B) = 0.35, and P(A and B) = 0.25, then P(A | B) is:

1.4
1.8
0.714
0.556

Which of the following is not a measure of central tendency?

Percentile

Quartile

Standard deviation

Mode

Random experiment can be repeated any no. of times under the…… conditions. :

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Different
Similar

The simultaneous occurrence of two events is called:
Select correct option:

Joint probability
Subjective probability
Prior probability
Conditional probability

In regression analysis, the variable that is being predicted is the

Dependent variable

Independent variable

Intervening variable

None of these

If any value in the data is zero, then it is not possible to have?
Hm
Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even?
3/4.
The variance of a sample of 81 observations equals 64 . The standard deviation of the sample equals
8
Which one of the following is less than mean for a symmetrical distribution:
4th
Variance is expressed in ..... units as the units of data set.
Square
According to Empirical rule, approximately how much values will fall within (Mean-3S.D, Mean+3S.D)?
100%
A tabular arrangement for classifying data into different groups is called:
Frequuency distribution
If we have mean=6,median =5 and standard deviation = 4 then the Pearson's Coefficient of Skewness will be:
0.75
What is the median for the following data:

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1,1,2,2,5,5,7,7,7
5
The classical definition of probability is not applicable when the assumption of ..... does not hold:
Mutually exclusive events

Which of the following method of skewness is a pure number/unitless?
bowleys
Sum of the absolute deviations of the values is least when deviations are taken from:
median
The population about which information is sought is called
Target population

In absolute measures of dispersion our answer is expressed in the ..... units as our original data set.
same
Which measure of central tendency takes into account the magnitude of scores?
median

In a probability distribution, the sum of the probabilities is equal to:
1
f the mean of 6 numbers is 41 than the sum of these numbers is
246
Positive correlation COEFFICIENT "r" will fall within the range:
0<r<1
The amount of hump of a distribution is called:
kurtois
From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?
756
Which of the following Measure of averages is affected by extreme (very small or very large) values in the data set?
Arithmetic mean

In the sequence of Five number summary which will be the last value:
xm

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MIDTERM EXAMINATION (Spring 2011)
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
A five number summery consist of :

► $X_0, Q_1, Median, Q_3, and X_m$ *(Page97)*

► $X_m, Q_1, Mean, Q_3, and\ X_0$

► $X_m, Q_1, Mode, Q_3, and\ X_0$

► $X_0, Q_1, Median, Q_2, and X_m$

Question No: 3 (Marks: 1) - Please choose one
in a multiplication theorem $P(A \cap B)$ equals:

►P(A) P(B)
►P(A) + P(B)
►P(A) * P(B|A) (Page 159)
►P(B\A) * P(B)

Question No: 4 (Marks: 1) - Please choose one
The probability of drawing 'White' ball from a bag containing 4 red , 8 black and 3 white ball is

►0
►3/15
►1/12
►1/2

Question No: 6 (Marks: 1) - Please choose one
If *A* and *B* are mutually exclusive events then *P*(*A* or *B*) equals:

►*P*(*A*) + *P*(*B*) - *P*(*A* and *B*)
►*P*(*A*) * *P*(*B*)
►*P*(*A*) + *P*(*B*) (Page 155)
►*P*(*A*|*B*) + *P*(*B*|*A*)

دنیا میں سب سے مشکل کام اپنی اصلاح اور سب سے آسان کام دوسروں پر نکتہ چینی کرنا ہے

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Question No: 7 (Marks: 1) - Please choose one
The simultaneous occurrence of two events is called
►Prior probability
►Subjective probability not confirmed
►Conditional probability
►**Joint probability** click here for detail

Question No: 8 (Marks: 1) - Please choose one
First moment about origin is always equals to:
►**Mean (Page 119)**
►Variance
►Standard Deviation
►Zero

Question No: 9 (Marks: 1) - Please choose one
Which one of the following measurement does ***not*** divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► **Standard deviations**

Question No: 10 (Marks: 1) - Please choose one
if first and third quartile are 21.16 and 56.36 respectively, then the quartile deviation is:
►**17.1 (Page 84)**
►34.2
►51.3
►50.5

Question No: 11 (Marks: 1) - Please choose one
The height of student is 60 inch. This is an example of……?
►Continues data
►Qualitative data
►Categorical data
►**Discrete data**

Question No: 12 (Marks: 1) - Please choose one
Which of the statement is true regarding a sample?
►**It is a part of population (Page 13)**
►It must contain at least five observations
►It refers to descriptive statistics
►It produces true value

خدا کے سوا کسی سے امید مت رکھو

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Question No: 13 (Marks: 1) - Please choose one
Which one of the following graphs is used for a time series data?

- ► Histogram
- ► Historigram
- ► Frequency curve
- ► **Frequency polygon** Click here for detail

Question No: 14 (Marks: 1) - Please choose one
Which of the following comes first to make frequency distribution

►No. of groups
►Class intervals
►**Rang** **(Page 28)**
►Tally marks

Question No: 15 (Marks: 1) - Please choose one
The average which is defines as reciprocal of arithmetic mean of the reciprocals of the values is called

►Geometric Mean
►**Harmonic mean** **(Page 77)**
►Mode
►Median

Question No: 16 (Marks: 1) - Please choose one
If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

►**Weighted mean** **(Not sure)**
►Harmonic mean
►Arithmetic mean
►Geometric mean



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MIDTERM EXAMINATION (Spring 2011)
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
The probability of drawing a 'jack card 'from 52 playing cards is:

- $\frac{1}{52}$
- $\frac{4}{52}$
- $\frac{13}{52}$
- $\frac{26}{52}$

Question No: 2 (Marks: 1) - Please choose one
If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient '**r**':

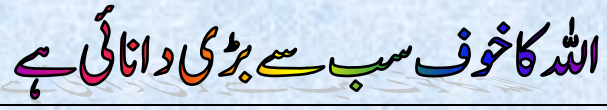
- $0 \leq r \leq 1$
- $r \geq 0$
- **r = +1**
- r=-1

Question No: 3 (Marks: 1) - Please choose one
If a curve has a longer tail to the right, it is called:

- **Positively skewed (Page 39)**
- Negatively skewed
- J-shaped
- Symmetric

Question No: 04 (Marks: 1) - Please choose one
Which one of the following is not included in measures of central tendency?

- **Quartile deviation (Page 82)**
- Harmonic mean
- Geometric mean
- Arithmetic mean



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Question No: 05 (Marks: 1) - Please choose one
Which of the following is not based on all the observations?

► Arithmetic Mean
► Geometric Mean
► Harmonic mean
► **Mode**

Question No: 06 (Marks: 1) - Please choose one
What is the Standard Deviation of 7, 7, 7, 7, 7, 7, 7

►49
►1
►**0** **Standard deviation will always be zero if all the values in data are same**
►7

MIDTERM EXAMINATION (Spring 2010)
STA301- Statistics and Probability (Session - 4)

Question No: 1 (Marks: 1) - Please choose one
10! =…….

► 362880
► **3628800**
► 362280
► 362800

Question No: 2 (Marks: 1) - Please choose one
If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

►1/52
► **13/52**
►4/52
►**26/52**

ایماندار کو غصہ دیر سے آتا ہے اور جلدی دور ہو جاتا ہے

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Question No: 3 (Marks: 1) - Please choose one
The probability of drawing a 'jack card 'from 52 playing cards is:

►1/52
►**4/52**
►13/52
►26/52

Question No: 4 (Marks: 1) - Please choose one
Which dispersion is used to compare variation of two series?

► **C.V. (Page 93)**
► Q.D.
► M.D.
► S.D.

Question No: 5 (Marks: 1) - Please choose one
If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient '**r**':

► $0 \leq r \leq 1$
► $r \geq 0$
► **r = +1 (Page 129)**
► r=-1

Question No: 6 (Marks: 1) - Please choose one
In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

►**Intercept a = 0**
► Intercept $a \neq 0$
►Slope b = 0
►Slope $b \neq 0$

Question No: 7 (Marks: 1) - Please choose one
When two coins are tossed the probability of at least one head is:

► 1/4
► **3/4 (HH, HT, TH, TT)**
► 2/4
► 4/4

زندگی میں کامیابی کا یہی راز ہے کہ پریشانیوں سے پریشان مت بنو

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Question No: 8 (Marks: 1) - Please choose one
Which one of the following measurement does ***not*** divide a set of observations into equal parts?

- ► Quartiles
- ► Deciles
- ► Percentiles
- ► **Standard deviations rep**

Question No: 9 (Marks: 1) - Please choose one
In the model Y = mX + a, Y is also known as the:

- ► Predictor variable
- ► Independent variable
- ► **Predicted (dependent) variable (Page 121)**
- ► Explanatory variable

Question No: 10 (Marks: 1) - Please choose one
According to empirical rule approximately 95% of the measurements will fall under which interval?

- ► $\bar{X} \pm S$
- ► $\bar{X} \pm 2S$ **correct**
- ► $\bar{X} \pm 3S$
- ► $\bar{X} \pm 4S$

Question No: 11 (Marks: 1) - Please choose one
Which one of the following is written at the top of the table?

- ► Source note
- ► Foot note
- ► Prefatory note
- ► **Title** **Click here for detail**

Question No: 12 (Marks: 1) - Please choose one
If a curve has a longer tail to the right, it is called:

- ► **Positively skewed rep**
- ► Negatively skewed
- ► J-shaped
- ► Symmetric



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Question No: 13 (Marks: 1) - Please choose one
Which one of the following is the class frequency?

- **The number of observations in each class**
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

Question No: 14 (Marks: 1) - Please choose one
If X is a discrete random variable, then the function $f(x)$ is

- A probability function
- A probability density function
- A density function
- **A distribution function (Page 173)**

Question No: 15 (Marks: 1) - Please choose one
Which one of the following graphs is used for a time series data?

- **Histogram**
- Historigram
- Frequency curve
- **Frequency polygon**

Question No: 16 (Marks: 1) - Please choose one
If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- Ogive
- Frequency polygon
- **Frequency curve (Page 38)**
- Historigram

Question No: 17 (Marks: 1) - Please choose one
If mean of the two observations is 10.5, then median of these two observations will be:

- 7.5
- 8.5
- 9.5
- **10.5 (Page 68)**



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Question No: 18 (Marks: 1) - Please choose one
Which one is the formula of mid range?

- ► $x_m - x_0$
- ► $x_0 - x_m$
- ► $\frac{x_0 - x_m}{2}$
- ► $\frac{x_0 + x_m}{2}$ **(Page 80)**

Question No: 19 (Marks: 1) - Please choose one
Which one of the following is not included in measures of central tendency?

- ► **Quartile deviation (page 82)**
- ► Harmonic mean
- ► Geometric mean
- ► Arithmetic mean

Question No: 20 (Marks: 1) - Please choose one
For the given data 2, 3, 7, 0, -8 G. M will be:

- ► Negative
- ► Positive
- ► **Zero (Page 75)**
- ► Undefined

MIDTERM EXAMINATION (Spring 2010)
STA301- Statistics and Probability

Question No: 1 (Marks: 1) - Please choose one
If a player well shuffles the pack of 52 playing cards, then the probability of a black card from 52 playing cards is:

- ►1/52
- ►13/52
- ► 4/52
- ► **26/52 rep**

عقل مند کہتا ہے میں کچھ نہیں جانتا جبکہ بے وقوف کہتا ہے کہ میں سب کچھ جانتا ہوں

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Question No: 2 (Marks: 1) - Please choose one
The probability of drawing a 'jack card 'from 52 playing cards is:

►1/52
►13/52
►**4/52 rep**
►26/52

Question No: 3 (Marks: 1) - Please choose one
In a regression line Y= a + bX, the value of the correlation coefficient will be zero if:

► **Intercept a = 0 rep**
► Intercept a ≠ 0
► Slope b = 0
► Slope b ≠ 0

Question No: 4 (Marks: 1) - Please choose one
Which one of the following measurement does ***not*** divide a set of observations into equal parts?

► Quartiles
► Deciles
► Percentiles
► **Standard deviations rep**

Question No: 5 (Marks: 1) - Please choose one
Which one of the following graphs is used for a time series data?

► Histogram
► Historigram
► Frequency curve
► **Frequency polygon rep**

Question No: 6 (Marks: 1) - Please choose one
If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

► Ogive
► Frequency polygon
► **Frequency curve (Page 38) rep**
► Historigram

خود کو تمھیں سے بڑھ کر کوئی اچھا مشورہ نہیں دے سکتا

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Question No:7 (Marks: 1) - Please choose one
Which one is equal to explained variation divided by total variation?

►Sum of square due to regression
► **Coefficient of determinant** Click here for detail
►Standard error estimate
► Coefficient of correlation

Question No: 8 (Marks: 1) - Please choose one
in the given series 1,2,1,1,2,2,2,3,4,5,3,2,3,1,4,2,3 mode of given is
► 4
► 3
►1
► **2**

Question No: 9 (Marks: 1) - Please choose one
True for the population,
► it must be large number of values
► It must refer to people
► **It is collection of individual objects or measurement** **(Page 12)**
► It is the small part of whole

Question No:10 (Marks: 1) - Please choose one
Data arrangement in ascending or descending order
► **Array data** **(Page 47)**
► Group data
► Ungroup data
► Raw data

Question No: 11 (Marks: 1) - Please choose one
What is the main objective of Descriptive statistics?
► To test population properties
► To describe the data we collected
► To infer something about the population
► **Making estimate** **Click here for detail**

Question No: 12 (Marks: 1) - Please choose one
Which measure of central tendency?
► Variation of distribution
► **Average of distribution** **(Page 51)**
► Scattering of distribution
► Dispersion of distribution

جو شخص ناکامیوں سے ڈر کر بھاگتا ہے کامیابی اُس سے ڈر کر بھاگتی ہے

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Question No: 13 (Marks: 1) - Please choose one
If a=4 b=2 estimate line (i.e y=a+bx) and independent variable has value 3 the value of dependent variable
► 6
► 9
► **10** **4+2(3)=10**
► 11

Question No: 14 (Marks: 1) - Please choose one
The number of ways in which 4 books can be arranged
► 4
► 6
► 12
► **24 (Page 142)**

Question No: 15 (Marks: 1) - Please choose one
If we plot paired observed (x,y)=1…….n on graph is called,
► Polygon
► Freasito diagram
► Scatter diagram
► Cumulative frequency diagram

Question No: 16 (Marks: 1) - Please choose one
The simultaneous occurrence of two events is called
► Descriptive probability
► Subjective probability
► Conditional probability
► **Joint probability**

Question No: 17 (Marks: 1) - Please choose one
Which one is the not measure of dispersion.
► The range
► **50th percentile**
► Inter quartile range
► Variance

Question No: 18 (Marks: 1) - Please choose one
In positively skew cure which relation is
► The mean, median and mode are equal
► **Mean is greater then median** **Click here for detail**
► Median is greater then mean
► Standard deviation must be greater then mean or median

جو لوگوں کے سامنے فخر کرتا ہے وہ لوگوں کی نظروں سے گر جاتا ہے

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Question No: 19 (Marks: 1) - Please choose one
When coin tossed we get only
►**1 outcome**
►2 outcomes
►3 outcomes
►4 outcomes

Question No: 20 (Marks: 1) - Please choose one
When mean is 25 and S.D is 5 then CV is

► 100%
► 25%
► **20%**
► 10%

MIDTERM EXAMINATION
Spring 2009
STA301- Statistics and Probability (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
For a positively skewed distribution m_3 will be:
► **Positive** **(Page 119)**
► Negative
► Zero
► 1

Question No: 2 (Marks: 1) - Please choose one
When data is labeled to identify an attribute of element, the measurement scale is:
► **Ordinal** **(Page 9)**
► Interval
► Nominal
► Ratio

عقل مند اپنے عیب خود دیکھتا ہے اور بیوقوفوں کے عیب دنیا دیکھتی ہے

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Question No: 3 (Marks: 1) - Please choose one

Suppose the estimated equation is $\hat{Y} = 5 - 2X$ has been calculated for a set of data. What is slop of the line:

- ► 0
- ► 2
- ► **-2 (Page 121)**
- ► 5

Question No: 4 (Marks: 1) - Please choose one

If P(B|A) = 0.25 and $P(A \cap B) = 0.20$, then P(A) is:

- ► 0.05
- ► **0.80 (Page 159)**
- ► 0.95
- ► 0.75

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

$$= P(A) = \frac{P(A \cap B)}{P(B \mid A)}$$

$$= \frac{0.20}{0.25} = 0.80$$

Question No: 5 (Marks: 1) - Please choose one

Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:

- ► Advance statistics
- ► Probability statistics
- ► **Descriptive statistics (Page 61)**
- ► Inferential statistics

Question No: 6 (Marks: 1) - Please choose one

In a sample of 800 students in a university, 160, or 20%, are Business majors. Based on the above information, the school's paper reported that "20% of all the students at the university are Business majors." This report is an example of :

- ► A sample
- ► A population
- ► Statistical inference
- ► **Descriptive statistics**

بدصورت چہرہ بدصورت دماغ سے بہتر ہے

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Question No: 7 (Marks: 1) - Please choose one
A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set (Page 134)**
- No of these

Question No: 8 (Marks: 1) - Please choose one
If X and Y are independent, then Var(X-Y) is equal to:

- $Var(X)-Var(Y)$
- $Var(X)+Var(Y)$
- $Var(X+Y)$
- Zero

Question No: 9 (Marks: 1) - Please choose one
Which of the following is the class frequency

- **The number of observations in each class**
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

Question No: 10 (Marks: 1) - Please choose one
How to construct the class interval:

- Divide the class frequencies in half
- Divide the class frequency by the number of observations
- **Find the difference between consecutive lower class limits**
- Count the number of observations in the class

Question No: 11 (Marks: 1) - Please choose one
Data in the Population Census Report is:

- **Ungrouped data (Page 11)**
- Secondary data
- Primary data
- Arrayed data

عقل مند آدمی اس وقت تک نہیں بولتا جب تک خاموشی نہیں ہو جاتی

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Question No: 12 (Marks: 1) - Please choose one
What is the range of -2,-3,-5,-10 :

► **-12 (Page 28)**
► 8
► -8
► 2

Question No: 13 (Marks: 1) - Please choose one
The algebraic sum of deviations from mean is:
► Maximum
► Minimum
► **Zero (Page 86)**
► Undefined

Question No: 14 (Marks: 1) - Please choose one
The sum of squares of deviations from mean is:

► Undefined
► Zero
► **Maximum**
► Minimum

Question No: 15 (Marks: 1) - Please choose one
Statistic is a numerical quantity, which is calculated from:

► Population
► **Sample (Page 7)**
► Data
► Observations

Question No: 16 (Marks: 1) - Please choose one
Which of the following is not based on all the observations?
► Arithmetic Mean
► Geometric Mean
► Harmonic mean
► **Mode (rep)**

انسان دکھ نہیں دیتے بلکہ انسانوں سے وابستہ امیدیں دکھ دیتی ہیں

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MIDTERM EXAMINATION
Spring 2009
STA301- Statistics and Probability (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
If any value in the data is zero, then it is not possible to have:
►Harmonic Mean
►Arithmetic Mean
►Median
►**Mode (rep)**

Question No: 2 (Marks: 1) - Please choose one
For a symmetrical distribution……is equidistance from median:
►X0 and Xm
►**Q1 and Q3 (Page 97)**
►X0 and Q1
►Xm and Q3

Question No: 3 (Marks: 1) - Please choose one
Which one of the following measure is not used in 'five number summery':
X_0
Q_3
$\overline{X}$ (*Page* 97)
Q_1

Question No: 4 (Marks: 1) - Please choose one
If Mean = 25 & S.D is 5 then C.V is
►100%
►25%
►**20% (Page 93)**
►10%

Question No: 5 (Marks: 1) - Please choose one
A coin is tossed 3 times then, the number of sample points ii the sample space is:
►3
►8
►**6 (Page 145)**
►4

بہترین تجربہ وہ ہے جس سے نصیحت حاصل ہو

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Question No: 6 (Marks: 1) - Please choose one
What is the difference between a permutation and a combination:
►**In a permutation order is important and in a combination it is not** Click here for detail
►In a permutation order is not important and in a combination it is important
►A combination is based on the classical definition of probability
►A permutation is based on the classical definition of probability

Question No: 7 (Marks: 1) - Please choose one
What we consider in simple correlation analysis:
►Several independent variables
►**Strength of the association between two variables (Page 128)**
►Intercept with the X-axis
►Intercept with the *Y*-axis

Question No: 8 (Marks: 1) - Please choose one
If all the values fall on the same straight line and the line has a positive slope then what will be the value of the correlation coefficient 'r':
►0 = r = 1
►r = 0
►**r = +1 (rep)**
►r=-1

Question No: 9 (Marks: 1) - Please choose one
Which of the following measure of dispersion is least affected by extreme values of observations in a data:
►**Range Page 82)**
►Quartile deviation
►Mean absolute deviation
►Standard deviation

Question No: 10 (Marks: 1) - Please choose one
The approximate number of observations in a set of data covered by the interval, $Median \pm Q.D$ are:
►**50 per cent (Page 84)**
►68.5 per cent
►95.4 per cent
►99 per cent

Question No: 11 (Marks: 1) - Please choose one
The dispersion which is calculated from all observed values is:
►**Standard deviation (Page 87)**
►Quartile deviation
►Rang
►Coefficient of Rang

خوبصورتی علم وادب سے ہوتی ہے لباس وحسن سے نہیں

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Question No: 12 (Marks: 1) - Please choose one
In a set of 20 values all the values are 2, what is the value of Geometric Mean:
►**2** **(Page 75)**
►5
►10
►20

Question No: 13 (Marks: 1) - Please choose one
In a set of 10 values all the values are 5, what is the value of 5th Decile:

►2
►**5**
►10
►20

Question No: 14 (Marks: 1) - Please choose one
If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30, which type of average will be used to find the average speed/hr:

►**W.M**
►H.M
►A.M
►G.M

Question No: 15 (Marks: 1) - Please choose one
In stem and leaf plot, data is measured on:

►Ratio Scale
►Interval Scale
►Ordinal scale
►**Nominal Scale**

Question No: 16 (Marks: 1) - Please choose one
In statistics, we deal with:

►Individuals
►Isolated items
►**Aggregates of facts** **(Page 7)**
►Qualitative data

تم اچھا کرو زمانہ تم کو برا سمجھے یہ اس سے بہتر ہے کہ تم برا کرو اور زمانہ تم کو اچھا سمجھے

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MIDTERM EXAMINATION
Spring 2009
STA301- Statistics and Probability (Session - 1)

Question No: 1 (Marks: 1) - Please choose one
Which of the following is a systematic arrangement of data into rows and columns:
►Component bar chart
►Classification
►**Tabulation**
►Bar chart

Question No: 2 (Marks: 1) - Please choose one
For any number k …….1, at least 1 – 1/k2 of the data-values fall within k standard deviations of the mean:
►**Greater than 1** **(Page 97)**
►Less than 1
►Greater or equal to 1
►Less or equal to 1

Question No: 3 (Marks: 1) - Please choose one
If Mean = 25 & S.D is 5 then C.V is
►100%
►25%
►**20%** **(Page 93)rep**
►10%

Question No: 4 (Marks: 1) - Please choose one When E is an impossible event,then P(E) is:
►**0** **(Page 146)**
►0.5
►1
►2

Question No: 5 (Marks: 1) - Please choose one
The data for an ogive is found in which distribution:
►**A cumulative frequency distribution** **(Page 43)**
►A relative frequency distribution
►A frequency distribution
►A joint frequency distribution

انسان کے لئے بری صحبت سے بڑھ کر بری کوئی چیز نہیں

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Question No: 6 (Marks: 1) - Please choose one
Which of the following statements is true regarding a sample:

►**It is a part of population (Page 13) rep**
►It must contain at least five observations
►It refers to descriptive statistics
►It produces True value

Question No: 7 (Marks: 1) - Please choose one
Which level of measurement is required for the median:

►Nominal
►**Ordinal**
►Interval
►Ratio

Question No: 8 (Marks: 1) - Please choose one
In a the regression line **Y= a + bX+ €** the variable which is non-random is:

►**X Page 122)**
►Y
►Both X and Y
►Neither X nor Y

Question No: 9 (Marks: 1) - Please choose one
The variable plotted on the horizontal or X-axis in a scatter diagram is called the:

►**Scatter variable** Click here for detail
►Independent variable
►Dependent variable
►Correlation variable

Question No: 10 (Marks: 1) - Please choose one
Which is the formula of range:
► $x_m - x_0$ (*Page* 28)
► $x_0 - x_m$
► $\frac{x_m - x_0}{2}$
► $\frac{x_m + x_0}{2}$

خاموشی غصے کا بہترین علاج ہے

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Question No: 11 (Marks: 1) - Please choose one
The descriptive measure of dispersion that is based on the concept of a deviation about the mean is:
►Range
►Inter quartile range
►Quartile deviation
►Standard deviation

Question No: 12 (Marks: 1) - Please choose one
Which branch of statistics deals with the techniques that are used to organize, summarize, and present the data:
►Advance statistics
►Probability statistics
►Descriptive statistics (rep)
►Inferential statistics

Question No: 13 (Marks: 1) - Please choose one
A population that can be defined as the aggregate of all the conceivable ways in which a specified event can happen is known as:
►Infinite population
►Finite population
►Concrete population
►Hypothetical population (Page 12)

Question No: 14 (Marks: 1) - Please choose one
First moment about origin is always equals to:
►Mean Click here for detail
►Variance
►Standard Deviation
►Zero

Question No: 15 (Marks: 1) - Please choose one
When two dice are rolled, the numbers of possible sample points are:
►6
►12
►24
►36 (6*6=36)

Question No: 16 (Marks: 1) - Please choose one
The correlation of coefficient lies between :
►0 to 1
► 0 *to* ∞
►- 1 to +1 (Page 128)
► +1 *to* ∞

جھوٹ رزق کو کھا جاتا ہے

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MIDTERM EXAMINATION
Spring 2009
STA301- Statistics and Probability (Session - 2)

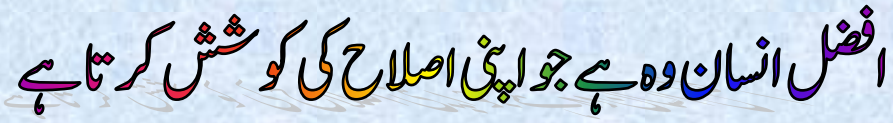
Question No: 1 (Marks: 1) - Please choose one
Median can be found only when:
►Data is Discrete
►Data is Attributed Data is continuous
►Data is continuous Data is attributed
►Data is arranged (Page 52)

Question No: 2 (Marks: 1) - Please choose one
From the following observations 2,3,4,5,4,6,4, the mode is:
►2
►3
►4
►5

Question No: 3 (Marks: 1) - Please choose one
How to construct the class interval:
►Divide the class frequencies in half
►Divide the class frequency by the number of observations
►Find the difference between consecutive lower class limits
►Count the number of observations in the class

Question No: 4 (Marks: 1) - Please choose one
How many elements are in the sample space of rolling one die:
►6
►12
►24
►36

Question No: 5 (Marks: 1) - Please choose one
When two coins are tossed the probability of at most one head is:
►1/4
►2/4
►3/4
►4/4



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Question No: 6 (Marks: 1) - Please choose one
If *A* and *B* are mutually exclusive events then *P*(*A* or *B*) equals:
►*P*(*A*) + *P*(*B*) - *P*(*A* and *B*)
►*P*(*A*) * *P*(*B*)
►***P*(*A*) + *P*(*B*)** **(Page 155)**
►*P*(*A*|*B*) + *P*(*B*|*A*)

Question No: 7 (Marks: 1) - Please choose one
In scatter diagram the variable plotted along Y-axis is:
►Independent variable
►**Dependent variable**
►Any one
►Undefined

Question No: 8 (Marks: 1) - Please choose one
Positive square root of variance of a distribution is:
►Rang
►Quartile deviation
►**Standard deviation** **(Page 91)**
►only (a) &(c)

Question No: 9 (Marks: 1) - Please choose one
When more values are lying at the start of the distribution, it is a:
►Symmetrical distribution
►Positively skewed
►**Negatively skewed**
►U shape figure

Question No: 10 (Marks: 1) - Please choose one
What is ' f_m ' in the formula of mode:
►First frequency
►Last frequency
►**Maximum frequency** **(Page 54)**
►Minimum frequency

Question No: 11 (Marks: 1) - Please choose one
If median = 7 and Mean = 5, what is the value of Q_2 :
►1
►3
►5
►**7** **(because median = Q2)**

اطمینان قلب چاہتے ہو تو حسد سے دور رہو

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Question No: 12 (Marks: 1) - Please choose one
The probability of drawing a king of spade from a pack of 52 cards is:

►1/4
►1/13
►1/26
►1/52

Question No: 13 (Marks: 1) - Please choose one
When referring to a curve whose longer tail is to the left, you would call it:

►U shape
►Skewed to the left (Page 39)
►Skewed to the right
►Symmetrical

Question No: 14 (Marks: 1) - Please choose one
In statistics,we deal with:

►Individuals
►Isolated items
►Aggregates of facts (Page 7) rep
►Qualitative data

Question No: 15 (Marks: 1) - Please choose one
When data is labeled to identify an attribute of element, the measurement scale is:

►Ordinal (Page 9)
►Interval
►Nominal
►Ratio

Question No: 16 (Marks: 1) - Please choose one
The distribution is mesokurtic if the Moment Coefficient of kurtosis b_2 is:

►Equal to 0
►Equal to 3
►Less than 3
►Greater than 3



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MIDTERM EXAMINATION
Spring 2009
STA301- Statistics and Probability (Session - 2)

Question No: 1 (Marks: 1) - Please choose one
A histogram consists of a set of adjacent rectangles whose bases are marked off by:
►**Class boundaries (Page 32)**
►Class limits
►Class marks
►Class frequency

Question No: 2 (Marks: 1) - Please choose one
Stem and leaf was introduced by:
►Francis Galton
►Friedman
►**John Tukey (Page 47)**
►Pearson

Question No: 3 (Marks: 1) - Please choose one
For positively skewed distribution
Mean…….Median…….Mode:
►=
►<
►> **Click here for detail**
►≠

Question No: 4 (Marks: 1) - Please choose one
For a positively skewed distribution:
►Mean<Mode<Median
►Mdian<Mode>Mean
►Mode>Mean>Median
►**Mean>Median>Mode** Click here for detail

Question No: 5 (Marks: 1) - Please choose one
${}^{5}C_{5}$ Equals :
►1
►5
►10
►25

ہر چیز کی ایک پہچان ہوتی ہے اور عقلمند کی پہچان غور و فکر کرنا ہے اور غور و فکر کی پہچان خاموشی ہے

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Question No: 6 (Marks: 1) - Please choose one
If a curve has a longer tail to the right, it is called :
►**Positively skewed (Page 39)**
►negatively skewed
►J-shaped
►Symmetric

Question No: 7 (Marks: 1) - Please choose one
In measures of relative dispersion unit of measurement is:
►Changed
►**Vanish (Page 82)**
►Does not changed
►Dependent

Question No: 8 (Marks: 1) - Please choose one
Data used by an agency which is originally collected by them is :
►**Primary data (Page 11)**
►Raw data
►Secondary data
►Grouped data

Question No: 9 (Marks: 1) - Please choose one
When we toss a coin , we get only:
►**1 outcome**
►2 outcomes
►3 outcomes
►4 outcomes

Question No: 10 (Marks: 1) - Please choose one
Which of the following is the class frequency
►**The number of observations in each class**
►The difference between consecutive lower class limits
►Always contains at least 5 observations
►Usually a multiple of the lower limit of the first class

Question No: 11 (Marks: 1) - Please choose one
Standard deviation is affected by the change of_____:
►Origin & scale
►Origin only
►**Scale only** Click here for detail
►Not origin & scale

اپنی مرضی اور اللہ کی مرضی میں فرق کا نام غم ہے

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Question No: 12 (Marks: 1) - Please choose one
For a moderately skewed distributions, the approximate percentage of case included between $\bar{X} - 2S$ *and* $\bar{X} + 2S$ is :
►99.7%
►68%
►95% (Page 95)
►50%

Question No: 13 (Marks: 1) - Please choose one
A die is rolled. What is the probability that the number rolled is greater than 2 and even:
►1/2
►1/3
►2/3 (greater than 2 = 3,4,5,6 = 4 numbers , 4/6=2/3)
►5/6

Question No: 14 (Marks: 1) - Please choose one
The probability of drawing any one spade card is:
►1/52
►4/52
►13/52
►52/52

Question No: 15 (Marks: 1) - Please choose one
Which of the following is not the type of frequency curves?
►The symmetrical frequency curve
►The extremely skewed frequency curve
►The U-shaped frequency curve
►Frequency polygon (Page 38)

Question No: 16 (Marks: 1) - Please choose one
If the third moment about mean is zero $(m_3 = 0)$, then the distribution is:
►Symmetrical
►Negatively skewed
►Positively skewed
►Mesokurtic

وہ لوگ مبارک ہیں جو الفاظ سے نصیحت نہیں کرتے بلکہ عمل سے کرتے ہیں

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MidTerm Papers Solved MCQS with Reference (1 to 22 lectures)

1. The probability of drawing a "white" ball from a bag containing 4 red, 8 black and 3 white balls is:

- o 0
- o $\frac{3}{15}$
- o $\frac{1}{12}$
- o $\frac{1}{2}$

The total probability of choosing white ball is 4 + 8 + 3=15, 3/15

2. If the probability of one event does not affected by the occurrence of another event then both events are:

- o Dependent
- o Not Mutually Exclusive
- o Mutually Exclusive
- o **Independent** **PG # 162 (Lec # 22)**

3. A die is rolled. What is the probability of getting a number that is an even number and greater than 2.

- 1/2
- 1/3
- **2/3**
- 5/6

2, 4, 5, 6: even or greater than 2. That means 2 out of 6 is 2/3

4. The collection of all the elements under study is called:

- **Population** **PG # 12 (Lec # 2)**
- Sample
- Data
- Registration

5. The mode value from raw data can be obtained by the help of

- **Dot plot** **PG # 53 (Lec # 6)**
- Stem and leaf plot
- Bar chart
- None of these

6. A student solved 25 questions from first 50 questions of a book. The probability that he will solve the remaining all questions is:

- 0.25
- **0.5**
- 1
- 0

7. The difference between the largest and the smallest data values is called the

- Variance
- Interquartile range
- **Range** **PG # 83 (Lec # 10)**
- Coefficient of variation

8. Which one is the formula of mid-quartile range:

- $Q_3 - Q_1$
- $\frac{Q_3 - Q_1}{2}$
- $\frac{Q_1 - Q_3}{2}$
- $\frac{Q_1 + Q_3}{2}$ **PG # 80 (Lec # 9)**

9. For the given data 2, 3, 7, 0, -8; G. M will be:

- Negative
- Positive
- Zero
- **Undefined**

Geometric mean cannot be calculated for negative values

10. If X and Y are independent variables then Var (X-Y) = _____

- **Var(X) - Var(Y)**
- Var(X) + Var(Y)
- Var(X+Y)
- Var(X) × Var(Y)

11. Rang measures the dispersion around the _____

- Extreme values
- Minimum value
- Maximum value
- **Mid-range value** **PG # 86 (Lec # 10)**

12. The variable plotted on the horizontal or X-axis in a scatter diagram is called the:

- Scatter variable
- **Independent variable** **PG # 121**
- Dependent variable
- Correlation variable

13. In a scatter diagram we plot:

- **X and Y** **PG # 121 (Lec #15)**
- X and $\hat{Y}$
- Y and $\hat{Y}$
- $\hat{Y}$ and X

14. If $P(A \cap B)=P(A|B)\times P(B)$ then A and B will be:

- **Independent**
- Mutually Exclusive
- Dependent
- Equally Likely

15. Which one of the following is the class frequency?

- **The number of observations in each class**
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

16. How to convert a frequency distribution to a relative frequency distribution?

- Find the difference between consecutive lower class limits
- Find the difference between consecutive lower limits
- Count the number of observations in the class
- **Divide the class frequency by the total number of observations PG # 31 (Lec # 4)**

17. In statistics, we deal with:

- Individuals
- Isolated items
- **Aggregates of facts** **PG # 7**
- Qualitative data

18. Pie chart consists of

- A rectangular
- A triangle
- A square
- **A circle** **PG # 23 (Lec # 3)**

19. When $Q_1 = 2$ *and* $Q_3 = 4$, what is the value of Median, if the distribution is symmetrical:

- o 1
- o 2
- o **3** **PG # 112 (Lec # 14)**
- o 4

20. A letter is chosen at random from the word STATISTICS. The probability of getting a vowel is:

- o **3/10**
- o 4/10
- o 5/10
- o 6/10

21. When two coins are tossed simultaneously, P (one head) is:

- o $\frac{1}{4}$
- o $\frac{1}{2}$
- o $\frac{3}{4}$
- o 1

Total Sample HH,TH,HT,TT Probability One Head TH,HT So 2/4=1/2

22. A value which has half of the observations above it and half of the observations below it; is known as……..

- o Rang
- o **Median** **PG # 64 (Lec # 7)**
- o Mean
- o Mode

23. In a set of 20 values all the values are 2, what will be the Geometric Mean?

- **2**
- 5
- 10
- 20

24. Which of the following measure of dispersion is least affected by the extreme values in a data?

- **Range** **PG # 83 (Lec # 10)**
- Quartile deviation
- Mean absolute deviation
- Standard deviation

25. Which is appropriate average for finding the average speed of a car:

- Mean
- Geometric mean
- **Harmonic mean** **PG # 78 and 79 (Lec # 9)**
- Weighted mean

26. What will be the value of the total amount of dispersion obtained by summing the $(X-\bar{X})$?

- **0** **PG # 88 (Lec # 11)**
- -1
- 1
- 100

27. The deviation of a distribution from symmetry is called:

- Kurtosis
- **Skewness** **PG # 86 (Lec # 10)**
- Dispersion
- Flatness

28. For the independent events A and B if P(A) = 0.25, P (B) =0.40 then P $(A \cap B)$= ……..

- 0.65
- 0.1
- 0.50
- 0.15

$P(A \cap B) = P(A) \times P(B) = 0.25 \times 0.40 = 0.1$

29. The variable plotted on the vertical or Y-axis in a scatter diagram is called the:

- Scatter variable
- Independent variable
- **Dependent variable** **PG # 121**
- Correlation variable

30. In a multiplication theorem P $(A \cap B)$ equals:

- P (A) P (B)
- P (A) + P (B)
- **P (A) ×P (B|A)** **PG # 162 (Lec # 21)**
- P(B|A) ×P(B)

31. For two independent events A and B, if P(A) = 0.2 and P(B) = 0.4 , then what is P $(A \cap B)$?

- 0.06
- **0.08**
- 0.02
- 0.04

$P(A \cap B) = P(A) \times P(B) = 0.2 \times 0.4 = 0.08$

32. The level of satisfaction for a certain facility which Govt. is providing is the example of:

- Constant
- **Qualitative variable**
- Variable
- Quantitative variable

Quantitative variable when a characteristic can be expressed numerically such as age, weight, income or number of children. On the other hand, if the characteristic is non-numerical such as education, sex, eye color, quality, intelligence, poverty, satisfaction, etc. the variable is referred to as a **qualitative variable**.

33. Quantitative variable is further divided into:

- Continuous variable
- Discrete variable
- **Continuous & Discrete variable** **PG # 9 (Lec # 1)**
- None of the above

34. If you connect the mid-points of rectangles in a histogram by a series of lines that also touches the x-axis from both ends, what will you get?

- Ogive
- Frequency polygon
- **Frequency curve** **PG # 32 (Lec # 4)**
- Historigram

35. Which type of data is collected in population census?

- Grouped data
- Secondary data
- **Primary data**
- Array data

36. Data obtained from the Bureau of Statistics, is an example of:

- None of these
- Primary data
- **Secondary data** **PG # 12 (Lec # 2)**
- Qualitative data

37. A set that contains all possible outcomes of a system is known as

- Finite Set
- Infinite Set
- **Universal Set** **PG # 134**
- None of these

38. In positively skewed distribution, which of the following relationship exists:

- The mean, median, and mode are all equal
- **The mean is larger than the median** **Click Here For More Detail**
- The median is larger than the mean
- The standard deviation must be larger than the mean or the median

39. For two mutually exclusive events A and B, P (A) = 0.2 and P (R) = 0.4 , then P $(A \cup B)$ is:

- o 0.8
- o 0.2
- o **0.6** **PG # 154**
- o 0.5

P (A ∪ B) = P(A) + P(B) = 0.2+0.4=0.6

40. When two coins are tossed the probability of at most one head is:

- o 1/4
- o 2/4
- o **3/4**
- o 4/4

41. In a set of 20 values all the values are 10, what will be median?

- o 2
- o 5
- o **10**
- o 20

42. Which one of the following measure is not based on all the observations?

- o Arithmetic Mean
- o Geometric Mean
- o Harmonic Mean
- o **Mode**

43. A five-number summary consists of :

- o X_0, Q_1 Median, Q_3 and X_m **PG # 97 (Lec # 12)**
- o X_m, Q_1, Mean, O_3, and X_0
- o X_m, Q_1 Mode, Q_3, and X_0
- o X_0, Q_1, Median, Q_2, and X_m

44. For what condition of k, at least $1 - 1/k^2$ of the data-values fall within k standard deviations of the mean:

- o **Greater than 1** **PG # 94 (Lec # 12)**
- o Less than 1
- o Greater or equal to 1
- o Less or equal to 1

45. Which of the following can never be taken as the probability of an event?

- o 1
- o 0
- o 0.5
- o **-0.5**

Probability always lie between 0 and 1

46. If E is an impossible event, then P(E) is:

- o 1
- o 0.15
- o **0**
- o 0.5

47. The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

- Geometric Mean
- **Harmonic Mean** **PG # 77 (Lec # 9)**
- Mode
- Median

48. An automobile is running, during the first 60 Km, at the rate of 10 Km/hr. During the second 60 Km at the rate of 30Km/hr, while during the third 60 Km its speed was 40 Km/hr. What method is more appropriate to calculate the average speed?

- Median
- Arithemetic mean
- **Harmonic mean** **PG # 78 (Lec # 9)**
- Geometric mean

49. What is the standard deviation of the following values? 7, 7, 7, 7,7, 7, 7

- 7
- 1
- 49
- **0** **Click Here For Reference**

50. Which one of the following is a meso-kurtic curve?

- Negatively skewed
- Positively skewed
- J-shaped
- **Normal** **PG # 114 (Lec # 14)**

The NORMAL curve is a curve which is neither very peaked nor very flat, and hence it is taken as A BASIS FOR COMPARISON. The normal curve itself is called MESOKURTIC.

51. Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly 5 years younger than themselves. Which of the following is correct?

- The correlation coefficient is -5
- The correlation coefficient is 5
- **The correlation coefficient is 1**
- The correlation coefficient is 0

The relationship is perfect, with no error

52. Which of the following comes first to make frequency distribution.

- Number of Groups
- Class interval
- **Range** **PG # 28**
- Tally marks

53. Data obtained from the Ministry of food, is an example of:

- Primary data
- **Secondary data** **PG # 12 (Lec # 2)**
- Qualitative data
- None of these

54. Total angle of a Pie Chart is:

- 90"
- 180°
- 270°
- **360°** **PG # 23 (Lec # 3)**

55. Which one is the formula of range:

- o $x_m - x_0$ **PG # 28 (Lec # 4)**
- o $x_0 - x_m$
- o $\frac{x_0 - x_m}{2}$
- o $\frac{x_0 + x_m}{2}$

56. If a player well shuffles the pack of 52 playing card, then the probability of a black card from 52 playing cards is:

- o 1 /52
- o 13/52
- o 4/52
- o **26/52**

57. Pearsons coefficient of skewness is not possible if standard deviation is:

- o 1
- o 1.5
- o **0** **PG # 109 (Lec # 13)**
- o 3

58. What is the difference between permutation and combination?

- o **In permutation order is important and in combination order is not important**

  Click Here For Reference
- o In permutation order is not important and in combination order is important
- o Only combination is based on the classical definition of probability
- o Only permutation is based on the classical definition of probability

59. If data is categorized about any characteristic, then which measurement scale should be used?

- Ordinal
- Interval
- **Nominal** **PG # 9 (Lec # 1)**
- Ratio

60. Which of the following scale has true zero point?

- **Ratio Scale** **PG # 9 (Lec # 1)**
- Interval scale
- Nominal scale
- Ordinal scale

61. In a simple bar chart:

- Width of bar is meaningful
- Height of bar is meaningful
- Height of bar is not meaningful
- **Height and width both are meaningful** **PG # 24**

A simple bar chart consists of horizontal or vertical bars of equal width and lengths proportional to values they represent

62. What is the name of following curve?



- **Positively skewed curve** **PG # 39**
- Symmetrical curve
- Normal curve
- Negatively skewed curve

63. If A = {1, 2, 3, 4, 5, 10} and B = {1, 3, 5} then $B \subset A$ means:

- A is less than B
- A is contained in B
- **B is contained in A.** **PG # 134 (Lec # 16)**
- B is less than A

64. The graph of cumulative frequency distribution is called:

- Frequency polygon
- **Ogive** **PG # 43 (Lec # 5)**
- Frequency curve
- Historigrarn

65. Given the series 1,2,1,1,2,2,2,2,3,4,5,3,2,3,1,4,2,3. Which one of the following is mode of the given series:

- 4
- 3
  3
- 2
  3
- 2
- 1
  3
  2^{*}

66. Which one of the following is equal to the 2nd quartile:

- P_{33}
- D_3
- **Median** **PG # 69 (Lec # 8)**
- Mode

67. The variance of a sample of 81 observations equal to 64. The standard deviation of the sample is equal to:

- o 256
- o 4096
- o **8**
- o 6,561

68. Under which condition standard deviation could assume a negative value:

- o When more than half of the data values are negative
- o If all the data values are same
- o **The standard deviation cannot be negative**
- o When all the data values are negative

Standard deviation cannot be negative because it is square rooted variance.

69. In a set of 10 values all the values are 5, what will be the D_5 ?

- o 2
- o **5** **Click Here For Reference**
- o 5
- o 20

70. When three, coins are tossed simultaneously, P (3 Heads) is:

- o 3/8
- o **1/8**
- o 4/8
- o 2/8

71. The distribution is mesokurtic if the moment ratio i. e. b_2 is:

- Equal to 0
- **Equal to 3** **PG # 119 (Lec # 14)**
- Less than 3
- Greater than 3

72. A set is any well-defined collection of

- Positive Objects
- Negative Objects
- Same Objects
- **Distinct Objects** **PG # 133 (Lec # 16)**

73. If an observation in the data set is negative, then the geometric mean is:

- Positive
- negative
- Zero
- **undetermined**

74. Mean deviation is least when it is calculated from_____.

- Mean
- Mode
- **Median**
- Geometric mean

Mean deviation is minimum if the deviations are taken from the median.

75. For a positively skewed distribution m_3 will be:

- o **Positive** **PG # 119 (Lec # 14)**
- o Negative
- o Zero
- o 1

76. Which of the following value could not represent a coefficient of correlation?

- o r = 0.93
- o **r=1.09** **PG # 128 (Lec # 15)**
- o r= -0.73
- o r=-1

r is a pure number that lies between – 1 and 1 i.e. $-1 < r < 1$

77. The conditional probability is P(A|B) is given by formula:

- o $\frac{P(A \cup B)}{P(A)}$
- o $\frac{P(A \cap B)}{P(B)}$ **PG # 159 (Lec # 20)**
- o $\frac{P(A \cap B)}{P(A)}$
- o $\frac{P(A \cap B)}{P(B)}$

78. If a curve has a longer tail to the right, it is called :

- o **Positively skewed** **PG # 39 (Lec # 5)**
- o Negatively skewed
- o J-shaped
- o Symmetric

79. Which of the following scale has not true zero point?

- Nominal scale
- Ordinal scale
- **Interval scale** **PG # 9 (Lec # 1)**
- Ratio scale

80. Mean is affected by the change of:

- Origin Only
- Scale Only
- **Origin and Scale**
- Not affected

Arithmetic Mean is affected due to the change of origin and scale.

81. What will be the mode for the data 2,3,4,5,4,6,4?

- 2
- 3
- **4**
- 5

The mode is the value which appears the most often in the data. It is possible to have more than one mode if there is more than one value which appears the most.

82. In a set of 20 values all the values are 5, what will be the mean?

- 4
- **5**
- 10
- 20

83. It is easy to find the median when data is……

- **Arranged**
- Discrete
- Attributed
- Continuous

The **median** of a set of data values is the middle value of the data set when it has been arranged in ascending order. That is, from the smallest value to the highest value.

Note: Give me a feedback and your Suggestion also If you find any mistake in mcqz plz inform me Viva Contact us Page on our Site. And tell me your answer with references.

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